DEMYSTIFIED PHYSICAL GEOGRAPHY

PAPER ONE P250/1

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INTRODUCTION TO PHYSICAL GEOGRAPHY P250/1

The term geography is derived from the two Greek words i.e. **"Geo"** meaning **Earth** and **"Graphia"** meaning **description.** From the above words, geography means the description of the Earth. However, geography can simply be defined as a subject that deals with the study of man and what surrounds him and how best he can live in harmony with his surroundings.

- Geography is one of the subjects taught at "A" level as a principal and being offered at principle level means it can be considered as an essential or relevant subject in particular fields of study that students wish to offer at university (degree) or tertiary institutions (diploma) level. For example, in a combination with other subjects, it is an essential subject in the fields of study such as meteorology, conservation forestry, environmental sciences, agriculture and rural innovation, horticulture, agri-business, business administration, etc.

BRANCHES OF GEOGRAPHY

There are two branches of geography i.e.

Human geography: this is a branch of geography which is concerned with the activities of man on earth's surface. It deals with how man utilizes the physical geography. Topics studied include; agriculture, settlement, urbanization, tourism, transport, mining, etc.

Physical geography: this is a branch of geography which deals with the study of the physical environment i.e. the study of evolution and effects of physical environment, such as mountains, valleys, climate, vegetation, soils, etc, the syllabus focuses on the following components of physical geography i.e. **geomorphology, climatology, pedology (soil geography) and vegetation**.

- **Geomorphology:** this is a branch of physical geography that deals with the study of the distribution of landforms, their origin, processes/forces causing changes over these landforms and how useful and problematic are the landforms produced to man and the environment. Geology provides basic information to the study of geomorphology. These landforms are either intrusive if they are formed from under the earth or extrusive if they form on the earth surface.
- **Geomorphic processes:** these are the processes which lead to the evolution of landforms on the earth. They fall into two categories i.e. Endogenic or Exogenic.
- **Endogenic processes:** are those that originate/generated from within the interior of the Earth/below the earth surface and are building processes that lead to the increase in elevation or relief. They include faulting, warping, folding and vulcanicity. Endogenic processes are further divided into two; i.e**. Epeirogenic** forces that operate vertically upwards or downwards in the earth crust like vulcanicity and warping while **diastrophic forces** operate horizontally like folding and faulting.
- Faulting is the breaking or fracturing of rocks involving the displacement of rocks on either side of the fracture or fault. Faulting results in the evolution of land forms such as block mountains, rift valleys, escarpments, tilt blocks, grabens and fault guided valleys.
- Folding is the slow but steady bending of rocks the earth's crust. It is a very gradual process whose effects are realized only after a long period of time usually hundreds of thousands of years. Folding is responsible for formation of the fold mountains of the world such as the Himalaya, Alps, Rockies, Andes and the cape Range.
- Warping is the deformation of the earth's surface by uneven upward or downward

movement. Like folding, it is an infinitely slow process where the movement is generally slight but likely to extend over a wide area. The Lake Victoria and Kyoga basins are examples of down warped basins in Uganda. (**Note**: The three processes of faulting, folding and warping collectively constitute **earth movements**)

- Vulcanicity or vulcanism is the movement of molten and igneous material from the earth's interior towards or onto the earth's surface. When material reaches the surface of the earth. It solidifies to give rise to distinctive features known as volcanic land forms. (**Note:** Earth move together with vulcanicity constitute **tectonic processes**. The term tectonic process of faulting, folding, warping and vulcanicity)
- **Exogenic processes** are those activities that operate on the earth surface and are divided into two i.e. denudational processes and depositional processes. These include; Erosion, mass wasting, weathering, wave action, glaciation and eustatism.
- Denudation is destruction or removal (wearing down) of the landscape by the work of weathering, mass wasting, erosion, wave action, glaciation and eustatism. Denudation leads to a reduction in elevation and relief.
- Weathering is the physical disintegration and chemical decomposition of rocks at the surface of the earth in situ. Weathering can be divided into those processes involving chemical reaction and formation of new minerals (**chemical weathering)**
- Erosion is the removal and wearing away of the land by the natural agents of rivers, waves, wind, glaciers, ice sheets and tidal currents. The same agents act as means of transport for the removal of the eroded material.
- Mass wasting is the down-slope movement of slope material under the influence of the gravitational force of the material itself and without the assistance of moving water, ice, air, or any other agent. The term mass movement if often regarded as synonymous with mass wasting although the latter is also used in broader sense to encompass all processes involved in the lowering of landscape.
- Deposition is the complement of denudation, while denudation destroys, deposition builds up landscape. It is the laying down of the transported material by rivers, waves, tidal currents, wind, glaciers and ice sheets.

- **Climatology** is another branch of physical geography that deals with the study of global and regional weather and climatic conditions by analyzing relevant data. Metrology provides basic information on the composition, structure and the changes in the atmosphere.
- **Pedology (Soil geography)** is another component of physical geography that deals with the study related to soil formation, soil profile, soil types, their fertility level and distribution. Soil erosion and conservation measures are also dealt with in this branch.
- **Vegetation**

PHYSICAL GEOGRAPHY SYLLABUS/COURSE OUTLINE

1. **Introduction to physical geography**
2. **The structure of the earth**
3. **The origin of continents**
4. **Tectonism**
5. **Rocks**
6. **Weathering and slope development**
7. **Mass wasting**
8. **Glaciation**
9. **lakes**

10. **Rivers and river systems**
11. **Coastal geomorphology**
12. **Climatology**
13. **Soils**
14. **Vegetation**
15. **Map Work**
16. **Photographic Interpretation**

MODE OF ASSESSMENT/ EXAMINATION STANDARD FOR PHYSICAL GEOGRAPHY PAPER ONE (P250/1)

This paper deals with evolution and effects of physical features and may be drawn from the whole world but more emphasized in East Africa.

- This paper will consist of three sections **A, B** and **C**. You will be required to answer four questions in all. Section **A** is **compulsory** and you will have to answer **two compulsory** questions from this Section **one** from Map work and **another** on photographic interpretation. You will also answer at least **one** question from **Section B** and **one** from **Section C.**

SECTION A

- Map work: this emphasizes aspects of physical geography and carries (25 marks)
- Photographic interpretation: this deals with the study and interpretation of a photograph of an area drawn in East Africa but still emphasizes aspects of physical geography. (25 marks)

SECTION B

- Three questions will be set and one question answered (25 marks) the section deals with geomorphology and covers topics such as; the structure of the earth, the origin of continents and ocean basins, continental drift, tectonism, vulcanicity, rocks, weathering, slope development, mass wasting, glaciation, rivers and drainage system, lakes, coastal geomorphology, etc.

SECTION C

- Three questions will be set and one question answered (25 marks). Three topics are covered in this Section i.e. climatology, vegetation and soil.

THE STRUCTURE OF THE EARTH

This topic covers three areas i.e. the origin of the earth, the structure of the earth and the shape of the earth.

THE ORIGIN OF THE EARTH

According to the latest scientific estimates, the age of the universe is put at 15billion years. This is a recent estimate but in the past people considered the origin of the earth and the stars to a catastrophic event because the level of science and technology limited the ability to think beyond that level. Currently, however, man has discovered the evidence that tries to explain the origin of the earth.

- The origin of the earth can be explained basing on two major groups of theories. Under the first group, it is suggested that the universe and the earth **formed gradually** while the second group suggests that the earth and the universe came into existence out of **a sudden and catastrophic event**.
- Under the gradual group, we have Thales and Emmanuel Kant's theories, the theories under the sudden and catastrophic event include: the big bang and the biblical theory.

THALE'S THEORY

- He was the first person to put up a theory about the origin of the earth based on observation but not traditional believes. He lived at the Nile delta from 1624 to 565 BC. He observed that at the Nile delta, there was constant deposition of silt that led to an outward growth of the delta into the sea.

This led him to conclude that water was the source of the earth's formation. Thales never explained the source of the silt that led to the development of the delta.

EMMANUEL KANT'S THEORY'

- He came up with this theory in 1755 and his theory is based on scientific research. According to this theory it is suggested that the earth was formed from a ring of hot gases that detached themselves probably from the sun. According to Kant, each of the rings of the hot gases moved away from the source and gradually condensed to form planets with sun at the center.

THE BIG BANG THEORY

- It was advanced by George Lemaitre in 1927. According to the theory, out of nothing a big explosion occurred (big bang) from which matter came into existence. The matter then began expanding to form the universe including everything in it as well as the earth. The big bang is estimated to have happened 15billion years ago and this is taken to be the age of the earth. However, the theory is doubted on the account of whether an explosion would occur out of nothing to form the earth.

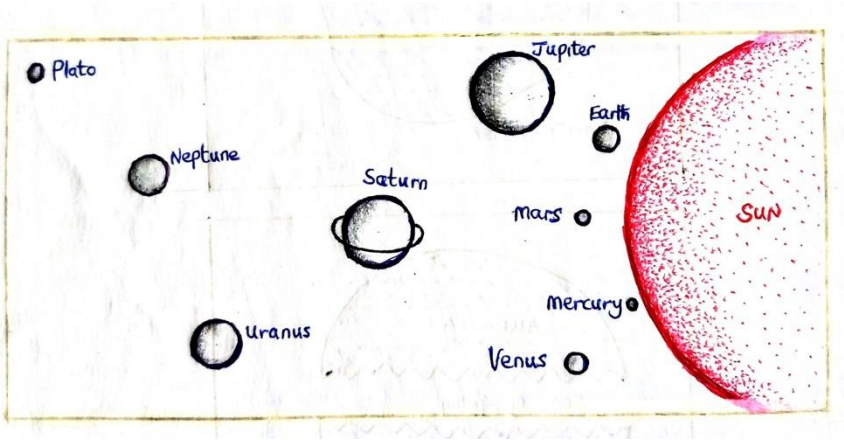
THE BIBLICAL THEORY

- This is based on the biblical book of Genesis where it is suggested in chapter 1-1 that the earth was formed by God. However scientists disagree with this theory. They based their arguments on rock dating. According to the scientific dating of rocks the earth's rocks are estimated to be 3.5 – 4.5 billion years yet is an estimation of the age of the earth. The universe is taken to be 6,000 years. It's worth noting that modern rock dating may also have set back i.e. rocks are affected by temperature, moisture, pressure, Earth movements, etc, which may affect their ages when scientifically dated.

THE EARTH AND SOLAR SYSTEM

The earth is one of the nine planets of the universe that constitute the solar system. It is the only planet where life exist i.e. fauna and flora. The earth is wrapped with an envelope of air (atmosphere). The earth surface is 71% covered by water (hydrosphere). The earth receives its natural energy from the sun. The sun is a combination of hot gases whose surface temperature average is 6,000°C. this sun is about 150 million km away from the earth. Its energy takes about 9 minutes to reach the earth surface at a speed of 3,000,000,000ms^{-1} (metres per second).

Illustration of the solar system



- As seen from the diagram, there are nine planets that revolve around the sun. The sun produces light which falls on them. Three planets i.e. Earth, Jupiter, Saturn, have satellite. The moon is a satellite for the earth.

- The earth is the third farthest planet an approximate distance of 148,800,000km from the sun. It has been estimated that a spacecraft travelling at an average speed of 28,000km/hr from the earth would take seven month to reach the sun. The farthest planet is Pluto at a distance of 5,850,000,000km. It would take the same spacecraft travelling at the same speed 24years to reach the sun.

MOVEMENT OF THE EARTH IN THE SOLAR SYSTEM

The earth and other planets revolve around the sun in ecliptic orbits. There are two movements made by the earth in the solar system: namely rotation and revolution.

- **Rotation of the earth** is the movement of the earth on its own axis from west to east. The makes a complete rotation every 24hours. This cause day and night as the sides facing the sun receive sun rays making day, the side away is in shadow and experiences night.
- **Revolution of the earth** is the movement of the earth around the sun. This occurs at a tilt of $23\frac{1}{2}^0$ and it takes $365\frac{1}{4}$ day to complete a revolution. As the earth revolves around the sun, some part of the earth may be tilted towards the sun than the other sides. This results into seasons.
- In June the northern hemisphere is tilted towards the sun resulting into a long warm days, a season called summer. At the same time, the southern hemisphere experiences cold conditions or winter.
- In December the southern hemisphere is tilted towards the sun leading to summer season in the south while in the northern would be experience winter.
- In March and September the earth is located midway between the north and the south leading to hot day temperature in the equatorial region and either spring or autumn in the northern and southern hemisphere respectively.

THE SHAPE OF THE EARTH

The earth is not flat. Its shape is described as spherical. In other words it appears flattened at the poles and bulges at the equator like an orange fruit. The first man to prove this notion was probably **Pythagoras** in about 530BC. Before this time people especially the sailors believed that the earth was flat. Later, **Magellan's** voyage around the earth proved that the earth is actually round when he sailed from Spain in 1522AD in an effort to move round the world.

THE STRUCTURE OF THE EARTH

The earth structure is divided into two external and internal structures.

External structure:

This comprises of the atmosphere, crust, biosphere and hydrosphere.

- **The atmosphere** is the gas and vapour content that surround the earth. It is estimated to cover about 600km. It is comprised of gases such as Nitrogen 78%, Oxygen 21%, Carbondioxide 0.03% and the rest are rare gases and also in the atmosphere dust particles exist. It is in this zone that climate and weather processes occur.
- **The crust is the earth's outer skin**. It is made up of the continental and oceanic crust. In these layers are found rocks, minerals and on the outer surface soil.
- **The biosphere** is composed of all living organisms i.e. flora and fauna. These exist in the air, water and soil.
- **The hydrosphere** is the water portion of the earth. It is composed of water vapour, water in Lakes, rivers, swamps, oceans, seas and water contained in the rocks. The hydrosphere forms the largest part of the earth covering for almost 71%.

Internal structure:

The earth formed billions of years ago, the earth comprises of several layers. Each layer has its own characteristics and purpose. The layers of earth include; **the crust** (continental crust and oceanic crust), **the mantle** (upper mantle and lower mantle) and **the core** (outer core and inner core).

1. THE CRUST;

- This is a thin outer shell of the earth accounting for less than 1% of the earth's volume. The crust ranges in depth from approximately 5-70 km. it is a cold, thin, brittle outer shell made of rock. The earth crust is very thin relative to the radius of the planet.
- There two different types of crust, each with its own distinctive physical and chemical properties. They are Continental Crust and Oceanic Crust.
- **The Continental Crust (sial)** is the outermost layer of the earth's lithosphere that makes up the planet's continents and continental shelves. It is about 65km in thickness. It is made up of mainly rocks that are light and rich in silica and aluminum minerals hence this layer is at times known as **SIAL.** The continental crust forms nearly the earth's entire land surface. The continental crust is broadly composed of granitic minerals with a density of about 2.7g/cm^3.
- **The Oceanic Crust (sima)** is the outermost layer of the Earth's lithosphere that is found under the oceans. It is formed at spreading centers on the oceans ridges, which occur at divergent plate's boundaries. The oceanic crust is thinner than continental crust, usually 5-10km thick. It is made up of mainly heavy rocks of silica and magnesium with an average density of 3g/cm^3. This layer is also known as **SIMA.** It is mostly composed of basaltic materials.
- Although particular rock types dominate the different crustal rocks, resent research has shown that the two layers have within them a large quantity of basaltic rocks. This could be a result of upwelling of basaltic materials from the mantle through fissures towards the surface.
- The crust and the upper mantle are separated by the **Moho-discontinuity**. This is a seismic and compositional boundary. It is a zone where earthquake waves travelling the earth suddenly change speed from 7.2km per second to 8.1km per second. This indicates that the rocks above are less dense than those below.
- The two layers of the crust and the rigid top part of the mantle are collectively called **Lithosphere.** It is for this reason that the crust is at times referred to as lithosphere.

2. THE MANTLE (MESOSPHERE)

This is a layer beneath the earth's crust. The mantle is about 2,900km thick and makes up about 84% of Earth's total volume. The mantle is also known as 'SIMA' layer. This layer is made up of rocks rich in magnesium and silicate materials. is a region of denser (semi plastic) rocks that are rich in silicate of iron and magnesium. These rocks have a density of up to 3.2gm/cc.

The mantle is divided into two i.e. the upper mantle and lower mantle.

- **The upper mantle** of the earth is a very thick layer inside the planet, which begins just beneath the crust and ends at the top of the lower mantle at 670km. The temperatures range from approximately 200℃ at the upper boundary with the crust to approximately 900℃ at the boundary with the lower mantle. The upper mantle is mostly solid, but it is separated from the lower mantle by a plastic layer known as the **asthenosphere.** This plastic region contributes to the tectonic activity. It is the origin of molten rock known as magma. The average density of the upper mantle is about 4.4g/cm^3.
- **The lower mantle** is made up of solid rock. It is also known as the **mesosphere**, which represents 56% of earth's total volume. The lower mantle extends from 670km to 2900km below Earth's surface. The average density of the lower mantle is about 5.6g/cm^3. The lower mantle is hotter and denser than the upper mantle. The temperatures reach about 4000℃. This temperature melts all rocks hence this region exists in the semi-liquid/molten form. The molten rocks generate

convectional currents which drive tectonic plates causing crustal movement such as earthquakes, faulting, warping and vulcanicity.

- The mantle is separated from the core by **Gutenberg discontinuity** where some earthquake waves are absorbed.

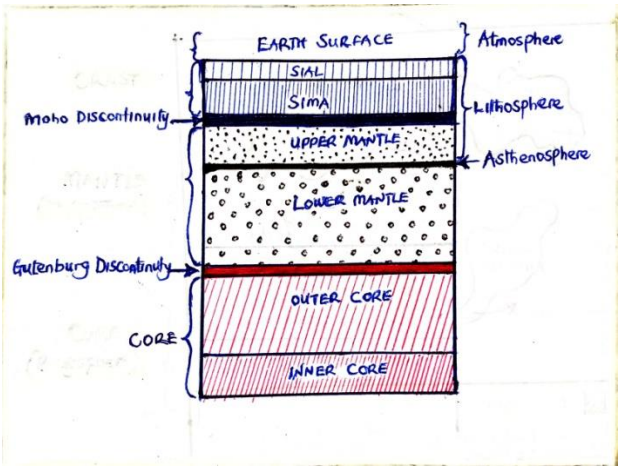
3. THE CORE (BERYSPHERE)

This is the inner most layer of the earth. It is a very hot and dense center of our planet. The ball-shaped core lies beneath the mostly-solid mantle. The core is almost entirely made up of metal minerals, especially, iron and nickel, hence a chemical name 'NiFe'. The core is divided into two namely; the outer core and the inner core.

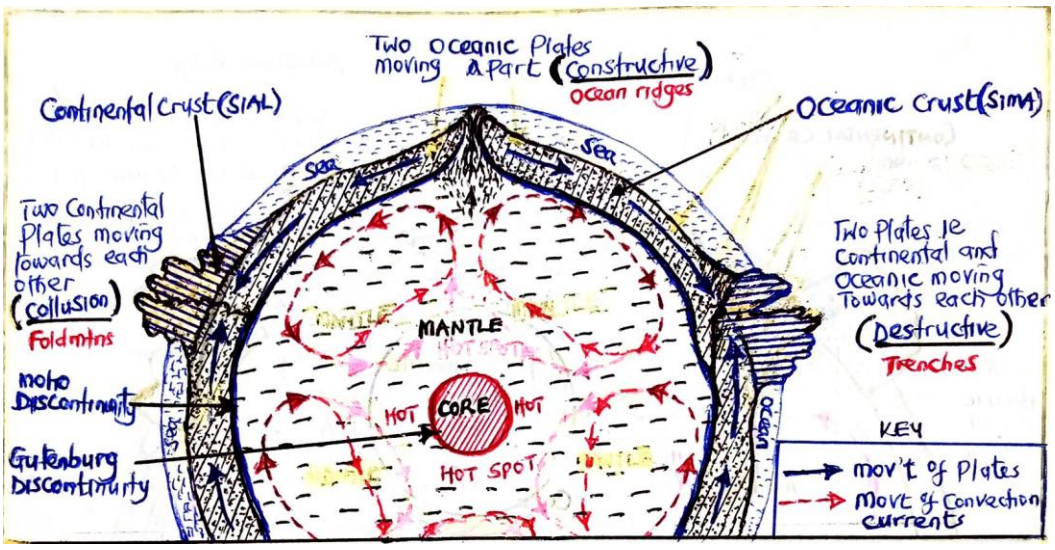
- **The outer core** is bout 2200km thick. It is made up of a semi liquid iron and nickel minerals. The outer core is very hot, between 4,500degrees and 5,500℃. The density of the outer core is about 12.6gl/cm^3
- **The inner core** is about 1,230km to 1,530km thick. It is made up of solid iron and nickel minerals. It exists in a solid state due to high pressure exerted by the overlying layers. The temperature in the inner core is about 6,000℃. The density of the inner core is between 9.9 to 12g/cm^3. The core extends up to a distance of 6,371km which is the centre of the earth.

ILLUSTRATION OF THE STRUCTURE OF THE EARTH

Vertical section through the earth



A CROSS-SECTION OF THE INTERIOR OF THE EARTH



Key points to note while studying the structure of the earth.

- *The identification of the different layers of different layers of the earth.*
- *The depth of each layer of the earth.*
- *The mineral content of the different layers of the earth.*
- *The density of different layers of the earth.*
- *The temperatures of different layers of the earth.*
- *Illustration of the earth showing different layers of the earth.*

REVISION QUESTIONS

Qn. *Describe the origin of the earth's interior*

Qn. *With illustrations, describe the structure of the earth's interior***.**

ORIGIN AND DISTRIBUTION) OF CONTINENTS AND OCEAN BASINS

- Continents can be defined as large continuous descript landmass ideally separated by expanse of water. Ocean basins are large depressions filled up with water surrounding the continents. The division of the earth surface into continents and ocean basins is a key feature of the earth's surface. The land continent occupies 29% of the earth's surface while oceans take the remaining 71%. In the world there are seven continents and four oceans. Continents are more concentrated in the northern hemisphere hence it is sometimes called a land hemisphere while the southern hemisphere is termed as the water hemisphere because much of it is covered with water.

The origin of continents and oceans

- Before the 19th century, people believed that continents and oceans always remained fixed throughout geological period in positions they occupy. This would suggest that no continental movement has taken place. In 1837 Dana expressed the view that continents and ocean basins have never changed positions and the general make up of the earth have remained stable.
- However, people like Francis Bacon in 1620, Placet in 1668 and Edward Forbes had already started arguments about a possibility of continents have changed position. **Francis Bacon** was impressed by the parallelism of the coast of Africa and South America and suggested that they might have been joined together. **Placet** in 1668 went further to suggest that before the big flood, the Atlantic continents might have been joined and might have only drifted apart. **Edward Forbes**, was unable to explain how certain plants and animals migrated from one mass land to another across a large ocean, he concluded that, there might have been a dry land to allow these plants and animals to cross.
- Because of these arguments about the distribution of continents and oceans basins, many scholars were stimulated to try and find out about their origin and distribution and this led to the emergence of several theories to try and explain the current distribution of continents and ocean basins.

THEORIES ABOUT THE ORIGIN AND DISTRIBUTION OF CONTINENTS AND OCEAN BASINS

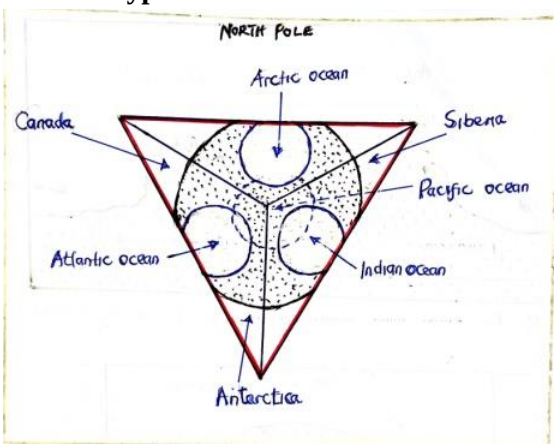
- A theory is a statement or group of statements established by reasonable arguments and intended to explain the occurrence of a particular part or event. The explanation still needs proof but nevertheless, it appears reasonable.
- Various theories have been advanced to explain the origin of continents and ocean basins. These theories fall under two categories namely: theories against/before continental drifting and the theories for continental drift theory.

Theories against / before continental drift:

THE TETRAHEDRAL HYPOTHESIS;

- This theory states that the distribution of continents and ocean basins is in resemblance of a tetrahedron. A tetrahedron is a 3-dimentional figure with 4 equilateral triangles. This theory was based on the belief that the earth was contracting. It was believed that the contracting earth would tend to shrink in a tetrahedron form. In this way the ocean basin would be depressed on the tetrahedron faces while the continental land mass would occupy the edges.
- Indeed the current distribution of ocean basins (The Pacific, Indian, Atlantic and the Arctic) correspond to the 4 faces of the tetrahedron while the continents were left to as elevated regions containing edges and corners. It's true that the present distribution of sea and land globally is similar to a tetrahedron structure. This theory attempts to explain the massive concentration of land in the North hemisphere and water in the South hemisphere.

Diagram to illustrate tetrahedron hypothesis



- The cause of the contraction of the earth was believed to be gravitational pull, which pulls the materials towards the earth's centre causing it to shrink/become smaller.
- However, the critics argue that the earth has got a uniform crust which could never shrink/go contrary to the theory of Isostacy which tends to keep the earth in a state of equilibrium.
- Another argument against the theory is that if the ocean depressions were a result of contraction, their floors would be made of sialic rocks which is not the case ocean floors are made of simatic rocks.
- They also argue that the tetrahedron contraction is possible with a small globe but not enormous globe like the earth.

THE MOON THEORY:

- This theory is based on the formation of the moon and the missing sialic crust in the ocean basins. According to this theory, the moon is made up of sialic materials (rocks) which are the very rocks that make up the earth crust. Hence it justifies the absence of sialic materials on some parts of the earth particularly in some ocean basins.
- It was suggested that early in the history of the solar system, the moon separated its self from the earth and carried with it the sialic crust that had hither formed on the earth. Osmond Fisher (1881), the proponent of the theory suggested that the huge Pacific Ocean depression may be the scar left behind when the moon detached its self from the earth.
- The theory has been supported by the fact that the moon deflects the sunlight in the same way as the powdered sialic material do. It's therefore possible that the shinny materials originate from the earth only separating during the great rapture.
- Secondly, the Pacific basin suggested by Osmond Fisher as a scar left behind when the moon detached its self from the earth, is the largest region from which Sial materials are missing. The

scar is about ⅓ of the earth's surface almost the same size as the moon. It's therefore possible that the sialic materials were carried by the separating moon.

However,

- The theory has also been criticized on the account that the moon could not have been formed from the crustal matter that separated from the earth unless the earth was subjected to strong/ immense solar tides which mechanically has been proved is not possible.
- The theory is also based on the missing sialic material in the Pacific Ocean alone and does not explain how other Oceans were formed.

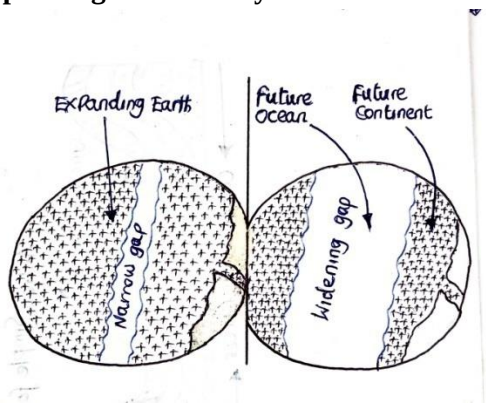
ISOSTACY

- As related to the origin of continents and oceans, the theory suggests that continents have not laterally moved away but only experienced vertical oscillation of the surface like uplift and down warping.
- According to the theory lateral rocks are uplifted to form continents while the heavier rocks collapse inside to form basins. Therefore continental masses correspond with lighter rocks that were uplifted while the basins overlie heavier rocks that collapsed inwards to maintain isostatic equilibrium.

THE EXPANDING EARTH THEORY

- According to this theory the earth was originally small with a thin sialic crust forming a continuous layer of the surface. As the interior of the earth expanded, a surface layer (sialic material) was stretched and broken into continental land masses. As the expansion continued more and more, continental blocks separated and basic material from the mantle flowed in the gaps between the blocks (continents) forming ocean basins.
- The cause/reason for the expansion of the earth was related to its aging. As the age of the earth increases, its gravitational force decreases/reduces and weakens hence limiting its ability to hold internal materials together. This resulted into its internal materials spreading outwards/expanding due enormous pressure from within than outside.

Diagram to illustrate the expanding earth theory



THECONTINENTAL DRIFT THEORY

Continental drift refers to large scale differential movement of sialic blocks towards each other or away from another. Under continental drift, it's believed that there was one original land mass which under went fragmentation yielding individual large masse/blocks of sialic crust. This was followed by moving apart of the individual blocks from one position to another until the blocks and ocean basins attained their current positions.

THEORIES JUSTFYING CONTINENTAL DRIFT

Several theories to justify continental drift have been advanced and they include: F.B. Taylor's theory, Alfred Wegner's theory, The Sea Floor Spreading theory and the Plate-tectonics theory.

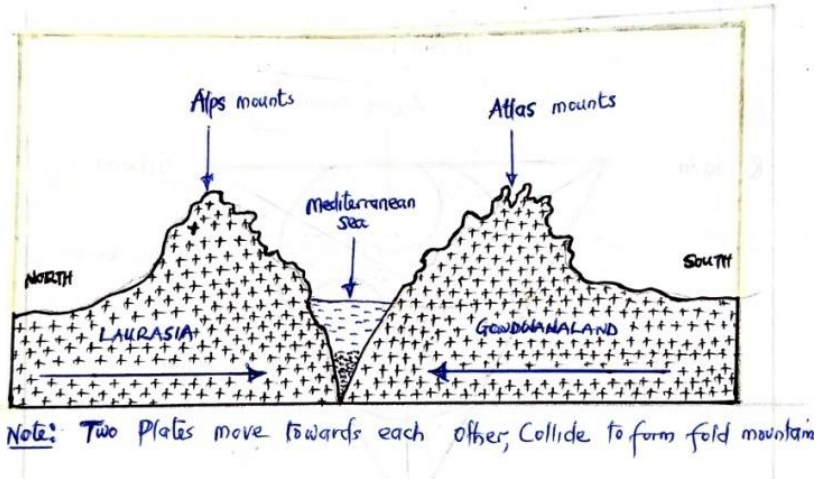
F.B. TAYLORS THEORY (1910)

- According to Taylor, there were originally two big land masses, namely, Laurasia and Gondwanaland. Laurasia was located near the present-day North Pole where it later drifted southwards towards the equator. Gondwanaland was located in the South Pole which also drifted towards equator.
- The movement of Laurasia and Gondwanaland is said to have occurred during the cretaceous period between 66 and 144million years ago.
- Taylor suggested that the movement was caused by the moon's gravitational pull/force. He contended that the moon came very close to the earth and exerted a powerful gravitational pull that caused Laurasia and Gondwanaland to move towards the equator away from their positions near poles.
- During the equator-ward movement, where the land masses encountered resistance, the crust would fold out or bulge to form mountains in front. According to Taylor when Laurasia and Gondwanaland met/collided along the equator, the formation of Fold Mountains occurred such as The Atlas in North Africa, the Alps in Europe and fold mountains in America.
- In some cases the crust stretched and split forming troughs that became ocean basins. Taylor suggested that the basins of the Black sea, Atlantic and the Indian oceans were stretches and broken regions left behind between the drifting continents.
- As the drifting continued, Laurasia split to form the northern continents i.e. North America, Asia, etc.

However, Taylor's theory is criticized on the following grounds;

- In the first place, he attributed the drifting of the landmasses to the moon's gravitational pull which is highly doubtable that the moon could ever exert adequate force to pull the gigantic landmasses (continents).
- Secondly it's suggested that if the moon came so close to the earth and then exert a dragging force on land masses, then the earth's rotation would come to a standstill/halt within a year, since this has not happened it therefore makes the theory doubtable.
- Taylor also attributed Fold Mountain building to collision of landmasses due to gravitational pull; however, the theory does not explain the formation of the earlier fold mountains like the Caledonian system of the siluro-devonian times and the Hercynian system of the mid-Permian and Carboniferous periods.

The formation of Atlas and Alps mountains according to Taylor.



ALFRED WEGNER'S THEORY (1915)

- Alfred Wegner, a German climatologist was the first scientist to suggest the theory of continental drift, in his book "Origin of continents and oceans".
- According to the theory, the present-day continents originated from one giant landmass known as "Pangea". This land mass is thought to have been located near the present south pole and was surrounded by a large expanse of water known as "Panthalassa".
- About 200million years before present, Pangea began drifting northwards. By about135years Pangea split into two supper continents to form Laurasia which drifted to the north and Gondwanaland which remained in the south. The two landmasses were separated by a narrow water body known as "The Sea of Tethys".
- Wegner believed that Laurasia lay across the Equator while Gondwanaland lay near the South Pole. Gondwanaland is believed to have experienced large scale glaciations during the carboniferous period, approximately 345-280million years ago.
- During the late Paleozoic and early Mesozoic eras, numerous cracks developed in the landmasses leading to separation and subsequent drifting. Finally, during the cretaceous period, approximately 137-70 million years ago the landmasses completely broke up.
- Laurasia broke up to give rise to North America, Europe, Asia and other land masses in the northern hemisphere such as Greenland and Ice land. Gondwanaland gave rise to the "southern continents" namely, Africa, South America, Antarctica, Australia, the sub-continent of India and the numerous islands in the southern hemisphere.
- During the drifting, the gaps between the continental blocks/landmasses became wider forming the present-day continental oceans and seas.
- About 65million years, India drifted northwards to join Eurasia. Wegner contends that the collision between the Indian sub-continental landmass and the Asian landmass led to the formation of Himalayas Mountains.
- In the south, Africa drifted to attain its present-day position astride the equator. South America drifted westwards and northwards towards the equator joining North America.
- Australia drifted eastwards away from Antarctica northwards, Eurasia drifted eastwards while North America continued to drift both north and west closure to the North Pole.
- Wegner suggested that the breaking-up and the northward drifting of the landmasses was due to gravitational attraction of the equatorial bulge or centrifugal force while the westward drift of landmasses was due to tidal attraction of the moon and the sun.

However, the theory has been criticized on the following grounds.

- Firstly, Wegner never bothered to explain the forces leading to the break-up of the landmasses before they could drift. Geophysicists argue that gravitational forces exist but it is rather too weak to drag large landmasses such as continents from their original positions.
- Wegner is also been criticized that he was a climatologist and not a geologist. Therefore he had minimal information and imperfect understanding of the field in which he wanted to meddle.
- Nevertheless, Wegner is credited for this was the first impressive theory in trying to explain the origin of continents and oceans. He marshaled a lot of evidence to support his theory especially from the southern hemisphere which attests to be the occurrence of continental drift.
- His theory was more exiting and it stimulated much more research into continental drifting giving rise to new theories such as the Sea Floor Spreading and Plate Tectonics.

Diagram of the supper continental Pangea

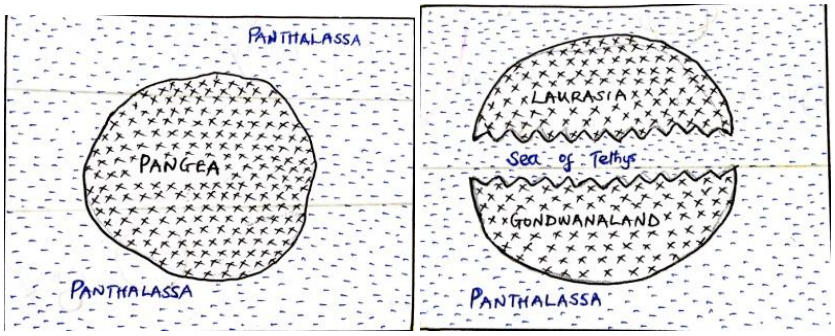
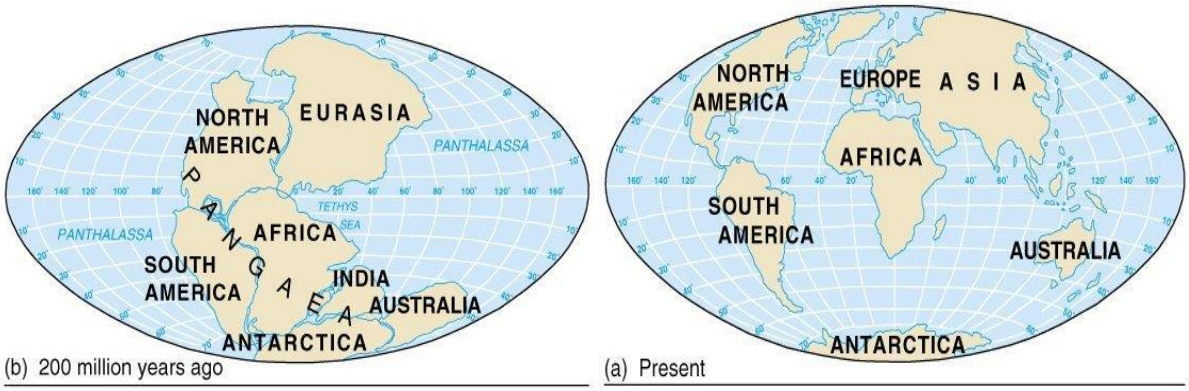


Diagram showing further split of Laurasia and Gondwanaland



REVISION QUESTIONS

Qn*. Explain the relevance of Wagner's theory of continental drifting to the understanding of the present day distribution of continents and ocean basins.*

Approach*: define continent and ocean basin, define continental drifting, explain Wagner's theory, and give the evidences to support the continental drifting.*

THE SEAFLOOR SPREADING THEORY

- This is one of the modern theories of the continental drift. This theory was propounded by geologist Harry Hess in the 1969's. Hess, who observed that in the mid Atlantic Ocean, there was a ridge whose rocks were extremely younger than the rocks away from the ridge, he attributed this to deposition from the mantle that would displace the old rocks sideways from the middle of the ocean floor.
- According to Hess, the interior of the earth is in molten (semi-liquid) state due to great/intense heat generated from radioactivity and geo-chemical reactions. This heat melts rocks and keeps the interior of the earth (mantle) in a hot semi-fluid state. This semi liquid rocks is highly mobile and tend to rise towards the surface inform of convection current. As they reach the base of the earth's crust, they flow horizontally and exert a drag force on the crust above in the direction of their flow. This causes movement of crustal layers of the continental landmasses/blocks hence continental drift.
- After Hess studying that the rocks in the mid ocean ridge were much younger than those away, he submitted that, as crustal layers moved apart, the semi-liquid mantle material (young rocks) rises through and fills up the gap between the moving crustal blocks hence sea floor spreading or expansion/widening of the ocean floor each time the new rocks are deposited at the mid ocean ridges.

However,

- Hess is criticized for he based his theory on the Atlantic Ocean floor only and he did not explain what was happening in other ocean floors.

Diagram to illustrate Sea Floor Spreading

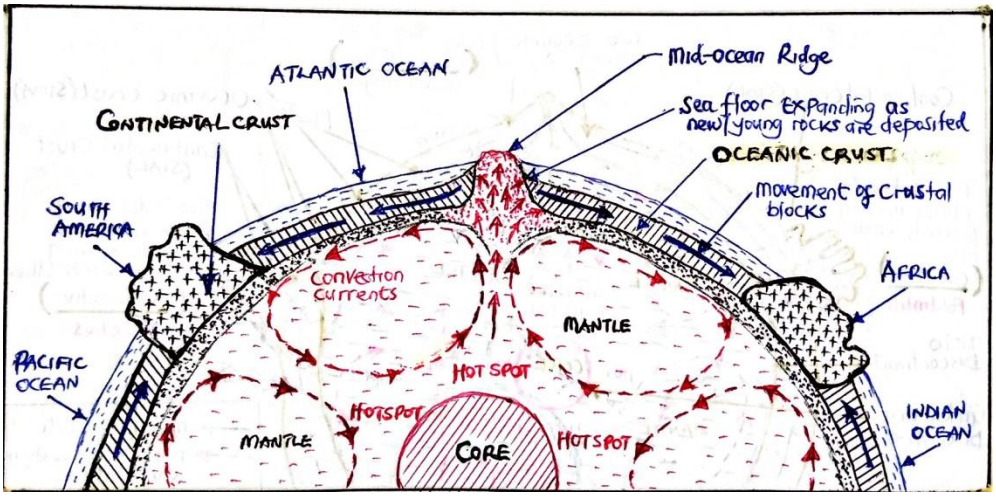
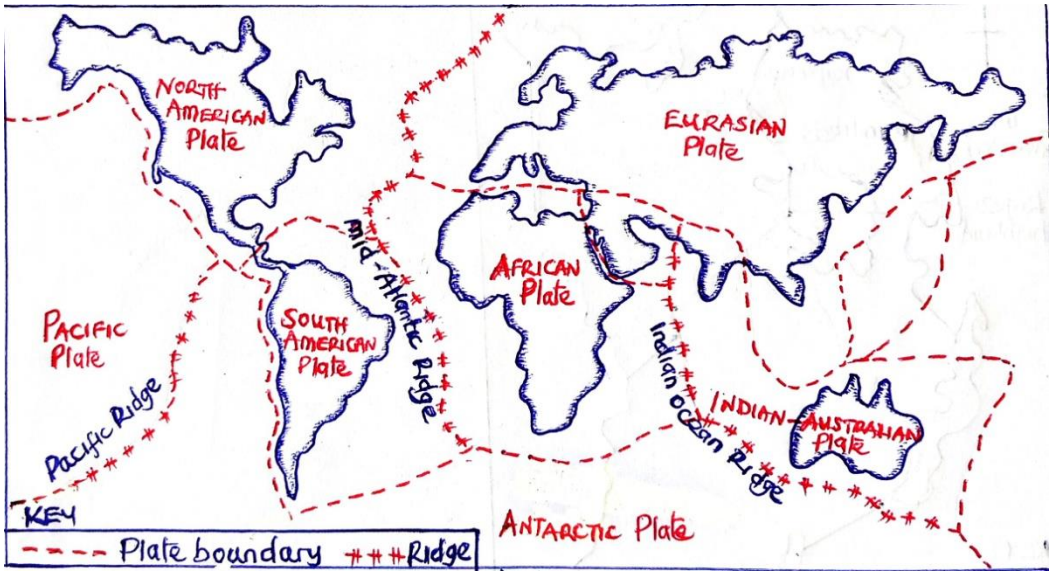


PLATE TECTONICS THEORY

Plate tectonics itself is the study of the movement of tectonic plates and the resultant landforms. Tectonic plate's theory assumes that the earth crust is not one piece but is divided into a number of separate parts called Tectonic plates.

- Tectonic plates are large, light-ridged blocks of the earth's lithosphere which are mobile and float on the molten rocks of the upper mantle like rafts as a unit due to convection current.
- Tectonic plates are divided into two namely; Continental plates and Oceanic plates. Continental plates are composed of lighter granitic rocks rich in SIAL and form basements for continents. Oceanic plates are made up of much denser and young basaltic rocks rich in SIMA and they form ocean basements
- In all there are seven major plates and about twelve minor ones. The plates include The African plate, The North American plate, The South American plate, The Indian-Australian plate, The Pacific plate, The Eurasian plate and The Antarctica plate.

The sketch map showing World's major tectonic plates:



MOVEMENT OF THE TECTONIC PLATES

- Tectonic plates are light and therefore float on a semi-molten mantle. The plates are also rigid so that if a force is applied to them, they move in the direction of the force as a unit. The force that drives the plates is known as convection current and is generated from the interior of the earth.
- In the interior of the earth there is great/intense heat generated from radioactivity, geochemical and geophysical reactions as well as pressure exerted by the overlying crustal layers. This heat has melted and kept the rocks of the mantle in a semi-liquid state. These molten rocks are highly mobile and produce convection current that drives the plates. The convection current may rise towards the crust where they are forced to flow uniformly and later sink downwards. This movement exerts a drag-force on the plates. As the plates move, continents on top also move following the direction of plate movement hence continental drift.
- The theory assumes that plates /continents generally moved northwards, others like The South American plate moved westwards. Others like Europe, Asian, and Australia moved north-eastwards. The Pacific plate is drifting north-westwards as well as North America landmass.

TYPES OF PLATES MOVEMENT AND THEIR EFFECTS ON PLATE BOUNDARIES

The most important regions are the plate's movements/boundaries where plates meet or move away. There are three types of plate movement that affects the distribution of continents, oceans and the resultant landforms. These are;-

Divergent movement; this refers to the movement of tectonic plates away from one another. It is caused by diverging convection current.

- If the divergent movement of pates occurs under a continental crust, it results into faulting, the associated volcanic activity and warping of landmasses.
- On the oceanic crust/sea floor, when plates move away from each other, constructive plate boundaries occur. This is because as molten rocks from the mantle rises to fill any possible gap, new crust is created on the ocean floors. This produces mid-ocean ridges and Islands e.g. the Mid-Atlantic ocean ridges and islands were formed as a result of drifting of the South American plate from the African plate. Ice land sits atop a divergent boundary where continental drifting occurs.

Diagram to illustrate diverging Oceanic Plates boundary

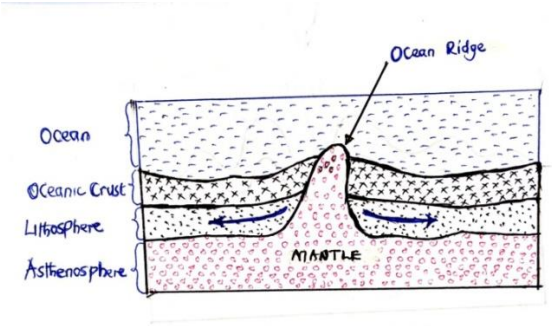
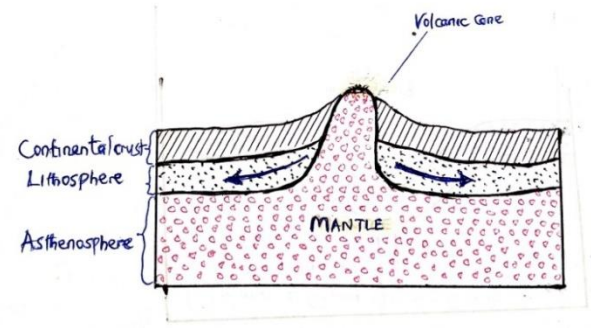


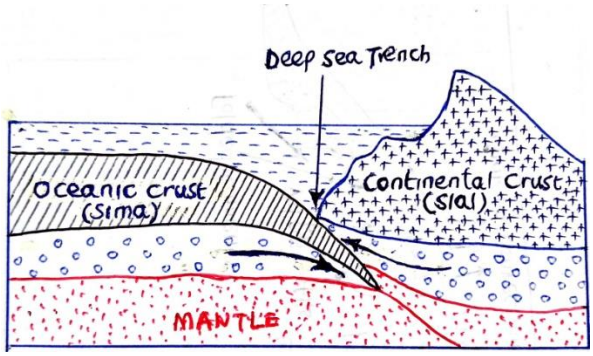
Diagram to illustrate diverging Continental Plate boundary



Convergent movement; this refers to the movement of plates towards one another.

- When continental plate moves towards the oceanic plate, destructive plate boundaries are produced or subdution of denser simatic rocks occurs as old rocks are destroyed. This results into the formation of trenches e.g. Java, Nazca and Tonga.
- Continental plates/landmasses may move towards each other and getting closer collision may occur since none can sink, the sediments between the plates are compressed to produce folding and formation of fold mountains e.g. the collision of the Indian plate and the Eurasia plate formed Himalayas mountains, others are Alps and the Atlas mountains in Europe and Africa respectively.
- Where dense oceanic plates move towards a less dense continental plate, the oceanic plate is forced to sink into the earth's interior where it becomes slowly heated and assimilated into the mantle. This forms deep sea trenches and Island arcs. E.g. the Nazca plate moved under the South American plate forming the Nazca trench and the Andes Mountain.
- When the oceanic plates move towards each other, the continents move nearer and the ocean basins are narrowed down. This leads to the formation of trenches and volcanic arcs. E.g. the Pacific and the Eurasian plates led to Mariana trench and Japan arc.

Illustration



Transform movement; this refers to the movement of plates sideways or past each other hence no collision or diversion.

- In the processes conservative/passive margins are created and new crust is either created or old ones destroyed instead. Transform fault line is where a section of land is off-set or created. E.g. St. Andreas fault in California was formed when North American plate moved against the Pacific plate causing off-setting of land to the west. Owen fracture near the Gulf o Aden and Romanche fracture off West Africa.

REVISION QUESTIONS:

Qn. (a) *What is meant by the term Plate Tectonics****? (b)*** *How does the theory of plate tectonics explain the present day distribution of continents and ocean basins?*

EVIDENCES TO PROVE/JUSTIFY THE OCCURANCE OF CONTINENTALDRIFT

- **The jig saw fit:** there is a closure (Jig saw) fitting of the continental coastline across the Atlantic Ocean, e.g. the coast of South America and west coast of Africa fit each other not only at the surface but also at a depth of 200 meters. This suggests that the two landmasses had been joined in a single landmass before fragmentation and drifting.
- **Similarity in animals and plants:** there is a similarity in the distribution of living organisms across large water bodies. E.g. the Mesosaurus Amphibians of Eastern South America is also found in Africa. The Congo rain forests in Africa resemble the rainforests in Brazil. This indicates that the landmass might have been joined at the time they experienced such environmental conditions that led to similar vegetation and animal species and later drifting occurred.

- **Similarity of rocks of the southern continents:** research has shown that there are remarkable similarities between rocks found in Ghana, Morocco, etc in Africa with those in Brazil in South America. This suggest that Africa and South America were formally one landmass but later drifted apart with the similar rocks drifting along on each of the content.
- **The widening gap between the Oceans**: it has been noted that the gap between Southern America and Africa i.e. Atlantic Ocean is widening at a rate of 2.5cm per year, the same is said of the East African Rift Valley which is widening at an average of 2cm per year. This indicates that the continents blocks are still drifting from each other every year.
- **The occurrence of earthquakes**: the occurrence of earthquakes, faulting, vulcanicity, are all points of geological disturbances due to convection current as explained under the Sea floor spreading and Plate tectonics theories. This is a proof of occurrence of continental drifting.
- **The proximity of continental landmasses to the North Pole than the South Pole** is another evidence to support continental drifting. For instance, according to Wegner's theory, the continental plates were pulled towards the northern hemisphere thus continental drifting.
- **Geological structure**: geologically the structure of the earth is that the SIAL floats on the SIMA, this floating point to occurrence of continental drifting as experienced under the Plate Tectonics theory.
- **Paleomagnetism evidence**: Paleomagnetism refers to the ancient or fossil magnetism in rocks. It is established from the geological research on the rocks of the earth crust that at the time of formation the magnetic properties of these rocks point to the earth's north-south poles. Further research on rocks in India, Australia and South America, reveal that the magnetic field lines in these rocks no longer point in the north-south direction as expected. This suggests that during the course of drifting, the rocks were twisted and as a result the magnetic field of these rocks changed their original positions hence supporting continental drift.
- **The glacial fossil evidence**: fossils are remains of living things that lived long ago. Studies have shown that thick deposits of fossilized glacial moraine discovered in Eastern Brazil, Paraguay and Argentina are exactly like those of South Africa and Australia. This implies that the continents were once joined together possibly during the carboniferous glaciations period before they drifted from each other.
- **The existence of laterite soils in the cold regions**: studies have found out that laterite soils exist in North America and certain parts of Europe and these are cold regions. This suggests that these regions at one time experienced tropical climatic conditions that facilitate laterites formation and drifted later. Laterites only form under hot, dry and wet conditions of the tropical lands.
- **Similarity of oil beds:** this is yet another evidence of continental drift. It has been found out, for example, that the oil beds of Brazil in South America are the same as those in Angola in Africa. This indicates that the two areas could have had a common origin and evolution of the oil generating organisms and the common oil beds that only drifted apart with each moving with what already formed hence a manifestation of continental drift.
- **Existence of salt evaporates in cold regions**; the salt evaporates/salt beds that are naturally in the tropics have been discovered in the cold regions of the world such as Britain, Germany, USA, South America and Russia. Salt evaporites are formed when water containing salts in solution evaporates leaving beds of salts called evaporites. They form under hot and dry conditions which currently are impossible in the cold regions. This indicates that they formed when those landmasses lay in the tropical areas and changed location when continental drift took place.
- **The existence of coral reefs in the cold regions**: the discovery of coral reefs in the cold Greenland in North America is another proof to justify the continental drift. Coral polyps and the resultant coral reefs only thrive in the hot, tropical, marine environment and not in the cold

conditions. This suggests that Greenland was at one time positioned in the hot tropical latitudes for corals to form and later drifted to its current location.

- **The marching of Fold Mountains (Orogenic) belts**: this is another evidence to justify the movement of plates and therefore continents. The alignment of zones of Fold Mountains across the zones of Africa and South America march e.g. the fold ranges in the flatland islands in Argentina are similar in age and structure to those in south-west Cape in South Africa; Appalachian mountains march UK Mountains, and ranges in Canada march those in Sweden and Norway.

Conclusively, several theories have been advanced to explain the continental drift, some of the have been criticized for some defects. Others seem to be reasonable and even have scientific proof. Modern evidence based on Paleomagnetism and plate tectonism will remain/stand the test of time.

REVISION QUESTIONS:

Qn. *Justify the occurrence of continental drifting.*

Approach: *define continental drifting; explain the theory and its origin; explain the evidence of continental drifting.*

Qn. *Account for the distribution of ocean basins and continents.*

Approach*: define continent and ocean basin; give examples of each; define and explain the theory of continental drifting; give other theories.*

Qn*. (a) What is a plate? (b) Distinguish between continental crust and oceanic crust. (c) Describe the landforms resulting from plate tectonism.*

Approach: *define continental and oceanic crust; define plate tectonics; give the origin of plate tectonism; explain the movement of plates and give the landforms.*

Qn. ***(a)*** *Distinguish between a converging plate boundary and diverging plate boundary.* ***(b)*** *Examine the influence of divergent plate boundary on the relief and landforms of East Africa.*

EFFECTS OF CONTINENTAL DRIFT ON RELIEF AND LANDFORM EVOLUTION OF EAST AFRICA

From the above evidences, one can easily see that the continental drift is a reality and some of the illustrations have been quoted from East Africa.

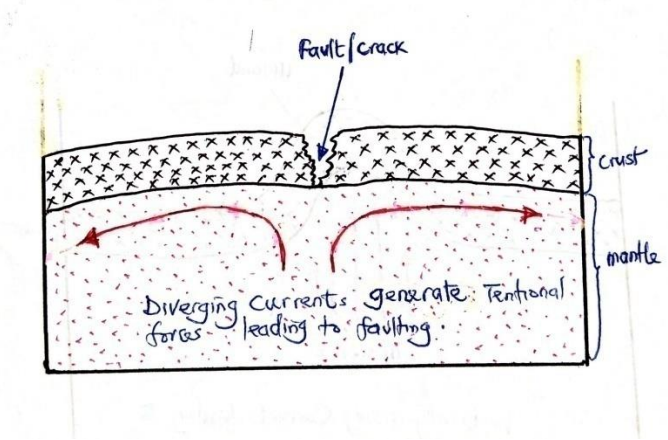
- Nearly all the major structural landforms and relief features of East Africa can be explained directly or indirectly using the continental drift theory. Under the continental drift's theories of Plate tectonics and Sea floor spreading, the forces that could have contributed to continental drift are explained. These are the same forces that account for the formation of landforms and relief features in East Africa.
- The theories are based on convection currents which move in different directions, generating forces like tension, compression, up-warping and down-warping.

DIRECTLY, continental drift theories are responsible for landforms evolution in East Africa as explained below:-

Continental drift and faulting landforms:

- The continental theories explain that **diverging convection current** (moving horizontally) pull the earth's crust apart forming tensional forces. Tensional forces are responsible for faulting in East Africa. The features/landforms resulting from faulting include; Rift valley, escarpments, fault scarps, block/horst mountains, Grabens, fault-guided rivers.

Diagram showing diverging currents:



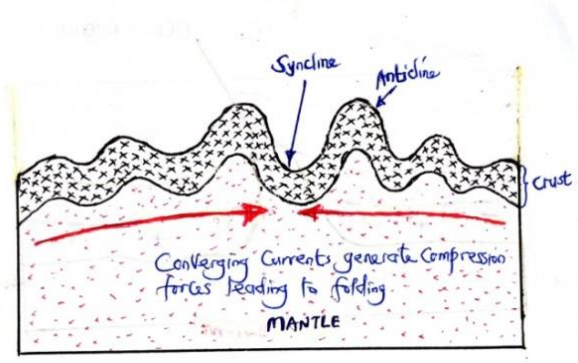
The continental drift and vulcanicity landforms:

- The continental drift theory explains that when rapture of the continental crust occurs as a result of the **diverging convection current**, magma may escape through the zone of rapture, which either cools and solidifies within the crust to form intrusive features like dykes, batholiths, sills, laccoliths, lapoliths, or cools on the earth surface to form extrusive features such as volcanoes, basalt domes, ash and cinder cones, craters, calderas, volcanic plugs, cumulo domes, lava plateaus, composite volcano cones e.g. mount Kilimanjaro, Elgon, Kenya etc.
- Geothermal features such as hot springs, geysers and fumaroles.

The Continental drift and folding landforms:

- Continental drift theories explain that Convection currents may also **converge** or move towards each other creating compression forces. These forces may occur in the landscape made up of young crustal rocks layers leading to folding and resultant landforms such as synclines and anticlines. An example of the folded landscape in East Africa is in Karagwe in Tanzania. Intense compression may also lead to reversed faulting forming escarpments with sharp edges.

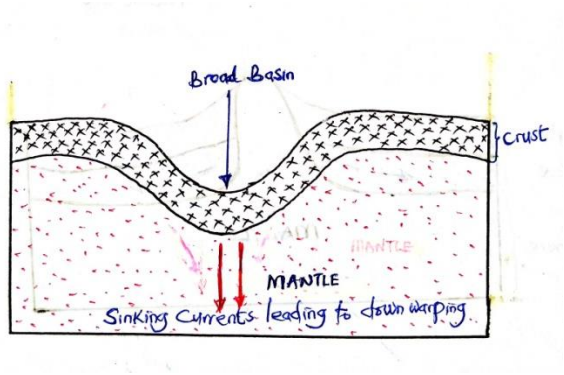
Diagram showing converging current;



The Continental drift and landforms of warping:

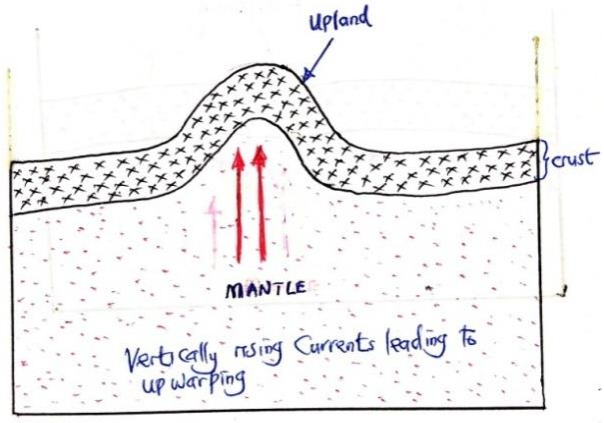
- The continental drift theory also explains that when convection currents **sink** into the mantle they may pull with them the crustal plates downwards resulting into a saucer-shape (down-warped) landscape on a broader scale. In East Africa this has been responsible for the formation of broad basins e.g. Victoria and Kyoga basins.

Diagram to illustrate the sinking current



- **Closely related to down warping**, when the convection currents **rise/move vertically**, they push an extensive area of the crust upwards leading to up-warping. This is responsible for the formation of the East African plateau and some uplifted highlands of western Uganda.

Diagram illustrate rising currents:



- Generally, the movement of convection currents underneath the crust is responsible for continental drift and it's the same forces that generate tensional, compressional uplift and down warp forces or earth movements which are in turn responsible for the formation of landforms in East Africa.

INDIRECTLY, continental drift theories especially the sea flow spreading and the plate tectonic explains the existence of other landforms in east Africa as below:

- **Continental drift and glacial landforms**: Glaciation refers to the movement of ice from the high ground. In East Africa glaciation occurs on the high mountains of Rwenzori (a block mountain), Mount Kenya and Kilimanjaro (both volcanic) whose formation is explained in relation to continental drift. The glaciers in these mountains are responsible for either erosional (e.g. cirques, arêtes, pyramidal peaks, hanging valleys, tarns, etc) or depositional (moraine, erratics, drumlins, outwash plains, eskers, kettle holes.etc.) glacial landforms.
- **Continental drift and the fluvial landforms:** Fluvial activity refers to the work of the rivers in terms of erosion and deposition and the features they produce. These rivers originate from highlands/mountains and their mouths end in lakes, seas and oceans. Mountains, oceans and lakes basins were a product of crustal plates movements already explained in the continental drift theories. Fluvial landforms are either erosional (e.g. waterfalls, gorges, interlocking spurs, rapids, potholes, etc) or depositional (e.g. deltas, deferred river tributary, alluvial fans, levees, flood plains, ox-bow lakes, braided channels.

- **Continental drift and mass wasting landforms:** Mass wasting refers to all forms of slope failure raging from slowest process up to the fastest landslides. Mass wasting like glaciation and fluvial, operate in highlands/mountains whose formation is directly linked to convectional currents, plate tectonics and continental drift. Being a denudational process its landforms include terraces, straight slopes, scars, slide-dammed lakes, etc.
- **Continental drift and weathering landforms:** Weathering processes occurs on the rocks. The genesis of the first-class rocks (igneous) is from vulcanicity and continental drift. The crustal layers crack creating gaps through which magma escapes, cooling down to form igneous rocks. Igneous rocks later yield sedimentary and metamorphic rocks. These rocks are weathered producing landforms such as exfoliation domes, inselbergs, stalagmites, stalactites, caves, pillars, etc.
- **Continental drift and the coastal landforms:** Oceans, seas and lakes interact with their coasts to produce various landforms. Oceans, lakes occupy depressions whose formation is linked to continental drift E.g. According to FB Taylor the Mediterranean Sea was formed after the collision of Gondwanaland and Laurasia forming a depression that became the Mediterranean Sea. According to Alfred Wegner, the Red sea emerged from sea of Tethys that developed when Laurasia was beginning to drift from Gondwanaland. Lake Victoria and Kyoga are basin lakes formed in areas of down-warping.
- Waves within these water bodies produce both erosional landforms (e.g. caves, geos, headlands, bays, etc) and depositional features (e.g. spits, bars, tombolos, beaches, lagoons, sand dunes, mad flats, etc). All these features can be seen along the East African coast and on the coast of internal lakes i.e. Lake Victoria.

REVISION QUESTIONS;

Qn. *Explain the relevance of plate tectonics theory in the evolution of landforms in East Africa.*

Approach: *explain the theory; outline the forces; explain the various landforms resulting from (faulting, warping, volcanicity, folding, etc)*

Qn. *To what extent can the theory of plate tectonics explain the present landforms in East Africa.*

Approach: *explain the theory; outline the forces; explain the various landforms resulting from faulting, volcanicity, etc, with illustrations and examples.*

TECTONISM / EARTH MOVEMENTS AND LANDFORM EVOLUTION

- Landforms are large surface features that have particular shape and sizes and are easily identifiable. In East Africa and indeed any part of the world, landforms are as a result of both endogenic and exogenic processes. The endogenic processes operate within the Earth's interior and so the tectonic forces.
- Tectonism refers to all disturbances of endogenic origin that leads to the formation of structural features on the earth's surface. The movement may be slow resulting into mountain building or may be rapid resulting into earthquakes. Tectonism involves both diastrophism and vulcanicity.
- Diastrophism or earth movements describe all forms of faulting, folding, down warping and uplift. The forces may be epeirogenic or orogenic. Epeirogenic forces act vertically i.e. uplift and down warping while orogenic forces act horizontally i.e. compression and tensional.

Origin of tectonic forces

- Tectonic forces have their origin in the interior of the earth. The core and the mantle area of the earth experience very intense heat/temperatures caused by radioactivity, geochemical geophysical

reactions and pressure exerted by the overlying crustal layers. This has kept rocks of the mantle in a semi-liquid form. These molten rocks are highly mobile and move inform of convection current.

- When convective currents **rise outwards** from the mantle, they push the crust above to cause the regional **uplift** and down warp. As the convective currents reach the outer mantle they spread beneath the crust and **flow laterally**. This causes tensional forces on the above crust that may result into **faulting.**
- When convectional currents **move towards each other**, compressional forces are generated which may contribute to faulting and **folding**. When the currents **sink/subducted** into the earth's interior, they exert a pulling force on the crust hence **down-warping** occurs.
- When the earth movements **create lines of weaknesses**, magma (molten rocks) from the mantle rises through the lines of weakness forming intrusive and extrusive **volcanic features**.

FAULTING

This is an endogenic process in which the earth's crust is fractured and the fractured rocks are relatively displaced.

Origin of faulting:

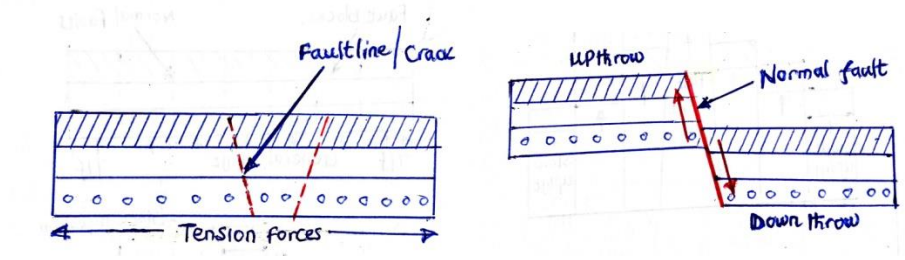
- Faulting has its origin in the interior of the earth. The core and the mantle layers of the earth experience very hot temperatures caused by radioactivity, geochemical reactions and the pressure exerted by the overlying crustal layers. This has kept the rocks of the mantle in a semi-liquid state. These molten rocks are highly mobile and move inform of convection currents. As the currents rise outwards from the mantle, they spread beneath the crust and flow laterally. This causes tensional forces and when they move towards each other, compressional forces are generated. The combination of tensional and compressional forces results into faulting.
- Faulting is a sign of crustal instability and is found in areas experiencing tectonic activities.

TYPES OF FAULTS

There are several types of faults/fractures along which crustal movement takes place and they include:- Normal faults, reversed faults, step faults, tear fault, trap door fault, and anticline fault.

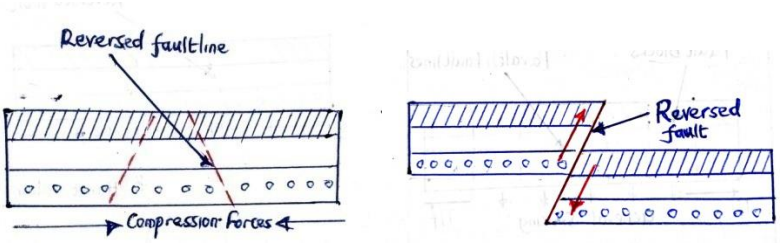
- **Normal faults**: this type of faults is formed when crustal rocks are stretched, fractured and displaced by tensional forces. Examples are Elgeyo and Mau in Kenya, Rukwa rift in Tanzania.

Illustration



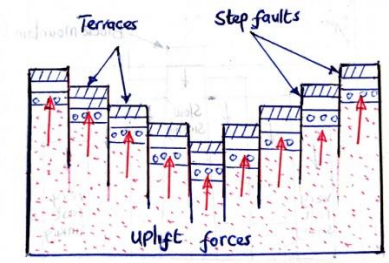
- **Reversed faults**: these faults are produced by compressional forces. They occur when compressional forces cause one block of the crust to thrust/over ride the other opposed block. As the block is compressed, a time will reach when one block will over ride the other. Example can be seen at the Albertine rift valley western Uganda.

Illustration

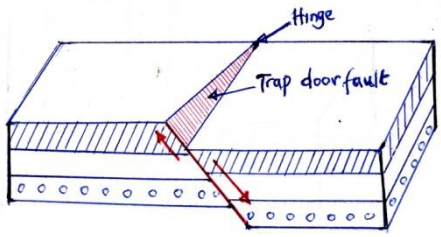


- **Step faults**: These faults are formed where there could be a general uplift on the land with already parallel faults but the rates of the uplift could be different such that some parts/blocks are lifted up faster than others. Several faults develop in the margins of the different uplifted blocks to form step-like landscape. Example can be seen at Kedony valley near Nairobi.

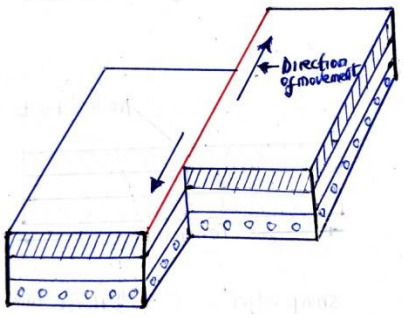
Illustration



- **Trap door / hinge fault:** these faults are produced by tensional forces which cause the crust to fracture but the fracturing does not cut through the whole rock, instead the scarp fault line ran for few meters/kilometers and disappear. This leaves one part of the block attached to the other like a hinge. Example of this fault is evident in the central Tanzania south of Eyasi and Manyara.

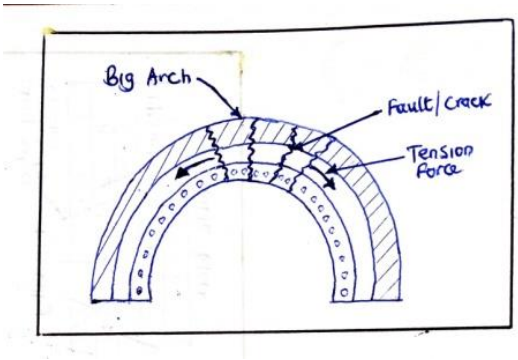


- **Tear faults:** this is formed when horizontal forces acting parallel to each other and opposed direction cause rocks on the either side of the fracture/fault to slide past each other.



- **Anticline faults**: this is formed when a certain region of the crust is subjected to up-arcing/up-bending. As up-bending takes place, tensional forces on the crust produce cracks that are called "Anticline faults".

Illustration



LANDFORMS DUE TO FAULTING

In East Africa, faulting has had a profound effect on the landform formation and it is probably more responsible for the main landforms than any other. The landforms of faulting include; - Block Mountains, Escarpments, Rift valley, Grabens, Tilt blocks, Fault guided valleys.

RIFT VALLEY:

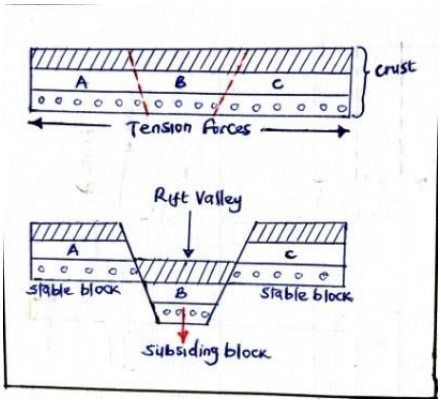
This is an elongated trough/depression bordered by in-facing fault scarps. It usually varies in width from a few hundred meters to several kilometers and it generally has a flat floor.

FORMATION OF THE RIFT VALLEY

The formation of the rift valley has been explained in several theories and these include: tensional forces, compressional forces, differential uplift, relative sinking theory, and basin and swell theories.

- **The Tensional Force Theory:** this theory was advanced by **G.W Gregory** and was based on the Kenyan section of East African rift valley.
- According to this theory, the tensional forces as a result of diverging convectional current pulled the earth's crust apart producing normal faults. This divided the crust into three sections/blocks. The central block subsided under its own weight to produce a rift valley. This is supported by the existence of Elgeyo and Mau faults in Kenya and Songwe fault in Rukwa in Tanzania.
- However the tensional force theory does not explain how the central block composed of light SIAL rocks could sink into the normally denser underlying SIMA rocks.

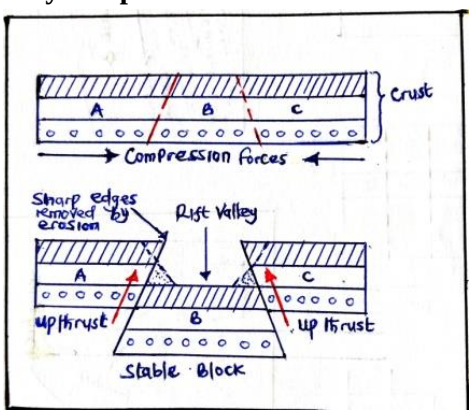
Illustration



- **The compressional force theory:** this theory was advanced by **E.J Wayland**.
- According to him, the earth crust was subjected to compressional forces which pushed the earth crust from either side creating reversed faults. This divided the crust into three block sections. The outer/side blocks over-rode the central block which remained stable to form a trough/rift valley. The over-hanging edges of the side blocks were later removed by erosion leaving a more clear rift valley.

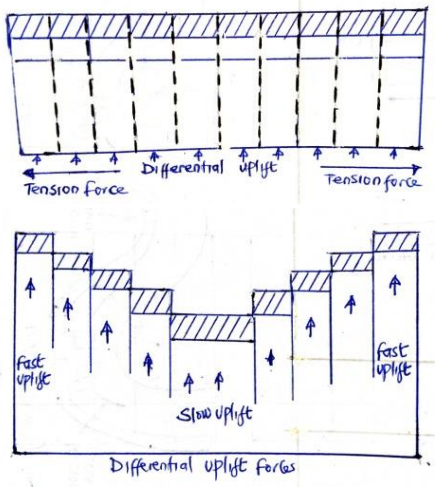
- Wayland's theory has been supported by the existence of folded sediments in areas around Lake Albert and Mont Rwenzori which show the evidences of reversed faults.

Illustration Rift valley formation by compression



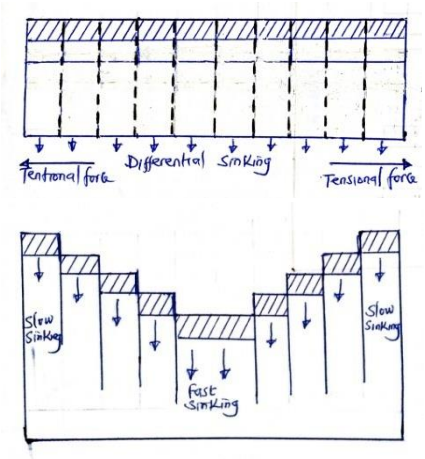
- However, the theory is disapproved on grounds that deep drilling of boreholes in Lake Albert floor has revealed the presence of normal faults. The theory is also disregarded in areas of Kenya and Tanzania where normal fault lines are found.
- **Differential uplift theory:** this theory was put forward by **Dixey** and later supported by **Troup.** The theory is attributed/based on the steep faulted part of the Eastern rift valley near Nairobi.
- According to them, tensional forces led to the formation of several parallel fault lines and this was followed by a general uplift of the faulted region. The uplift was probably caused by the rising convection currents from the mantle. Crustal blocks on the either sides of the central block were uplifted faster and at different rates while the central block lagged behind or remained stable to form a rift valley.

Illustration



- The theory has been supported by the existence of a step-faulted region like Kedony in Kenya however, most parts of the rift valley have step fault scarps which validates the theory.
- **The relative sinking theory**: According to this theory, tensional forces led to the formation of multiple/several faults on the crust creating several blocks. This was followed by a general downward pull of the crustal layers. The downward pull was caused by the sinking convectional currents in the mantle. The central block sunk faster than the side and the extreme blocks to form a step/ terraced rift valley.

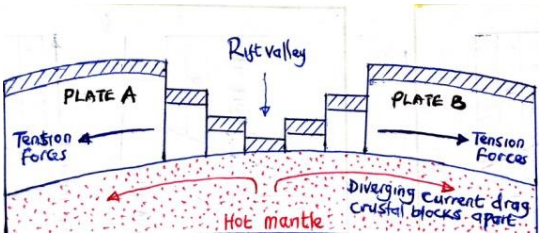
Illustration



- **Seafloor spreading theory**: this theory was advanced by **Girdler** basing on Henry Hess's theory of crustal separation and seafloor spreading.

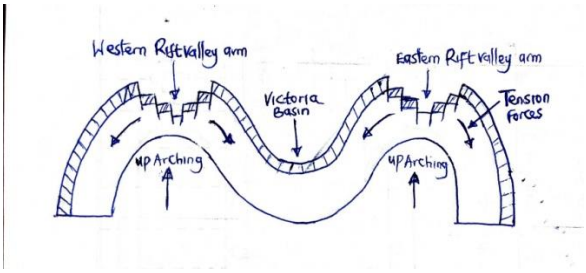
- According to this theory, the crust is made of solid blocks called tectonic plates which float on molten rocks of the mantle. The convective currents in the mantle caused by radioactivity and geochemical reactions may rise up towards the crustal plates and move horizontally dragging with them the crustal plates in the direction of their flow.
- This theory further suggests that, the East African rift valley was formed due to north-east drift/movement of the east African plate.
- It also suggested that the formation of the East African rift valley represents the first stage in the breaking away/separation of the East African plate from the African landmass. This theory is supported by the widening of the east African rift valley at a rate of 2cm per year.

Illustration



- **The basin and swell theory**: this theory was advanced by **L.C King**.
- According to him, the East African rift valley was produced by the process of up-arching /bending/swelling caused by the vertical forces to which Central and East Africa was subjected during Pleistocene period. As the uplift forces created big arches on the crustal layers, tensional forces developed at the top of the arches/anticlines leading to cracks on the arches. As the up-arching continued, the cracks also continued widening and finally formed a rift valley.

Illustration

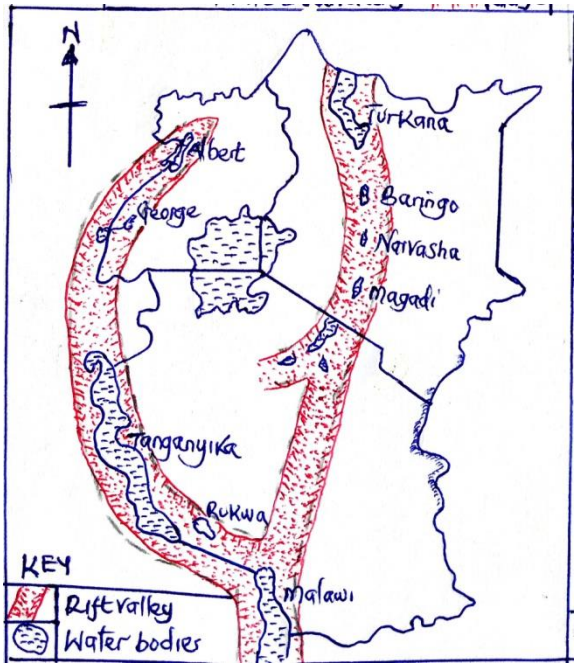


NOTE: despite the differences and similarities in the theories advanced by different geomorphologists, they all agree that the rift valley has greatly been modified by denudation, vulcanicity, secondary faulting to form its current shape/appearance.

THE EXTENT OF THE EAST AFRICAN RIFT VALLEY

The East African Rift Valley is the most impressive in the world with a width varying from between 40-80km and a distance of about 5,600km. It extends from Lake Turkana in the north and runs southwards until it reaches Lake Malawi. At Malawi, it divides it self into the western arm which runs northwards and ends out at the northern shores of Lake Albert.

The sketch map of East Africa showing the extent of the Rift Valley



SIGNIFICANCE OF THE RIFT VALLLEY

- The rift valley provides attractive scenery to the tourists e.g. lakes within the rift valley floor such as Albert, etc, escarpments, etc. these earns foreign exchange into the east African countries. This is used to develop other infrastructure like roads, hospitals, schools, etc.
- The rift valley and other associated features are sources of minerals such as soda ash extracted from Lake Magadi, salt from Lake Katwe, oil drilling in Lake Albertine region in western Uganda, limestone at Muhokya in western Uganda. All these promote the mining activities.
- The rift valley provides fishing grounds in the rift valley lakes such as Albert, Edward, George, Tanganyika, etc, from where fish species such as Tilapia, Nile Perch, Dagaa, etc, are caught. Fish provide food to the people and is also a source of income to the local people when sold.
- Large rift valley lakes such as Turkana, Tanganyika, etc, influence the climate of the surrounding areas through the facilitation and formation of convectional rainfall and land and sea breezes. This promotes crop cultivation to provide food to the local people.
- Graben lakes within the rift valley especially along the borders have facilitated transport and communication between the East African countries and her neighbouring countries such as Uganda and DRC through lake Albert, Tanzania and DRC through lake Tanganyika, Tanzania to Malawi and Mozambique through lake Malawi.

- The flat relief floor with fertile soils facilitates large scale farming through the use of machines and irrigation farming e.g. Mobuku prison farm is maintained by Mobuku irrigation scheme in western Uganda where cotton and other crops are grown.
- The rift valleys are grounds for research. A rift valley is one of the current indicators of continental drift. It is used as a reference in studying earth movements and continental drift hence promoting research in primary, post primary and higher levels of learning institutions,
- The rift valley escarpments which receive heavy rainfall favour the growth of forests giving rise to a flourishing forestry industry e.g. the Rwenzori forests in western Uganda.
- Hot Springs and Geysers are commonly found in the rift valley due to their association to faulting. These are harnessed for geothermal power production. In Kenya these are harnessed to produce geothermal power at Ol-karia valley.
- Similarly, rivers flowing along escarpments before plunging down into the rift valley floor forms water which can be harnessed to generate hydro power e.g. Murchison and Karuma falls along river Nile.
- The flat section of the rift valley floor offer an advantage to easy construction of buildings hence attracting settlement and urbanization e.g. Kasese town in western Uganda stands on the rift valley floor at the foot hills of mount Rwenzori.
- The rift valley provides habitat to wildlife conservation e.g. Queen Elizabeth and Rwenzori national park are all in the western rift valley in Uganda. In Kenya, the national parks in the rift valley are; Sibiloi near Lake Turkana, central and south island in Lake Turkana, Amboseli, Lake Bogoria, etc.

However,

- The Steep escarpments bordering the rift valley restrict transport facilities for example the Uganda railway had stalled in at Nairobi near the rift valley. The Butiaba escarpment near Lake Albert and the steep escarpment along the lake have hindered the development of transport and communication making the area remote.
- The rift valley areas are infested with pests such as tsetse flies which transmit Nangana disease to cattle and sleeping sickness to man. Examples of such areas infested are Bundibugyo, Hoima, Bwera and the Semiliki in western Uganda.
- The rift valley is an area of crustal instabilities since earthquakes often strike those areas e.g. Kasese and Bundibugyo. Volcanic eruptions are also associated to these regions. This causes loss of live and property incase any occurs.
- Landslides and general mass movements are also common in the rift valley areas. These are promoted by the steep fault scarps along the rift valley. These are common in Bundibugyo in Uganda, Kerio valley, Lokori and Kajaido in Kenya destroying settlements, transport and communication networks, etc.
- The rift valley areas are associated with poor soils which are not suitable for crop farming. The rift areas lie at a low altitude hence discouraging rainfall formation. Low rainfall limits agricultural activities resulting into persistent famine
- Flooding around the rift valley lakes is a common feature especially around lakes such as Albert, George, Edward and Malawi. This affects fishing activities and displacing settlement along these lakes.

REVISION QUESTIONS

THE BLOCK/HORST MOUNTAIN

Block Mountains are some times called Horst Mountains. A block/horst mountain is an up-land bordered by fault scarps on one or all of its sides. It generally stands above the surrounding land with sharply defined edges. The horst mountain is associated with a rift valley and probably they may be formed together.

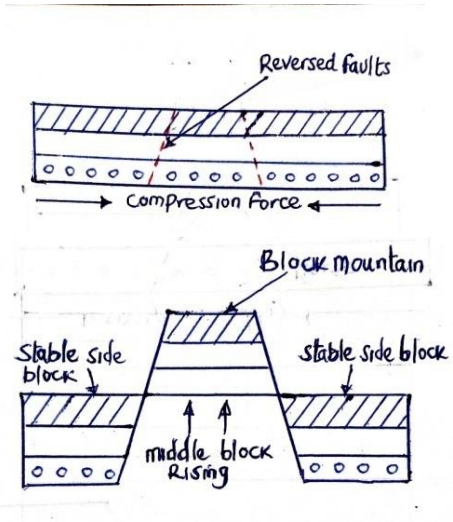
- Examples of Horst Mountains in East Africa are: Rwenzori, Buhweju hills, Bunyarunguru uplands, Apac hills, Masindi hills, Bunyoro-Toro hills all in Uganda, Usambara, Ulunguru, Pare, Mbeya range Kungwe, Iramba, Iringa and Ufipa in Tanzania and Aberdare ranges, Ndoto, Nyiru, Mathew ranges, all in Kenya.

FORMATION OF A HORST MOUNTAIN

Horst Mountains are generally as a result of faulting brought about by the action of convectional currents operating in the mantle of the earth. As the currents rise upwards from the mantle, they exert a pushing force on the crust above which may result into **differential uplift.** When currents move away from each other, **tensional forces** are created and when currents converge it leads to **compression forces**. As the currents are subdued/sink into the mantle they pull with them the crust above leading to **relative sinking**.

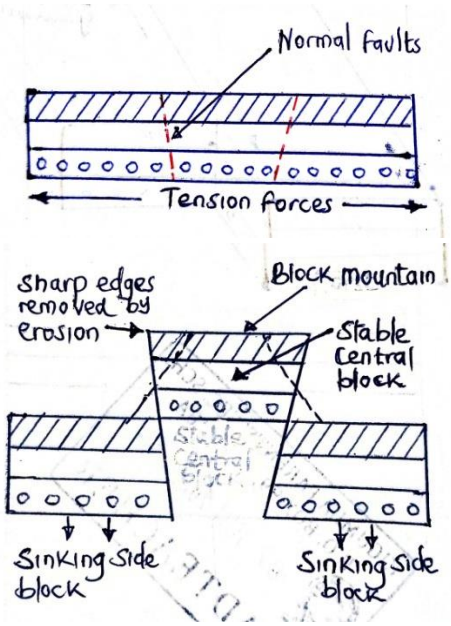
Therefore there are four major processes responsible for the formation of Horst Mountains. These theories are: compression forces, Tensional forces, Differential uplift theory and the Relative sinking theory.

- **Compressional forces theory**: this theory suggests that compressional forces as a result of converging currents, divided the crust into three blocks. As the action of compression continued, the central block was forced to uplift to ever-ride the side blocks forming a block/horst mountain while the side blocks remained stable below.



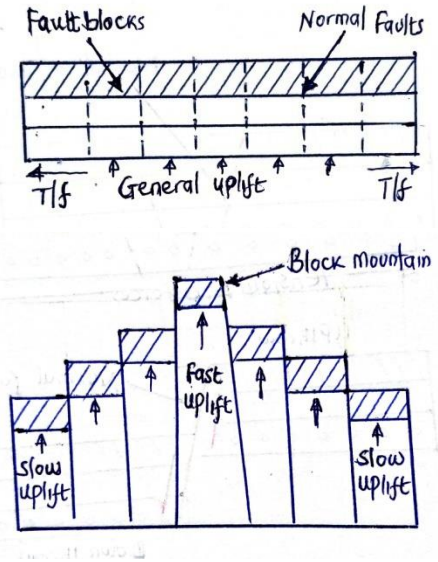
- **Tensional forces theory**: this theory suggests that the tensional forces as a result of diverging convectional currents pulled the crust a part to produce normal faults. These faults divided the crust into three blocks. As tensional force continued it led to the subsidence/ sinking of the side blocks leaving the central block stable and standing above as a horst mountain.

Illustration



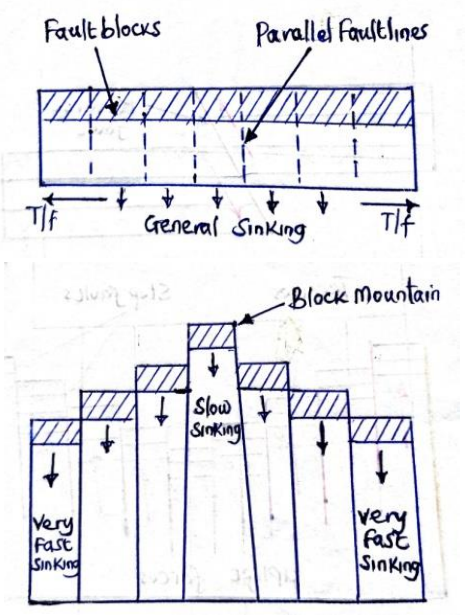
- **Differential uplift theory**: according to this theory, tensional or compressional forces led to the occurrence of parallel multiple faults on the crust. This created a series of crustal blocks of varying sizes and densities. This was followed by the general uplift of the faulted area probably caused by the rising convective currents. But the uplifting forces acted on the crustal blocks with varying strengths. The strongest force pushed the central block higher than any side blocks to form a horst mountain. The side blocks remained below in steps to form the sides of the horst.

Illustration



- **Relative sinking theory**: this theory suggests that tensional or compressional forces led to the development of multiple parallel faults that divided the crust into several blocks. General sinking of the fault blocks probably due to sinking convective currents occurred but at different rates. The side blocks sunk faster and deeper while the central block remained relatively higher to form a horst mountain.

Illustration



SIGNIFICANCE OF HORST MOUNTAIN

(Case study on mount Rwenzori in western Uganda)

- Rwenzori mounts has got magnificent scenery which attract tourists to East Africa. These attractions include 37 glaciers, Margherita, and Bakers peaks; glacial features such as Bujuku valley, cirques, arêtes, tarns, etc. all these earn foreign exchange to Uganda later used to develop infrastructure.
- Rwenzori Mountains provide a home for wildlife conservation especially within the Rwenzori Park. This is important for protection of biodiversity for future generation to see.
- Rwenzori mountains is a source of important rivers such as Mubuku, Mpanga, Bujuku, etc. these rivers are a source of water for domestic and industrial use in urban centres of Kasese, Kilembe and Hima cement industries.
- Rwenzori mountain is a source of rivers which have been harnessed to provide HEP e.g. river Mobuku has a power station. This power station has facilitated the growth of industries such as Kasese Cobalt plant in Hima cement factory.
- Rwenzori Mountain is rich with minerals such as copper and cobalt in Kasese. This has promoted mining activities e.g. cobalt recycling at Kasese cobalt plant along the slopes of mount Rwenzori. Plans are underway to resume copper mining at Kilembe because copper prices in the world market are rising.
- Rwenzori Horst Mountains contains tropical rainforests at the lower slopes e.g. the Rwenzori forest reserve. This has promoted forestry industry and some lumbering under controlled levels.
- Rwenzori Mountains influences the formation of heavy rainfall received on the windward side by trapping the moist Congo air stream. Much of this rainfall is received west of the mountain in the DRC side and south of the mountain for Uganda side. This has favoured forestry and arable farming of the surrounding areas such as Bwera and Nyabirongo.
- The lower slopes of mount Rwenzori horst have fertile soils partly due to alluvium deposited by rivers and general soil erosion from the top of the mountain. This fertile soil supports crop growing e.g. cotton, vegetables, rice, in Mobuku, Kasese, Muhokya, etc.
- Rwenzori is used for research and study purposes such as field work, geophysics, seismology, etc, which promotes education.

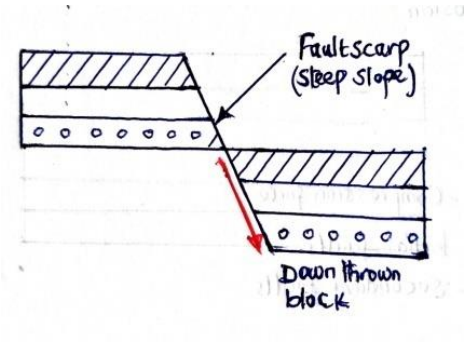
However,

- Mount Rwenzori has steep slopes which limit construction of transport and communication lines. This has created accessibility problems in areas around Kasese, Nyabirongo, Bwera,
- Rwenzori creates a rain shadow effect in Kasese, Semiliki, and the rift valley region because the winds which descend on the leeward side are dry and promote aridity.
- The steep slopes of mount Rwenzori are prone to landslides which have destroyed lives and property in areas of Bwera, Nyabirongo and Kasese.
- Forested slopes of mount Rwenzori have been used for along time by rebels to encourage political instability across the borders of Uganda, DRC, Rwanda, Burundi for example the ADF rebels use the forests as their hiding bases, the Intarahemwe from DRC use the western side of the mountains.
- The Rwenzori mounts have steep slopes with rugged soils which reduce land for cultivation and settlement e.g. areas around Kilembe, Bundibugyo, etc.
- Accumulation of snow/ice and glaciers on mount Rwenzori has caused very cold conditions which discourage settlement and coldness is associated with diseases such as pneumonia.

FAULT SCARP

This is a steep feature/slope produced when one of the fault blocks on one side of the fault line sinks downwards along a line of fault to form a steep slope which is fresh and not distorted by weathering and erosion. Fault scarps occur in areas of resent faulting. Examples of fault scarps are; Butiaba and Kichwamba in western Uganda, Elgeyo and Nyando scarp in Kenya and Chunya scarp in Tanzania.

Illustration



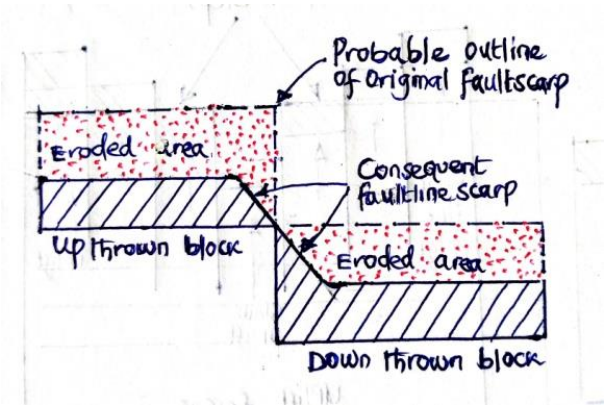
FAULTLINE SCARP/ESCARPMENT

This is a steep slope produced when the ancient/old fault lines have been greatly distorted by the forces of denudation (weathering and erosion) on resistant and less resistant rocks of either sides of the faults. It usually happens when faulting brings rocks of different resistance into a close just a position so that denudation will concentrate on the line of fault to form a new steep slope known as fault line scarp.

There are two types of fault line scarps namely: - normal/consequent fault line scarp and reversed/obsequent fault line scarp.

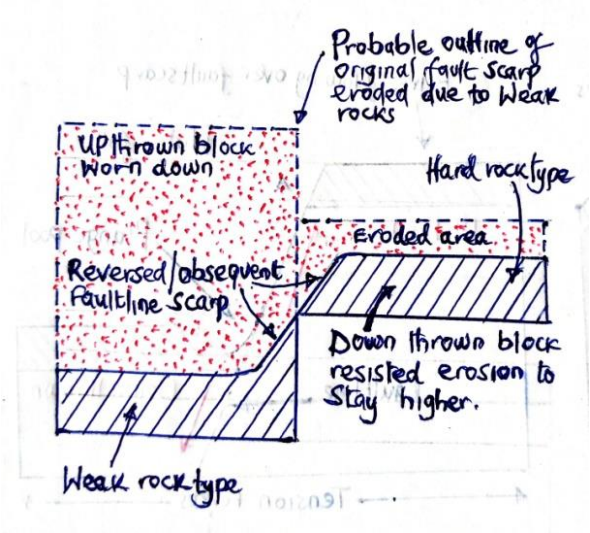
- **Consequent fault line scarp** is produced when the original fault line scarp is modified by weathering and erosion to almost a level surface with a new rock type and the scarp remains facing the same direction as the original fault scarp.

Illustration



- **Obsequent fault line scarp** is produced when sometimes rocks on the down thrown block is much resistant to erosion that the up thrown block is eroded faster to form a lowland. This produces a scarp facing the opposite direction to the original fault scarp.

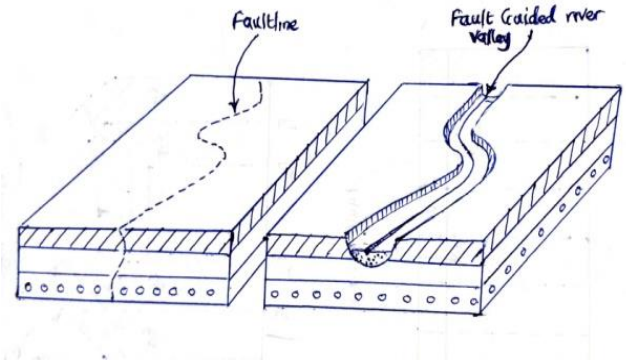
Illustration



FAULT GUIDED VALLEY:

As the name suggests these are valleys which develop where already fault lines exist such that they follow the pattern of the fault lines. As faulting takes place, rocks are displaced along the fault lines. This makes the rocks on the shattered zone to become relatively weaker hence are easily weathered and eroded. With time, a valley is formed following the pattern of the fault line. Often such fault guided valleys are occupied by rivers e.g. river Aswa, Wassa valley between mount Rwenzori and lake Albert in Uganda Ruaha valley in Tanzania, Kerio valley, river Ewaso, Melawa valley in Kenya.

Illustration



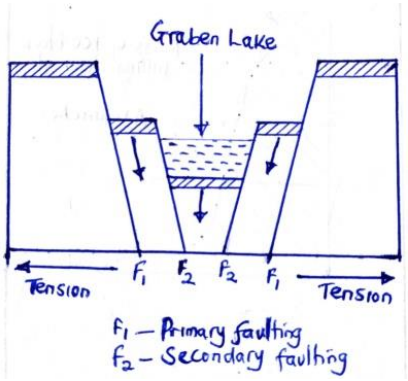
GRABENS/RIFT VALLEY BASINS (LAKES):

Grabens are depressions found at the rift valley floor. These basins are usually narrow, deep, regular in shape and elongated. They are formed due to tensional, compressional and differential uplift forces.

Formation of the Graben lakes through tensional forces

- Under tensional forces theory, the diverging convective currents pulled the crust apart producing normal faults that divided the crust into three blocks. As tensional forces continued, the middle block sunk under its own weight to form a rift valley. Later further actions of tensional forces caused secondary faulting on the floor of the rift valley and formed a secondary depression known as a Graben.
- These Grabens/depressions were later filled with water from rain, springs and rivers to form rift valley lakes. Examples include lakes such as Albert, George, Edward in Uganda Tanganyika, Rukwa, Malawi in Tanzania, Turkana, Baringo, Magadi, etc in Kenya. These lakes follow the pattern of the depressions and are sometimes salty.

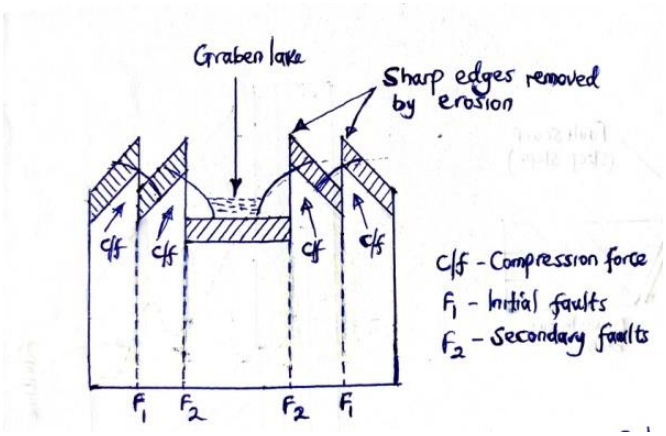
Illustration



Formation of Grabens through compressional force:

- According to compression theory, converging convective currents pushed the crust from either side creating reversed faults. Reversed faults divided the crust into three blocks. As the compressional forces continued the side blocks were forced to over-ride the central block to form a rift valley. Further action of compressional forces led to secondary faulting on the floor of the rift valley producing a graben. These Grabens were later filled with water from rain, rivers etc to form rift valley lakes.

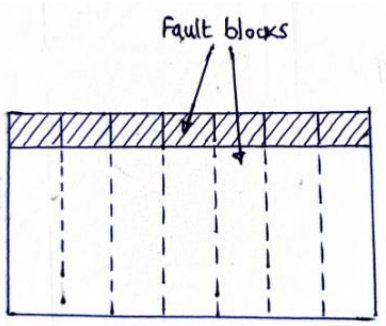
Illustration



FAULT BLOCKS:

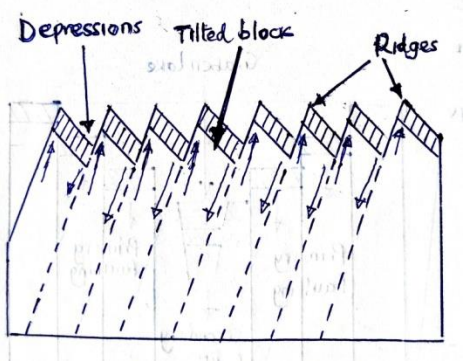
These are formed when faulting results into the earth's crust being divided into individual elevated and sub-divided masses of blocks.

Illustration



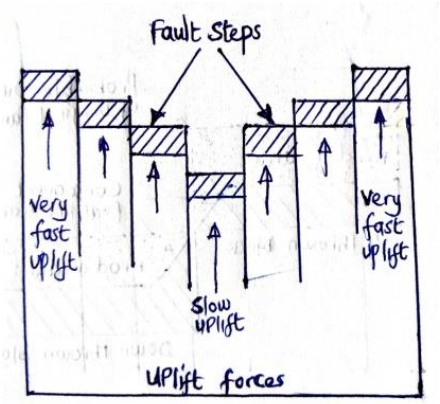
TILT BLOCKS/ TILTED LANDSCAPE:

A tilt block is an Up-land of inclined crustal blocks with angular ridges and depressions. They are formed when tensional or compressional forces generated by convectional currents create multiple/several parallel fault lines on the crust. This is followed by general uplift of the faulted region. However, the uplift is not uniform on the either side of each fault block such that one side of each block rises higher than the other. As a result, blocks form different heights. If many blocks are tilted in the same area, it forms a tilted landscape. Examples can be seen in Aberdare ranges in Kenya, Kichwamba in Uganda.



FAULTSTEPS/TERRACES:

These are step-like landscapes that occur on the edges of the rift valley. They are formed due to differential uplift of the faulted landscape. A good example is at Kedony near Nairobi



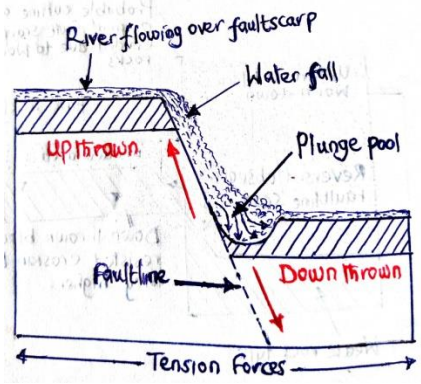
INFLUENCE OF FAULTING ON DRAINAGE IN EAST AFRICA

Drainage refers to all surface water like lakes, rivers, swamps, etc. In East Africa faulting has had a great significance on rivers and other and other drainage systems as seen below:

- **Faulting and water falls**: a water fall is a sharp break along a river channel where water falls from a higher elevation to a lower one at a high speed. Most of the water falls in East Africa were formed where faulting has occurred. Tensional or compressional forces may cause one of the fractured and jointed blocks of the river bed to be down thrown leaving the other one at a higher

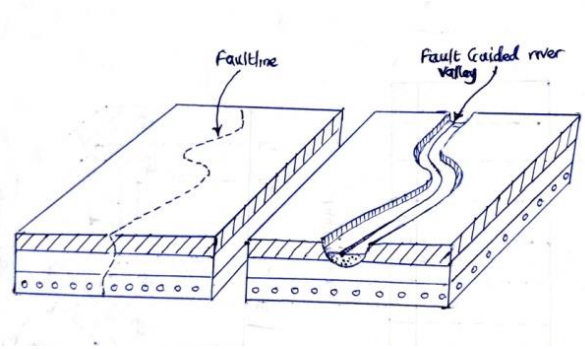
elevation. At this point water falls from a fault scarp to a lower surface forming a water fall. E.g. Murchison falls, Kizisi falls, etc.

Illustration



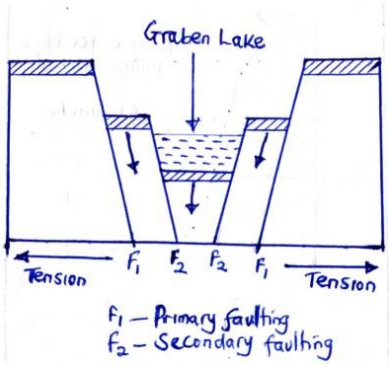
- **Faulting and guided river valleys**: faulting has also led to the development of guided river valleys. During faulting, the faulted zones became more prone to weathering and erosion creating a valley following the fault line. Later water passed through the river valley forming fault guided river valley. Examples are river Aswa in northern Uganda.

Illustration



- **Faulting and the Grabens**: Faulting has also led to the formation of the graben/rift valley lakes. Grabens are formed due to secondary faulting on the rift valley floor resulting into a secondary depression called a graben. This is later filled with water to form a graben/rift valley lake. Examples of such lakes are Albert, Tanganyika, Rukwa, Malawi, Eyasi, Natron, in Tanzania, Turkana, Magadi, Baringo, Naivasha, in Kenya.

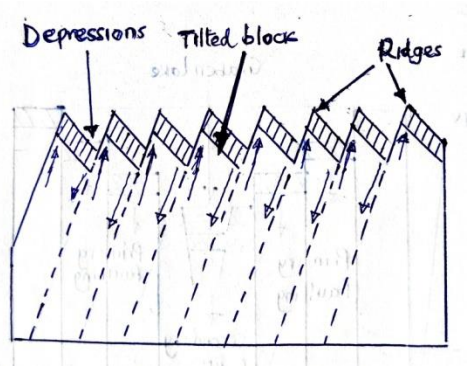
Illustration



- **Faulting and tilt blocks lakes**: Faulting has also resulted into the formation of tilt block lakes. A tilted block is an up-land of inclined/tilted crustal blocks with angular ridges and depressions. It's formed when one side of the fault block is uplifted higher than the other. When a series tilted blocks occur in the same faulted region they form a tilted landscape with ridges and depressions.

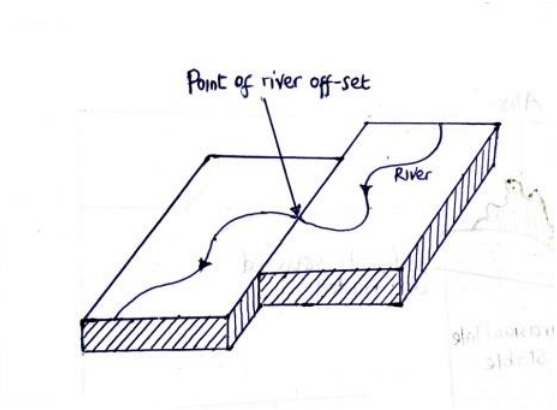
The depressions may later be filled with water to form tilt block lakes. Example is Lake Ol-bolsat in Aberdare ranges in Kenya.

Illustration



- **Faulting and river valley rejuvenation**: River rejuvenation refers to the renewal of erosive activity of a river. Faulting has also influenced river rejuvenation in that if one part of crust of the river bed slips down in relation to the other, a river flowing on such a landscape will experience a renewal of its erosive activity hence rejuvenation. The result is the formation of rejuvenated landscape with features like gorges, valley in valley river terraces, etc.
- **Tear faulting and river off-sets:** Faulting has also led to horizontal/lateral movement of faulted blocks across a river valley causing a river to be off-set at a point it crosses the fault lines.

Illustration



- **Faulting and radial drainage**: Faulting is indirectly responsible for radial drainage on Mountains through the formation of Block Mountains. These mountains experience frequent orographic rainfall and are sources of many rivers and streams. Mount Rwenzori is characterized with radial drainage pattern with many rivers radiating from its top and such rivers are; Mubuku, Mpanga, Sebwe, Nyamwamba.
- **Faulting and river reversal**: Faulting is also responsible for river reversal in the western part of Uganda. During faulting, the western Uganda under went uplift while the central plateau down warped to form Victoria basin. Rivers that were flowing westwards to the Congo basin such as Kagera, Katonga, and Kafu were reversed to flow into Victoria basin to form Lake Victoria. The slight uplift in the east forced rivers flowing eastwards to reverse into Victoria basin and these rivers are Mara and Nzoia.

IMPORTANCE OF FAULTING TO MAN

(**Do research on the following**)

- Supports tourism – earns foreign exchange thru attractions e.g. Horst Mountains etc,
- Supports mining - soda ash on lake Magadi and salt in lake Katwe,
- Supports wildlife conservation – Tsavo N/P, Queen Elizabeth N/P, etc,

- Supports pastoralism – Masai in Kenya,
- Potential sites for HEP – Murchison falls, etc,
- Supports fishing – Dagaa fish in Lake Tanganyika, etc,
- Supports water transport – Lake Tanganyika, Lake Albert, etc
- Supports research – mount Rwenzori, continental drift, etc,
- Support settlement – flat areas of the rift valley floor e.g. Kasese town, etc,
- Influences climate modification – easterlies from eastern Congo – support agriculture and forestry,
- Steep slopes - hinder construction of transport and communication lines,
- Faulted areas prone to earthquakes – loss of life and property,
- Steep slopes - promote soil erosion – infertile soils,
- Steep slopes - promote landslides – destruction of life and property,
- Causes rain shadow effect – aridity – Queen Elizabeth, Semiliki region in western Uganda.

REVISION QUESTIONS

FOLDING

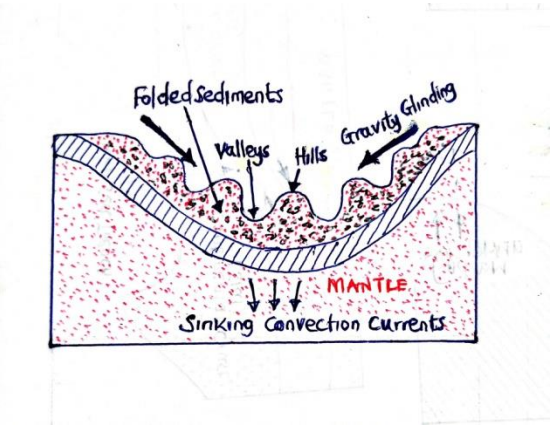
Folding is the bending or crumpling of rock strata caused by compressional forces. It occurs mainly in the sedimentary rocks since they are young and exhibit some elastic state i.e. they can bend without breaking when compressed.

Origin of folding

Folding is basically caused by tectonic compression and gravity gliding**.**

- **Tectonic compression** forces have its origin in the interior of the earth. The core and the mantle experience intense heat generated by radioactive and geochemical reactions which keeps the rocks of the mantle in a semi molten state. The molten rocks are so mobile and move inform of convection currents. Convectional currents rise upwards from the mantle and when they reach beneath the crust they flow laterally before sinking back into the mantle. When convectional currents move towards each other, they create compressional forces within the crustal plates. This results into folding of the crustal layers as long as they are composed of young rocks and elastic
- Secondly, **gravity gliding** may have led to folding. Gravity gliding occurs when sinking convectional currents in the mantle pull the crustal plate downwards forming a down warped basin. The rocks strata on an inclined surface slides down slope in a down warped surface. Due to friction and resistance during sliding, the rock strata folds to form anticlines (hills) and synclines (valleys).

Illustration



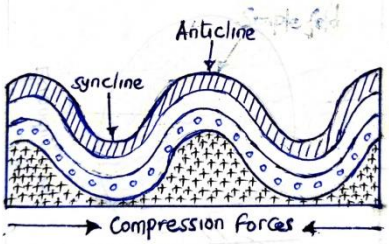
In east Africa folding occurred in Ankole, Karagwe, Toro and Buganda (hills), in Tanzania it occurred in Dodoma, northern Tanzania and in Kenya it occurred in Nyanza – Kavirondo area.

TYPES OF FOLDS

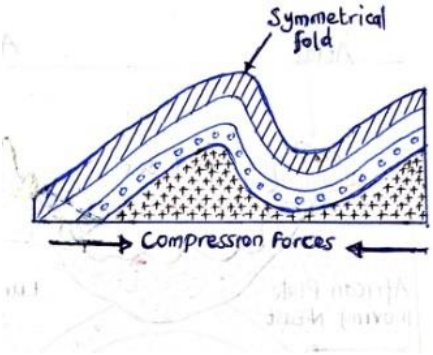
The different types of folds are caused by the difference in the intensity of convectional forces subjected to the crustal layers namely:

- **Simple folds (symmetrical folds);** these are folds that are produced by gentle compression that forces the rock strata/layers to develop simple up-folds/anticlines and down-folds/synclines.

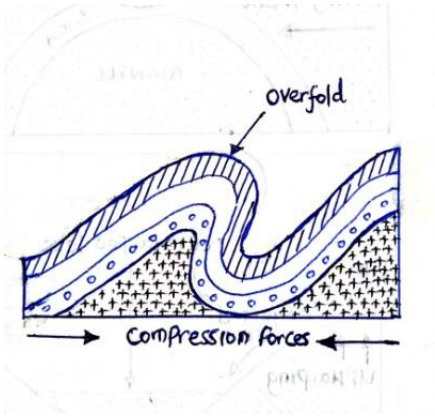
Illustration



- **A symmetrical folds**; these are produced by an increase in the degree of compression and especially when the pressure on one side is greater than it is on the other side. This creates one side of the up-fold/anticline steeper than the other hence symmetrical folds are not uniform. E.g. folds in Kigezi, Kampala-Mityana road, Arua, Buganda-Toro rocks, etc

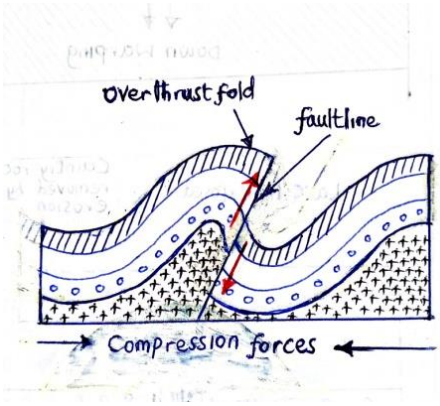


- **Over fold**; these are produced when the degree of compression is very high due to high pressure such that the folds so much that the anticline/crest is pushed over the syncline.



- **Over thrust fold**; these is produced when the degree of compression is very intense than in the over fold such that faulting occurs in the over riding anticline/crest. The anticline/crest is separated and lifted over the other.

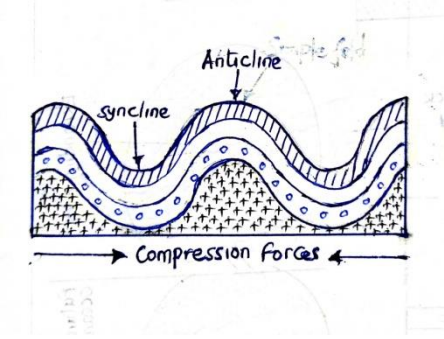
Illustration



FEATURES OF FOLDING

- **Anticlines and synclines**; these are features formed by a small scale and gentle compressional forces. Anticline is the crest/upland/hill while syncline is the depressed part/valley. These features are commonly found in Ankole and Karagwe.

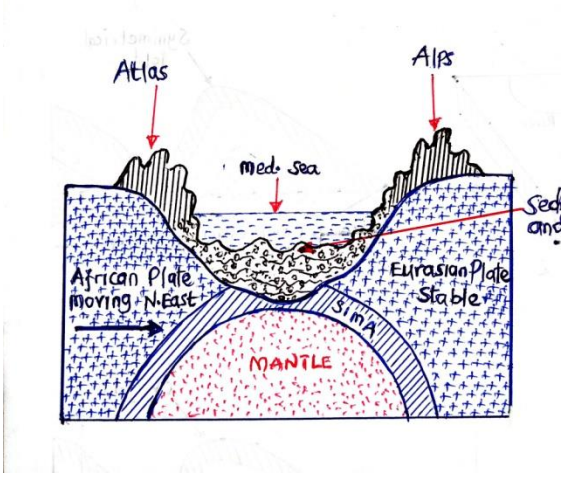
Illustration



- **Fold mountains**; these are formed as a result of compressional forces acting on young sedimentary rocks. Rivers and glaciers over the years deposit sediments into the seafloor whose weight causes the sea floor to subside and form a deep basin called geo-syncline. When continental plates move towards each other or oceanic plates move towards each other as a result of converging convectional currents, compressional forces are generated this squeezes sediments accumulated in the geo-syncline into Fold Mountains. Fold Mountains have been formed through the above processes world over.

- The Atlas Mountains in Africa and the Alps mountains in Europe were formed as a result of collusion between Eurasian plate and African plate that was advancing northwards. The sediments that were accumulated in between the geosynclines (Tethys) were squeezed and compressed into Fold Mountains.

Illustration



- Elsewhere in the world, the Andes in South America were formed through the collision of the South American plate with the Nazca plate,
- The Himalayas in India were formed due to the compression of sediments after the collusion of the Indian-Australian plate and Eurasian plates, the Juu-De-Frun sunk under the North American plate and formed the Rocky Mountains in North America.

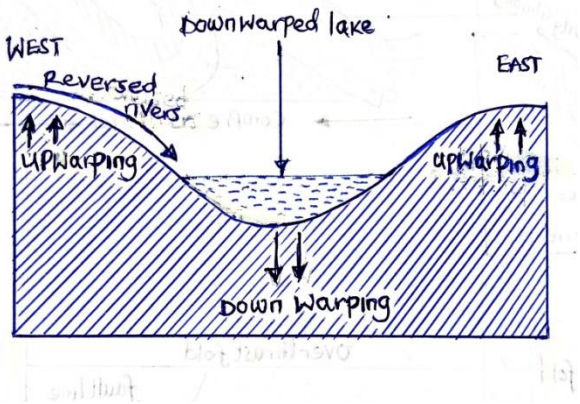
WARPING

This is the distortion of the earth's surface by an even upward and downward movement of the crustal layers. The forces involved are too small that the movement or uplift leads to formation of plateaus/uplands while the downward movement results into forming depressions/basins.

Origin of warping

- Warping has its origin in the interior of the earth. The core and the mantle layers of the earth experience very intense temperatures caused by the radioactivity, geochemical and geophysical reactions together with the pressure exerted by the overlying crustal layers. This has kept the rocks of the mantle in a semi-liquid state. The molten rocks are highly mobile and move inform of convection currents. When convection current move upwards from the mantle, they push the crust above to cause regional uplift/up warping. When the convectional currents are being subdued into the earths interior they exert a pulling force on the crust to form depressions/basins.
- Down warping took place in central Uganda during the formation of the Great east African rift valley. The eastern and the western axes were uplifted which left the land between the central Uganda plateaus greatly down warped.
- The lowest parts of the depression were filled with water from reversed rivers and other small streams that previously flowed westwards to form Lake Victoria and Kyoga. Lake Victoria was formed due to the reversal of river Katonga and Kagera while Lake Kyoga was a result of back ponding of river Kafu. The filling up of Lake Kyoga continued and gradually the water over flooded to form the Nile river system.

Illustration



- Down warping on the East African plateau was also responsible for the formation of the Tana plains of Kenya, a region now with sediments and the Wambare down warped region with its fold margins in Tanzania.
- Down warping and sediments in-filling has also occurred in Lake Amboseli basin and in the upper Pangani river valley.
- Down warping has also caused flat plains which are now drained by tributaries of Wani and Kilombero rivers.

VULCANICITY

Vulcanicity refers to the total process by which molten rocks (magma), solids and gaseous materials are intruded into the earth interior or extruded onto the earth's surface forming intrusive and extrusive features respectively.

Origin of vulcanicity

- Vulcanicity has its origin in the interior of the earth where radioactivity and geochemical reactions produce intense amount of heat. This heat is additionally generated by friction between moving rocks or collusion of plate boundaries and also from the weight exerted on the mantle by the underlying rocks. The enormous/intense heat generated contributes to very hot temperatures of over 4,500 degrees Celsius which practically keep the mantle rocks in either a liquid or semi-liquid state but under high pressure. When the pressure is relieved due to crustal instabilities such as faulting or other earth movements, the supper heated rocks begin to move upwards through lines of weaknesses(fault lines) created by faulting on the crust.
- As the magma flows upwards, it expands and becomes lighter. When magma reaches the earth surface, it loses its gaseous contents and forms lava which later cools down to form extrusive or volcanic features. When the magma cools in the crust before reaching the earth's surface, it forms intrusive/plutonic features which may later be exposed by denudational processes.

THE NATURE OF VULCANIC LANDFORMS/ FEATURES

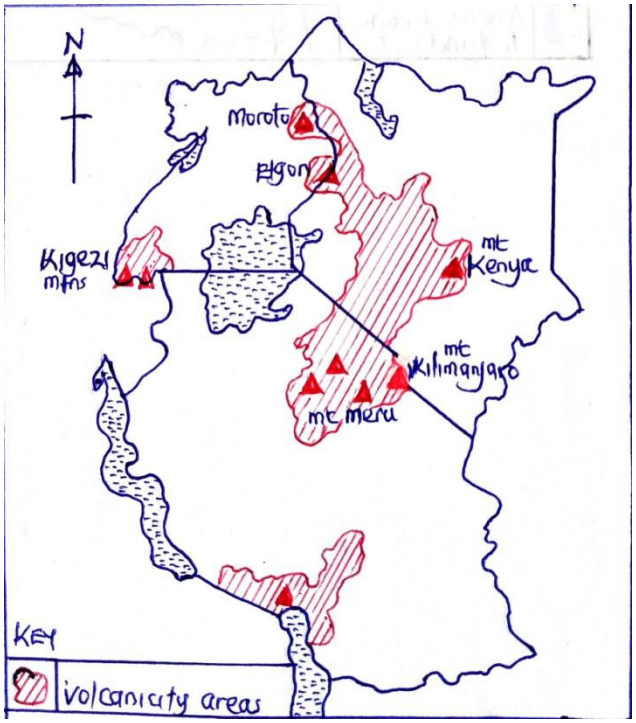
- The nature of volcanic landforms, type, their shape and characteristics depends on the magma/lava ejected. Magma/lava may be acidic or basic and the degree of acidity and basisity depends on its amount of silica content.
- Basic lava is poor in silica content but rich in magnesium and iron. This lava is very fluid, light and cools at low temperatures. On the surface the lava tends to flow covering a large area and later cools down forming gentle rolling plateau or gentle sloping volcanoes.
- Acidic lava is rich in silica content and solidifies at very hot temperatures. On the surface it cools down immediately forming steep-sided volcanic features.
- Acidic and intermediate lava, because of their viscosity may block fissures which may result into very violent eruption. The characteristics of lava types can be summarized as below.

Type of lava	% of silica content	Degree of mobility	Example of rocks formed
Acidic lava	66	Extremely viscous and immobile, solidifies at very hot temperatures	Granite, Rhyolite
Intermediate lava	52-66	Fairly viscous unable to flow for long distance before solidifying	Diorite, Trachyte, Andesite
Basic lava	45-52	Very fluid and mobile, able to flow for long distance before solidifying	Basal, Gabro,

VULCANICITY IN EAST AFRICA

East Africa is a region whose landscape has been seriously affected by vulcanicity and the area stretches from Ethiopian highlands through Kenya and Tanzania as well as Madagascar. The most important period/time East Africa experienced large-scale vulcanicity was from Miocene (25-50million years ago) up to Holocene (recent). The last eruption occurred in 1966 from mount Oldonyo Lengai in Tanzania. However, out side East Africa, there are other more recent eruptions e.g. Nyamulangira (1981) and Nyirangongo (2002) in eastern DRC.

SKETCH MAP OF EAST AFRICA SHOWING THE DISTRIBUTION OF VULCANICITY



LANDFORMS OF VULCANISM

Volcanic landforms are divided in to two namely; intrusive and extrusive landforms.

EXTRUSIVE /VOLCANIC LANDFORMS

Extrusive/volcanic landforms are formed through volcanicity process. Volcanicity is a total process through which lava; other solids and gasses are ejected from the interior and deposited on to the earth's surface forming extrusive volcanic landforms. These landforms vary in sizes from small cones to big mountains and wide spread lava plains.

The formation of these landforms depend on the following factors

- **The nature of the lava ejected**: Basic lava has low silica content between 45-52% and is freely mobile. It flows for along distance before solidifying, resulting into landforms which are gently sloping. Acidic lava on the other hand has thigh percentage of silica content above 66% and cools rapidly at very hot temperatures resulting into landforms characterized by steep slopes.
- **Secondly the nature of eruptions**: This can either be vent or fissures eruptions. Vent eruptions are highly explosive and results into the formation of high volcanic features. The erupted materials usually accumulated around the vent or the central opening to build steep landforms. Fissures eruption occurs when magma flows through several lines of weaknesses. This yields landforms with relatively large basal diameter of a low altitude.
- **Thirdly the number/times of lava emissions**: this can either be successive eruptions or single eruption. Successive eruptions lead to the formation of complex composite volcanoes like Muhavura and Mount Kilimanjaro. Single eruptions lead to formation of small volcanic cones.

Extrusive volcanic features/landforms include the following:

VOLCANOES:

A volcano is a hill or mountain formed by eruption and pilling up of magma around a central opening or a vent until a highland is built. At the exit of the vent a volcano usually the volcano continues to build as long as the supply of magma is guaranteed but when pressure reduces magma flows through alternative routes to build parasite cones on the sides of the main volcano. Magma may also solidify within the vent to form a volcanic plug.

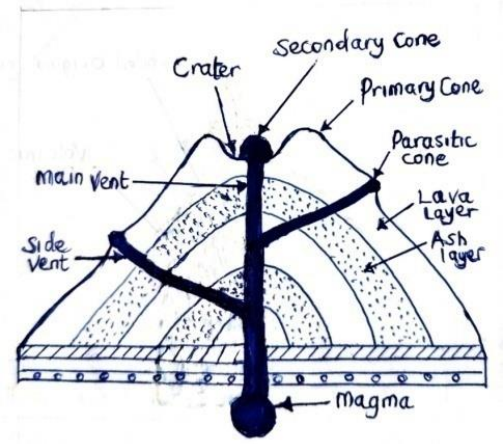
Volcanoes are classified according to their present state of activity and the nature of materials that form them. According to the state of activity we have extinct, dormant and active volcanoes.

- **An extinct volcano** is that which shows no sign of further eruption and much of its original structure is or has been destroyed e.g. mount Moroto, Kenya and Elgon.
- **A dormant volcano** is that which has not been known to erupt and yet is not extinct. Usually they show limited activity inform of fumaroles e.g. mount Kilimanjaro.
- **Active volcano** are those which are still erupting or has erupted in the recent years e.g. in the rift valley areas of east Africa mounts such as Longonot, Oldonyo Lengai and Meru.
- According to materials that form them, volcanoes include the following. Note that these are also the different extrusive volcanic features.

Composite/Strato volcano: this is a large cone composed of alternating layers of ash and lava.

- It is formed through repetitive eruption. It begins with great violent eruption due to gaseous content which blows lava into great heights breaking it into small fragments that fall back and build a layer of ash. As eruption becomes less violent, the gentle lava settles on top of the ash to form a layer of lava. Secondary eruption may occur which blows off the top of the volcano leaving a funnel-like depression called a crater. When pressure reduces to push the magma up the vent, it may flow through fissures to build parasitic cones on the sides of the volcano.
- Thus composite volcanoes are characterized by numerous cones e.g. Kilimanjaro has Kibo and Mawenzi while Mufumbiro ranges has Muhavura and Sabinyo. Other composite volcanoes are Mount Kenya, Elgon, Meru, etc

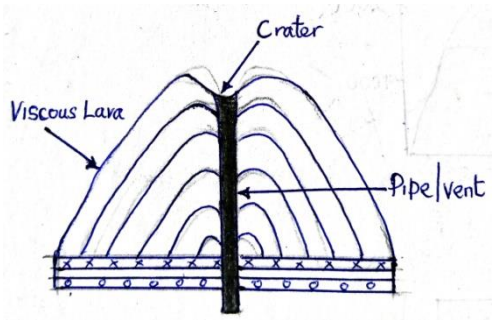
Illustration



Acidic lava cone; this is a cone with steep sides, a narrow base but very high in altitude.

- It is formed when eruption of viscous lava with high silica content solidifies very quickly to form a cone.
- Examples of acidic cones are mount Mgahinga western Uganda and mount Rungwa in Tanzania.

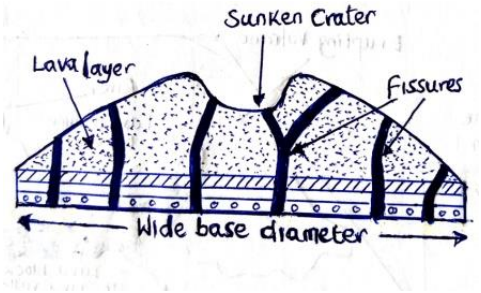
Illustration



Basalt dome/Shield volcano/Lava cone; this is a large convex-topped dome formed from basic lava.

- It is formed when fluid lava flows for along distance before solidifying and through fissures rather than a single vent. The result of this is the formation for a gentle but also broad and almost a flat-topped convex dome known as basalt lava dome. Basalt domes may also have sunken craters and small subsidiary cones along the numerous fissures.
- Examples are Virunga ranges between Lake Kivu and Edward, Muhavura, Nyamulangira on the Uganda-DRC border, Longonot and Mount Kenya.

Illustration



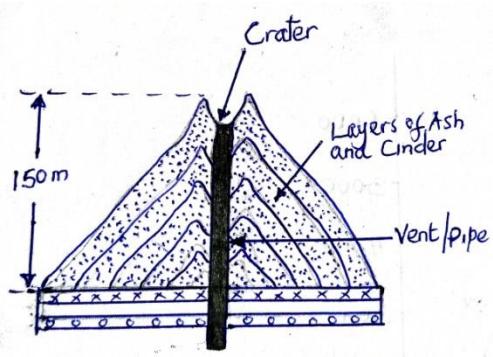
Ash and cinder cones: These are small and symmetrical steep-sided hills composed of pyroclasts and are usually less than 150m high.

- They are formed when violent eruption of acidic lava is ejected high up into the air in a liquid form and falls back to the ground as semi-solid, cooled pieces/fragments called pyroclasts. This pyroclasts accumulate around the vent in successive layers to build a steep sided volcanic hill.

Each layer represents a phase in eruption. Ash and cinder cones also have large craters which almost cover a whole summit because of the pressure with which the lava is ejected.

- Examples in east Africa include Teleki, Likaiyu, and Nabuyatom south of Lake Turkana in Kenya, Muganzi, Shozi, sagitwe and Bisalo in Kisoro district.

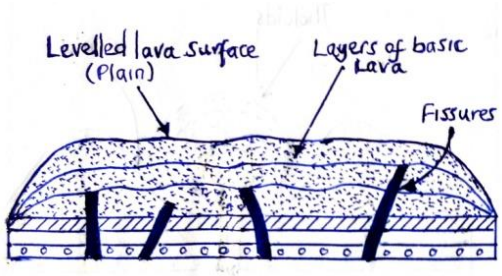
Illustration



Lava plateau/lava plains: this is an upland with a flat-topped area composed of different successive layers of lava.

- It is formed by quiet eruption of basic lava on to the surface through many fissures. The lava reaches the surface quietly and then spreads out over older lava flows. As a result the depth of the lava may be 100th of km thick, completely covering any original landscape of hills or valleys.
- Examples are Laikipia, Yatta, Kano plains, Loita, Athi plains, Kapti phonlite, Uasin Gishu in Kenya, Kisoro lava plains in Uganda, etc,

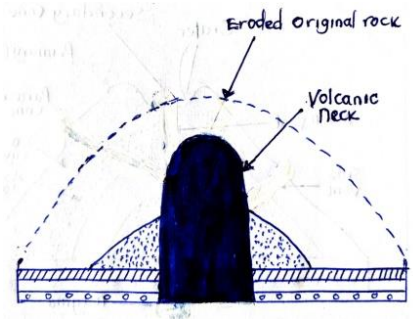
Illustration



A volcanic plug/neck: this is a hard and resistant cone of solidified magma which stands out prominently above the ground.

- It is formed when viscous magma quickly solidifies within a vent forming a solid mass of rigid and resistant rock. Overtime erosion removes the surrounding softer parts of the volcano leaving a hard resistant cone.
- Examples are Tororo rock in Eastern Uganda, Alekilek in Napak in Karamoja, Nelion and Batian peaks on Mount Kenya, Mawenzi peak on Mount Kilimanjaro.

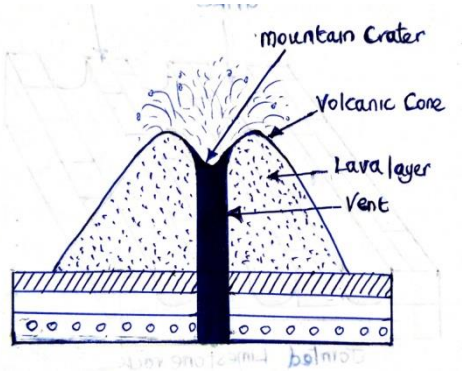
Illustration



Mountain crater: this is a flat floored circular depression usually on top of a volcano. It is generally 1 mile in diameter.

- It is formed when violent eruption of magma and gases under intense pressure from the ground blows off the top of the volcano leaving a depression called a crater.
- Examples are Kilimanjaro, Elgon, Sabinyo, Suswa, Menengai, Shomble, Etc

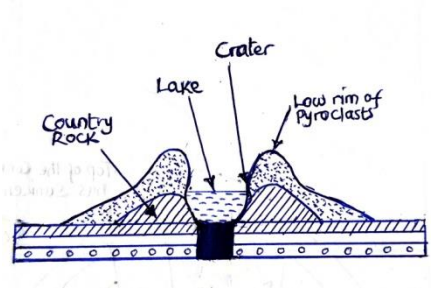
Illustration



Explosion/Ring crater: this is a shallow flat floored depression surrounded by a low rim of pyroclasts and country rock (local rocks).

- It is formed from eruption caused by vigorous escape of gases such as sulphur-dioxide, hydrogen-sulphide, etc, from the ground. Due to the violent explosion, rocks at the surface are blown off leaving a depression surrounded by broken rock particles (pyroclasts) forming a rim/ring. Explosion craters are usually small in size and rounded. Others are dry while others contain water and are called explosion crater lakes.
- Examples are Lake Katwe, just close to Lake Edward, Nyamunuka, Nyamsingire, Nkugute, Kasenyi and Kyamwiga in south western Uganda, Lake Saka, Kyengere, etc, in Tanzania are Galama and Ndoto craters N.E of Singida.

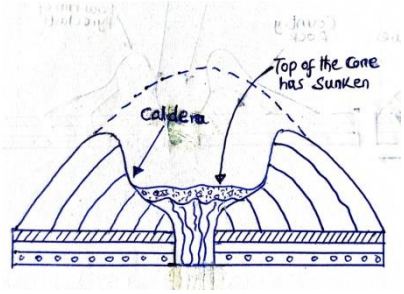
Illustration



A caldera: this is a large circular depression on top of a volcanic mountain exceeding 1km in diameter.

- It is formed either through violent secondary eruption or cauldron subsidence.
- **Secondary violent eruption**; according to this, after the first eruption, the vent may properly be blocked and gases and magma may build up intense pressure such that when secondary explosion occurs, it blows off the top of a volcano leaving a large circular depression.
- **Cauldron subsidence/ basal wreck**; according to this, a caldera is formed when during first eruption, a lot of magma is poured out due to intense pressure which reduces supply of magma underneath the volcano creating a big empty space (chasm) beneath the volcano. Later the weight of the overlying cone becomes great, develops cracks and the top of the volcano collapses/sinks/subsides down into the chasm leaving a wide circular depression on top.
- Examples in East Africa include Napak on mount Moroto which is thought to have been formed through cauldron subsidence process; others are Ngorogoro and Ebangai calderas in Tanzania, Longonot and Suswa in Kenya.

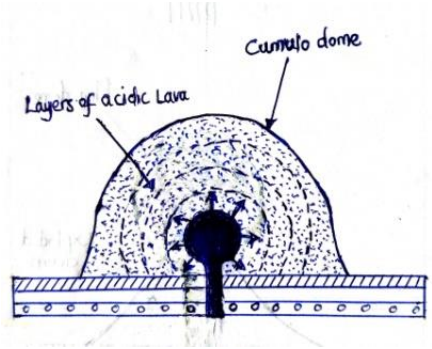
Illustration



Cumulo domes: this a circular and rounded dome made of acidic lava.

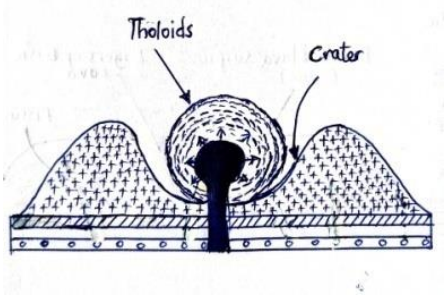
- It is formed when moves out of the vent under low pressure. Once the dome is formed its surface quickly hardens while the interior is fluid. Further uprising of lava forces the outer layer to expand outwards, forming a more or less rounded feature.
- Example are Ntumbi dome east of Mbeya in Tanzania, Gombe and Nakasongola domes in Uganda and in Kenya they are found in Tsavo national park.

Illustration



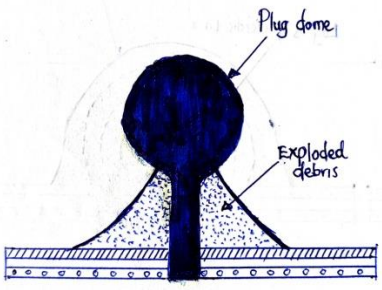
- When cumulo domes form in craters or calderas, they are called **Tholoids** or Wizard islands e.g. on mount Rungwe in Tanzania.

Illustration



Plug dome:

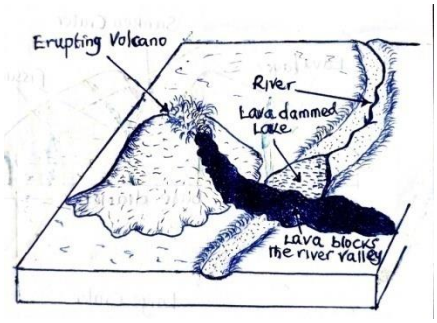
- This is formed by violent eruptions that force out acidic lava in a cylindrical form. Sometimes the lava is so viscous that it may completely block the vent.
- Example is Huggar Mountains of Algeria.



WATER FEATUIRES OF VOLCANICITY

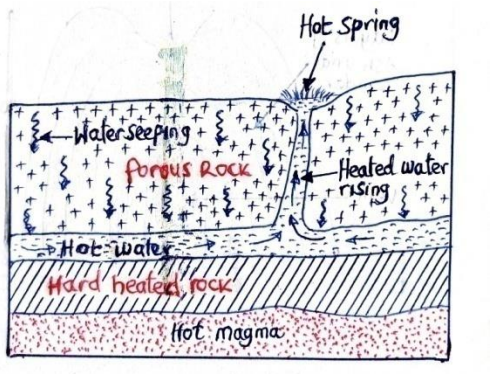
- **Lava dammed lakes**: these are formed when lava flowing from an erupting volcano blocks a river valley causing back ponding of the river; the water in the river becomes stagnant forming a lava dammed lake. Examples are Lake Mutanda, Bunyonyi, Muhehe in Kisoro and Kabale south western Uganda,

Illustration



- **Crater lakes**: (refer to the formation of craters)
- **Calderas lakes**: (refer to formation of calderas)
- **Hot springs**: this is a feature of hot natural water that constantly flows out of the ground. Hot springs are formed when rain water seeps/percolates into the ground through layers of porous rocks and comes onto contact with hot rocks. When this water accumulates on the hot rocks, it will be heated up. The supper heated water is less dense than the cold water sinking around it so it will flow to the surface through fissures as hot water to forming a hot spring. This gashing water may be clear but is often rich in mineral salts dissolved from rocks it passed through. Examples are Sempaya in Bundibugyo, Kibiro on Lake Albert shores, Kitagata in Bushenyi, Ihimbo and Kisizi in Rukungiri in Uganda, Maji moto at Lake Manyara national park in Tanzania and Lake Bogori, Maji yamoto, Njorwa and lake Hanington in Kenya.

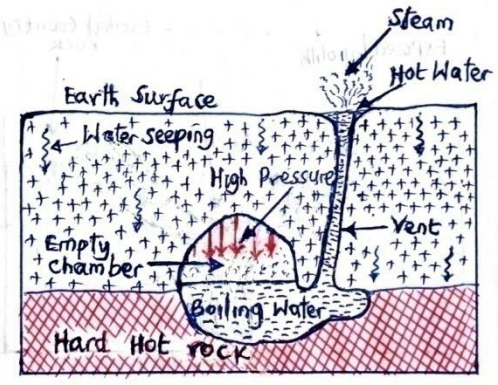
Illustration



Note: cold water seeps/percolates through porous rocks into the hard supper heated rocks. Convection current is created forcing hot water to rise up through/flows off to the surface as hot spring.

- **Geysers**: these are natural water fountains that throw up jets of hot water and steam at regular intervals through a vent. They are formed when rain water seeps/percolates through rocks and drains deep into an empty chamber/cavern near hot rocks. The intense heat from the rocks boils the water and turns it to steam. As the steam bubbles increases inside the chamber, it builds up strong pressure that pushes/pumps water and steam out as a jet through the vent with violence. When the steam reduces the pressure and the jet dies down and the chamber refills with new water and the process repeats/continues.

Illustration



- **Fumaroles:** The term fumarole is derived from a Latin word "fumu" which means "smoke". The formation of fumaroles is similar to the above two; however, the only difference is that with fumaroles, its only steam escapes continuously under low pressure. Fumaroles describe the emission of steam and gases such as Carbondioxide and hydrogen-sulphide from an opening in the crust. Such gases exist under high pressure due to the weight exerted on them by the overlying rocks. When pressure is suddenly released due to crustal instabilities, the gases are emitted as fumaroles. Where fumaroles are strongly sulphurous, they are called Solfatara.
- In east Africa, fumaroles are evident on the sides of Lake Hannington area, between Lake Elementaita and Lake Naivasha, Menengai and Eburi area, Likaiyu south of Lake Turkana mount Longonot in the rift valley of Kenya, craters of Mount Kilimanjaro.

INTRUSIVE LANDFORMS

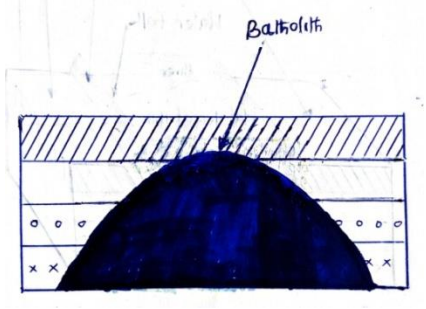
These are formed when magma is ejected from the interior of the earth and solidifies before reaching the earth's surface. The features formed depends on the nature of magma whether basic or viscous and the character of the bending planes, joints or cracks. The features formed include Batholith, Dykes, sills, lapoliths, laccolith, Phaccolith, etc.

Though intrusive landforms are buried down at the time of their formation, these features may be exposed to the surface by denudation processes such as weathering and erosion which affect the surface morphology and relief. Therefore they affect relief only after being exposed.

Batholiths: this is a very large dome-shaped intrusive volcanic feature of generally granitic rocks formed at great depth.

- It is formed when a large mass of magma solidifies very slowly at great depth to produce far large rock. It covers 100th of km across and rises covering several layers of the earth crust.
- Examples are Mubende batholiths, Nakasongola, Luwero and Singo, Kachumbala in Kumi, Labwor, Parabongo, Kalongo, Otuke, Aloi and Akia hills in northern Uganda, batholiths are found between Mwanza(Bismarck rock), Kakamega in Kenya, and Iringa in Tanzania.

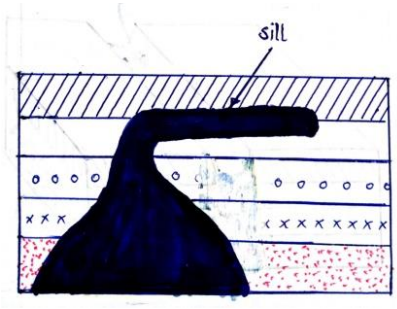
Illustration



Sill: this is a horizontal sheet of magma intruded and solidified along the bedding planes (layers) of sedimentary rocks either by partial melting of the beds or by forcing them apart.

- Sills may also extend 100th of km. When a sill lies across a river, water falls may be formed.
- Examples in East Africa include Sezibwa falls, Bujagali falls, and Karuma falls in Uganda, the Thika falls in Kenya and others outside East Africa are the Three Sister sills in South Africa, the great Whin sill in northern England.

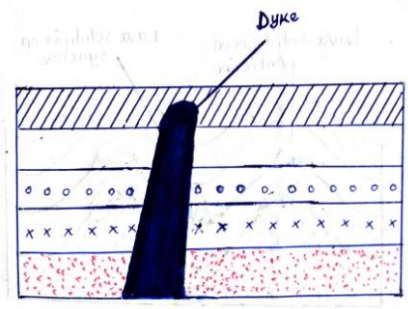
Illustration



Dyke: this is a vertically or steeply inclined sheet of igneous rock structure.

- It is formed when magma intrudes through vertical fissures and solidifies before reaching the earth surface as a discordant wall. Dykes may also split from a base of a volcano if the outlets are blocked to form several dykes (groups). When dykes occur in groups they are called "Ring dykes" or "Swarms".
- Examples in East Africa are; around western Lake Turkana, Rungwa area east of Kisumu, in Kenya, Tororo ring dykes in the Sukulu hills. Other examples are in the Jos plateau in Nigeria, Malawi and in Lesotho. When dykes come across a river valley the form waterfalls e.g. Thika falls, Sipi falls and Sisiyi falls on Mount Elgon.

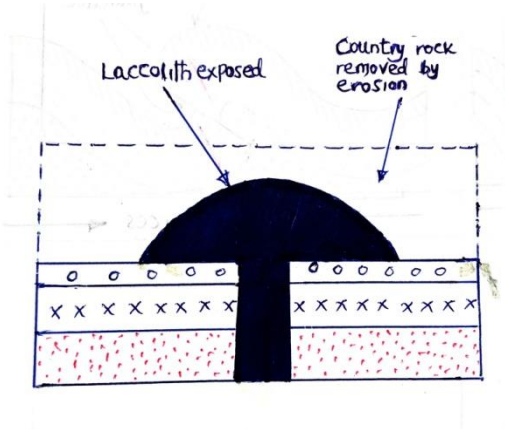
Illustration



Laccolith: this is a dome shaped volcanic intrusion but more or less a flat-topped floor/base.

- It is formed when viscous magma rises and solidifies within the crust before reaching the surface. So because viscous magma is thick and does not spread far, its accumulation and solidifying in the strata forces the overlying rock layers to up-arch in a dome shape. When denudation occurs and the laccolith is made up of resistant rocks than the surrounding, may result in to an upland.
- Examples in East Africa can be seen along Lyantonde-Mbarara road, in Voi in Kenya, Funjay and Ambereny in Madagascar and on the Algerian coast.

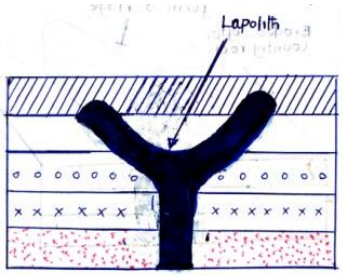
Illustration



Lapolith: this is a very large saucer-shaped volcanic intrusion.

- It is formed when magma forces its way into the crust and spreads horizontally between the rock strata/layers. The heavy weight of the overlying rocks depresses the magma into a saucer-shape as it solidifies. When exposed by denudation, the up-turned edges form out-facing scarps while the "saucer" shape forms a shallow basin called Arena.
- Examples can be seen along Kabale-Kisoro road, there is the Rubanda Arena, in Singo and Mubende hills, others are found in Belingwe in Zimbabwe and Bushveld basin of Transvaal in the Republic of South Africa.

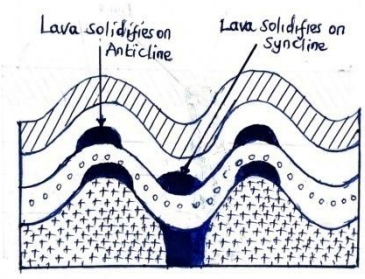
Illustration



Phaccolith: These are intrusive features found only in areas which have experienced folding.

- Phaccoliths are formed when magma rises through the crest, spreads and solidifies, on the crest of the anticlines and bases of the synclines in folded rocks.

Illustration



THE INFLUENCE OF INTRUSIVE FEATURES ON RELIEF

Intrusive features influence the relief after denudation has removed the overlying layers exposing them to the surface. Among these intrusive landforms that have been exposed to the surface include; batholiths, sills, dykes, lapoliths, laccoliths, etc.

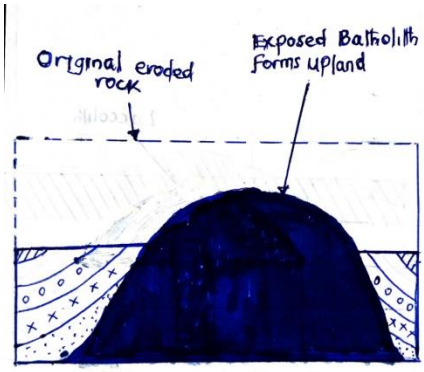
Batholiths influence on relief:

- A Batholith is a great intrusion with steep sides formed inside the earth crust. It is formed from very viscous magma, a reason a batholiths therefore very harder than the surrounding country rocks. When denudation sets in, the less resistant country rock is removed. This exposes the

batholiths gradually inform of massive **upland/residual hills/inselbergs**/surrounded by low-lying relief/plain.

- Examples can be seen in Mubende, Nakasongola, Singo, Labwor and Parabongo hills in northern Uganda, Maragoli area of western Kenya and in Embu.

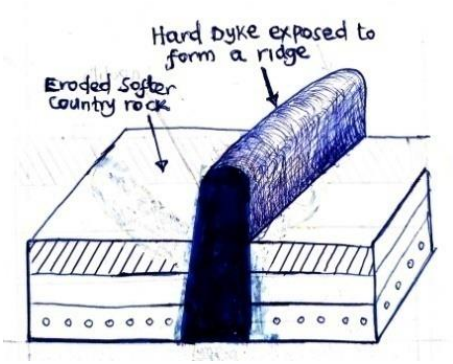
Illustration



Dyke influence on relief:

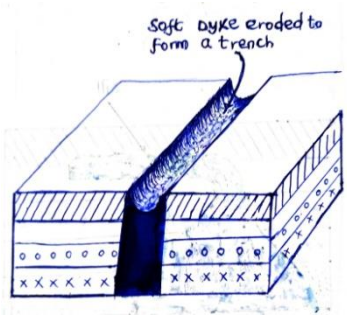
- This is a vertically or steeply inclined rock sheet lying vertically across the bedding planes. Dykes are made up of materials whose resistance to erosion varies depending on the nature of magma forming them. If the dyke is made up of a resistant (harder) material than the surrounding country rocks, the erosion attacks the weaker surrounding rocks than it does to the dyke. Therefore the dyke is exposed to the surface as an **elongated hill/ridge**.
- Examples can be seen in Busia, Sukulu, Isingiro, Kisumu, Rungwa complex, etc,

Illustration



- On the other hand if the dyke is made up of soft rocks than the surrounding country rocks then it will be weathered down faster than the country rocks. These results into small depressions called **Trenches.**
- Examples of linear Trenches associated with dykes can be seen west of Lake Turkana and in Rungwa south east of Kisumu.

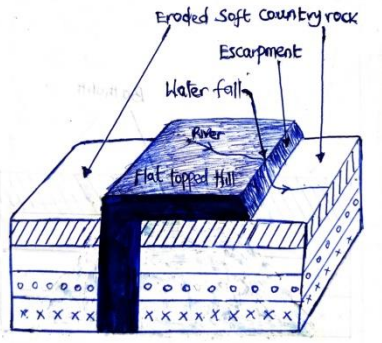
Illustration



Sills influence on relief:

- These are horizontal intruded igneous features lying along the bedding planes. Magma from the interior of the earth, rises, spreads and solidifies horizontally in the bedding planes before reaching the surface.
- If the material forming the sill are resistant than the surrounding country rocks, erosion eats up the surrounding country rocks faster exposing the sill as an **Escarpment.**
- On the other hand if the escarpment crosses the **river valley,** a water fall may be formed. Example is the Thika falls in Kenya.
- And also if the upper side of the sill is flat, the prolonged denudation may expose it to form **a flat-topped hill or cliffs.**

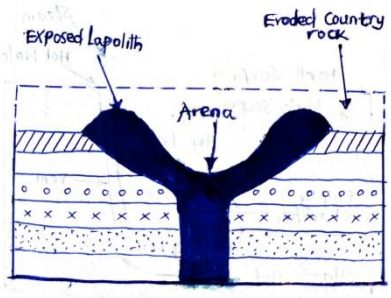
Illustration



Lapoliths influence on relief:

- The lapoliths forms a saucer shape and when the overlying rocks are removed by denudation, a shallow but wide saucer-shaped depression likened to an **Arena** is formed.
- Example is The Rubanda arena on the Kabale-Kisoro road.

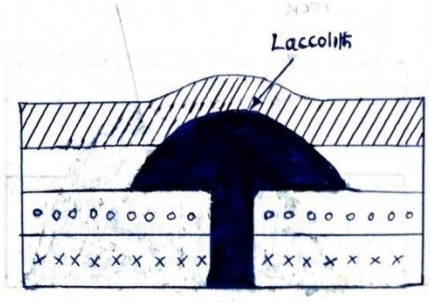
Illustration



Laccolith influence on relief:

- This is a dome shaped feature which forces the overlying layers of rocks to up-arch. When the overlying rocks are exposed by denudation the Laccolith appears like a **dome-shaped hill/upland.**
- A good example can be seen in Kitui and Voi areas of Kenya, in Madagascar.

Illustration



SIGNIFICANCE OF VOLCANICITY

- Volcanic rocks/highlands weather down to produce fertile soils which support cultivation of crops e.g. mount Kilimanjaro slopes support coffee and tea crops, mount Elgon slopes support Arabica coffee and bananas, Kenya highlands support pyrethrum, etc.
- Volcanic intrusions are associated with valuable minerals some of which occur in igneous rocks e.g. diamonds at Mwadui in Tanzania, the diamonds of Kimberly in the Republic of South Africa. This boosts the mining industry. Others include salt mining in Lake Katwe in Uganda; limestone mining in Hima and Tororo are all associated with volcanicity.
- Igneous rocks can be quarried. These rocks can be excavated and crashed to produce aggregates that are used for building and construction industry tarmacking roads e.g. Muyenga quarry in Kampala in Uganda.
- Volcanic features such as hot springs and geysers produce steam jets that can be harnessed to generate geothermal power to run industries and for domestic purposes, e.g. this has been at Olkaria in Kenya. Hot springs are used as a source of hot water to houses e.g. in Iceland and New Zealand.
- Volcanic features such as high mountains provide magnificent scenery which attracts tourists where foreign exchange is generated used to develop infrastructure. Such mountains include Kilimanjaro, Kenya, which are some of the highest mountains in Africa and are of volcanic origin.
- Some volcanic mountains are sources of many rivers and streams which facilitate irrigation farming e.g. rivers such as Manafwa, Sironko, Malaba, Namatala and Turkwel originate from Mount Elgon. Some of these rivers like Manafwa have supported Doho irrigation scheme in eastern Uganda.
- High volcanic mountains directly influence the formation of orographic rainfall when they force warm and moist winds to rise, cool and giving rise to heavy rainfall on the windward side. This rainfall support crop farming on the windward side.
- Volcanic mountains are colonized by montane vegetation due to heavy rainfall on the windward slopes. Forests such as tropical rainforests facilitate forestry industry and lumbering e.g. on Elgon mountain, etc,
- Volcanic features like sills when they cross rivers, they result into water falls which are potential points for hydro electric power generation e.g. Bujagali falls and Karuma falls along river Nile.
- Volcanic mountains are sites for wildlife conservation e.g. mount Kenya national park, mount Elgon NP, etc, this protects wildlife for our young generation to see and above all attracts foreign tourists who bring foreign exchange used for infrastructure development.
- However,
- Volcanic eruptions are destructive to settlement, life and property. For example in 2002, Nyamulangira volcano in DRC erupted and covered a half of Goma destroying property and killing people.
- Volcanic mountains induce rainfall formation on only one side(windward side) while the leeward side is left dry, e.g. the Eastern side of mount Elgon in the western parts of Kenya are largely dry hence discouraging agricultural development in this area.
- Steep slopes of volcanic mountains are prone to landslides which also destroy property, settlement and killing people. E.g. Bududa mudslide of 2010 in Uganda killed many people while others were internally displaced.
- Steep slopes of volcanic mountains accelerate soil erosion which affects the development of agriculture e.g. in Kigezi highlands soil erosion has retarded the development of agriculture due to steep slopes.

- Steep slopes of volcanic mountains limit the development of transport and communication network. Roads are difficult to construct along steep slopes and if its to be done involves high costs in the constructions. E.g. Kigezi region is not well served with transport infrastructure.
- Steep slopes and rugged terrain of regions of volcanic activities limit modernization of agriculture. Machines such as tractors, combine harvesters, etc is practically impossible to use in hilly areas such as Kigezi region in south western Uganda.
- The mountain tops of some of the volcanic mountains limit settlement because they are so high above sea level and reach the snowline. These mountain top are so cold and sometimes glaciated. These temperatures are too cold to favour settlement. E.g. top of Mount Kilimanjaro and Mount Kenya.
- Young volcanic mountains are very porous and have contributed to scarcity of underground water in such as Ntungamo and Kisoro in south western Uganda.

REVISION QUESTIONS

ROCKS/GEOLOGY OF EAST AFRICA

Geology deals with the study of rocks and the mineral compounds it contains. A rock is an aggregate of minerals that compose/form the earth crust and exists in a solid state. Rocks are made up of minerals or elements. Some rocks are composed of single element/minerals while most rocks are made up of a complex combination of elements and minerals e.g. gold, quartz, granite, bauxite, clay and sand. The main elements found in minerals that form rocks are Potassium, calcium, iron, aluminum, sodium, silicon, oxygen, magnesium and phosphorus. Therefore minerals can have one or more elements.

- Rocks can therefore vary depending on their structure and mineralogical composition. As per structure some rocks vary depending on colour, texture, porosity and physio-chemical composition.

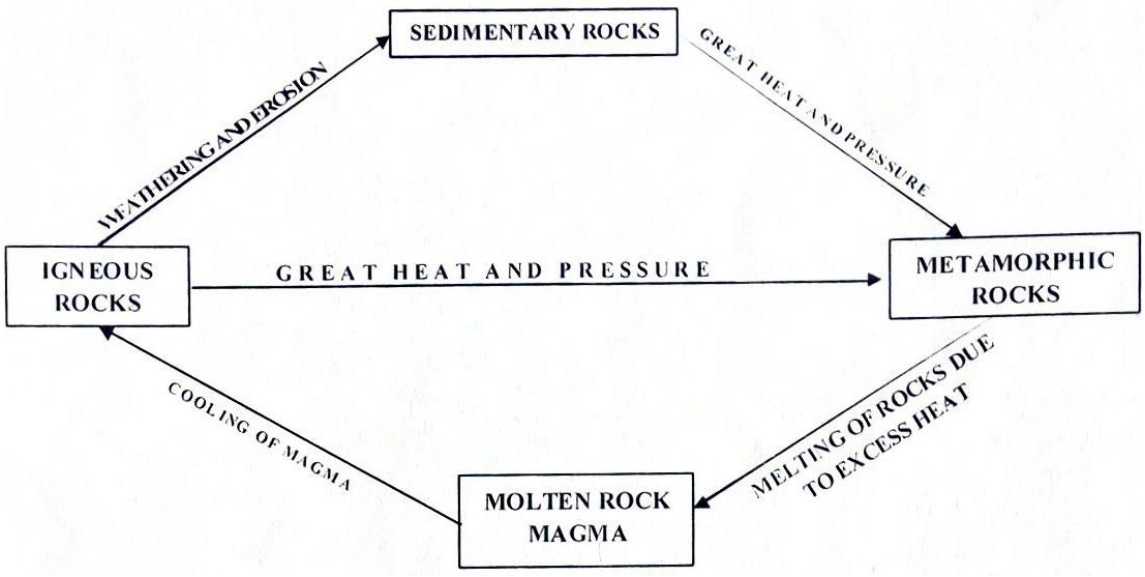
CLASSIFICATION OF ROCKS

Rocks of the earth crust are classified in two major ways i.e.

- **According to geology age,**
- **According to the mode of formation**;

Classification according to the mode of formation is the most commonly used and according to this mode rocks are of three major types namely **Igneous rocks, Sedimentary rocks** and **Metamorphic rocks**

THE ROCK CYCLE:



IGNEOUS ROCKS

The word "igneous" comes from a Latin word "Ignis" which means fire. Therefore igneous rocks are fire formed rocks. Igneous rocks are mainly formed through the process of vulcanicity.

- Vulcanicity is the process by which molten rocks (magma) under intense heat and pressure from deep in the interior of the earth is ejected/flows out through a vent/fissures or burns its way through lines of weaknesses, cools down, crystallizes and solidifies either within the Earths crust to form intrusive rocks or onto the Earths surface to form extrusive rocks.

Characteristics of igneous rocks

- They are fire-formed in structure, they are crystallized rocks because are formed under heat, they are formed as a result of cooling and solidification of magma, they do not have fossils, they do not occur in layers/strata, they are hard and resistant to soil erosion, they have large crystals when formed deep under the ground and they are fine grained when formed on the surface such as basalt, some igneous rocks are spongy such as pumice, some are dark coloured such as basalt and glassy in appearance like basalt, some are acidic, basic and intermediate in chemical composition.
- **Examples of igneous rocks include**; Quartz, Gabbro, Pumice, Basalt, Obsidian, Dolerite, Diorite, etc.

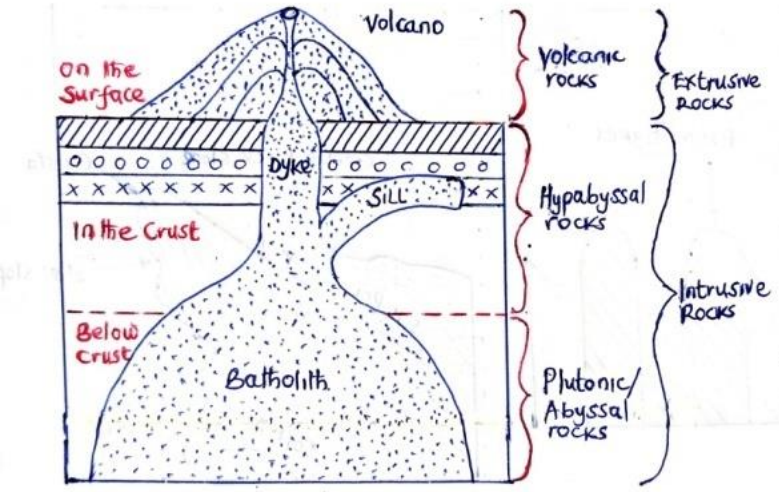
Classification of igneous rocks

Igneous rocks are classified according to a number of ways i.e. the rate of cooling, depth and chemical composition. The most commonly used is that based on the rate of cooling, level of depth, and size of the crystals after solidification. Under this they include:

- **Volcanic rocks/extrusive igneous rocks:** These are rocks formed from very fast cooling of lava extruded onto the Earth surface due to its exposure to air. These types of rocks solidify on the earth's surface hence forming small sized crystals. Examples are Basalt, Obsidian, Rhyolite and Andesite. Some of the volcanic rocks are spongy in nature due to the enclosure of gases at the time of cooling of lava, e.g. pumice. Some volcanic rocks are glassy in appearance such as obsidian, other volcanic rocks are dark coloured such as Basalt. Volcanic rocks are mainly found in features such as the typical volcanic mountains such as Elgon, lava plateaus like Yatta, Etc. Also other examples which are large extrusions include; Lava, ash, ignimorites and imorites. The smaller extrusions can include typical volcanoes, lava flows and ash beds.

- **Hypabyssal/ intermediate rocks**: these are formed from magma that cools and solidifies near the surface for the earth/shallow depth. The rate of cooling of magma is moderate causing it to solidify at a shallow depth/near the earth hence forming medium sized crystals. These rocks can be exposed by denudational processes. Examples are; quartz, dolerite, porphony, they are found in features such as sills, dykes, in the Turkanaland, etc.
- **Plutonic/abyssal rocks**: these are formed from solidification of magma at great depth hence are some times called deep-seated rocks. The rate of cooling of magma at great depth is very slow due to the existence of hot temperatures and lack of contact with air hence forming rocks with large crystals such as Granite, Gabbro, Diorites, Synite, etc. at their origin.
  These rocks are found in features such as exposed Batholith at Mubende, Singo, etc.

Illustration



NOTE:

- ***Abyssal/Hypabyssal rocks can generally be called intrusive igneous rocks. Major intrusions include: batholiths, laccolith, associated with a number of basses and stocks. Minor intrusions include; sills, bysmalith, thacolith cone sheet and dykes.***
- ***There are a number of minerals found in igneous rocks. There are two ways of grouping the minerals in these rocks i.e. Looking at the:***
- ***Colour of these minerals i.e. light coloured and dark coloured minerals. Light coloured minerals are mica, quartz, felspher. Dark coloured include biorites, dolomites and diorites.***
- ***Texture or composition of the rocks; on texture some igneous rocks are course textured where as others are fine textured e.g. granites are course textured.***

IMPOTANCE OF IGNEOUS ROCKS

- Igneous rocks break down to yield fertile volcanic soils capable of supporting the growing of various crops e.g. coffee, bananas, etc, like in the case of Mount Elgon, Kilimanjaro, etc.
- Igneous rocks have formed volcanic highlands which are sources of rivers that can be used for irrigation e.g. Mount Elgon is a source of river Manafwa that provides irrigation water to Doho rice scheme.
- The volcanic highlands from igneous rocks provide favourable conditions for the growth of mountain forests which has boosted lumbering e.g. on mount Elgon.

- Igneous rocks are made up of various minerals which have boosted mining in East Africa e.g. Mwadui near Shinyanga in Tanzania, diamonds is extracted from volcanic pipes which are composed of igneous rocks.
- Igneous rocks such as granite and basalt can be broken down into small aggregates which are used for making concrete during construction e.g. Muyenga quarry in Kampala.
- Igneous rocks have formed several relief features e.g. volcanic mountains, calderas, inselbergs, craters, etc, these constitutes tourist attraction features on East African countries hence earning foreign exchange used to develop infrastructure.
- Hot igneous rocks beneath the earth crust have formed hot springs, fumaroles and geysers. These have been harnessed to produce geothermal energy e.g. at Ol-karia valley in Kenya.
- Some igneous rocks have been intruded along the river courses forming water falls. These can be harnessed to generate hydro electric power e.g. along river Sezibwa, there is Sezibwa falls and Bujagali along the river Nile.
- Impermeable granite rocks under the ground trap rain water which seeps into the ground. This acts as a reservoir for underground water for domestic use in rural areas.
- Highlands formed by igneous rocks have influenced rainfall formation on the windward sides of mountains hence promoting agriculture e.g. the western side of Mount Elgon.
- Volcanic mountains formed from igneous rocks have booted wildlife conservation and tourism e.g. Mount Elgon.
- Igneous rocks provides firm foundations for bridges and dams for example the Owen falls dam was sited where the underlying rocks are strong and firm.

However,

- Igneous rocks are formed under violent eruptions which are destructive to human plant life as well as properties.
- Igneous rocks from volcanic highlands are prone to landslides which kill people and destroy crops and settlement, e.g. Bududa landslides on Mount Elgon in 2010.
- Exposed granites are a setback for agricultural development as they hinder mechanization. This is a serious problem in Kabale and Kisoro districts in south western Uganda.

QN. Explain the process responsible for the formation of igneous rocks in East Africa.

Approach:

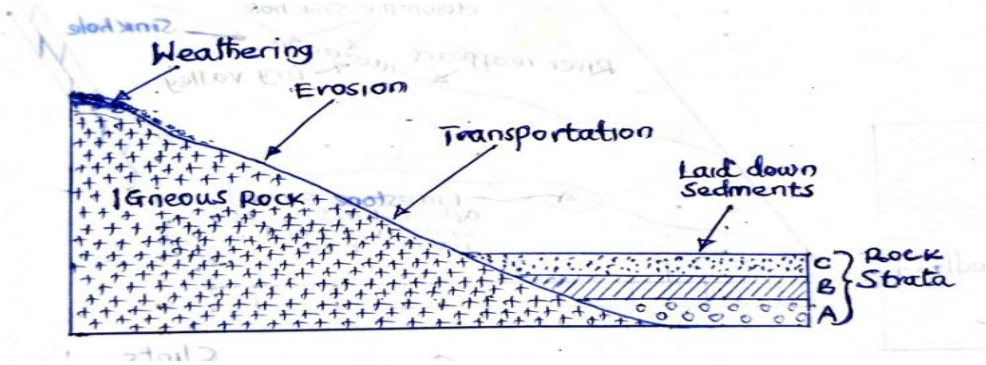
- *Define igneous rocks, Identify process of formation; Give the origin of the process, Identify and describe the types of igneous rocks.*

SEDIMENTARY ROCKS

These are rocks that are composed of particles of rocks that have been deposited in layers by wind, water and moving ice.

- They are formed from rocks broken down by weathering, eroded, transported by winds and running water or other denudational processes and finally deposited in layers (strata) and later compacted or cemented to form sedimentary rocks. The deposition normally takes place in the lowland areas or in the valleys. These deposited sediments accumulate and becomes cemented and compact under pressure by solution of calcareous or silicerous materials which determine the colour of the rocks.

Illustration



Characteristics of sedimentary rocks

- They are layered/stratified, the layers are separated by bedding planes, the layers/strata are either horizontal, gently sloping or steeply dipping, they contain fossils (remains of dead plants and animals), they are non crystalline because they are not formed under heat, they are laid down rock wastes, they are made up of the sediments, some are porous/permeable and they are soft.
- **Examples include**; Sandstone, Mudstone, Clay, Rock salt, Coral limestone, Coal, Dolomites, Soda ash, Gypsum, Iron ores, Iron stone, etc.

Classification of sedimentary rocks:

Sedimentary rocks are classified further into their modes of formation namely: mechanically, organically and chemically formed processes.

1. **Mechanically formed sedimentary rocks**: these are sedimentary rocks which form from pre-existing rocks like igneous volcanic mountains. Due to their exposure, they are disintegrated by denudational processes to produce debris which is later laid down in lowlands. The debris is later `on consolidated to form a rock after a period of time. These rocks vary inform of texture depending on the agent of erosion for example, river deposit form alluvial, wind deposit forms loess, glacier deposits form moraine.

- Examples resulting from mechanically formed sedimentary rock are Sand stones, Clay, Shale, gravel, boulder clay, mud stones, etc.

2. **Organically formed sedimentary rocks:** these rocks are formed from remains of dead plants and animals known as fossils.

- The plants accumulation during the carboniferous period resulted into the compression of sediments of plants to form coal of different types i.e. lignite, peat and brown coal e.g. coal is found in Ruhulu valley in southern Tanzania.
- Coral reefs came from the skeletons of polyps which are rich in limestone. When polyps die, their skeletons fall and accumulate at the base of oceans where they are cemented to form limestone rocks. Examples can be seen at Kilifi, Mombasa, Tanga, Dar-es-salaam and Mtwara at the coast.
- The accumulation of the ferruginous materials resulted into the formation of iron and ironstone. Other examples include chalk, natural gas.

3. **Chemically formed sedimentary rocks**: these are formed from the precipitation and or evaporation of mineral substances from a solution. They form when minerals dissolved in water begin to precipitate out of the solution and deposit at the base of the water body. Some sea/lake water contains salts in solution so when the water evaporates, beds of salts known as evaporites are left behind.

- E.g. rock salts, dolomite and soda ash from Lake Katwe and Magadi were formed due to continued deposition and accumulation of the salt crystals as a result of Precipitation/Evapouration.

- The laterites were formed from leaching and oxidation of iron and aluminum and this are found on Buganda hill tops.
- The deposition and accumulation of limestone after Evapouration has resulted into the Stalagmites and stalactites. Limestone rocks are found at Nyakasura, Tororo area, along Mombasa and Tanga at the East African coast.
- Other rocks are gypsum and potash.

IMPORTANCE OF SEDIMENTARY ROCKS

- Mechanically formed sedimentary rocks are a collection of various rocks materials hence weathered down to produce very fertile soils e.g. alluvium of the Rufigi delta, along swamps, rivers and lakes such as Victoria.
- Some sedimentary rocks are a source of raw materials for industries e.g. coral limestone is a raw material for the manufacture of cement at Bamburi in Kenya, potash used for making fertilizers, rock salts for the manufacture of common salt.
- Sedimentary rocks have boosted the mining sector which contributes to East Africa's economic development e.g. sand mining at the coast of East Africa and at Kasenyi on the shores of Lake Victoria.
- Clay which is a form of sedimentary rock has boosted the construction sector in that it's a raw material for the making of bricks and tiles e.g. at Kajansi and in most swamps in East Africa.
- Sedimentary rocks in swamps and along rivers are colonized by hydrophytes some of which are raw materials used in the craft industry e.g. papyrus.
- Coral reefs at Mombasa, Tanga, Dar-es-salaam, etc, at the coast together with karst sceneries at Nyakasura in Fort portal are tourists attraction which earn foreign exchange used for development.
- Sedimentary rocks are sources of medicine as well as food for man and livestock e.g. rock salts.
- Sedimentary rocks contain fossils which were squeezed for a long time to produce oil. Oil and natural gas are important sources of fuel. The presence of oil in the Albertine region in Uganda is due to the existence of sedimentary rocks in the region.
- Lagoons enclosed by coral reefs provide breeding grounds for fish hence boosting the fishing industry e.g. at Mombasa at the coast.
- Sedimentary rocks like corals also provide good sheltered harbor hence promoted the shipping industry e.g. at Mombasa.

However,

- Sedimentary rocks are made up of very fine rock materials such as silt and clay, these are not porous hence are poorly drained during rainy season and are prone to flooding.
- Sedimentary rocks like coral reefs are barriers to shipping. Under high tide they may be concealed (hidden) under water and not spotted by sailors. This may result into devastating accidents.
- Besides, coral reefs are also a hindrance to fishing as they may tear nets and obstruct navigation
- Some sedimentary rocks like clay are poorly drained and are highly leached. This makes them infertile for crop cultivation e.g. laterite soils in Buganda region.
- Sedimentary rocks tend to form immature young soils which are unproductive for agriculture e.g. the Semiliki valley and in the coastal areas of Mombasa.
- Areas such as swamps that are dominated by sedimentary rocks are breeding grounds for insects' vectors like mosquitoes that transmit malaria to the people.

Metamorphic rocks

The term "**meta**" is a Greek word which means **change**. This implies that metamorphic rocks are changed rock type. They are formed from the pre-existing rocks (igneous or sedimentary) that have been subjected

to intensive heat or pressure or both, causing change in the physical and chemical composition of the original rocks to form metamorphic rocks.

- The heat may be as a result of hot rocks/magma being in contact with the surrounding igneous or sedimentary rocks. Such rocks are generally more compact and harder than their original rocks.
- **Examples are**; Gneiss from granite, Quartzite from sand, Marble from limestone, Slate from clay.

Characteristics of metamorphic rocks

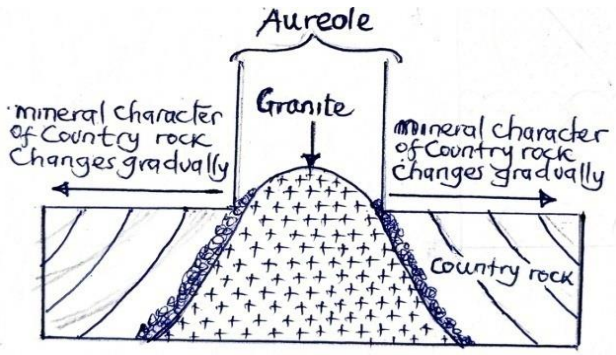
- They are changed rocks due to heat and pressure, their chemical and physical composition are different from original rock, they are very hard rocks, they are compacted rocks due to pressure and heat, they contain precious minerals, such as diamonds and marble, and they form basement with the continental crust.

Classification of metamorphic rocks:

Metamorphic rocks are classified into two, namely;

- **Contact /thermal metamorphic rocks**: these are formed when the pr-existing rocks are changed by great heat but less/limited pressure. This may arise from the interior of the earth crust where temperatures are very hot. This forces the rocks in close proximity to change their characteristics to metamorphic rocks.
- Alternatively some rocks may come into contact with the rising magma and rocks are melted. Re-crystallization and re-mineralization takes place. The process where the mineral character changes is known as a metamorphic "**aureole**".

Illustration



- **Dynamic metamorphism**: these are formed due to great pressure and less heat exerted to the pre-existing rocks by earth movement on large scale/region such that the earth movement may be inform of folding produced by compressional forces. When convergent forces push country rocks from opposite direction towards each other they create immense pressure hence changing country rocks into metamorphic rocks, e.g. Sand may be changed to Slates or Schist, Limestone to Marbles, Granite to Gneiss, Sandstone to Quartzite, etc.

NOTE: ***dynamic metamorphism is called regional metamorphism because it covers a large area such as a region.***

IMPORTANCE OF METAMORPHIC ROCKS

- Metamorphic rocks are harder than the original rocks as such they are quarried to provide aggregates for construction purposes. Besides marble chips are used in the decoration of houses while slate can be used for roofing.
- Minerals such as copper, gold, diamonds often occur in metamorphic rocks. Thus the rocks have boosted the mining in East Africa e.g. the diamonds mining in Mwadui in Tanzania, copper at Kilembe and gold in western Kenya.

- Landforms built by metamorphic rocks like Block Mountains, fault scarps, create impressive sceneries that attract tourists hence fetching foreign exchange that boost the economic development.
- The metamorphic rocks are sources of raw materials to the industries e.g. phosphates from Tororo are obtained from the previously metamorphosed igneous rocks and it is a raw material for making fertilizers.
- Some of the landforms built by metamorphic rocks such as mount Rwenzori have promoted nature conservation i.e. such areas have been gazetted into national game and forest reserves.
- Metamorphic rocks are generally compact and resistant to erosion so soils formed from such rocks are very resistant to erosion
- These rocks tend to be hard and compact and as such they are natural water reservoirs. Man is able to get water from these rocks through springs or sinking boreholes.
- In rural areas, some of the metamorphosed granite rocks are used as grinding stones.

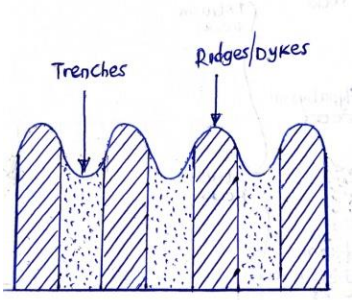
REVISION QUESTIONS

THE INFLUENCE OF ROCK STRUCTURE AND ROCK TYPE ON RELIEF

Rock structure refers to the physical and chemical composition of a rock. In simple terms it is all the ways in which rocks differ from each other in their physical and chemical characteristics. These characteristics /differences can be inform of colour, permeability, hardness, softness, jointing, mineral composition, age, porosity, faults, folds, etc. due to these differences rocks react differently to endogenic and exogenic processes. In East Africa, there are many landforms that illustrate the relationship between lithology (physical) characteristics of rocks and landforms as below:

- **Rock hardness**: hard rocks are resistant to weathering and erosion and as a result they stand out as residual hills such as inselbergs. An inselberg is an isolated hill that stands above a level plain. It is formed when erosion removes the top most layer of the crust exposing a massive resistant granite intrusion. Examples include Mubende hills, Nakasongola, Labwor and Parabongo hills in northern Uganda, Maragoli hills in Kenya.
- **Rock softness**: if the granitic rocks are less resistant to the surrounding country rocks, the granitic rocks are eroded to a much lower level leaving behind a depression called arenas e.g. Rubanda arena on the Kabale - Kisoro road.
- **Adjacent hard and soft rocks**: this is a situation where hard rocks are emerging with soft rocks. In such a landscape soft rocks will wear down to form trenches while resistant rocks will form ridges. This is because of the different rates of erosion experienced by both hard and soft rocks.

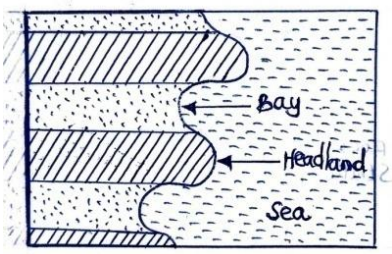
Illustration



- **Alternating hard and soft rocks**: the existence of alternating hard and soft rocks at the coast has led to formation of bays and headlands. Soft rocks are fast eroded by wave action producing bays

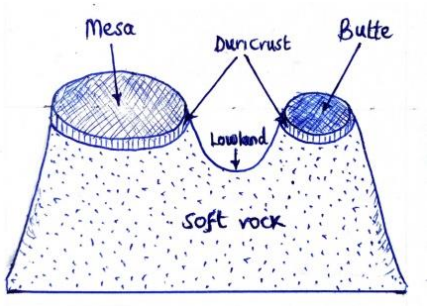
while the hard and resistant rocks remain protruding into the water body as headlands. Examples can be seen at Kasenyi in Lake Victoria, Mombasa at coast, etc.

Illustration



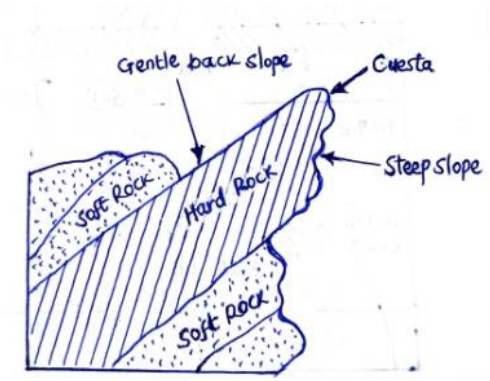
- **Overlying hard and soft rocks**: where hard lateritic rocks (duricrust) overlie soft rocks, Mesa and butte are formed. These are prominent flat topped tabular hills usually capped with a resistant rock stratum or duricrust. The duricrust protects the softer rocks underneath from erosion and weathering as the softer surrounding racks are eroded to form lowland. The
- flat-topped hills are mesa or butte. A mesa being more extensive than a butte e.g. Mubende hills, etc.

Illustration



- **Tilted overlying hard and soft rocks** can also result into the formation of a ridge with a steep slope and a gentle slope back slope known as cuesta. A cuesta is formed when tilted soft and hard rocks are subjected to weathering and erosion. The weaker rocks are worn down leaving the more resistant rocks to form cuestus with a moderately dipping back slope.

Illustration



- Other landforms related to rock hardness and rock softness include; volcanic plugs, flat topped hills from erosion of country rocks surrounding sills, water falls, etc.
- **Rock jointing:** rocks with joints/cracks allow water into them which facilitate easy weathering and erosion of such rocks resulting into a number of features such as cliffs, caves, blow holes, etc, along the coast. A cave is a hole drilled in a cliff face. It is formed due to coastal erosional processes of aggression and hydraulic action of waves which erode a joint on a cliff face. The process starts slowly with a joint and gradually widened into a cave.

- Joints have influenced the formation of other landforms such as granite tors, cirques and karst sceneries, etc
- **Existence of faults in rocks**: rocks with faults in them have influenced the formation of fault guided valleys. Such areas along faults are always weak and so easily weathered down to produce depressions. These depressions in many cases for river valleys e.g. Aswa river valley.
- The existence of faults in rocks has also facilitated the formation of escarpment. An escarpment in steep face that over looks the rift valley. They are formed when faulting has taken place and one part of the crust slips down in relation to the other e.g. Butiaba and Kichwamba escarpments in Uganda.
- **Rock solubility**: this has influenced the development of landforms for example soluble limestone rocks dissolve in weak carbonic acid to form karst landscapes with features such as stalagmites, stalactites, pillars, dorlines, sinkholes, etc. these features are common in areas such as Nyakasura in fort portal, Tanga in Tanzania and Kilifi in Kenya.
- **Rock permeability**: this is the ability of a rock to allow water to sip/percolate through it. Rocks that allow water to percolate reduce the rate of runoff hence less erosion. As a result these rocks tend to form uplands e.g. carbonate rocks forming the Sukulu hills in Tororo, coral reefs along the coast of East Africa form raised landforms such as Bamburi and Mombasa
- **Rock mineral composition**: some minerals are composed of minerals that are soluble in water like rock salts. This are massively eroded to form lowlands while rocks composed of resistant minerals like granite rocks give rise to uplands.

DENUDATION

The term denudation is derived form a Latin word "denudare" which means to strip bare, uncover or remove. The term includes all processes which work together to wear down the earth's surface. It includes the combine effect of weathering, mass wasting, erosion and transportation. These processes operate at or near the earth's surface and are therefore referred to as sub-aerial or exogenic processes. They wear down highlands with ultimate goal of creating a uniformly flat landscape.

WEATHERING OF ROCKS

Weathering refers to the gradual breakdown, disintegration or decay of rocks either physically or chemically in situ (i.e. in one place with no movement of regolith/weathered materials involved). The earth's surface is continuously undergoing destruction where by rock materials are wearing off.

Weathering is generally divided into two i.e. Physical weathering and Chemical weathering.

- **Note;** ***this division of weathering into two types may not imply that these types of weathering are exclusively independent. There is always interdependence as well as complementarity between these different types. However, certain weathering type predominates in certain environmental conditions e.g. physical weathering is dominant in arid regions while chemical weathering is dominant in humid regions.***

PHYSICAL WEATHERING

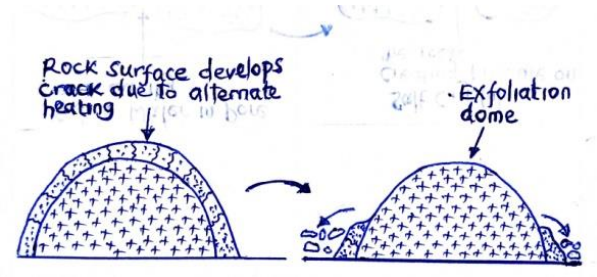
This is also known as Mechanical weathering. It involves mechanical breakdown of rocks into smaller particles in situ without change in the chemical composition of the rock. It is mainly brought about by temperature changes, influence of plants, animals and man. The conditions favouring its occurrence mainly prevail in arid regions and these include; prolonged dry season, hot day and cold night temperatures and hence a large diurnal temperature range. The arid areas also experience low humidity, low cloud cover, low rainfall and scanty vegetation.

- In East Africa, physical weathering is common in areas such as Karamoja, Turkana land, Ankole–Masaka corridor, Semiliki and Albert flats and central Tanzania.

PROCESSES OF PHYSICAL WEATHERING

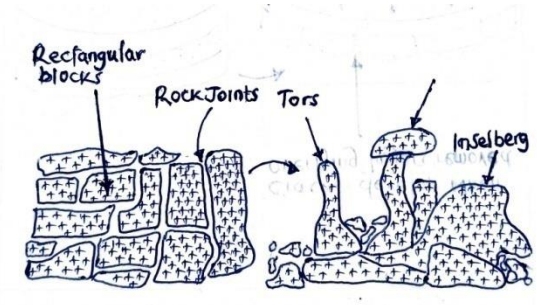
- **Exfoliation**: this is also called insalation or onion weathering. It is common in the arid areas with large diurnal temperature range. The process involves the heating up and expansion of the outer rock surface during day and cooling and contraction of rock layers during the night. Repeated heating and cooling creates internal stress within the top few centimeters of the rock. These results into cracks on the rock surface which later results into peeling off leaving behind a smooth landform known as Exfoliation dome.
- Examples are Panyjok rock in Apac, Kalongo hills in northern Uganda, Nakasongola hills, Kumi in Uganda, Marsabit in northern Kenya, and Hell's-gate at Naivasha in Kenya, Serengeti, and Dodoma in Tanzania.

Illustration



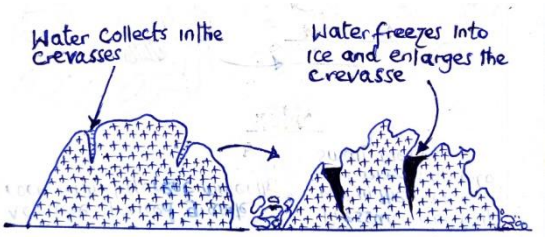
- **Block disintegration**: this is also brought about by temperature changes and is common in well jointed rocks. This process occurs when differences in temperatures and cooling leads into contraction and expansion of jointed rocks. As this continues, rock joints widen and with time rocks break into rectangular blocks. This process also leads to the formation of **Tors**. When only vertical joints appear in a rock, **finger type tors** are produced. If horizontal and vertical joints appear on rock then **castellated tors** are produced.
- Examples are Kachumbala rock in Kumi, Mubende tors in western Uganda, Bismarck rock in Mwanza in Tanzania, Kit Mikai in Sema in Lukenya in Kenya.

Illustration



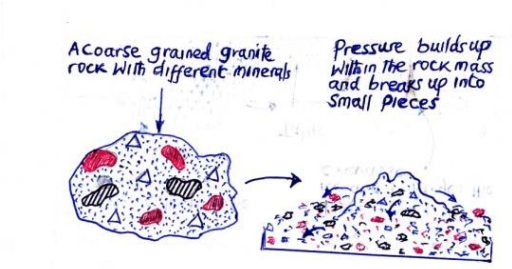
- **Frost shuttering**: this process is the disintegration of rocks mainly due to the impact of glacier. It occurs in rocks that contain crevasses and joints, with limited vegetation cover and where temperatures fall below zero (0^0) degrees like the mountains of Rwenzori, Kenya and Kilimanjaro.
- During the day, ice melts and water enters the cracks/joints of the rocks but at night this water freezes again and exerts pressure within the cracks (Frozen water occupies 10% its liquid volume). The continuous freezing and thawing widens the cracks consequently making rocks to shutter. It is sometimes called frost thaw. Incase shuttering occurs in steep slope angular rocks collect at the foot hills as Scree and on a gentle slope it's called felsedrmen.

Illustration



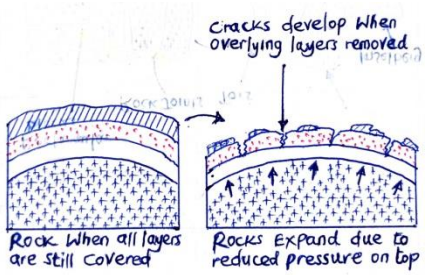
- **Granular disintegration**: this is a form of weathering which occurs in heterogeneous rocks or rocks composed of different minerals. As a result, such rocks absorb heat differently and therefore expand and contract differently. This leads to cracking of such rocks into granules (small particles) in a process called granular disintegration. For instance, granite rocks have several minerals like quartz, mica, etc. in deserts due variation in rock minerals, on heating, pistol-like shots can be heard to represent cracking.

Illustration



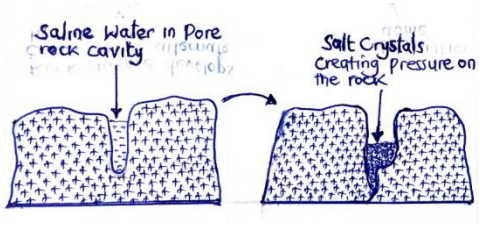
- **Pressure release**: this is a weathering process common in igneous and metamorphic rocks because they were formally under high pressure and heat. It occurs when the overlying weight on the parent rock is removed by denudational processes or human activities. This releases pressure on the exposed rocks and so they begin to expand as they adjust to the new conditions. Cracks then develop parallel to the rock surface. Due to repeated expansion the joints widen and the surface rock layer peels off in a process known as spolling/sheeting.
- Examples can be seen Mayanjo near Kampala, Panyjok in Apac, in Karamoja region, Mwanza, Iringa, Serengeti and Rukwa in Tanzania, Turkana region in Kenya.

Illustration



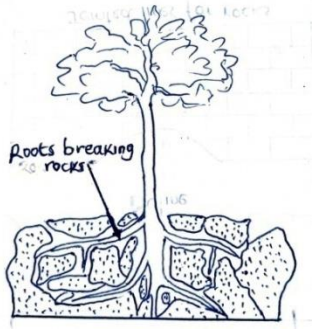
- **Salt crystallization**: in this process, saline water may collect in rock cavities or in pores between rock particles. As the water evapourates due to hot temperatures, salt crystals become larger and they exert pressure on rock cavities causing the rocks to break.
- In East Africa it is common in areas with carbonates and other salt bearing rocks e.g. around lakes like Nyamunuka, Magadi, Katwe, Natron, Albert, etc.

Illustration



- **Aridity shrinkage**: this is a physical weathering process in which there is alternate wetting and drying of porous rocks especially in swamp clays. During the wet season, the clays swell because of absorption of water. As the dry season sets in, Evaporation occurs and the rocks shrink hence disintegrating. Evidence of this type of weathering is the appearance of cracks during dry season on bare compounds and dry swamp beds of seasonal swamps.
- **Spheroidal weathering**: this is a physical weathering process in which once it rains the outer layer of the rock absorb water and in the process the rocks surface swells and forms concentric layers like an onion. The penetration of rain water forces the outer layer to peel-off from the underlying rock and this loosens the rock. e.g. in limestone rocks.
- This is common in areas with heavy rainfall for example around Lake Victoria basin.
- **Biological-physical weathering**: this is a physical weathering in which living things break rocks physically. Tree roots growing in rock joints and spaces between the rock particles widen the joints and spaces eventually causing the rocks to disintegrate. Also animal's hooves walking on rocks, burrowing animals such as rodents, rabbits and rats, and human activities lake agriculture, mining, quarrying and road construction also cause rocks to disintegrate physically.

Illustration



CHEMICAL WEATHERING

This is the decomposition or decay of rocks in situ/one place resulting into change in their chemical composition. This is brought about by the reaction between atmospheric gases such as Carbondioxide, oxygen and hydrogen in solution and rock mineral elements resulting into a change in the chemical composition. The chemical reactions usually take place on the exposed surfaces of the rock to water and may also take place inside the rocks if water can enter through pores/joints/cracks.

- This type of weathering is common in humid regions which are characterized by hot temperatures, high humidity, thick cloud cover and heavy rainfall.
- These areas in East Africa include; the lake Victoria basin, Kigezi highlands, Kenya highlands, southern Tanzania, the coastal areas of Kenya and Tanzania and the slopes of mountains such as Rwenzori, Kenya and Kilimanjaro.

PROCESSES OF CHEMICAL WEATHERING

This processes include; oxidation, carbonation, solution, hydration, reduction, hydrolysis, chellation and Spheroidal.

- **Oxidation**: this is the addition of oxygen to another element. Rain water passes through the atmosphere and absorbs gases such as oxygen. When rain water sinks into the ground, the dissolved oxygen oxidizes elements in the rocks such as iron, sulphur, calcium and magnesium to form oxidized rust on top of the rock surface. The oxidized rocks can easily crumble since the rock structure has been transformed. E.g. oxygen in the atmosphere changes iron to iron oxide.

  $4Fe(s)+O_2(g) \longrightarrow 2Fe_2O(s)$.

- Another form of oxidation involves Ferrous oxide which is oxidized to form hematite.

  $4FeO+O_2 \longrightarrow 2Fe_3O_2$

NOTE: oxidation is responsible for the formation of laterite soils which are common in Buganda. These soils are reddish-brown (latosols).

- **Carbonation:** this involves the reaction of Carbondioxide and rain water to form a weak carbonic acid which eats up calcium carbonate/limestone rocks.

  $CO_2(g) + H_2O (l) \longrightarrow H_2CO_3 (aq)$.

- When calcium carbonate comes into contact with the carbonic acid, it produces a soluble calcium bi-carbonate which is carried away in solution by underground water. This results into rapid decomposition and disintegration of rocks.

  $CaCO_3+H_2CO_3 \longrightarrow Ca(HCO_3)_2$

NOTE: carbonation has led to the formation of limestone valley, ditches, caves, etc, evident in Kajansi on Entebbe road, Nyakasura in western Uganda, Kilifi on the Kenyan coast,

- **Hydration**: this is a process by which certain rocks absorb/take up water and in the process, they expand/swell such rocks like mica, iron and sandstone. When water is taken up by the rock its volume and stress increase and eventually weakens the internal bonding of rock particles causing rocks to fracture, disintegrate and decompose giving rise to new compounds. E.g. when calcium sulphate in the rocks absorbs water, it changes into calcium hydrogen sulphate (gypsum) common in the coastal areas of East Africa.

  $2CaSO_4+2H_2O \longrightarrow 2Ca(HSO_4)_2$

- **Reduction:** this involves the removal of oxygen from a substance and addition of hydrogen to it. This process occurs in the saturated conditions i.e. under conditions of abundant water such as in waterlogged swamps where clay can change colour due to alternate wet and dry season. Under reduction Ferric hydrate may be reduced to iron. Reduction occurs in swampy and waterlogged areas such as Kajansi, Aswa valley, Tana, Pangani, Athi, etc.
- **Hydrolysis**: this process involves the reaction between water and mineral elements of a rock. The reaction takes place only when hydrogen ions of water and the ions of minerals come together. E.g. in the decomposition of felspher, the felspher breaks down to produce potassium hydroxide and acids. The clay mineral which have silica acid and potassium hydroxide, react with Carbondioxide to produce potassium carbonate. The potassium carbonate is removed in solution leaving silica and residual clay materials as the end product. This process is common in areas with clay soils such as Kajansi in Wakiso. Zanna, etc,

  $2KALSiO_8 + 2H_2O + CO_2 \longrightarrow Al_2Si_2O_5(OH)_4 + K_2CO_3 + 4SiO_2$ (felspher).

- **Solution**: this is a process by which soluble rocks minerals are dissolved in water and carried away in solution leaving behind cracks or widened holes in the rock. This weathering process in common in areas with limestone and rock salt rocks and such areas include Tanga in Tanzania, lake Magadi, Kilifi on the Kenyan coast, and Lake Katwe, Nyakasura in fort portal in Uganda.
- **Organic /biological weathering (chellation**): this process involves the decomposition of rocks due to the action of organic organisms e.g. rodents, man, plants, etc. as organic matter decomposes, it releases organic acid such as urea, lactic acid, humic acid, etc. these acids are known as chelating agents attack rocks leading to their decomposition. The rock minerals become

soluble and are then transported downwards by chelating agents in a process known as chelluviation.

- Also plants extract nutrients from the soils such as potassium, magnesium, calcium, sodium, and alluminium and in the process rock are weakened and they disintegrate. The roots release fluids which affect rocks minerals forcing them to breakdown chemically.

REVISION QUESTIONS

INTERDEPENDENCE OF PHYSICAL AND CHEMICAL WEATHERING

As earlier pointed out physical or chemical weathering do not operate in isolation one aids the other in certain times. Biological weathering, however, is either physical or chemical; hence no single weathering type can form an independent category of its own.

How physical weathering aids chemical weathering

- Physical weathering involves the expansion and contraction of rocks which leads to creation of cracks and joints. Joints/cracks/weaknesses found in rocks allow deeper penetration of water in to the rock hence enhances chemical weathering processes like hydrolysis, oxidation, hydration and carbonation
- The physical process of frost shuttering in areas above 4,700m, opens up and widens cracks on rocks and when water occupies this crack, dissolved Carbondioxide will cause carbonation to take place.
- Roots of plants which expand within the bedding planes of rocks create cracks on rocks thus increasing permeability of rocks. This provides favourable conditions for chemical weathering processes. Plants roots also produce Carbondioxide which when dissolved in soil water promotes carbonation weathering process.
- The burrowing animals such as rodents, worms, are good agents of physical weathering. These animals drill holes in rocks which allow water to enter these rocks hence accelerating chemical weathering processes such as oxidation and hydrolysis. These animals as they burrow into the soils release Carbondioxide which when dissolved into soil water promote carbonation weathering process.
- Human activities such as road construction, crop cultivation, mining and quarrying results into physical break down of rocks and exposure of new rocks to the surface. This increases the surface area for chemical weathering processes to operate e.g. hydration, oxidation and reduction.

How chemical weathering aids physical weathering

- Chemical weathering processes such as hydration involves the absorption of water by the rock which makes the rock to swell/expand. Such rocks when temperatures become hot and cause water to evaporate, they will eventually contract resulting into aridity shrinkage/ granular disintegration which are physical weathering processes.
- Some rocks such as chlorides, carbonates, sulphates, rock salts, which are a result of chemical weathering processes like solution process, may undergo precipitation to form crystals in a process of crystallization. The crystals may crystallize in rock cavities which will exert pressure that will cause physical disintegration in a process called salt crystallization.

- Chemical weathering processes like hydrolysis, hydration, carbonation, change the molecular structure of rocks thus weakening the rocks internal bonding. This facilitates physical weathering processes such as frost shuttering and exfoliation.
- When vegetation and animals die they decompose into organic matter through chemical processes which later enhances vegetation growth. Vegetation growth eventually increases root action as a form of physical disintegration.

In conclusion, physical and chemical weathering processes do not work in isolation although certain weathering type may pre-dominate in certain environmental conditions. In all cases one is aided by the other hence complimentary to one another.

FACTORS AFFECTING WEATHERING

1. Climate:

Through its elements of climate i.e. rainfall and temperature, climate is regarded as the most important factor in rock weathering. It influences both the type and the rate of weathering.

- Rainfall is responsible for the chemical decomposition of rocks through processes like hydration, hydrolysis whereas temperature change is responsible for the physical disintegration of rocks. Therefore, equatorial region e.g. around the Lake Victoria basin experiences high rate of chemical weathering due to heavy rainfall received and hot temperatures experienced.
- In arid and semi-arid regions such as Karamoja, Turkana-land, and central Tanzania with low and unreliable rainfall but with very hot temperatures, high rates of physical weathering are experienced. The very hot temperatures lead to rocks expansion while the very cold night's temperature leads to rock contraction. Repeated expansion and contraction of rocks leads to their physical disintegration resulting into processes like exfoliation and block disintegration.
- In cold climatic areas such as snowcapped peaks of mount Rwenzori, Kenya and Kilimanjaro frequent freezing and thawing lead to an increase in the rate of weathering in a process called frost thawing or shuttering.
- Climate also indirectly influences weathering through its effects on living organisms like plants which depend on climate for their metabolism. This increases on the rate of physical, chemical and biological weathering. Plants for example use water and sunlight for their proper growth which aspect leads to expansion of roots which causes physical disintegration.

REVISION QUESTION

Qn: *Climate is the most important factor affecting the rate and nature of weathering in East Africa. Discuss.*

Approach*: define weathering, describe the various types of weathering, talk about climate and its elements that affect weathering and show how they influence weathering, give other factors and conclude systematically.*

2. Influence of living organisms:

- Plants and animals also affect the rate and the nature of weathering e.g. plants aid physical weathering by making roots to disintegrate rocks under the pressure of their growing root system leading to weathering processes such as block disintegration.
- Chemically plant roots add chemicals such as nitric and citric acids as well as Carbon dioxide into the rocks. This increases the rate of chemical decomposition of rocks aiding processes such as carbanonation.
- Animals on the other hand contribute to physical weathering in many ways e.g. hooves of large animals like elephants physically dislodge rock particles as these animals walk on them. Small animals like rodents physically break rocks through their burrowing activities like digging holes.

- Dead and decaying animals are also important in the chemical decomposition of rocks. When decaying animals release organic acids such as urea, ammonia, lactic acid reacts with the rock minerals to cause chemical decomposition of rocks.
- Man, through his activities like bush burning, farming, construction, mining and quarrying, etc, breakdown rocks physically.

3. Nature of parent rock/lithology:

- Rocks influence weathering through their mineralogical composition, texture, colour, joints, permeability and hardness.
- In terms of rock colour, dark coloured rocks are weathered faster than light coloured rocks. Dark coloured rocks like granite absorb more heat than light coloured rocks which can reflect some heat into the atmosphere. This means that when subjected to temperature change, dark coloured rocks expand and contract through exfoliation more than the light coloured rocks as such they are more susceptible to physical weathering than light coloured rocks.
- As per mineralogical composition e.g. in igneous and metamorphic rocks when exposed to the surface, they are highly affected by physical weathering because they were formed at great heat and pressure which conditions are different from those on the earth surface resulting into a physical weathering process of granular disintegration.
- Also, limestone rocks containing calcium carbonate breaks down faster on reacting with water containing Carbon dioxide in the process called carbonation.
- Rocks containing iron, magnesium, Sulphur, calcium, etc, weather faster when exposed to oxygen in the process called oxidation.
- Rock hardness or softness also affect weathering. rocks vary in terms of hardness. The harder rocks such as granite or quartzite are for example resistant to weathering processes than soft rocks like clay. Therefore, harder rocks slowdown the rate of rock disintegration whether through physically or chemically.
- The degree of rock jointing is also another factor to consider. Rocks with joints or faults allow water to easily penetrate them hence enhancing chemical weathering process such as solution. Also, physical weathering processes such as frost shuttering, block disintegration are facilitated by the presence of cracks/faults/joints.
- As per texture the roughness or smoothness depends on the rock particles or grains. Coarse textured are weathered more quickly than fine grained rocks. This is because large grains contract and expand faster than the fine-grained rocks. Also, in coarse grained rocks that has several minerals, weathering of minerals has a big effect on the fine-grained rocks of the same nature of minerals.
- Permeable rocks allow the penetration of water into the mass of the rocks hence promoting chemical weathering process easily.
- Age of the parent rock influences weathering process in that rocks of resent formation have not been exposed to agents of weathering hence are less weathered while those rocks formed long ago are more exposed to weathering hence are more weathered.

REVISION QUESTION

QN: *To what extent does the rate and nature of weathering depend on the nature of the parent rock/lithology?*

Approach: *define weathering, give types of weathering, describe weathering processes, define the nature of rock/lithology, and show how it affects weathering, give other factors.*

4. Topography/relief:

The nature of landscape influences weathering through the following ways

- Hilly and mountainous areas influence the distribution of rainfall. Rainfall is usually received on the windward side of mountains thus accelerating chemical weathering process such as hydration which require water to operate. On the leeward side because of arid conditions and low rainfall, usually physical weathering processes like exfoliation, etc, take place.
- Also, mountains with an average height of 4,700 meters above sea level like Rwenzori, Kenya and Kilimanjaro experience rain water and humidity turn into snow. The existence of snow leads to freeze-thaw action/frost shuttering which is a physical weathering process.
- On the steep slopes, weathered materials are removed by erosion thereby exposing fresh rocks surfaces to further weathering hence increasing the rate of rocks disintegration. Weathered rocks materials are deposited into lowlands thus covering the exposed rocks and protecting them from further weathering.
- In the lowlands water tends to collect there hence facilitating chemical weathering processes like hydrolysis, carbonation, etc. Poorly drained areas like the lowlands and flood basins also encourage reduction and solution as chemical weathering processes.

5. Time:

- Although it can be regarded as less important factor, time is important in as far as rock weathering is concerned. The longer the time weathering operates the greater and the deeper the rate of weathering and vice-versa.
- Time can also be considered in terms of rocks age, old rocks` like granites are very hard and so slowdown the rate of weathering as compared to young and soft rocks.

6. man's activities.

REVISION QUESTIONS

LANDFORMS OF WEATHERING

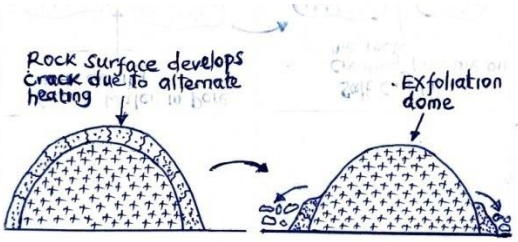
Weathering like other earth processes produce distinct landforms/features. These landforms are categorized into those due to physical weathering and chemical weathering processes.

PHYSICAL WEATHERING LANDFORMS

Physical weathering is the disintegration of rocks by physical means without changing the composition of the rock. It is brought by temperature changes, influence of plants, animals and man. It is accomplished through processes such as exfoliation, block disintegration, pressure release, granular disintegration, etc. it is therefore responsible for the formation of various landforms as seen below.

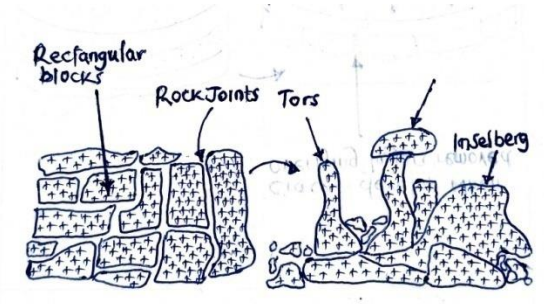
- **Exfoliation dome**: this is a smooth and round topped hill formed due to alternating hot day and cold night temperatures. During day rock surfaces expand and contract at night when temperatures drop. Repeated expansion and contraction result into the peeling-off of the outer surface of the rock leaving behind an exfoliation dome.
- These features are common in arid areas such as the Chalbi desert in Kenya, Moroto and Kotido in Uganda.

Illustration



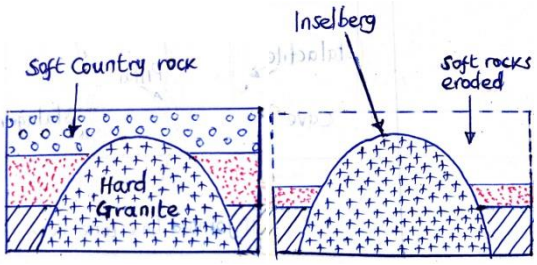
- **Tors**: these are pillars of rounded weathered boulders rooted in the bed rock. They are formed through the physical weathering process called block disintegration. When rocks that have vertical and horizontal joints undergo repeated expansion and contraction during the day when temperatures are hot and nights when temperatures are cold, joints are enlarged and pieces of rocks fall off leaving up-standing pillars rooted on the earth called tors.
- Examples include Kachumbala tor in Kumi eastern Uganda, Mubende tor, Bismarck rock in Mwanza in Tanzania, Kitmikaye near Seme in Kenya.

Illustration



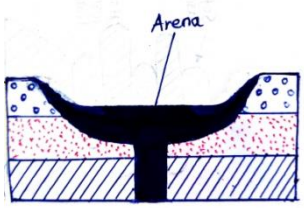
- **Inselbergs**: these are hills of hard and resistant granite rocks usually in a level plain. They are formed when weathering and erosion removed away surface softer rock layers leaving the hard-resistant granite rock exposed as a hill.
- Examples can be seen at Mubende, Nakasongola, Kangole in Moroto, Seme near Kisumu, Koma near Nairobi, and Bismarck rock near Mwanza in Tanzania.

Illustration



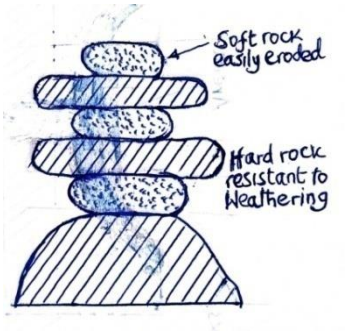
- **Arenas**: these are large circular depressions on the earth surface. They are formed in areas of alternating hard and soft rocks. The soft rocks are attacked by weathering and erosion leading to the formation of a basin/depression while the hard/resistant rocks stand prominently around the basin. An Arena may be filled with water to form ponds.
- Example can be found at Ntungamo and Mbarara. And Rubanda arena along Kabale-Kisoro road

Illustration



- **Rock Pedestals**: these are tower-like structures of various rock sizes standing prominently on a flat landscape in desert and arid areas. They are formed where the rocks are composed of soft and hard rocks. The softer rocks are attacked by physical weathering and erosion and shaped into a tower-like structure.
- Example is the Devils rock in Niger.

Illustration



- **Boulders/Hard core stones**: these are generally boulders and rocks stones left behind when weathered material that sounded them are removed. They often appear isolated when left in a totally different landscape.
- They are often found near tors and can be found in areas such as Mubende, Nakasongola, Mwanza, etc.

CHEMICAL WEATHERING LANDFORMS

Chemical weathering is process in which rocks are decomposed as a result of the reaction between atmospheric gasses in solution water with the rock mineral elements resulting into change in the chemical composition of the rocks. It is influenced mainly by heavy rain water and is common in the humid areas.

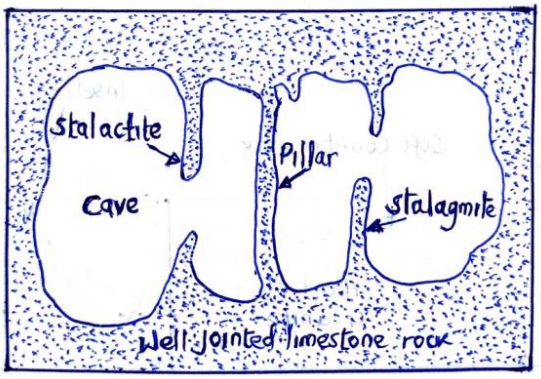
- It is a accomplished through hydrolysis, oxidation, Solution, carbonation, etc.

Landforms due to chemical weathering

- **Karst landscapes**; in limestone areas, carbonation and solution processes have led ton the formation of karst landscape with features such as stalagmites, stalactites, pillars, caves, dolines, limestone pavements, sink holes, etc. when rainwater mixes with Carbondioxide gas it forms a weak solution of carbonic acid. The acid dissolves limestone rocks resulting into the formation of several features below.
- **Underground Caves**: these are naturally underground chambers in limestone rocks areas usually joined to the land surface by a system of inter-connected shaft and pillars. They are formed when limestone rocks are dissolved away through carbonation and solution weathering. Sometimes mechanical erosion by underground streams after rock collapse may also produce empty chambers/cavity hence the formation of caves.
- Examples are found at Nyakasura in Fort Portal.
- **Stalactites:** these are columns of calcium carbonate protrusions formed on the roof of a limestone cave. They are formed when carbonic acid dissolves limestone rocks to form calcium bicarbonate solution. When the solution reaches the roof of the cave it starts dripping. As water evapourates, dripping stops and calcium carbonate is deposited and dries to form protrusions called stalactites.
- Examples are found at Nyakasura in western Uganda, Tanga in Tanzania.
- **Stalagmites**: these are whitish or grey protrusions on the floor of the cave in limestone areas. They are formed from the accumulation of the dripping calcium bicarbonate after solution weathering process. As water evapourates, a dry and compact mass of calcium carbonate protrusions are formed standing at the floor of the cave as stalagmites.

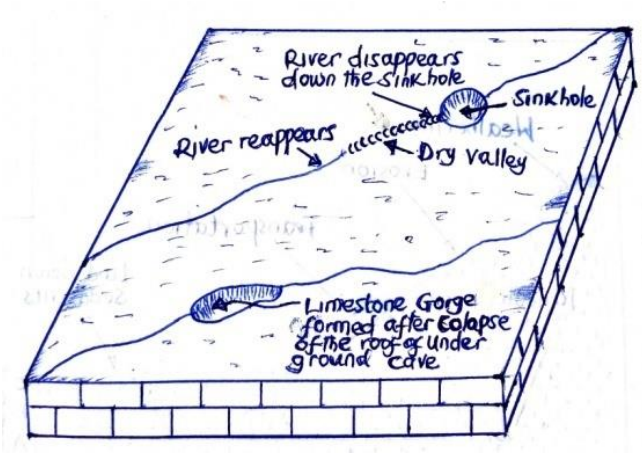
- Examples are at Nyakasura in Uganda.
- **Pillars**: these are vertical stands of calcium carbonate which form after stalagmites and stalactites grow and meet to form a stand or a pillar.
- Examples are found at Tanga and Nyakasura.

Illustration



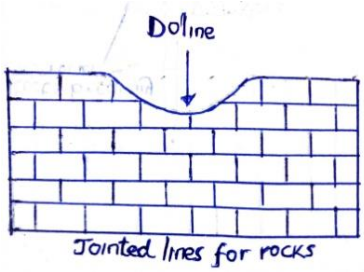
- **Sink holes/swallow holes**: This is a deep hole with nearly vertical sides leading into an underground cave. They are formed through process of carbonation and solution. Limestone rocks from the surface up to the cave may be dissolved and removed in solution to leave behind deep holes which penetrate into the caves. Alternatively, they are formed due to surface collapse.
- **Dry valley**: This is a valley with no permanent stream and often with steep sides. Rivers disappear underground through a sink hole on entering a limestone region and reappear on the surface at a junction of the limestone rock and impermeable underground rock. The resulting surface valley is the one termed as a dry valley
- Example on river Malopo in South Africa.

Illustration



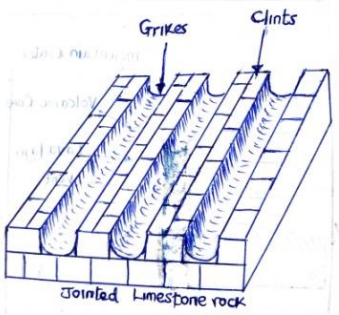
- **Limestone gorge**: this is a deep narrow steep sided valley which is formed when the roof of an underground cave collapses. Sometimes it is formed when a river cuts across and down into a limestone region. Example can be seen at the caves of river Manambolo in central Madagascar.
- **Duricrust**: This is a hard crust formed at the earth's surface or just below the ground due to oxidation. They are formed when weathered layers become mixed with iron solution due to leaching. When the overlying layer is removed laterites harden to form duricrusts e.g. the flat topped hills of Buganda.
- **Dolines:** These are shallow depressions or hollows with gently sloping sides generally circular or oval in shape. They are formed at the intersection of major joints where there is concentration of chemical reactions which quickly dissolve limestone rocks to form small basins.

Illustration



- **Limestone pavements**: This is a bare surface which is cris-crossed by many grooves where different carbonation weathering has worked along joints. They are formed as a result of carbonation on limestone areas with rocks of different chemical composition. Carbonation weathering may weather down the less resistant limestone rocks to form hollows/deep narrow grooves called **Grikes.** While some relatively resistant limestone rocks may form flat/round topped ridges called **Clints**. Clints may form between two grikes.
- These can be found Karasuka region of Karamoja, western side of Tororo rock, Tanga region of Tanzania, Kilifi in Kenya.

Illustration



- **Uvala:** This is a large depression intermediate between a doline and a polge. They are formed when dolines are enlarged or joined due to the solvent action of carbonic acid on limestone rocks. Example can found at Nyakasura.
- **Polge**: This is a shallow but very large steep-sided depression with a generally flat bottom. It covers 100ths of km. they are formed when Uvalas or Dolines are enlarged and joined by solution weathering. Polges may get filled with water to form solution lakes.

REVISION QUESTIONS

WEATHERING IN THE DIFFERENT REGIONS OF EAST AFRICA

In equatorial climatic regions:

- Under equatorial climatic conditions, chemical weathering is predominant due to mostly heavy rainfall (up to 2,000mm) hot temperatures (up to 25 degrees) and high humidity throughout the

year. These conditions are mainly experienced in the Lake Victoria basin. Some of the weathering processes include hydration, oxidation, reduction, hydrolysis, solution, etc.

- However, the type of climate is also favourable for living organisms e.g. plants and animals thus leading to biological-physical and biological-chemical weathering processes.

REVISION QUESTIONS

Qn*: Examine the weathering processes taking place in Equatorial climatic regions.*

Approach*: define weathering, describe the characteristics of equatorial climatic regions, explain the chemical weathering process and how it is dominant, and give some physical weathering process of bio-physical weathering.*

Qn: *examine the weathering processes taking place in Lake Victoria region.*

Qn: *Examine the weathering processes taking place in humid regions*

In the tropical/savannah climatic region:

- This region is characterized by wet and dry conditions. Rainfall ranges between 700mm-1,200mm. Rainfall is seasonal, temperature hot and hence during the wet/rainy season chemical weathering dominates such as oxidation, solution, etc.
- Physical weathering is active in the dry season e.g. Exfoliation weathering is active during the drier seasons due to hot temperatures and cold nights which cause expansion and contraction of rocks. Other weathering processes during dry season are aridity shrinkage, block disintegration.
- The nature of the savannah climate influences pastures which support a wide range of game animals for survival for example in Murchison Falls' national park. These animals are responsible for the occurrence of biological-physical and biological-chemical weathering processes.

REVISION QUESTION

Qn: *Examine the weathering processes taking place in the tropical region.*

Approach: *define weathering, describe the characteristics of the savannah climatic region, relate the characteristics and the weathering processes there, explain the physical and chemical processes taking place during both the dry and wet seasons.*

In the arid and semi arid areas:

- These regions are characterized by low rainfall high rate of Evapouration, large diurnal ranges of temperature, cloudless sky and besides there is also limited vegetation cover and bare surfaces in some places due to low rainfall. As a result of these conditions physical weathering dominates.
- At night some little vapour from air is deposited on a rapidly cooling rock surfaces as dew thus allowing some chemical weathering process like hydrolysis, oxidation, etc, to occur. This occurs in areas of Karamoja, Turkana land, central Tanzania, etc,

REVISION QUESTIONS

Qn: *Examine the weathering processes taking place in the arid and semi arid regions*

Approach: *define weathering, give types of weathering, give the characteristics of arid and semi arid areas relate the characteristics to the processes, explain the physical weathering processes, conclusion.*

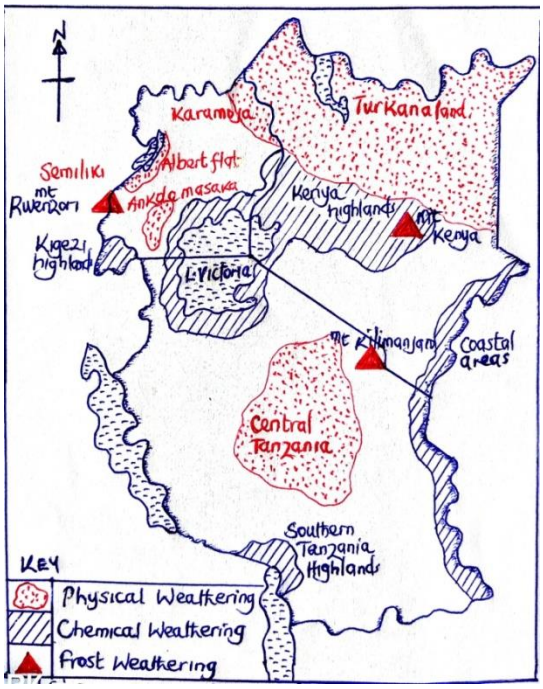
In Montane climatic regions:

- Mountainous areas especially at higher altitudes above 4,700m, temperature fall bellow 0°C, snow/ ice is common. This condition facilitates the dominance of frost shuttering weathering thus physical weathering is common in high altitudes. These areas are mounts such as Kilimanjaro, Rwenzori and Kenya.
- The windward sides of mountains experience heavy rainfall which induces chemical weathering processes such as carbonation, hydrolysis, etc while the leeward slopes are dry and in the rain shadow, this regions influence physical weathering processes.

NOTE: though some weathering processes seem to dominate in certain regions they can not work in isolation in fact they are interdependent.

REVISION QUESTIONS

SKETCH MAP OF EAST AFRCA SHOWING THE MAJOR WEATHERING REGIONS



IMPORTANCE OF WEATHERING

- Weathering is the first and very important process in the soil formation. Volcanic rocks have weathered down to provide very fertile soils in East Africa e.g. at the slopes of Mount Elgon, Kenya highlands, etc, and this has supported crop cultivation.
- Weathering is responsible for the creation of magnificent features that attract tourists such as caves, stalagmites, tors, etc. These features earn foreign exchange through tourism which is used to develop the economies of the East African countries.
- Weathering is responsible in the formation of clay soils through hydrolysis which support brick and tiles making industry in East Africa. For example the tiles and brick making at Kajansi along Entebbe road hence encourage building and construction industry.
- Weathering has boosted the mining industry by exposing underground mineral ores such as diamonds at Mwadui in Tanzania. In the same way sand mining is carried out in many areas along the shores of Lake Victoria.
- Weathering has also exposed granite rocks which are quarried to provide aggregates for the construction industry e.g. Nakasongola rocks were quarried for the construction of Kampala– Gulu highway.
- Weathering is responsible for the creation of flat surfaces on rocks which makes construction activities easy like the constructions on the flat topped hills in Buganda.
- Weathering is also responsible for lateritic soils formation through oxidation of iron and alluminium. These rocks are importantly used in the construction industry such as roads and houses in Buganda region.
- Weathering is responsible for the formation of various landforms such as caves in the limestone rocks, tors, etc which are very important for research and study hence promoting education sector.

However,

- Weathering on the steep slopes leaves behind loosened rock materials that are prone to erosion by running water. This has increased erosion of soils on slopes of Mount Kilimanjaro in Tanzania, Kigezi in south western Uganda, etc.
- Inselbergs pose a problem towards road construction. They need blasting before the road can be constructed and this is very expensive.
- Weathering in limestone areas has left rugged terrain of cliffs, caves, grinks, which has limited crop cultivation and settlement. Such areas are Nyakasura in Uganda, Tanga in Tanzania, and Kilifi in Kenya.
- Physical weathering results into the cracking of buildings and their eventual collapse. This increases costs of construction.
- Weathering in some areas has resulted into the formation of very hard and infertile soils which do not support crop cultivation such as the lateritic soils in Buganda.
- Weathering leads to disintegration of rocks hence aiding downhill movement of soils/landslides which destroy lives and properties.

GLACIATION IN EAST AFRICA

This is the process by which glaciers shape the earth's surface. Or it can be defined as the activities of a glacier.

In east Africa, glaciation can only take place on very high mountains that rise above the snow line at 5000m namely mountain Rwenzori, Kenya and Kilimanjaro. On mountain Rwenzori, there are 37 glaciers e.g. stanely, speke, Bujjuku, Mubuku, Edward, Backer, Moore etc.

Mountain kenya has the second largest coverage of glaciers and best examples here include lewis, Gregory, krapf, dawin, tyndal, ford, diamond, heim etc.

On mountain Kilimanjaro glaciers include kibo, great penk, furt wangler etc.

A GLACIER

A glacier is a mass of moving ice which moves down slope along a pre-existing river valley under the influence of gravity. Glaciers move continuously from higher to lower ground and is enclosed within valley walls.

FORMATION OF GLACIERS.

When the temperatures in an area fall below 0^0, water vapor in the atmosphere condenses to form snow. The snow accumulates in pre-glacial hollows. Under pressure, the air in it escapes melting and freezing occurs. Further compaction expels all the air and the bottom layers are compacted into ice. With time under the influence of gravity, the ice moves out of the hollow and is now known as a glacier.

TYPES OF GLACIERS.

a) **Valley glaciers**; these are formed on highland or mountain peaks and upper valleys of mountain ranges. Valley glaciers move by sliding over the rock surfaces. Friction with the ground produces heat that in turn melts ice and helps to lubrication hence speeding up the sliding process.
b) **Lowland glacier**; this is a glacier found in colder and higher latitude areas mainly between 60° latitudes and the poles. They are mostly found in Greenland, Arctic and Antarctica.
NB There is limited glacier coverage in East Africa due to the following reasons;

Latitudinal location of east Africa. Latitude refers to the distance from the equator. East Africa lies along or astride the equator at approximately 5 degrees north and south of the equator. For this reason, whether the apparent overhead sun shifts north or south, the angle of incidence of the solar rays is still small. So the sun rays strike the earth's surface at a wide angle and at a short distance hence ensuring maximum solar heating throughout the year. This implies that the temperatures in east Africa are high throughout the year. Therefore, its only in highland areas which rise above the snowline where temperatures drop to below freezing point that enables glaciers to form. However, these are very few areas where temperatures can drop below zero degrees hence limited coverage.

Altitude. Much of East Africa lies at a low altitude of less than 2000m above sea-level and this limits glacier formation because low altitude areas are associated with hot temperatures thus limiting glaciers to very few areas in East Africa rise to the current snowline of 4800m above sea level.

Precipitation. East Africa generally receives moderate rainfall of between 1000 to 1500m per annum. This is just sufficient to sustain life. However, glaciers form in areas where temperatures drop below freezing point and with torrential rainfall.

Global warming. The world temperatures have been rising with time and this is attributed to human activities like burning fuels e.g. coal, oil and natural gas which increases carbon-dioxide concentration in the atmosphere and other greenhouse gases. These gases absorb a lot of heat from the sun during day and prevent heat loss at night (they act as a blanket). This accounts for the increase in the global temperatures by 2.5°C. this increase in temperature has led to the melting away of glaciers to the point of extinction of some glaciers e.g. the sempaya glaciers on mountain Rwenzori.

Aspect. This refers to the angle at which the slope receives the sun's insolation. East Africa lies between the tropics and therefore experiences direct over sun throughout the year. This implies that all slopes of the mountain are subjected to at least 10 hours of great sunshine a day without any sheltered slope or obstructed slope to enable glacier formation.

Impact of volcanism. Some of the high mountains where glaciation would have taken are volcanic in nature. The temperatures in these mountains are high due to the hot interior which brings about the warming effect on the surroundings. This is therefore preventing glacier formation.

GLACIAL ACTIVITIES.

Erosion

Transport

Deposition

GLACIAL EROSION

Glacial erosion occurs through the following processes;

Plucking or exaration.

It is the tearing away of the blocks of rocks which have become frozen into the sides and bottom of a glacier (glacial depressions). It is more effective where there well jointed rocks subjected to freeze and thawing.

Abrasion.

It's the wearing a way of rocks underneath a glacier by the swirling of rocks emborded in glaciers. In this process, the glaciers use the materials being carried or transported as the grinding tool which is used to break up rocks along the floor and sides of the glacial trough.

Basal Sapping.

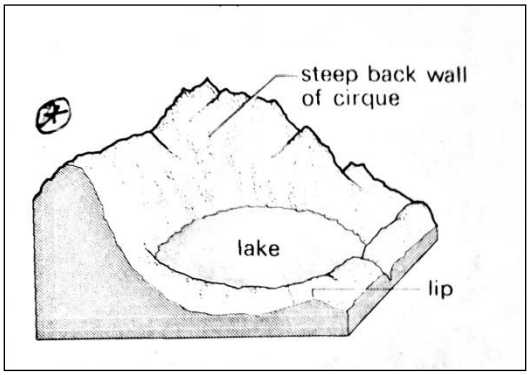
This involves the breakup of rocks by alternate freezing and thawing of water in rock joints/cracks that exist on the sides and floor of a valley or the side of a mountain.

GLACIAL EROSION FEATURES / GLACIAL UPLAND FEATURES

1. ***CIRQUES/CORRIE/TARNS.***

These are semi-circular steep sided rock basins which have been cut into the sides of a glaciated mountain. They are formed when water enters the rocks and freezes breaking them down. The joints become enlarged. Abrasion drags the debris over the rock floor deepening the depression; back wall cutting or recession also enlarged the depression as well as steepening the sides hence a Cirque/Corrie. When filled with water, it forms a tarn (lake) such as teleki and mawenzi on mountain Kenya and Kilimanjaro respectively. Others are Lac du Noir, Lac du Catherin and Lac du vent on mountain Rwenzori.

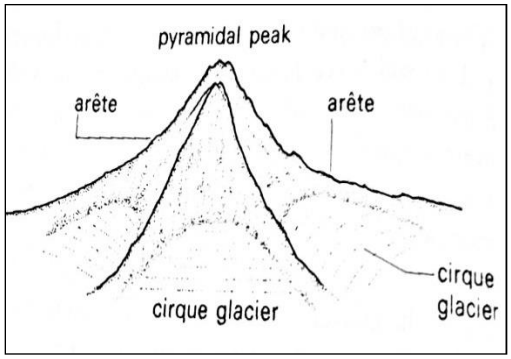
illustration.



2. ***Pyramidal peak /horn***

This is a jagged peak with a series of radiating aretes. It is formed by back wall recession of two or several cirques which lie on the sides of a glaciated mountain. Two or more cirques cut back into the original mountain sides leaving an isolated peak called a pyramidal peak. These peaks become shaped by frost action. Examples include kibo, mawenzi on mountain Kilimanjaro and margarita peak on mountain Rwenzori.

Illustration.



3. Arête.

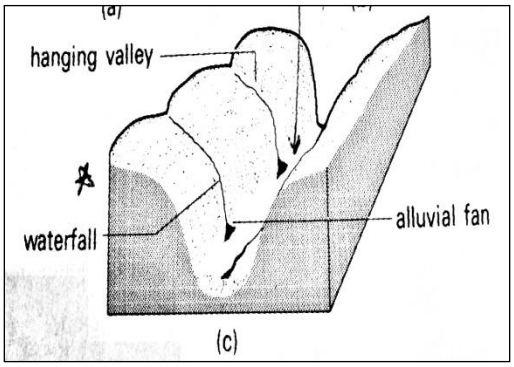
This is a sharp knife like edged feature or narrow ridge in between two or more cirques on the mountain sides. It's formed when two or more cirques erode backwards (back to back) through the process of head Ward erosion. Best examples can be seen on my Rwenzori from Mount Backer up-to bujuku Valley and point John and midget peaks on mountain Kenya.

4. U-shaped valleys or glacial troughs.

These are broad flat bottomed steep sided valleys with an open u shape in the cross profile. These were usually former river valleys which are filled by glaciers. Through the process of abrasion, plucking and sapping, the valley floor and sides were worn away hence deepening and widening the valley. So the valley is changed from a V-shape to an open U-shape hence a U shaped valley/trough.

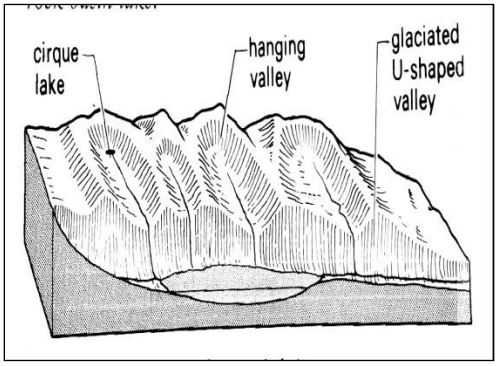
Examples of u shaped valleys include Mubuku valley, Mugusu Valley and komusonso valleys on mountain Rwenzori, hobley and Teleki valleys on mountain Kenya and the karanga through on mountain Kilimanjaro.

Illustration



5. Hanging valley

These are small narrow short and V-shaped valleys laying above the main the main valley found in highland areas. They normally descend steeply into the main valley by a water fall. For example, Nithi river is joined by the little Nithi from a hanging valley on mountain Kenya.



6. Truncated spurs

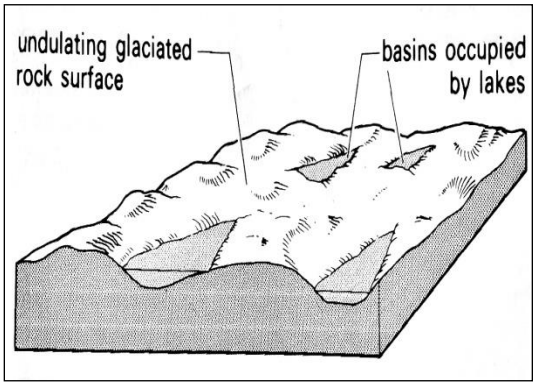
These are ridges with steep edges formed when former Interlocking spurs of a former river valley are cut off by lateral erosion resulting from accumulated glaciers in the valley forming a U-shaped valley with Truncated spurs instead of interlocking spurs for example around Mubuku and bujuku Valleys on mountain Rwenzori.

7. Rock basins

These are irregular depressions on the bed of the glacier trough. They are formed by un equal glacial erosion of the bed rock when out rock protrusions along the trough are removed to form hollows called rock basins.

Similarly, the path of the glacier may have rocks of different types and resistance to weathering and erosion where a band of soft rocks Alternate with bands of hard rocks, the soft rocks will be worn out to form a depression which is filled by water to form a basin lake. Best examples include lac Michaelson on mountain Kenya, Lac noir and Lac vert in the kamusoso valley on mountain Rwenzori.

Illustration



8. Rock steps.

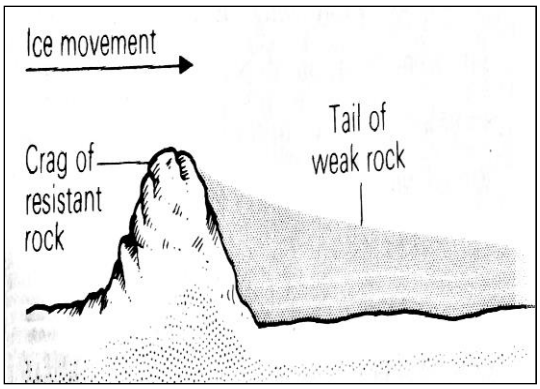
These are step-like hard rock projections in the glacial valley formed as a result of differential erosion due to difference in rock resistance and variation in glacier thickness. The increased ice in the main valley makes it possible to cut deep in the valley creating step like features known as rock steps for example at Vivian falls on mountain Kenya

illustration. (research)

9. Crag and tail.

This is an elongated rock mass with a steep slope on the upstream which protects the softer lee ward rock from being eroded or worn out by the glacier. Its formed when advancing ice meets a resistant rock protecting a weaker rock downstream from erosion leading to the formation of an elongated tail on the side of the weaker rocks while the resistant rock mass forms a crag with a steep slope.

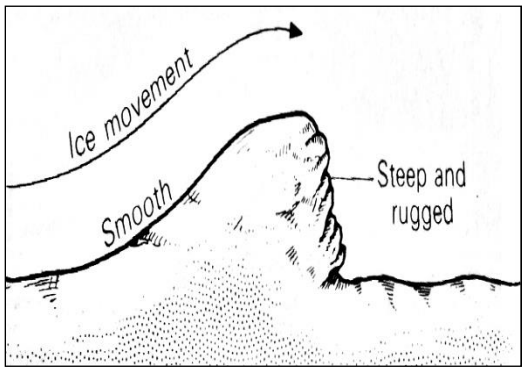
Illustration.



10. Roche montannes.

These are resistant rock masses smoothened on the upstream side by abrasion of ice and steepened on the downstream side. The downstream side is worn by plucking of ice making it rough. Examples can be found at the mobuku valley along the slopes of mountain Rwenzori and along the George valley on mountain Kenya.

Illustration



11. Stratitions

These are thin parallel scratches on glaciated areas formed by abrasion when large fragments of angular rock debris embodied in the ice tend to leave a series of parallel scratches. Such scratches can be seen in Stanley valley on Mt Rwenzori and near Peters hut on Mt Kilimanjaro.

Question

1. ***Account for the formation of upland glacial landforms in east Africa.***

   Approach

   - Define glaciation
   - Identify areas where glaciation is taking place in east Africa
   - Factors or conditions for the formation of glaciers.
   - Processes of glaciation
   - Identify and describe the landforms mainly erosional with clear illustrations.

2. ***Describe the processes responsible for the formation of glacial erosion landforms in east Africa***

   Approach

   - Define glaciation

- Identify areas in east Africa where glaciation is taking place
- Explain the condition for the formation of glaciers/factors for the formation of glaciers.
- Explain the processes of glaciation
- Identify and describe the landforms resulting from glacial erosion in east Africa with clear illustrations and examples.

GLACIAL DEPOSITION – LANDFORMS FORMED DUE TO GLACIAL DEPOSITION

Glaciers transport a variety of materials ranging from fine rock flour to large boulders. All these materials are deposited in the valleys just beyond the point of melting and can modify the landscape markedly. Glacial depositional landforms are mainly found in the lowlands in the lower slopes of the mountains I.e. the valleys. The moraine transported and deposited by the glaciers results into the formation of depositional landform and these include:

1. Moraine

This refers to the rock debris that is eroded by flowing ice from the sides and base of the valley and deposited in the low-lying areas. Moraine are large and small fragments that are detouched, transported and deposited when ice losses it's capacity to transport the materials.

Terminal moraine, it's a ridge like feature formed by accumulation of unsorted fragments of all sizes extending across the valley as a belt of low hills for many kilometers. It's formed by extensive deposition along the snout of an ice sheet and may cover a reasonable distance and height of 50 meters. It builds up when the glacier is static. Examples can be seen in the low-lying areas of Rwenzori i.e. bujuku, Mubuku valley kimberly plains of South Africa etc.

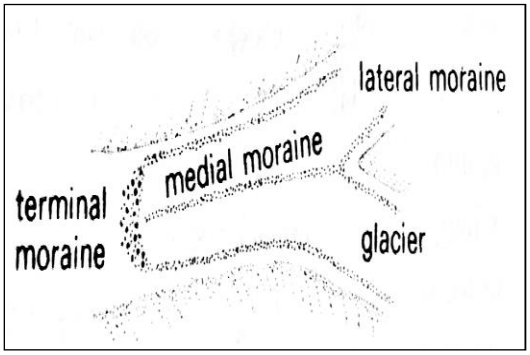
Lateral moraine, this is an elongated ridge or moraine formed along the valley glacier sides. The deposition is as a result of friction between the moraine and the valley sides. Lateral moraine can be seen along Bujjuku, Mubuku and Kamsuso troughs on my Rwenzori and at the point where little Nithi river joins the main stream of river Nithi in the Teleki valley on my Kenya.

Medial moraine

When two glaciers meet, lateral moraine of the inner sides of both glaciers joins to form medial moraine. Medial moraine forms at the center of the glacier when a glacier retreats and drops its moraine.

Ground or basal moraine. This is formed when moraine is deposited at the bed of the glacier. It's composed of mainly fine glacial moraine dropped by retreating ice. Ground moraine almost covers the whole width of the valley.

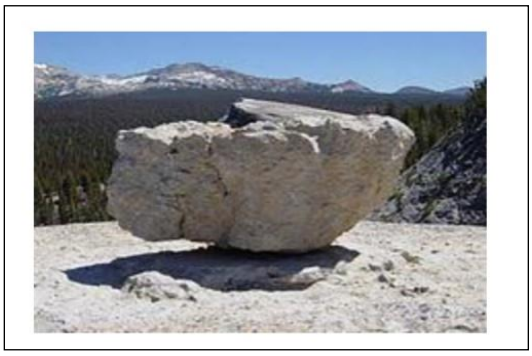
Illustration



2. Erratics

These are rock fragments of varying sizes which are eroded, transported and deposited in areas where they are far different from the surrounding rocks where it's deposited. They are useful in determining the direction and source of ice movement. Best examples can be seen at kamusoso and bujuku Valleys on mountain Rwenzori in western Uganda.

Illustration



3. Till plains.

These are extensive areas of monotonous relief or landscape formed when moving Ice transports boulders, sand and clay hence burying former valleys and protruding rocks to form an almost leveled landscape. e.g. Teleki valley on mountain Kenya and Mubuku valley on mt Rwenzori.

Illustration (research in ssekalega abas ssentamu)

4. Eskers

These are long winding steep sided ridges of sand and gravel lying parallel to the direction of ice movement. They are more than 30 meters high and several kilometers long. Sometimes they are formed by rivers flowing beneath or within ice. they develop on areas of stagnant ice where rivers maintain sub glacier tunnels.

It's process of formation is attributed to stagnant ice sheets under hydrostatic pressure hence the ice develops melt water streams flowing within the ice following permanent sub glacial tunnels thereby deposition taking place within the tunnels. When the ice finally melts, it leaves behind long and steep sided ridges known as Eskers.

Illustration (research from ssekalega abas ssentamu)

5. Drumlins

These are low, rounded and elongated hills composed of bolder and clays lying parallel to the direction of ice movement. They are usually about 30m high and cover an area of up-to 1km. Drumlins occurs till plains in large groups or swarms aligned to the direction of the former ice advance. They are mostly formed when fragments or ground moraine are compressed by ice movements. Best examples are to be found in the teleki valley on mountain Kenya.

Illustration.

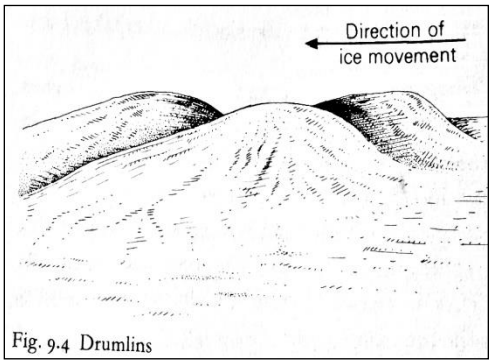
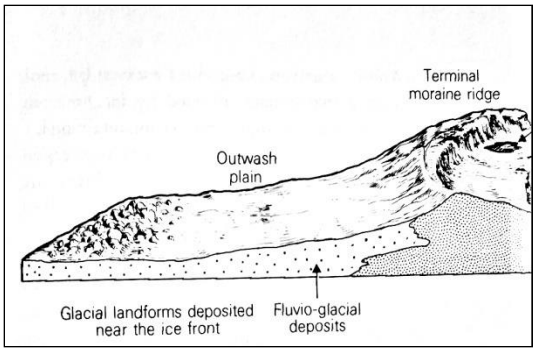


Fig. 9.4 Drumlins

6. Out wash plains

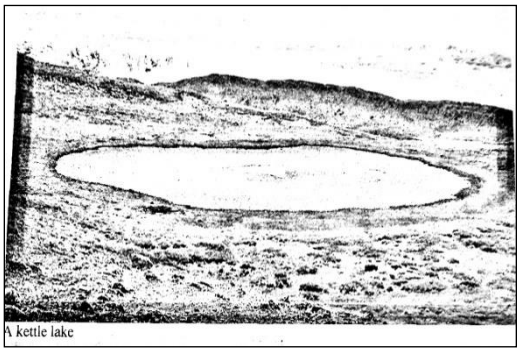
These are wide gently sloping plains of gravel, fine sand and silt. They are formed when volumes of melt water redeposit sorted materials further down slope. Best examples can be seen at kibo and mawenzi on mt. Kilimanjaro, Mubuku and bujuku Valleys on mt. Rwenzori.



7. Kettle holes

these are circular holes in fluvial glacial deposits formed when an ice block is detached from the main glacier as it retreats. Such a block is then buried within the till by fluvial glacial deposits. The subsequent melting of Ice blocks leaves behind depressions to form kettles. When water collects in these depressions, they form kettle lakes.

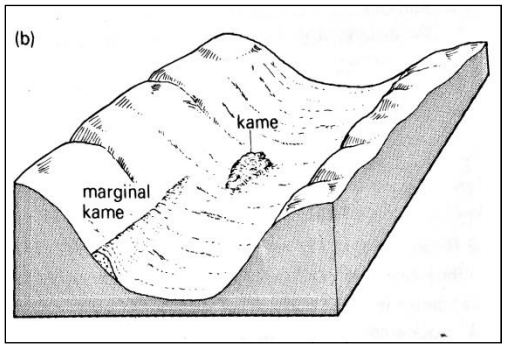
Illustrations



8. Kame moraine

These are ridges of sand and gravel on the sides of the glacial valley. They are laid down by streams melt water depositing materials of the same weight and size on the sides of the trough. Examples are found in Mubuku valley on mt Rwenzori and Hobley valley on mt Kenya.

Illustration



ECONOMIC IMPORTANCE OF GLACIATION OR GLACIATED REGIONS TO THE PEOPLE OF EAST AFRICA.

- Melt waters from glaciers form important sources of rivers used for various uses e.g. the chagga on the slopes of mountain Kilimanjaro use it for irrigation River Sebwe provide water for Mobuku irrigation scheme in Kasese.
- The water falls formed at the hanging valleys are potential sites for hydroelectric Power generation e.g. at Corinne falls on mountain Kenya, and mobuku power station in kasese.
- The beautiful scenery provided by glaciated landforms such as arêtes, pyramidal peaks etc. attract tourists who bring in foreign exchange, employment to people such as tour guides who earn income thereby improving their standards of living and promoting infrastructural development.
- The moraine deposited contains sand and gravel which can be extracted for building and construction purposes e.g. in kasese.
- The boulder clay plains in the glaciated low lands are sometimes very fertile and suitable agriculture especially arable farming e.g. in the mobuku valley on the lower lands of mountain Rwenzori.
- Glaciated areas may act as recreation and Sports centers for example mountain climbing and ice skating on mt. Rwenzori, mt Kenya and Kilimanjaro
- The U-shaped valleys form natural routes or ways for mountain climbers on the mountainous lands.

- ❖ Glacial lakes such as lac-Gris, lac-Catherine are potential fishing grounds on mt Ruwenzori.
- ❖ Glaciation has led to Modification of the micro climate through lowering of temperature and rainfall formation suitable for settlement and farming on the gentle slopes of the mountain Kenya, Kilimanjaro and Rwenzori.
- ❖ It has enhanced research and study purposes leading to acquisition of skills and knowledge in different education institutions like universities and secondary schools.
- ❖ Fertile soils like the boulder clay soils on the plains support crop cultivation such as Mobuku irrigation Farm in Kasese
- ❖ Glacial landscapes/features are used for filming and photography industry, advertisement in East Africa.
- ❖ The U – shaped valleys are also used for settlement and communication passes for roads for mountain climbers and railways because they are flat.

Negative

- ❖ Glacial regions are associated with avalanches which move downhill leading to extensive loss of property and life e.g. on the low lands of Ruwenzori
- ❖ Glacial regions like mountain Rwenzori and Kenya are associated with cold temperatures which limit agriculture and settlement.
- ❖ Upland glaciation has removed most of the fertile top soil making the region unstable for agriculture eg on mts Ruwenzori and Kenya.
- ❖ The melting of the glaciers leads to the river flooding of the lower areas hence destruction of farm land and property.
- ❖ Massive deposition of rock boulders leads to infertile soils hence limiting agricultural activities.
- ❖ Rock falls/ice falls from the mountain cause accidents, death and destruction of property in the lower areas of the glaciated mountains.
- ❖ Inaccessibility due to glacial activities for example glacial erosion has led to the form of gullies.
- ❖ Stagnant water on the lower parts of the mountains becomes breeding areas for the diseases vectors like mosquitoes.

SLOPES AND SURFACES/LANDSCAPE DEVELOPMENT

Most of the land surfaces is made up of slopes which are generally convex in the upper parts, concave in the lower valley floors and to the level of the sea at their alternate depth. Only every small part of the land can be said to be truly flat, vertically or horizontally. The development of landscapes has been explained by several theories and these theories include: Peneplanation, Pediplanation and

PENEPLANATION

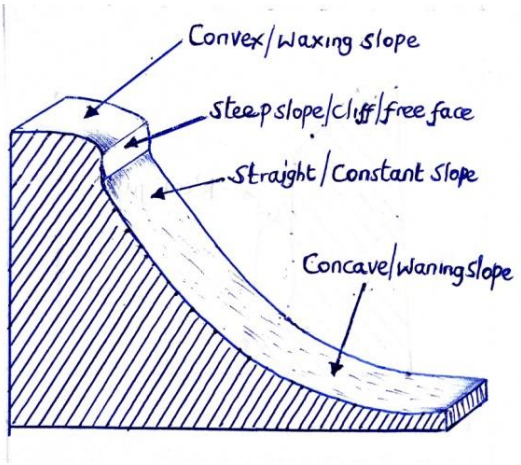
- This is a cycle of land form evolution propounded by William Moris Davis and also referred to as Davisian Cycle of erosion/geomorphic/erosion cycle/slope decline.
- Generally the theory explains that the physical landscape undergoes modification as a result of natural agencies in an orderly progressive sequence or stages starting from the uplift of land into upland to an ultimate low, and future plain.
- According to Davis, landscapes develop through a progressive sequence of stages i.e. the youthful stage, mature stage and the old stage and then rejuvenation all again.

- The cycle begins with the uplift of the land leading to the building of highlands or uplands. These uplands become the youthful stage and the process could be due to earth movements like faulting, folding, warping and volcanicity.
- The uplift has to be simple and fast enough that it does not experience significant erosion during this period.
- The uplifted land then undergoes erosion/down cutting by rivers, glaciers, winds, running water and the weathering processes.
- Rivers in the uplifted slopes quickly erode and deepen the valley, erosion decreases the slopes and the valleys become gentler, lateral erosion is more dominant and v-shaped valleys become broader and broader, this is the mature stage.
- As the relief becomes flat due to constant cutting down, flood plains begin to form. This is then the old stage where materials eroded from the youthful and the mature stages are deposited.
- As the relief continues to flatten and lowland finally develops, deposition results in meandering of rivers which Davis termed as a peneplain.
- When the peneplain stage is achieved, uplift occurs so that the cycle starts again. Davis refers to this as a return to a level or point of origin.
- In conclusion, the Davisian cycle points out to the value of structures, process and stage or time in landform evolution.

The relationship between rock structure and slope formation

- Hard rocks are more resistant than soft rocks hence form upland areas with convex or waxing slopes on top examples of such rocks include quartzite, granite and gabbro.
- Below the waxing slope is the free face which is very steeply sloping. This is due to erosion, weathering and mass movement.
- From the free face slope is the constant or straight or even slope that is gently sloping resulting from uniform weathering or erosion of land or soft rocks and deposition from the free face.
- Soft rocks like clay and shale form lowland that highly deeply weathered or eroded to form concave slope or waning slopes. Such slopes may also result from deposition from the upper slopes.

Illustration



REVISION QUESTIONS

Qn: *"Land evolution is a result of complex processes" With reference to examples in East Africa, illustrate this statement.*

Qn: *"Landform evolution is a result of structure, process and time". Discuss*

***Qn**: Using Davisian cycle of erosion concept, examine the relationship between rock hardness and slope formation.*

MASS WASTING/MASS MOVEMENT

This is the downhill/slope movement of weathered rock materials/ regolith under the direct influence of gravity. It is a large scale movement in which the stability of the slope has failed. It is sometimes called slope failure. It involves the creeping, flowing, sliding, falling of rocks and weathered materials/debris from the higher to lower slopes of the hill under the influence of gravity.

- Rain water is very important in mass wasting because it helps in overcoming resistance as it increases weight and also acts as a lubricant between the slope and the weathered materials.

Types of mass wasting

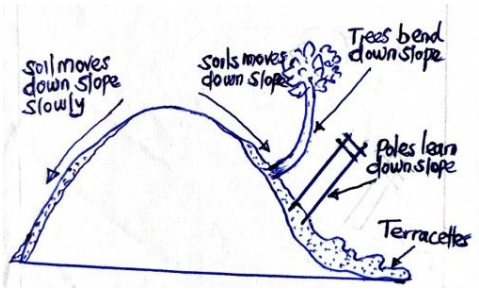
There are various ways in which the weathered materials move down the slope under gravity and they are categorized into two i.e. the slow movement and the rapid movement.

SLOW MOVEMENT PROCESSES

These movement processes are very slow and are only discovered over along time of observation. Under these the processes include; soil creep, talus creep, rock creep and solifluction.

- **Soil creep**: this occurs on gentle slopes with unconsolidated weathered materials. It involves slow movement of massive soils down slope under the influence of gravity. It is facilitated with heavy rains and the soils get saturated and heavy. This is followed by a slow movement down slope under the influence of gravity. It is normally detected after seeing tilted trees and poles which tend to bend due to excessive weight exerted by the creeping soils. In 1995 mass wasting took place in Tukuyu valley in Tanzania. This was very massive that the soils blocked the river valley thus creating a barrier lake which lasted for eight hours.

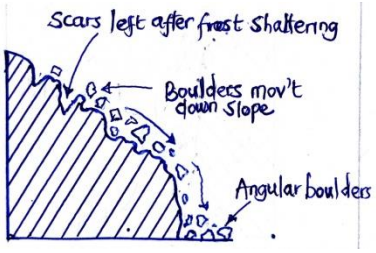
Illustration



Note: soil creep creates terrace-like features called terracettes on the hill slopes. These features are common in Shema and Kajara counties of Bushenyi.

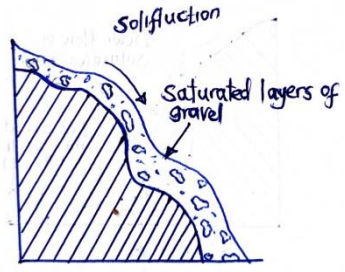
- **Rock creep**: This involves the movement of rock debris slowly down slope. Such rocks may move under soil creep. E.g. they can be found in Kapchorwa and Sironko in eastern Uganda.
- **Talus creep**: this is the slow movement of angular rock debris of different sizes down on moderate slopes at moderate. It's common in areas where freeze-thawing are frequent. Frost shuttering, tears out rock particles from the surface such that they move down slope under the influence of gravity and thawing. Ice provides a lubricant.

Illustration



- **Solifluction**: It involves slow movement of fractured gravel and weathered rocks over the frozen sub soils and rocks. During freezing there is less movement but with thawing/melting of snow/ice, the gravels tend to creep for a short distance. It is limited to glaciated regions like the high altitude and the mountains of East Africa such as Kilimanjaro, Rwenzori and Kenya.

Illustration



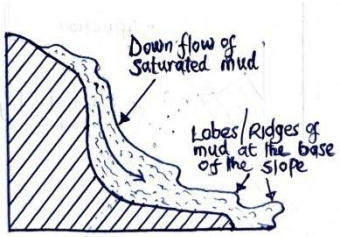
PROCESSES RAPID MOVEMENT /LANDSLIDES

These are generally called landslides. Landslides are defined as the sudden and rapid movement of rocks and soil debris down the slope under the influence of gravity.

Types of landslides

The major types of landslides include; earth flows, mudflows, slumping, rockslide, rock fall and avalanches. In East Africa landslides are common in the highland areas such as Kigezi highlands, Mount Elgon, and Rwenzori. Other areas include Bundibugyo, Bududa, Bulamuli and Kapchorwa. All the types of landslides are explained below.

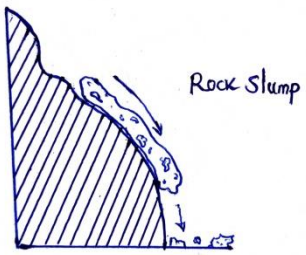
- **Talus creep**: this is the slow movement of angular rock debris of different sizes down on moderate slopes at a fast speed. It's common in areas where freeze-thawing are frequent. Frost shuttering, tears out rock particles from the surface such that they move down slope under the influence of gravity and thawing. Ice provides a lubricant.
- **Earth flows**: this is a rapid down-slope movement of water saturated soil materials and other debris on moderate slopes. When it rains, slope materials may be soaked in small amounts of water and are saturated. The water acts as a lubricant and adds weight to the materials. A sudden movement of materials takes place producing short flow tracks and small bulging lobe or turn at the feet of the slope. It is estimated that it takes place at a rate of about 15km per hour. It is similar to mud flow except it's relatively drier.
- **Mudflows**: this involves the movement of semi-liquid mud mixed with gravels and boulders on a moderate to steep slopes at a fast speed. When it rains, the soils/materials absorb water and become supper saturated, lubricated and slippery and therefore move down-slope under gravity influence. Some materials may move down slope at a high speed of over 15km per hour. For example in March 2010, a mudflow occurred on the slopes of Mount Elgon killing an estimated 450 people in Bududa district in eastern Uganda.



- **Rocks slump/slumping**: this involves an actual tearing away and very fast movement of large masses of rock-material and debris on over steepened slopes, cliffs/scarps/road cuttings at a fast speed as a whole mass leaving behind a fresh scar. The sliding rock block doesn't breakup into

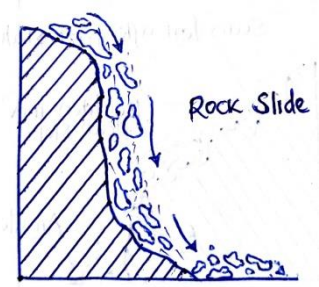
particles but moves rotationally on a curved plane. Slumping is caused by massive impermeable rocks overlying soft rocks.

Illustration

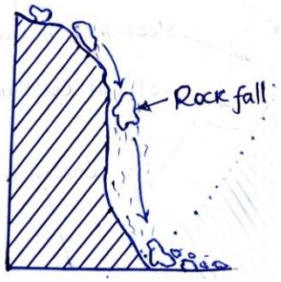


- **Rock slide**: this involves the dislocation and rapid movement of large masses of rock particles/debris rolling down along a slippery and over steepened slope/escarpment/road cutting at a fast speed. It takes place where slopes are very steep and therefore high intensity of gravity. Its rate of occurrence and frequency is accelerated by earthquakes. This is common in Usambara and Pare ranges in Tanzania and Butiaba escarpment in western Uganda.

Illustration

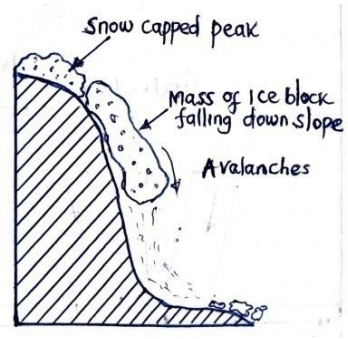


- **Rock fall**: this involves a free fall of individual rocks and boulders down very steep slopes/vertical slopes at very fast speed. This is the fastest type of landslides. This could occur on a mountain side or a cliff. Therefore where as in the **rock slide** the rocks literally slides/rolls down along a steep slope at a fast speed, in **rock fall** the detached individual rock drops clear of the surface before finally resting on the ground below where it may cause enormous destruction. The ideal conditions for rock fall are the vertical or nearly vertical slopes in mountainous areas where there are loose rocks produced by weathering. This can be traced at mount Rwenzori and Sironko.



- **Avalanches**: this refers to sudden down fall of large masses of rock materials embedded in snow and ice on a steep slope/ mountain side due to gravity. This is usually restricted to areas with permanent snow/glacier e.g. in East Africa it occurs in the snow capped mountains of Kilimanjaro, Rwenzori and Kenya.

Illustration



CAUSES OF MASS WASTING/SLOPE FAILURE/ROCK COLAPSE/LANDSLIDES

Mass wasting/landslides are caused by several factors which are both physical and human as below.

Physical factors:

The relief determines the rate of downward movement of the weathered materials in that;

- Very steep slopes encourage faster mass wasting types such as rock fall, rock slide, slumping e.g. in slopes of Rwenzori, Kigezi highlands and mount Muhavura.
- The over steepened scarps, cliffs and road cuttings encourage rockslide, rock slump and avalanches.
- Steep slopes encourage rock slide, mudflow and avalanches.
- On the other hand moderate slopes favour the occurance of talus creep and mudflows
- While gentle slopes encourage soil creep, earth flows and solifluction.

Climate

- Heavy rainfall avails water that saturates and lubricates weathered materials on the slopes. It also increases the weight of weathered materials and loosens attachment of rocks/soils resulting into soil creep, earth flows and mudflows.
- Pounding effect of heavy rainfall also destabilizes surface layers of rocks hence triggering the movement of rock and weathered materials under the influence of gravity.
- Temperature changes that lead to alternate freezing and thawing makes the ground slippery and the water made available increases the weight of debris leading to mudflow, soil creep, and avalanches.
- Any disturbance by nature such as thunder will destabilize the accumulated rock debris which eventually flows or falls down slope due gravity influence.

Nature of rock:

- Permeable weathered rocks overlying impermeable rocks absorb water; get saturated, overloaded and lubricated hence encouraging mudflow and rockslide and vise-versa.
- Well jointed rocks may encourage landslides inform of rock fall, rock slide and avalanches, etc.
- **Nature of soils may also cause mass wasting.** Heavy wet and massive clay soils encourage landslides when they absorb water, the weight increases, become lubricated are much slippery thus leading to mudflow.
- **Freezing and thawing** in cold environments also leads to mass wasting. When temperatures rise, ice melts and water is added into the weathered materials. This increases the weight of weathered materials/debris and weakens the friction between the rock surface and the debris. More over water from the melting ice contributes to the circulation of weathered materials thus being able to move on the frozen surface a process termed solifluction.
- **Tectonic movements** such as faulting, warping, etc create steep slopes, cliffs and escarpments that encourage landslides like rock fall, rockslide, etc. while

- **Volcanic eruption** may also trigger mass wasting. As eruption takes place, it shakes the ground hence loosening the rock particles. This may result into rock slide, slumping and rock fall. Eruption also yields particles of different sizes such as cinder that when lubricated may flow down slope inform of mudflow
- **Earth movements** like earthquakes and Earth tremors also cause landslides. Once an earthquake occurs the ground is shaken and ground vibrations loosen the rocks hence triggering off landslides such as rock fall, rockslide and slumping. These are common in Bundibugyo and Kabarole in Uganda.

  The nature of rocks also leads to the occurrence of mass wasting.
- In areas where permeable rocks overlie impermeable rocks, and it rains, permeable rocks get soaked in water which can easily slide off inform of rock slide and slumping.
- Similarly when jointed rocks overlie hard rocks with slippery surfaces they tend to loosen from the mother rock thus triggering off mass movement. Such rocks easily peel off and slide off or fall down slope due to the influence of gravity.
- Alternate layer of hard and soft rocks also cause landslides inform of rock slump and rockslide, etc.
- **Overloading/accumulation** of rocks cause mass wasting: The accumulation of rock debris in large masses on a steep slope trigger off mass movement. Additional weight from materials or rain water promotes overloading hence rock slump, rock fall, and mudflow.
- **Living organisms** like burrowing animals loosen the soils/rocks allowing water into the soils hence overloading and lubrication encourages rock slump, rock slide and mudflow. The movement and trampling by animals causes vibrations that encourage rock slides, rock slump etc.

Human factors/causes:

- **Road construction and Settlement** on steep slopes/highland /hilly areas lead to the creation of cliffs that encourage rock slump, rock fall and rock slide.
- **Deforestation along the slopes** leaves the soils bare and the binding force disappears and soil particles become loose and unconsolidated. The roots of trees and other vegetation types bind the rocks particles and soils together. Once it rains such rock particles become saturated and slippery and may flow down slope in form of earth flows, mudflows, etc.
- **Overstocking and overgrazing** on the steep slopes also off-sets mass wasting; large herds of grazing animals such as cattle, buffalos, elephants, sheep and goats, trample and loosen exposed soil particles/rock particles along the slopes leading to soil creep, earth flows and mudflows.
- **Mining and quarrying** contribute to mass wasting. Mining weakens rocks through the use of explosives that cause vibrations hence triggering off landslides e.g. vermiculate mining in Mbale, gold mining in Kabong, tin and gold mining in Bushenyi in Uganda. Quarrying on the hillside creates steep slopes which enhances landslide occurance.
- **Movement of heavy vehicles** such as Lorries, trailers, buses, trains, etc; cause vibrations along the road cuttings and steep slopes. These vibrations destabilize unconsolidated rock materials resting loosely on the slope causing them to start rolling down slope inform of rockslides and rock falls. Example is common along Ntungamo – Kabale road, Kabale-Kisoro road, Simu-Kapchorwa road, etc.
- **Poor farming methods** like ploughing up and down the slope aids the movement of unconsolidated soils and weathered materials down slope. Over cropping results into soil exhaustion and erosion, this promotes the occurance of earth flows, mudflows and soil creeps.

LANDFORMS DUE TO MASS WASTING/LANDSLIDES

- **Formation of terracettes**; slow movement such as soil creep usually create steep-like features called terracettes. These are common in the hills of Ankole and Kigezi.

- **Formation of patterned grounds/polygons**: these are stones which are arranged in polygons shapes in glaciated areas. They are formed from solifluction or soil creep due to frost shuttering or heaving.
- **Formation of scars**: A scar is a hollow or space found on the hill side where rock debris has been detached.
- **Formation of lobes and langues**: these landforms are formed where debris eventually rests after being removed from a scar on a hill side.
- **Formation of straight slopes**: soil creep and slope wash fills up cavities or irregularities along the slope resulting into straight slopes e.g. on some slopes in Ankole.
- **Drainage blockage**: landslides and slumps on the side of the river valley may block river courses to create temporal lakes. For instance one of the Tukuyu landslides in Tanzania blocked river Mbaka cresting a lake which lasted for eight hours.

EFFECTS OF LANDSLIDES/MASS WASTING

Mass wasting has a number of effects on human activities which are both positive and negative

Negative effect of landslides

- Landslides cause loss of life. Several landslides especially those associated with aspects of earthquake, heavy rainfall, rock flows and mudflows destroy settlements and kill people living down slope. In between 1997 to 2001, the Bulucheke landslide on the slopes of Mount Elgon killed over 100 people while the Bundibugyo landslide of 1966 destroyed almost a half of the town. In Bududa, the landslide of March 2010 killed estimated number of 450 people.
- Landslides often result into the destruction of agricultural land and crops in areas prone to landslides. On the steep slopes, soils may be carried away and deposited in the valleys e.g. in Kigezi highlands and the southern highlands of Tanzania. Crops may also be carried down by the creeping soils and destroyed by the landslides. This in the long run may cause problems like famine.
- Landslides also destroy forest recourses' for example the Bulucheke landslides on Mount Elgon destroyed a large section of Mount Elgon forests. Eventually processes like soil creep result into the bending of tree trunks hence reducing the timber quality.
- Landslides may also results into the destruction of infrastructure. Electric transmission and telephone lines are usually destroyed while roads and railways may bee blocked or destroyed by slides or creeps. E.g. in 1989, telephone and road links in Bundibugyo were cut off by land slides for almost a year. Earlier in 1995, Kisoro–Kabale road was also blocked by landslides. Besides landslides creates engineering problems and construction along the steep slopes is made very difficult and dangerous by creeps and slides. The poor network in Kisoro and Kapchorwa is partly because of this factor.
- Mudflows and soil creep may result into flooding when they block river valleys and creating a barrier-lake hence flooding the adjacent areas. For example in 1995, river Mbaka within the Tukuyu valley in Tanzania was blocked by mass wasting. On mount Rwenzori Lake Bujuku was created through damming of river valleys by mudflows.

However, landslides may have positive effects too as discussed below

- Continuous removal of surface soil/rock materials from the hill slopes by mass wasting may expose underlying minerals. This contributes to the development of the mining industry e.g. gold in Bushenyi at Kyamhinga is often exposed by landslides.
- Mass wasting may also lead to the formation of fertile soils on the lower slopes of highlands. The soil particles removed from the upper slopes in the processes like soil creep and solifluction are deposited at the lower levels where they can be used for cultivation. In addition mass wasting

facilitates soil formation as new rock surfaces are exposed to weathering processes such as exfoliation.

- Mass wasting leave behind landforms that create beautiful scenery on the hill slopes such as scars, terracettes, etc. These features are appealing to curious people hence attracting tourists who generate foreign exchange.
- Landslides are used for education and research purposes for example students and other researchers learning more about landslides usually go to Bundibugyo and Mount Elgon areas for field study research.
- Mudflows also fill pits and depressions left after mining. For examples in Bushenyi pits left after tin mining are being filled up by mudflows. These depressions would be breeding grounds for mosquitoes which are dangerous to man's life.

STEPS BEING TAKEN TO CONTROL LANDSLIDES

- There are various steps that are being put in place to curb/control the effects of landslides. These measures are aimed at ensuring slope stability and maintenance of slope equilibrium and they include;
- A forestation and reforestation are being done along areas prone to mass wasting and landslides. This helps to bind the soils once again, reducing the impact of raindrops and soil saturation e.g. in Kigezi highlands, the Kenya highlands and Kiisi areas of Kenya.
- Better farming methods are being introduced such as terracing in Kigezi, mount Moroto, etc, this reduces the down slope movement of debris.
- Forest reserves and national parks are being created in highland areas to reduce deforestation and quarrying on mountains. This eventually reduces landslides. E.g. are mount Mgahinga, Rwenzori, Elgon, Kadam and Morungole in Kidepo valley.
- Mining and quarrying are being done in Uganda following safety provisions by the National Environmental Management Authority (NEMA). NEMA as an organisation is mandated to carryout environmental impact assessment before mining and quarrying is authorized to be done.
- Hill slope draining is being done to reduce excess water that causes overloading and lubrication. Deep trenches have been dug on hill slopes on highlands especially the Kenya highlands and Kigezi highlands. In Kigezi it takes the form of land reclamation. Another way in which excess water is being drained is by planting eucalyptus trees in the valleys and other water logged areas.
- Resettlement of excess population is being undertaken. The people in densely populated areas have been resettled in less populated areas in order to minimize landslides. People from Kigezi have been resettled in Kasese, Kibale and Kamwenge areas. People from Bududa have resettled in Kiryandongo refuge settlement camp while people from Kenya highlands have been resettled in Kiisi, Mwea-Tebere, Galole, etc.
- Settlement on steep slopes is being discouraged e.g. in Kenya highlands, Kigezi highlands and Elgon slopes. This is to ensure that slopes remain naturally intact.
- Environmental education is being done to create awareness about the dangers of clearing vegetation on the highlands. People living mountainous areas are also being taught better farming methods like crop rotation, contour ploughing, mixed cropping, etc.
- Concrete walls are being built along steep faces so as to control mudflows and soil creep. Such walls are common at Kololo, Muyenga, and Naguru in Kampala.
- Other measures that may control landslides include controlled grazing i.e. reducing the number of animals grazing on the slopes prone to landslides,
- Law enforcement where environmental education fails and this could involve strict punishment for illegal felling of trees.

- Family planning which is aimed at reducing population pressure on land is also being encouraged and advocated by government policies,

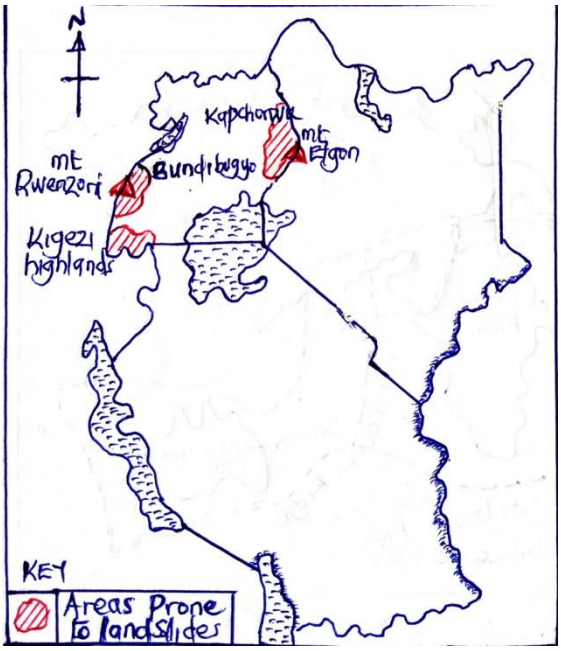
Note: Like all natural calamities/disasters, landslides can not be completely stopped but rather their impact can be minimized.

REVISION QUESTIONS

QN*: With reference to either Kigezi or Bugishu highlands, discuss the causes of landslides and their effect to human activities.*

Approach*: Define landslides, mention the types of landslides, draw a sketch map showing areas affected by landslides, explain both physical and human causes, and explain the effects both positive and negative.*

SKETCHMAP OF EAST AFRICA SHOWING AREAS AFFECTED BY LANDSLIDES



GLACIATION

This is the process by which glaciers shape the earth's surface. Or it can be defined as the activities of a glacier.

In east Africa, glaciation can only take place on very high mountains that rise above the snow line at 5000m namely mountain Rwenzori, Kenya and Kilimanjaro. On mountain Rwenzori, there are 37 glaciers e.g. stanely, speke, Bujjuku, Mubuku, Edward, Backer, Moore etc.

Mountain kenya has the second largest coverage of glaciers and best examples here include lewis, Gregory, krapf, dawin, tyndal, ford, diamond, heim etc.

On mountain Kilimanjaro glaciers include kibo, great penk, furt wangler etc.

A GLACIER

A glacier is a mass of moving ice which moves down slope along a pre-existing river valley under the influence of gravity. Glaciers move continuously from higher to lower ground and is enclosed within valley walls.

Formation of glaciers.

When the temperatures in an area fall below 0^0, water vapor in the atmosphere condenses to form snow. The snow accumulates in pre glacial hollows. Under pressure, the air in it escapes melting and freezing occurs. Further compaction expels all the air and the bottom layers are compacted into ice. With time under the influence of gravity, the ice moves out of the hollow and is now known as a glacier.

Types of glaciers.

a) **Valley glaciers**; these are formed on highland or mountain peaks and upper valleys of mountain ranges. Valley glaciers move by sliding over the rock surfaces. Friction with the ground produces heat that in turn melts ice and helps to lubrication hence speeding up the sliding process.
b) **Lowland glacier**; this is a glacier found in colder and higher latitude areas mainly between 60° latitudes and the poles. They are mostly found in Greenland, Arctic and Antarctica.
NB There is limited glacier coverage in East Africa due to the following reasons;

Latitudinal location of east Africa. Latitude refers to the distance from the equator. East Africa lies along or astride the equator at approximately 5 degrees north and south of the equator. For this reason, whether the apparent overhead sun shifts north or south, the angle of incidence of the solar rays is still small. So the sun rays strike the earth's surface at a wide angle and at a short distance hence ensuring maximum solar heating throughout the year. This implies that the temperatures in east Africa are high throughout the year. Therefore, its only in highland areas which rise above the snowline where temperatures drop to below freezing point that enables glaciers to form. However, these are very few areas where temperatures can drop below zero degrees hence limited coverage.

Altitude. Much of East Africa lies at a low altitude of less than 2000m above sea-level and this limits glacier formation because low altitude areas are associated with hot temperatures thus limiting glaciers to very few areas in East Africa rise to the current snowline of 4800m above sea level.

Precipitation. East Africa generally receives moderate rainfall of between 1000 to 1500m per annum. This is just sufficient to sustain life. However, glaciers form in areas where temperatures drop below freezing point and with torrential rainfall.

Global warming. The world temperatures have been rising with time and this is attributed to human activities like burning fuels e.g. coal, oil and natural gas which increases carbon-dioxide concentration in the atmosphere and other greenhouse gases. These gases absorb a lot of heat from the sun during day and prevent heat loss at night (they act as a blanket). This accounts for the increase in the global temperatures by 2.5°C. this increase in temperature has led to the melting away of glaciers to the point of extinction of some glaciers e.g. the sempaya glaciers on mountain Rwenzori.

Aspect. This refers to the angle at which the slope receives the sun's insolation. East Africa lies between the tropics and therefore experiences direct over sun throughout the year. This implies that all slopes of the mountain are subjected to at least 10 hours of great sunshine a day without any sheltered slope or obstructed slope to enable glacier formation.

Impact of volcanism. Some of the high mountains where glaciation would have taken are volcanic in nature. The temperatures in these mountains are high due to the hot interior which brings about the warming effect on the surroundings. This is therefore preventing glacier formation.

GLACIAL ACTIVITIES.

Erosion

Transport

Deposition

GLACIAL EROSION

Glacial erosion occurs through the following processes;

Plucking or exaration.

It is the tearing away of the blocks of rocks which have become frozen into the sides and bottom of a glacier (glacial depressions). It is more effective where there well jointed rocks subjected to freeze and thawing.

Abrasion.

It's the wearing a way of rocks underneath a glacier by the swirling of rocks emborded in glaciers. In this process, the glaciers use the materials being carried or transported as the grinding tool which is used to break up rocks along the floor and sides of the glacial trough.

Basal Sapping.

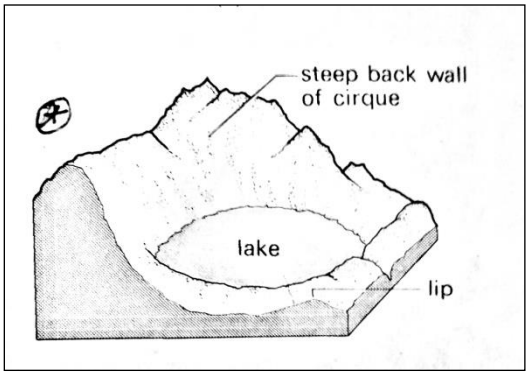
This involves the breakup of rocks by alternate freezing and thawing of water in rock joints/cracks that exist on the sides and floor of a valley or the side of a mountain.

GLACIAL EROSION FEATURES / GLACIAL UPLAND FEATURES

CIRQUES/CORRIE/TARNS.

These are semi-circular steep sided rock basins which have been cut into the sides of a glaciated mountain. They are formed when water enters the rocks and freezes breaking them down. The joints become enlarged. Abrasion drags the debris over the rock floor deepening the depression; back wall cutting or recession also enlarged the depression as well as steepening the sides hence a Cirque/Corrie. When filled with water, it forms a tarn (lake) such as teleki and mawenzi on mountain Kenya and Kilimanjaro respectively. Others are Lac du Noir, Lac du Catherin and Lac du vent on mountain Rwenzori.

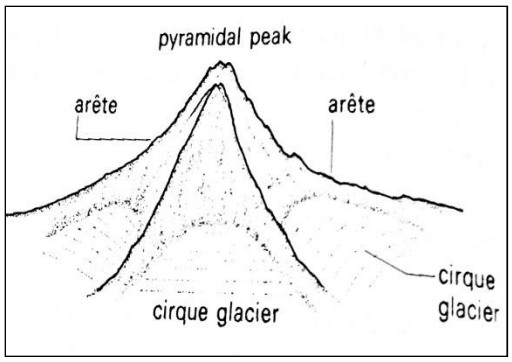
illustration.



Pyramidal peak /horn

This is a jagged peak with a series of radiating aretes. It is formed by back wall recession of two or several cirques which lie on the sides of a glaciated mountain. Two or more cirques cut back into the original mountain sides leaving an isolated peak called a pyramidal peak. These peaks become shaped by frost action. Examples include kibo, mawenzi on mountain Kilimanjaro and margarita peak on mountain Rwenzori.

Illustration.



Arête.

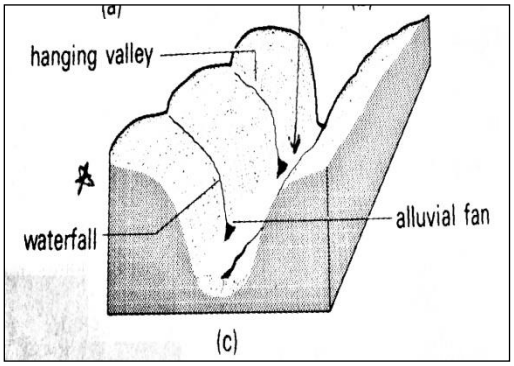
This is a sharp knife like edged feature or narrow ridge in between two or more cirques on the mountain sides. It's formed when two or more cirques erode backwards (back to back) through the process of head Ward erosion. Best examples can be seen on my Rwenzori from Mount Backer up-to bujuku Valley and point John and midget peaks on mountain Kenya.

U-shaped valleys or glacial troughs.

These are broad flat bottomed steep sided valleys with an open u shape in the cross profile. These were usually former river valleys which are filled by glaciers. Through the process of abrasion, plucking and sapping, the valley floor and sides were worn away hence deepening and widening the valley. So the valley is changed from a V-shape to an open U-shape hence a U shaped valley/trough.

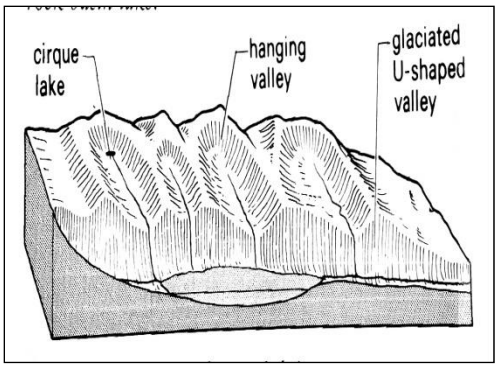
Examples of u shaped valleys include Mubuku valley, Mugusu Valley and komusonso valleys on mountain Rwenzori, hobley and Teleki valleys on mountain Kenya and the karanga through on mountain Kilimanjaro.

Illustration



Hanging valley

These are small narrow short and V-shaped valleys laying above the main the main valley found in highland areas. They normally descend steeply into the main valley by a water fall. For example, Nithi river is joined by the little Nithi from a hanging valley on mountain Kenya.



Truncated spurs

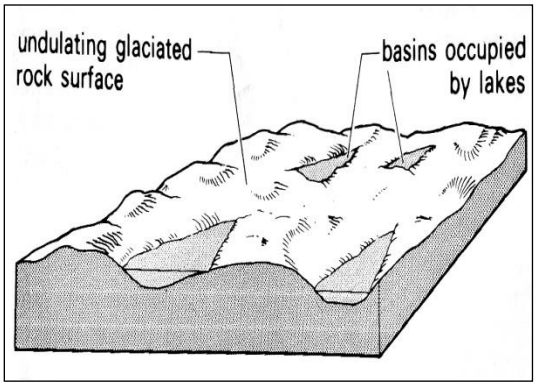
These are ridges with steep edges formed when former Interlocking spurs of a former river valley are cut off by lateral erosion resulting from accumulated glaciers in the valley forming a U-shaped valley with Truncated spurs instead of interlocking spurs for example around Mubuku and bujuku Valleys on mountain Rwenzori.

Rock basins

These are irregular depressions on the bed of the glacier trough. They are formed by un equal glacial erosion of the bed rock when out rock protrusions along the trough are removed to form hollows called rock basins.

Similarly, the path of the glacier may have rocks of different types and resistance to weathering and erosion where a band of soft rocks Alternate with bands of hard rocks, the soft rocks will be worn out to form a depression which is filled by water to form a basin lake. Best examples include lac Michaelson on mountain Kenya, Lac noir and Lac vert in the kamusoso valley on mountain Rwenzori.

Illustration



Rock steps.

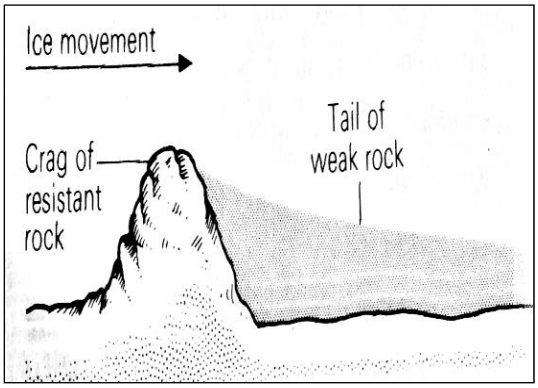
These are step-like hard rock projections in the glacial valley formed as a result of differential erosion due to difference in rock resistance and variation in glacier thickness. The increased ice in the main valley makes it possible to cut deep in the valley creating step like features known as rock steps for example at Vivian falls on mountain Kenya

illustration.

Crag and tail.

This is an elongated rock mass with a steep slope on the upstream which protects the softer lee ward rock from being eroded or worn out by the glacier. Its formed when advancing ice meets a resistant rock protecting a weaker rock downstream from erosion leading to the formation of an elongated tail on the side of the weaker rocks while the resistant rock mass forms a crag with a steep slope.

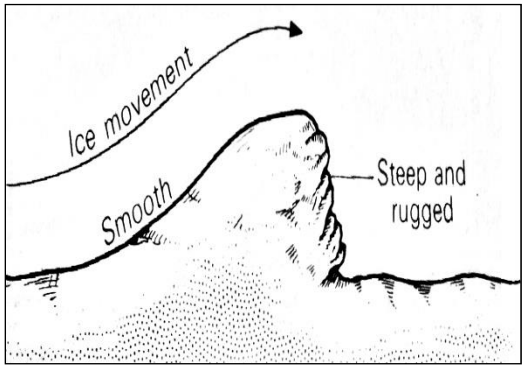
Illustration.



Roche montannes.

These are resistant rock masses smoothened on the upstream side by abrasion of ice and steepened on the downstream side. The downstream side is worn by plucking of ice making it rough. Examples can be found at the mobuku valley along the slopes of mountain Rwenzori and along the George valley on mountain Kenya.

Illustration



Stratitions

These are thin parallel scratches on glaciated areas formed by abrasion when large fragments of angular rock debris embodied in the ice tend to leave a series of parallel scratches. Such scratches can be seen in Stanley valley on Mt Rwenzori and near Peters hut on Mt Kilimanjaro.

Question

9. Account for the formation of upland glacial landforms in east Africa.
   Approach
   Define glaciation
   Identify areas where glaciation is taking place in east Africa
   Factors or conditions for the formation of glaciers.
   Processes of glaciation
   Identify and describe the landforms

10. Describe the processes responsible for the formation of glacial erosion landforms in east Africa

Glacial deposition – landforms formed due to glacial deposition

Glaciers transport a variety of materials ranging from fine rock flour to large boulders. All these materials are deposited in the valleys just beyond the point of melting and can modify the landscape markedly. Glacial depositional landforms are mainly found in the lowlands in the lower slopes of the mountains I.e. the valleys. The moraine transported and deposited by the glaciers results into the formation of depositional landform and these include:

Moraine

This refers to the rock debris that is eroded by flowing ice from the sides and base of the valley and deposited in the low lying areas. Moraine are large and small fragments that are detouched, transported and deposited when ice losses it's capacity to transport the materials.

Terminal moraine, It's a ridge like feature formed by accumulation of unsorted fragments of all sizes extending across the valley as a belt of low hills for many kilometers. It's formed by extensive deposition along the snout of an ice sheet and may cover a reasonable distance and height of 50 meters. It builds up when the glacier is static. Examples can be seen in the low lying areas of Rwenzori i.e. bujuku, Mubuku valley kimberly plains of South Africa etc.

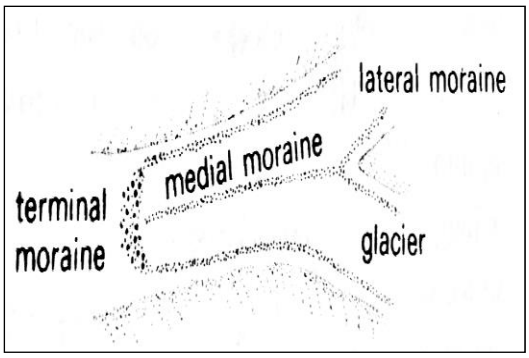
Lateral moraine ,This is an elongated ridge or moraine formed along the valley glacier sides. The deposition is as a result of friction between the moraine and the valley sides. Lateral moraine can be seen along Bujjuku, Mubuku and Kamsuso troughs on my Rwenzori and at the point where little Nithi river joins the main stream of river Nithi in the Teleki valley on my Kenya.

Medial moraine

When two glaciers meet, lateral moraine of the inner sides of both glaciers joins to form medial moraine. Medial moraine forms at the center of the glacier when a glacier retreats and drops its moraine.

Ground or basal moraine.This is formed when moraine is deposited at the bed of the glacier. It's composed of mainly fine glacial moraine dropped by retreating ice. Ground moraine almost covers the whole width of the valley.

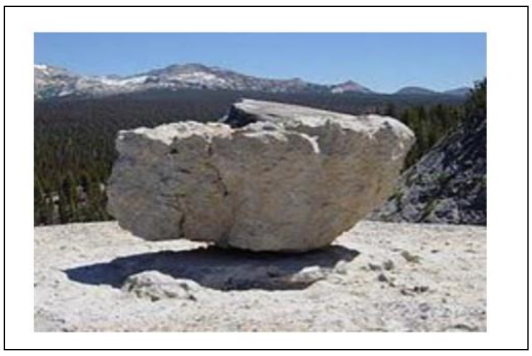
Illustration



Erratics

These are rock fragments of varying sizes which are eroded, transported and deposited in areas where they are far different from the surrounding rocks where it's deposited. They are useful in determining the direction and source of ice movement. Best examples can be seen at kamusoso and bujuku Valleys on mountain Rwenzori in western Uganda.

Illustration



Till plains.

These are extensive areas of monotonous relief or landscape formed when moving Ice transports boulders, sand and clay hence burying former valleys and protruding rocks to form an almost leveled landscape. e.g. Teleki valley on mountain Kenya and Mubuku valley on mt Rwenzori.

Illustration

Eskers

These are long winding steep sided ridges of sand and gravel lying parallel to the direction of ice movement. They are more than 30 meters high and several kilometers long. Sometimes they are formed by

rivers flowing beneath or within ice. they develop on areas of stagnant ice where rivers maintain sub glacier tunnels.

It's process of formation is attributed to stagnant ice sheets under hydrostatic pressure hence the ice develops melt water streams flowing within the ice following permanent sub glacial tunnels thereby deposition taking place within the tunnels. When the ice finally melts, it leaves behind long and steep sided ridges known as Eskers.

Illustration

Drumlins

These are low, rounded and elongated hills composed of bolder and clays lying parallel to the direction of ice movement. They are usually about 30m high and cover an area of up-to 1km. Drumlins occurs till plains in large groups or swarms aligned to the direction of the former ice advance. They are mostly formed when fragments or ground moraine are compressed by ice movements. Best examples are to be found in the teleki valley on mountain Kenya.

Illustration.

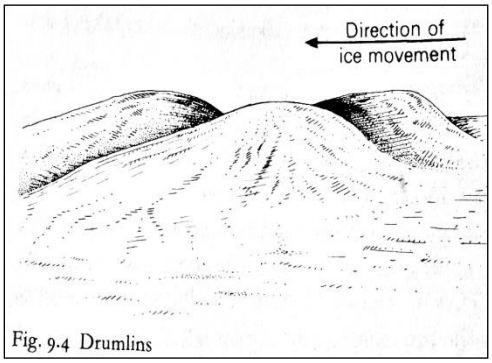
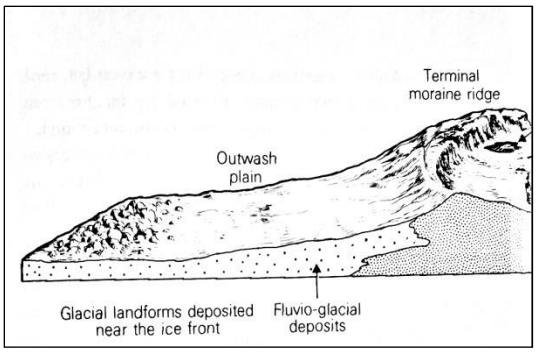


Fig. 9.4 Drumlins

Out wash plains

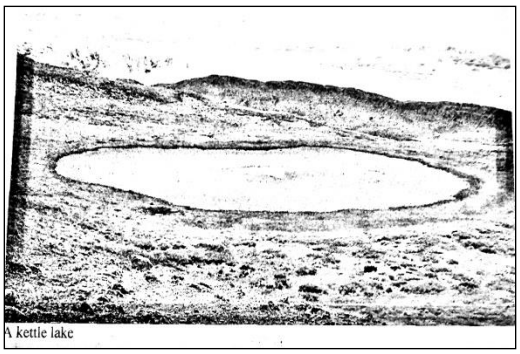
These are wide gently sloping plains of gravel, fine sand and silt. They are formed when volumes of melt water redeposit sorted materials further down slope. Best examples can be seen at kibo and mawenzi on mt. Kilimanjaro, Mubuku and bujuku Valleys on mt. Rwenzori.



Kettle holes

these are circular holes in fluvial glacial deposits formed when an ice block is detached from the main glacier as it retreats. Such a block is then buried within the till by fluvial glacial deposits. The subsequent melting of Ice blocks leaves behind depressions to form kettles. When water collects in these depressions, they form kettle lakes.

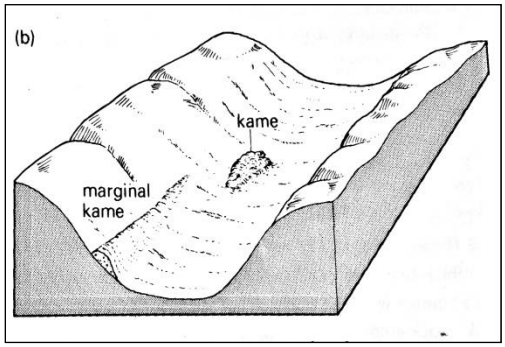
Illustrations


A kettle lake

Kame moraine

These are ridges of sand and gravel on the sides of the glacial valley. They are laid down by streams melt water depositing materials of the same weight and size on the sides of the trough. Examples are found in Mubuku valley on mt Rwenzori and Hobley valley on mt Kenya.

Illustration



ECONOMIC IMPORTANCE OF GLACIATION OR GLACIATED REGIONS TO THE PEOPLE OF EAST AFRICA.

Melt waters from glaciers form important sources of rivers used for various uses e.g. the chagga on the slopes of mountain Kilimanjaro use it for irrigation.

The water falls formed at the hanging valleys are potential sites for hydroelectric Power generation e.g. at Corinne falls on mountain Kenya.

The beautiful scenery provided by glaciated landforms such as arêtes, pyramidal peaks etc. attract tourists who bring in foreign exchange.

The moraine deposited contains sand and gravel which can be extracted for building and construction purposes e.g. in kasese.

The boulder clay plains in the glaciated low lands are sometimes very fertile and suitable agriculture especially arable farming e.g. in the mobuku valley on the lower lands of mountain Rwenzori.

Glaciated areas may act as recreation and Sports centers for example mountain climbing

The U-shaped valleys form natural routes or ways for mountain climbers on the mountainous lands.

Glacial lakes such as lac-Gris, lac-Catherine are potential fishing grounds on my Ruwenzori.

Negative

Glacial regions are associated with avalanches which move downhill leading to extensive loss of property and life e.g. on the low lands of Ruwenzori

Glacial regions are associated with cold temperatures which limit agriculture and settlement.

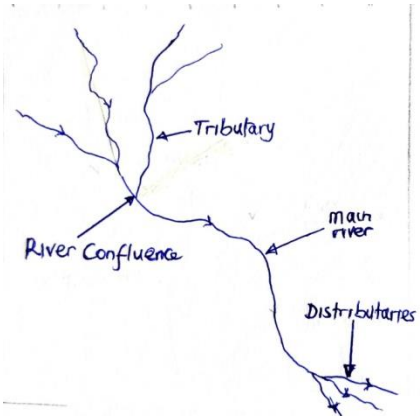
Upland glaciation has removed most of the fertile top soil making the region unstable for agriculture eg on mts Ruwenzori and Kenya.

RIVERS AND RIVER SYSTEM

A river is a mass of water flowing over the surface of land in a definite channel from its source to the mouth. A river originates from a point known as its "source".

- **The source of the river** may be a stream, swamp, glaciated melt water, collected surface run-off from rain water, a spring or a lake. It is also known as a river head.
- **The mouth of the river** is a point where rivers "ends" like a sea, lake, ocean, swamp or a forest. Some rivers simply disappear in arid regions or in sink holes as in limestone areas. For instance many rivers that originate from northern Kenya simply disappear in the arid areas of northern Kenya and Somalia.
- The source of river Nile for example is Lake Victoria while that of river Mobuku on the slopes of mount Rwenzori is melt water from glaciers and the high rainfall. The river Nile ends in the Mediterranean Sea while river Omo pours its waters to Lake Turkana.
- **A river system** refers to the total arrangement of the main river with its tributaries and sub tributaries. It is also known as the **drainage system**. Once a river is established it begins to lengthen, deepen and widen its channel. This creates secondary slopes on the valley sides on which tributaries to the main river develop. The main stream is called **Consequent River** and the tributaries are called **Subsequent stream**.
- The tributaries soon cut slopes against which other tributaries to the subsequent stream develop.

Illustration of a river system

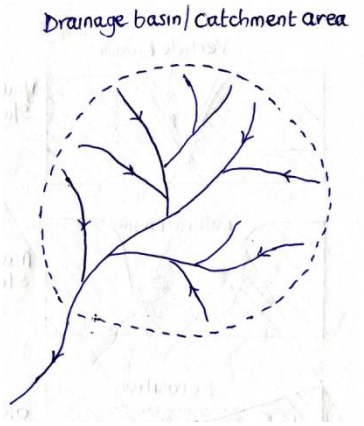


CONCEPTS USED IN RIVER DESCIPTIONS

- **A river course:** this refers to a path taken by a river.

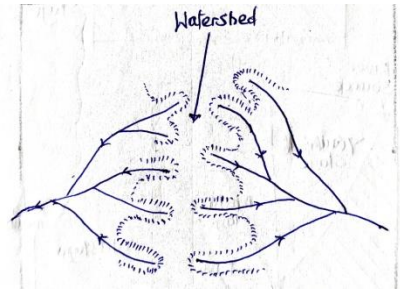
- **Drainage basin/catchment area**: this is a region which drains all the rainwater that falls on it into a river/stream and then carries the water to the sea or lake.

Illustration



- **Watershed/divide**: this is a highland region which separates two drainage basins or different river systems/basins. For example the uplift of western Uganda during the formation of the western rift valley resulted into the formation of a watershed. Kafu River became Kafu-Nkusi. Kafu River flows to L. Kyoga while Nkusi River flows to L. Albert.

Illustration

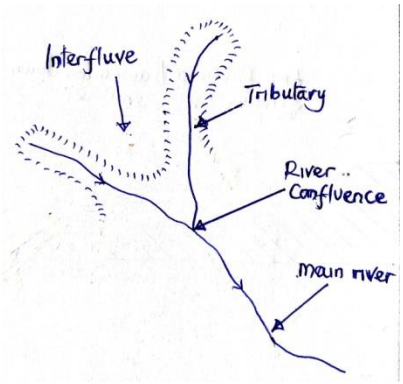


- **River regime**: this refers to the seasonal variation in the volume of water in a river. It partly depends on seasonal variation in rainfall. During dry season water in the river channel reduces while during the heavy rainy season water rises. There are types of regimes i.e. **Simple regime** is where a river has one period of high water volume and one period of low water volume. **A double regime** is where a river has two periods of high water volume in a year such kind of rivers are

found in places of double maxima rainfall. **Complex regime** is where a river has more than two periods of high water volume in a year e.g. Nyamwamba and Mubuku rivers.

- **Interfluve:** this is a highland area separating two tributaries of a river system.

Illustration



- **River confluence**: this refers to a point where a tributary joins/meets a main river or a tributary joins another tributary.
- **A river discharge**: this refers the quantity of water flowing down stream.
- **River profile**: there are river profiles i.e. the longitudinal profile of a river which is a section of a slope of the river from its source to its mouth while the cross profile of a river is a section showing the vertical shape of a river from bank to bank.
- **River channel**: this is elongated depression in which the river flows.
- **River flows** refers into the way in which water flows in a river channel. There are two types of river flow i.e. **Lamina flow** where water flows in a river channel without obstacles i.e. there is smooth flow and **Turbulent flow** where a river flows in a channel with a lot of obstacles such as potholes, falls, rapids, rock outcrops, etc.

THE GEOLOGICAL WORK OF A RIVER

As a river flows, it performs three tasks/functions i.e. erosion, transportation and deposition. Through these tasks, the rivers modify the surface features of their basins resulting into the formation of erosional and depositional features.

As rivers flow in their basins, they wear/wash away the surface land, erode valleys and it is for this reason that rivers have been referred to as important agents of land sculptures.

- **Erosion by rivers** refers to a progressive/continuous removal of rock materials from the river bed/floor and the sides of the river channel. Erosion on the river bed is called vertical erosion while erosion on the valley sides is known as lateral erosion.
- **River transportation** refers to the movement of eroded materials/debris by suspension, solution or dragging them along the river bed.
- **River deposition** refers to the progressive accumulation of the transported materials/debris/particles on the river bed, river bank, flood plain, or at the bed of the water bodies where the river pours its water such as lake, sea or oceans.
- The ability of a river to carryout the work of erosion, transportation and deposition depends majorly on its energy/competence. The greater the river's energy/competence, the more the ability the river have to erode and transport and vice-versa.

FACTOR THAT INFLUENCE THE RIVER'S ENERGY/COMPETENCE

Volume of water: the higher the volume of water a river has, the greater the river's energy and vice-versa. The volume of water of a stream/river normally increases from the source to the mouth. This is mainly

because as the river flows/proceeds down stream it is joined by numerous tributaries except where a river flows through an arid area there are few tributaries or where it crosses a porous rock.

Velocity/speed of water: the higher the speed of water, the greater the river energy and vice-versa. The speed of depends on the following factors.

- **The shape of the channel**; if the stream/river is flowing through a wider and shallow channel, its speed will be very low because it has a large surface area and as such, it has to overcome more friction and hence remains with less energy for erosion and transportation.
- If a stream/river is flowing through a narrow deep channel, its speed will be greater and as such it has more energy to erode and transport the eroded materials.
- **Roughness of the river channel**; if the river has a rough channel it is characterized by turbulence flow i.e. water moves in a circular form. This decreases the speed of water because turbulences cause friction along the rough sides and the bed of the river channel. A decrease in speed means a decrease in the river energy. Illustration
- **Smoothness of the river channel**; if the river is flowing through a smooth channel, it is characterized by a laminar flow i.e. water moves in layers parallel to the river bed. Such flow increases the velocity and ultimately the energy of such a river will be greater. This is because such a river has low friction along its bed and the sides. illustration
- **The load of the river**; the speed of the river will depend on the materials it is carrying or transporting. If the river is fully loaded with materials, it has less energy to erode and transport. This is because the load carried does not allow free movement of water which would help in the erosion of the river channel.
- **The river gradient**; the higher the river gradient, the greater the speed at which the river flows. This is because under such circumstances, gravity acts with more intensity. Therefore rivers flowing over steeper gradients have a greater ability to erode and transport. On the other hand, if the gradient is low, the rivers speed/velocity is also low and therefore limiting its ability to erode and transport.

RIVER EROSION

River erosion refers to the progressive removal of rock particles/materials from the river bed and river banks. Rivers erode in various ways depending on the nature of the channel, nature of the materials, etc. and this occurs through processes as below.

PROCESSES OF RIVER EROSION

- **Hydraulic action**: this is a process by which the force of moving water is able to remove loosened rock particles from the sides as well as the bed of the river channel. Such materials may include poorly consolidated gravel, sand, slit and other weathered materials. This force of moving water is able to loosen solid rocks by penetrating into cracks in the rocks especially the well bedded rocks leading to breakage of solid rocks. It involves the scooping out of rock materials from the valley sides in a process known as bank cavitations/caving.
- **Attrition**: this is the process by which the load itself is worn down as its being transported in the river channel. This occurs due to the fact that the rock fragments which make up the load are in constant collision with each other and with the river bed. Through this process there is progressive reduction/wearing in the size of the load materials and as it moves down stream it becomes easier to transport.
- **Abrasion/Corrasion**: this process involves the wearing down of the sides as well as the bed of the river channel by the load being carried. In this way walls and the bed the river channel are scooped and excavated by the load being carried leading to the scooping away of materials. This occurs

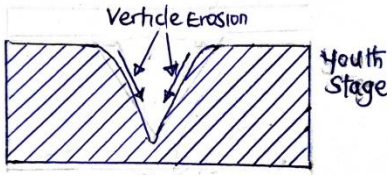
when the particles carried by the river strikes/knocks the bed of the river channel leading to the removal rocks. The rocks removed are added to the load and in turn they also become corrosive agents.

- **Solution /corrosion**: this is a process where a river dissolves the rocks over which it flows. This process is very common in areas where river flow over limestone rocks which are hard but dissolve quickly due to the action of carbonic acid dissolved in the river water.

TYPES OF RIVER EROSION

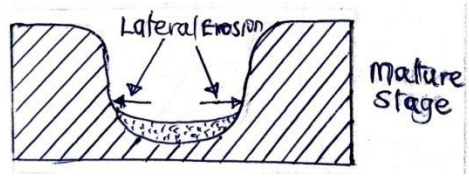
- **Vertical erosion**: this is erosion which removes/wears down the rock particles in the river bed. It's the type of erosion through which a river deepens its channel particularly through abrasion. It mostly occurs in the youthful stage of a river profile or where the gradient is steep.

Illustration



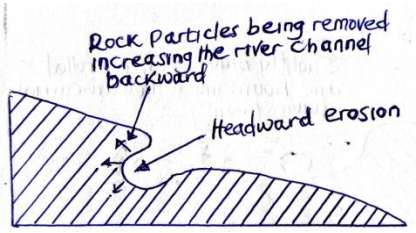
- **Lateral erosion**: this is erosion through which a river wears down the sides/banks of the channel. It is usually effective along the outside banks of the meanders. This means lateral erosion takes place in the mature and old stage of a river profile. It results into widening of the river channel

Illustration



- **Head ward erosion**: this is erosion in which the river undercuts backwards to its source. Its through this erosion the river is able to increase its length. It is concentrated on the steep ranges of the channel such as where water falls occur. It can be caused by rain-wash, gullying, spring sapping and soil creep.

Illustration



RIVER TRANSPORTATION

River transport refers to the movement of eroded materials/particles/debris along a river channel. All materials that a river transports are referred to as river load. The load is derived from the products of weathering, glacial deposits (on tributaries) or from the river's own erosive activity. The amount of load the river would transport depends on mainly the river's volume and velocity. The greater the volume of the river, the more the materials it is able to carry and the faster the river, the more the load it can carry or move along with. The measure of the ability of a river to carry a load is known as river competence. The river load of the river can be carried in many ways.

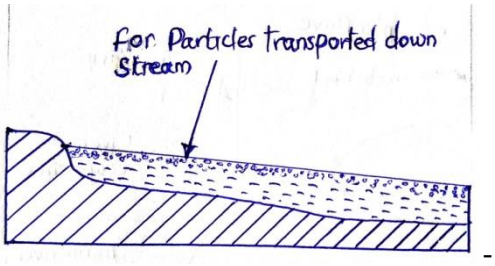
WAYS IN WHICH A RIVER TRANSPORTS ITS LOAD

- **Solution:** this is the movement of materials dissolved in water in the river channel. This process is common in rivers which flow over soft and soluble rocks such as limestone rocks. Water dissolves

rock materials in solution and transports it invisibly inform of chemical ions. The load carried in solution is called solution load.

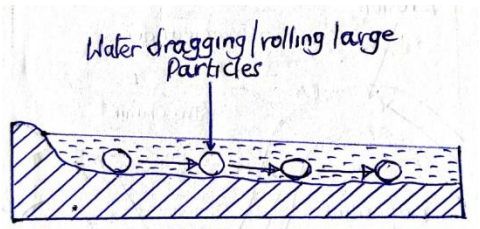
- **Suspension**: this is the movement of very small and fine particles held up by turbulences of water. The fine particles are swirled along in suspension. Clay and silt are normally carried in this process. The load carried by suspension is known as suspension load.

Illustration



- **Traction**: this process the dragging or rolling of large particles along the river bed. The materials moved in this way are called traction load.

Illustration



- **Salutation**: this process involves the rolling and bouncing of small particles along a river bed. It is a common way pebbles are moved. The load moved in this form is called salutation load.

Illustration



RIVER DEPOSITION

River deposition occurs when a river has insufficient energy to carry its load. At this stage the river is said to have a low competence and capacity to erode. Normally larges boulders and heavy particles are deposited first followed later by fine pebbles of silt and clay. The general name given to all materials deposited by a river is alluvium. Deposition takes place it any point along the river course however it is greatest at the lower course.

CONDITIONS UNDER WHICH RIVER DEPOSITION OCCUR

- Deposition may occur during flooding when the channel becomes too small to hold the discharge. Here deposition takes place at both the bed and the sides resulting into the formation of flood plains.
- Deposition may take place due to the decrease in gradient. This reduces the speed and the energy of water as well as its erosive capacity. The transported particles are therefore deposited in the valley.
- Deposition may also take place due to high rate of evapouration. In arid areas, if evapouration rate is higher than inflow, then the volume of water will reduce and this definitely will reduce the

energy of the river. Heavy particles will be deposited as the river will not be competent to continue transporting them.

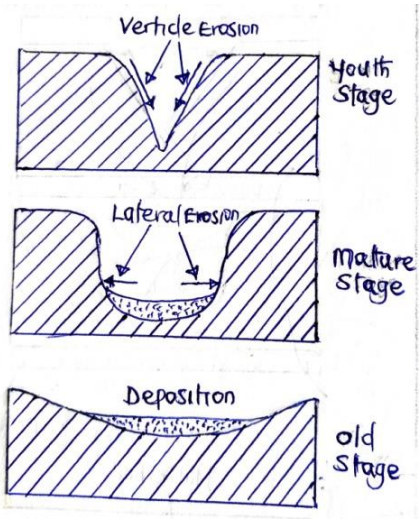
- Deposition also occurs when there is a sudden increase in the channel width of a narrow valley into a wide/ broad valley. If a river is flowing in a narrow valley and then all of a sudden the same valley widens, the energy and the velocity of the river will be reduced and particles will be deposited.
- Deposition may also occur due friction from rough surfaces along a river channel. Some river beds and banks are composed of rough surfaces inform of protrusions like rock outcrops and depression like potholes. These increases friction and enable some of the transported load to be trapped and laid down.
- Deposition may occur due to a sudden increase in size and the amount of the load being transported. Landslides inform of large boulders falling from a mountain side may find their way into the stream/river and if the river is not competent enough to transport this load for long distance it is laid down.
- Deposition may also occur where a river enters a region having porous rocks. Here much water percolates/sips into the underlying layers leading to a reduction of water volume in the river channel. This will reduce the energy of the river hence deposition will occur.
- Large scale deposition of river load usually takes place where a river enters into stationary waters of a sea, ocean or a lake leading to the formation of Lacustrine deltas or marine deltas.
- Apart from the effect of the stationary sea water, deposition at the sea coasts occurs also because fine clay particles coagulate and settle down when they mix with salty water of the sea. This process is called flocculation.

THE CROSS-PROFILE OF A RIVER

This refers to a section showing a vertical shape of a river from one bank to the other. As a result it shows the width attained by a river valley at a given point. The cross-profile is characterized by the following;

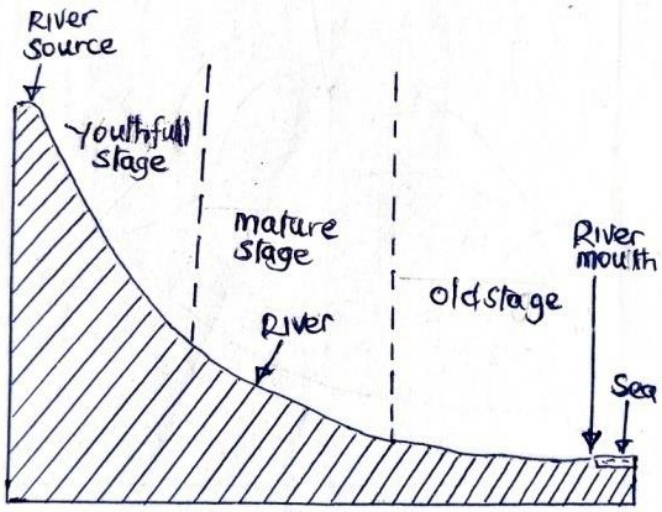
- A narrow valley at the youthful stage because vertical erosion is more pronounced than lateral erosion in the youthful stage. A wide valley in the old stage because lateral erosion is more pronounced than the vertical erosion hence the valley is wide. The cross-profile is V-shaped in the youthful stage due to increased vertical erosion which deepens the valley. While a U-shaped in the old stage due to reduced vertical erosion and more deposition which makes the valley shallow.

Illustration



THE LONG PROFILE OF A RIVER

This refers to the slope of a river from its source to the mouth. It's explained in terms of a distance covered by a river from its source to the mouth.



DEGRADING AND AGGRADING OF THE RIVER PROFILE

The river profile is never static i.e. rivers always change their profiles/slopes. When a river enters any point in its profile without a full load, it erodes its bed adding materials eroded to its load and this is known as **degrading** of the bed. However, erosion decreases as the load increases. On the other hand if a river enters any point along its slope where gradient is very gentle, the rivers velocity decreases and deposition begins to take place. This deposition on the river bed is called **aggrading**.

CAUSES OF AGGRADING AND DEGRADING ON THE RIVER BED

- **Earth movements**: this causes the up and down movement of the crustal landscape. Where the land is uplifted, the river rejuvenates causing degrading if the profile. Where a depression is formed there will be aggrading through deposition such that the depression is filled or bridged with the deposited materials.
- **Climate change**: this brings about the change in the volume of water in the channel. During the wet season the volume of water and its power to erode increases. This results into degrading of the profile. At the same time the eroded materials are deposited in the low-lying areas resulting into aggrading of the river profile.
- **Rock structure**: hard rocks are more resistant to erosion and weathering than soft rocks. The soft rocks when eroded form depressions while the hard rocks form outcrops. The river therefore degrades the outcrops and aggrades the depressions.
- **Presence of a lake**: when a lake exists along a given river profile, it may result into aggrading or degrading. Since all lakes occupy depressions, such depressions are abnormalities in the river profile which have to be removed by degrading or aggrading. The river may hinder the (degrade)of its base level until it removes the lake or it may deposit materials in the lake until the basin is bridged up by the deposits.
- **Human activities**: in areas where river water is used for irrigation or where a dam is constructed, there is decrease in the volume of water in first case and an increase of in the second case there is therefore depression where irrigation is practiced thus leading to aggrading of the river profile. In the case of damming, the river's volume of water increases due to controlled flow of the river and this leads to degrading.

STAGES OF A RIVER PROFILE

As a river moves from its source to the mouth, it moves through three stages. These stages can be explained by W.M. Davisian theory of Peneplanation. According to Davis, all landscapes pass through

three stages i.e. the youthful, mature and the old stage. A river usually starts as a rushing torrent from a high altitude source (youthful structure) but develops into maturity in the middle of its course and ends sluggishly (old stage) at its mouth hence the three stages.

THE YOUTHFUL STAGE

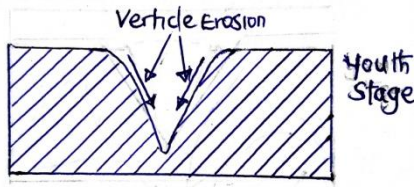
This stage is also referred to as the **torrent, upper** or the **juvenile** stage. This is the stage near the source of the river. It is characterized by very steep gradient, very fast flowing water, high volume of water, intensive vertical erosion that results into valley deepening and formation of a v-shaped valley, presence of interlocking spurs, water falls and rapids.

FEATURES OF THE YOUTHFUL STAGE

These features include; V-shaped valley, potholes, interlocking spurs, gorges, rapids, water falls, etc.

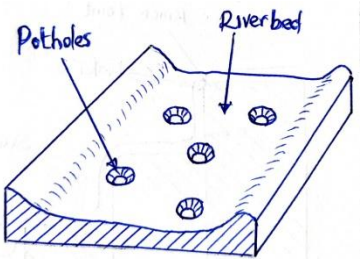
- **V-shaped valley**: this is a narrow and very deep valley. It is formed due to continuous vertical erosion of fast flowing rivers over a steepened gradient. This results into the continuous undercutting of the base of the river leading to a narrow deep valley. Example can be seen at river Mobuku as it starts.

Illustration

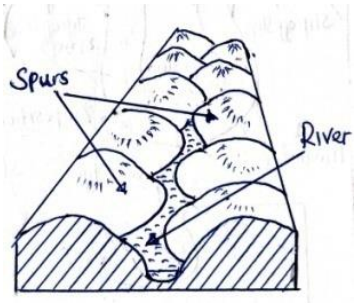


- **Potholes**: these are circular depressions on the river bed. They are formed due to alternate soft and hard rocks. The river cuts holes in the areas of soft rocks along a river bed. They can also be formed when moving water removes loosely consolidated rocks at its bed or when the river undercuts and removes resistant rock outcrops leaving behind a depression at the river bed forming a pothole. As potholes gets over deepened and larger, they may form water falls e.g. Kamsasa valley on mount Rwenzori, river Mobuku on the slopes of Rwenzori and on river Manafwa on the slopes of Mount Elgon.

Illustration

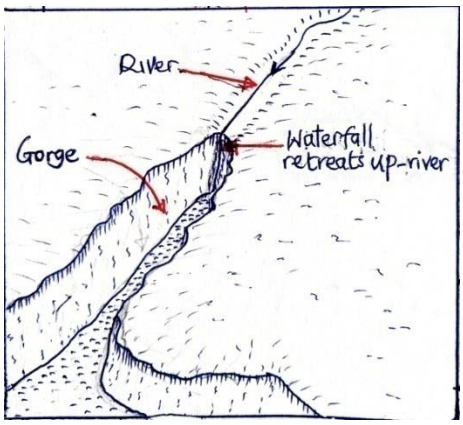


- **Interlocking spurs**: these are highlands which interlock/alternate on the either side of a river. They are characteristic features of rivers flowing in the highland areas. They are formed as a result of river erosion on rocks that offer alternating resistance. The softer rocks usually on the concave slope are easily eroded to form river cliffs while the concave slopes with resistant rocks experience less erosion but some deposition. As this continues, the bends along the river become more pronounced which later causes projections of highlands called spurs which alternate on both sides of the river. Examples can be seen on river Semiliki, river Nile and river Mobuku.



- **Gorge**: this is an elongated steep-sided narrow valley with a stream/river. It develops where vertical erosion cuts down more rapidly.

Illustration



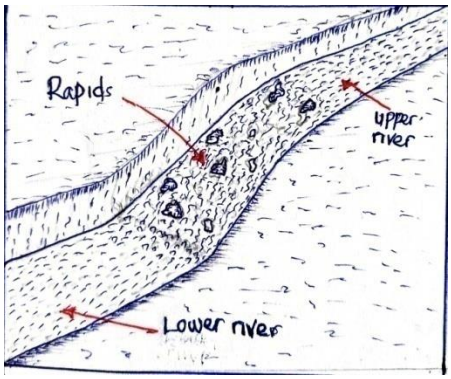
A gorge may develop due to the following factors;

- A gorge may occur in areas where a river flows over a long distance of a band of soft rocks. Vertical erosion on the soft rock deepens the valley to form a gorge.
- A gorge may also form where a river is superimposed on a zone of hard rocks under-laid by soft rocks. The under cutting of hard rocks leads to rapid erosion of soft rocks thus forming a gorge known as a superimposed gorge such as the Sabaloka gorge which is 100m deep
- A gorge may also be formed by a water fall retreating up stream as it is below the Victoria Falls on river Zambezi.
- A gorge also forms where a river has cut across a zone of rocks which are being slowly lifted such as the great Ruaha River in Tanzania. This is called an antecedent gorge.
- A gorge can be formed in regions of limestone where vertical erosion exceeds lateral erosion e.g. in Eastern DRC.

Note: big gorges are called canyons e.g. the fish river in Namibia and the grand canyon on river Colorado in the USA.

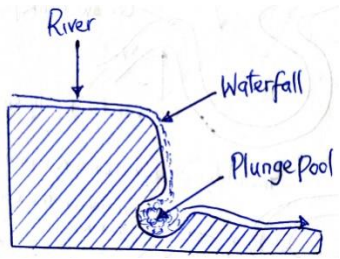
- **Rapids:** these are sections of fast flowing water in a river channel where the river bed is rough and rocky. Unlike in the case of water falls, there are no definite banks or slopes in rapids. They therefore extend several km along the river profile. They develop where a resistant rock overlying a less resistant rock/soft rock lies gently down the river. It can also be caused where a water fall has been steadily eroded until it can no longer cause vertical falls such as on river Mbaira in Tanzania and a section of river Nile near Karuma.

Illustration



- **Water falls:** these are sharp breaks or steps in a river bed over which water falls from a higher elevation to a lower ground. The bed on which the water falls may be over deepened by hydraulic force to form a plunge pool such as in Karuma falls, Bujagali falls, and etc.

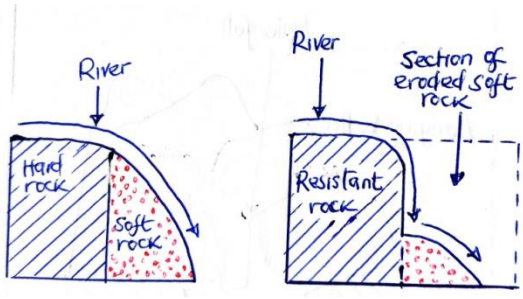
Illustration



CONDITIONS WHICH INFUENCE THE FORMATION OF WATER FALLS

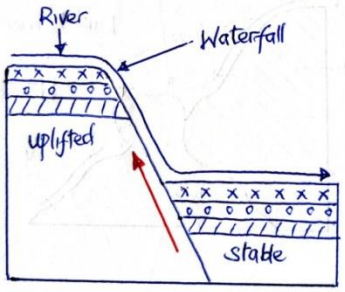
- Difference in rock hardness: where hard rock layers alternating with soft rocks, lie across a river course, the soft rocks are easily eroded causing a steep gradient or a sharp break in the long profile of the river thereby creating water falls e.g. the Nalubale falls on river Nile.

Illustration



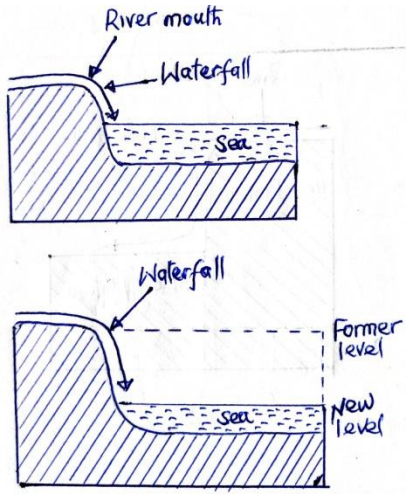
- Faulting across a river bed; a fault with weak rocks down stream, can lead to the erosion of weak rocks leading to the formation of water fall at the fault line e.g. Kalambo falls in Tanzania.
- Faulting may also lead to formation of a water fall at appoint where a river flows on the edge of a fault scarp e.g. Kabalega falls/Murchison falls on the Butiaba fault scarp in Bulisa western Uganda.

Illustration



- A water fall may also form near a river mouth where a cliff is formed by wave erosion more so by a destructive wave.

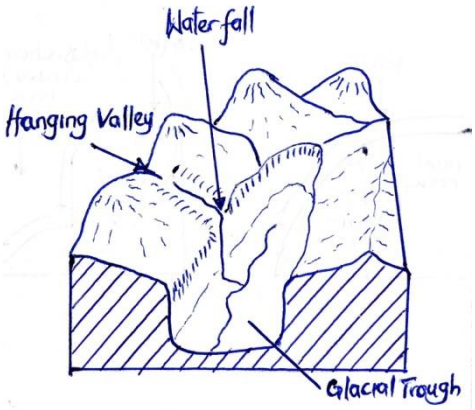
Illustration



- Water falls may also form where the sea level changes/ eustatism such that the river channel remain hanging as it is for the Lobi falls in Cameroon.

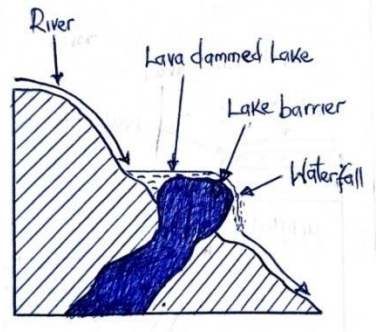
- A water fall may also form at a point where a tributary on a hanging valley joins the over deepened glaciated trough/ U-shaped valley to form a water fall. Examples include a series of water falls joining the Kamsasa valley on the slopes of mount Rwenzori.

Illustration

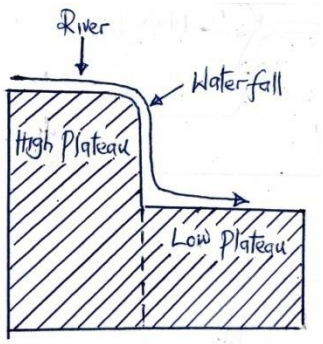


- A water fall may also form through lava damming across a river. After the lava dammed lake is formed, water falls develop at a point where leaves the lake to flow over the lava barrier. Water falls such as on Lake Mutanda in south western Uganda was formed in this process.

Illustration

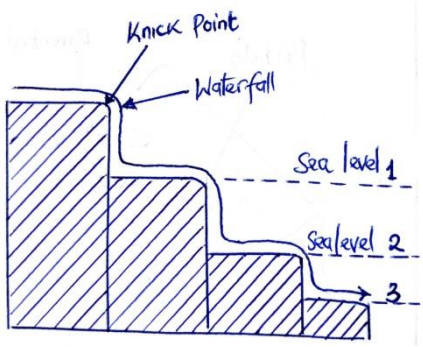


- A water fall can also be formed where a river falls over a plateau edge i.e. from a higher plateau to a lower one. E.g. Augrabies falls along river Orange in the Republic of South Africa was formed in this way.



- Water falls may also form where there is river rejuvenation. Rejuvenation produces sharp points in the long profile of the river valley known as knick points and at the knick point a water fall develops.

Illustration



SIGNIFICANCE OF WATER FALLS

- Water falls are sites of hydro electric power generation for domestic and industrial use such as the Nalubale water falls at Owen falls dam, Bujagali falls, along river Nile serves power to Uganda. Karuma falls is under construction.
- Water falls act as tourist attractions hence a source of foreign exchange used for economic development e.g. at Bujagali falls the Europeans come for annual rafting, diving, Murchison falls, Karuma falls all on river Nile attract tourists.
- Water falls promote infrastructure development such as roads to link the tourists to such falls like to Murchison falls, hotels such as Para lodge near Murchison falls, recreation grounds, etc.
- Water falls are used for research and study purposes for geography students, geologists, etc thus promoting education sector.
- Water falls may attract urban development especially when it is exploited for hydro power generation and tourism. Jinja town partly grew because for this reason, Karuma Township, etc.
- Water falls indirectly has created employment to the people such as guards to the hydro power stations, those working in the tourist sector. These workers earn income they use to improve their standards of living.
- Water falls promotes the sports sector especially water sports such as diving, water rafting, skewing, skating, etc.
- **However,**
- Water falls hinder navigation along rivers which limits inland water transportation e.g. along the river Nile. This explains partly why most African rivers are not used for navigation.
- Waterfalls cause accidents due to high velocity and steepness thus leading to loss of lives.

- Waterfalls cause noise pollution to the nearby people.

REVISION QUESTION

Qn*: Account for the formation of water falls*

Qn*: explain the economic importance of water falls*

Approach: *Define a water fall, give examples of water falls, explain the conditions for their formation, and explain the positive and negative importance of water falls.*

THE MATURE STAGE

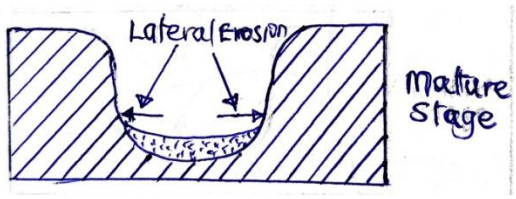
This stage is some times called a **middle** stage. It is a stage of the river between the youthful stage and the old stage. The mature stage is characterized by; the valley changes into a U-shaped, the gradient is gentler compared to that of the youthful stage, erosion is mainly lateral and deposition begins to take place due to the reduced velocity/energy.

FEATURES OF THE MATURE STAGE

At this stage the features include; U-shaped valley, meanders,

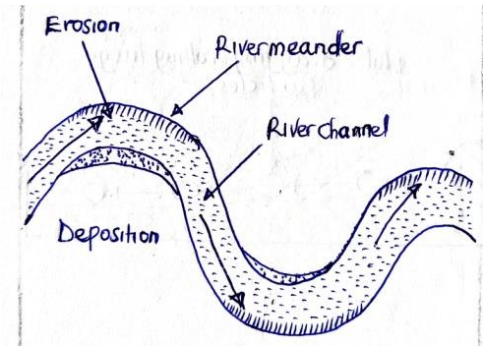
- **U-shaped valley**: this is a large broad valley within which a river is flowing through. It is formed due to the dominance of lateral erosion as compared to vertical erosion. Due to that the river cuts the lower valley sides facing themselves to form an open U-shaped valley.

Illustration



- **Meanders**: these are curved bends of a river channel or it is a section of a river channel where a river begins to move in a series of curves/bends. Constant erosion on the concave banks produces a river cliff while deposition on the convex banks produces a slip-off slope. Examples are found on rivers such as Tana in Thika, Nzoia in Tanzania, Rwizi in Mbarara in south western Uganda, Athi, Mpanga, etc.

Illustration



THE THEORIES OF RIVER MEANDERS

- **Coriollis force**: according to Ferrell and other geographers, they believed that all objects rotate towards equator. So to them rivers also meander due to Coriollis force caused by the earth's rotation. In other wards, because the earth rotates, the river rotates too. However, the question about this theory is that why do rivers around the equator meander while Coriollis force is negligible or almost to non-existence at the equator.
- **Existence of obstacles**: some geographers forwarded that rivers meander because of the existence of obstacles in their channels which prohibit a straight course and that the river flows while a

voiding the obstacles hence developing meanders. However, this theory has been criticized in the way that if obstacles were responsible for river meandering, then rivers must have been of the same shape and category in all the rivers of the world.

- **Meandering occurs at a mature or old stage of the river course**: it's true that many meanders are found in the mature stage or flat areas because of reduced gradient, river competence, etc, but up to today, there is no scientific theory that stops meandering on the steep slopes as referred to by Davis. Therefore meandering takes place at any stage.
- **Helical flow of fluids**: it states that all fluids assume a helical flow/spiral flow that is when fluids are in the channel and as a result of that, rivers do not move in a straight channel thus meandering. However, the theory does not enlighten on the mechanism/forces behind the helical flow and does not explain why it takes place therefore the idea has been found lacking.
- **Erosion and deposition**: some geographers argue that meandering is due to erosion and deposition. That erosion helps in steepening and deepening but deposition diverts the channel hence river meandering. However it has also been found that meandering also occurs in plastic channels where erosion and deposition do not occur.
- **Gradient/slope**: geomorphologists argue that meandering is purely due to gradient. It is said that rivers tend to adjust their slopes to materials they carry. When the load is low the river tends to develop gentle slopes as compared to when the load is high. In other words, if the slope exceeds the required energy to transport the load, then the river will stretch which may affect the meanders. However, the criticism to the theory is that it does not clearly indicate how rivers meander as they tend to adjust themselves to the materials they carry.
- **Riffles and pools theory**: **riffles** are shallow sections due to deposition of the river while **pools** are deep sections due to erosion of the river. Erosion tends to create a straight course of a river while deposition tends to cause deflection of the river channel flow. The alternate ripples and pools along the rivers profile induce the river to meander.

REVISION QUESTION

Qn: Critically examine the conditions which have led to the development of river meanders in East Africa.
Qn: account for the formation of meanders along a river channel.
Approach: define a meander, show where they are found, outline the conditions in the mature and old stages of a river, outline the conditions for their development and explain the theories for their development

THE OLD STAGE

This is also called the **lower, plain, senile** stage of the river profile. This is the last section/part of the river long profile next to the river mouth where the river ends in to a lake, sea or ocean. The stage is characterized by; broad/wide valley, the valley is flat/plain, the gradient is very gentle, deposition is dominant, the river load is mainly silt and other fine materials, the river flows sluggishly due to low gradient, etc.

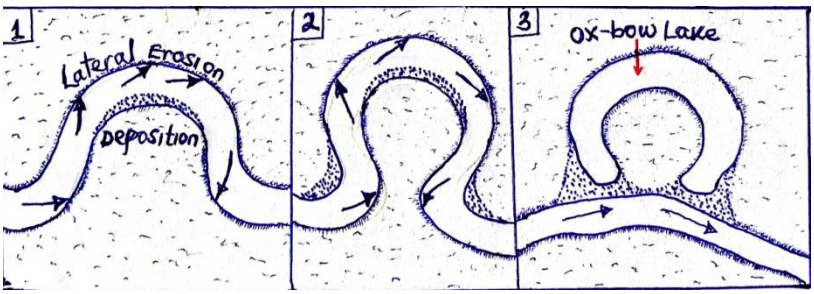
FEATURE OF THE OLD STAGE

At this stage the features include; **well developed meanders, ox-bow lakes, cliffs or bluffs, slip-off slopes, flood plains, levees, braided channels deferred junction, an alluvial fans, deltas, distributaries, etc.**

- **Well developed meanders**: (refer to the mature stage for its description)
- **Ox-bow lakes:** these are horse-shoe shaped lakes formed near the end of the river course. They are formed from river meanders. During times of flooding there will be an increase in the speed and volume of a river and so, when a river comes into a tight meander that it can not go round the

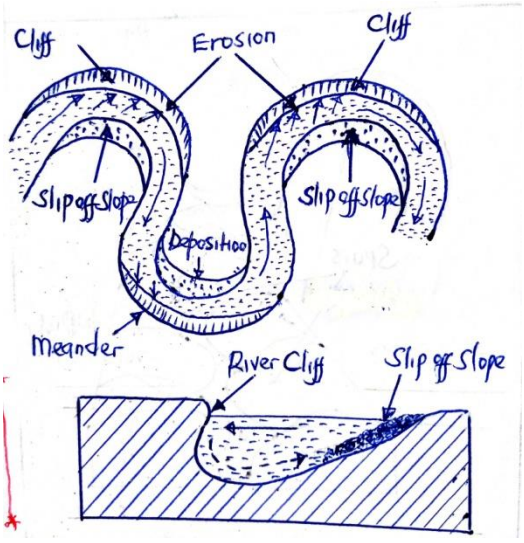
meander, it simply bursts its banks and cuts through the meander neck. At this time water flowing around the meander is slow and so deposition will take place. With time, deposition cuts off the meander from the main river to form an ox-bow lake. Examples can be found along rivers such as Nzoia and Tana in Kenya, Semiliki in western Uganda and Rufigi in Tanzania.

Illustration



- **A meander scar:** this is a scar left after an ox-bow lake has been filled up with alluvium from later floods and later dries up.
- **River Cliffs/bluffs**: this is a steep rock surface found on the outer/concave bank of the river meander. It is formed by fast flowing water (lateral erosion) that erodes more/undercuts the outer/concave bank of the river forming a steep surface called a river cliff. **Bluffs** are the cut-spur ends due to lateral erosion that extend down into the river valley.
- **Slip-off slope**: this is a gently sloping surface found at the inner/convex bank of a river meander. It is formed due to deposition of eroded soil particles from the outer/convex bank of a river meander. The deposition is due to the reduced velocity of water flow.

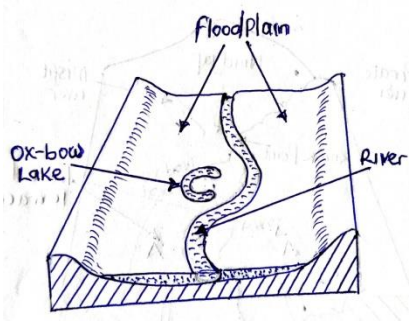
Illustration



- **Floodplain:** this is a wide plain of alluvium covering the floor of a river valley across which a river flows in a meandering channel. It is formed from deposits of sediments carried down by a river. The gradient is low and the river has excess load. When the river overflows its banks due to flooding, the water spreads over the entire width of the plain hence the name floodplain. Successive flooding and deposition leads to building up of a platform of alluvium called a floodplain. Examples of floodplain can be seen on rivers such as Manafwa in Eastern Uganda, on river Ngaila on the Kano plains of Kenya, R. Nyando, R. Tana, and R. Rufigi.

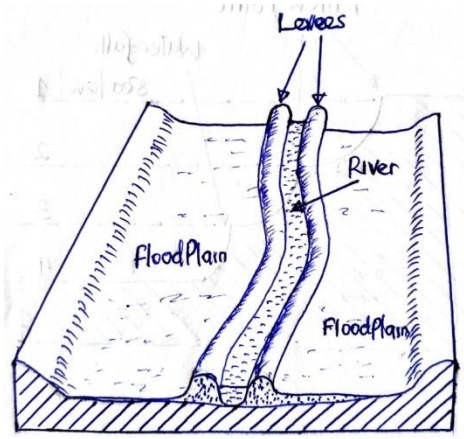
Note: floodplains have fertile soils hence often used for cultivation.

Illustration



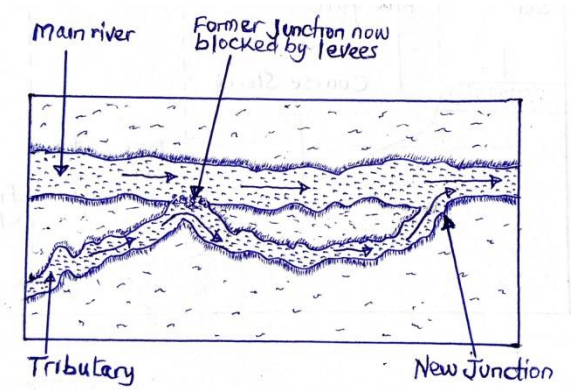
- **Levees:** these are raised ridge-like embankments composed of sediments deposited on the sides of a river channel. They rise above the level of the adjacent floodplain. Levees are formed through successive flooding and deposition of sediments by the river. Deposition is greatest nearest the river channel because as the water floods out of the main channel, its speed is immediately reduced by friction with the banks and the heavier sediments are deposited first. Examples of levees can be seen on the lower R. Omo, where it joins L. Turkana, on R. Malaba and R. Yala and Kano plains of Kenya.

Illustration



- **Deferred junction/tributary**: this is a tributary of a river that is forced to flow a alongside the main river for some distance before joining it. It is formed when river flooding builds up levees across the original confluence of the two rivers hence blocking the tributary from joining the main river for some several kilometers. Therefore the river junction is deferred before joining hence called deferred junction.

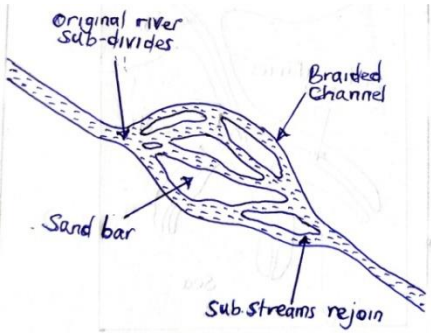
Illustration



- **Braided channel**: this is a wide shallow channel in which a stream subdivides into a series of inter-connecting small channels separated by deposits. They are formed when a river carrying large quantities of load reduce its competence. This leads to deposition of the load on the river bed

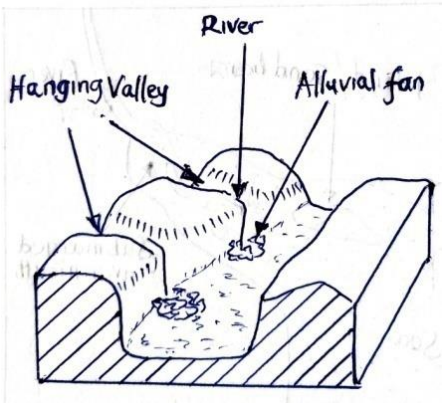
in series. Once these accumulate, they form sand bars which separates the river into numerous small streams. However these streams later rejoin to form a big river. Examples are found on R. Nile between Malakal and Khartoum, on R. Zaire, R. Kilombero and R. Tana.

Illustration



- **Alluvial fan**: this is a cone shaped heap of sediments (sand and gravel) produced by deposition of the load at the foot of the highland by heavily loaded mountain torrent. They are formed when mountain torrent deposits its load at a point where it enters into a main valley. Examples are found on Kilombero valley on Mahenge and Uchungwe highlands in Tanzania. In Uganda there is Lume fan on R. Lume on the Semiliki valley.

Illustration



- **Delta:** this is a large flat, low-lying plain of river deposits such as sand, silt, boulders, gravel and pebbles laid down where a river enters/flows into the lake, sea or ocean. Deposition of river load at the river mouth forces the river to divide into several channels called Distributaries. Examples of deltas are found on R. Omo as it enters L. Turkana in Kenya, Nile delta in Egypt, Niger delta in Nigeria, etc
- Many deltas as being in a low-lying plain are covered with swampy vegetation and other types of plants and may also have lagoons. Deltas bear triangular shapes.

CONDITIONS THAT FAVOUR THE FORMATION OF DELTA

- A river must have a large amount of load/sediments transported and deposited by a river,
- Reduction in the gradient and speed of the river thus reduced river competence,
- Presence of a sheltered coast such that deposits laid down are not washed away,
- Low tidal currents such that deposits are swept away by sea waves,
- Difference in salinity of water between the river and the sea thus enabling deposition,
- Absence of obstacles/barriers like dykes at the mouth of the river and
- Presence of shallow adjoining area/continental shelf along the sea where the deposits are laid to accumulate thus gradually forming a delta.

FORMATION OF A DELTA

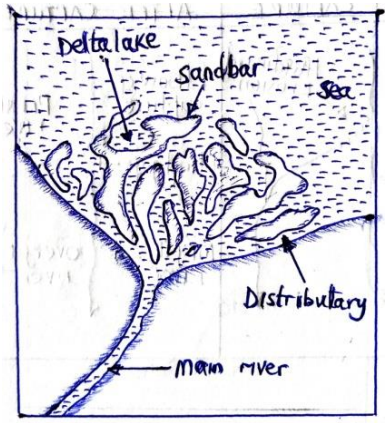
- The formation of a delta begins with a river with a large load being forced to deposit its load near the river mouth due to reduction on its velocity.
- This is followed by deposited sediments blocking the existing river channel causing new distributaries to form and with the growth of vegetation, distributaries are encouraged to form.
- With more deposition, spits bars and lagoons are built within the delta. Their formation is associated with accumulation of deposits in ridge like features which at times blocks off water to create lagoons.
- The older parts of the delta take on solid appearance.
- Lagoons are eventually filled with sediments and later colonized by swamp vegetation. The swamps later disappear leaving dry land where agriculture, settlement, etc, can be practiced.

TYPES OF DELTAS

There are three types of deltas which can be formed and their characteristics depend on the nature and quantities of sediments. These deltas are:

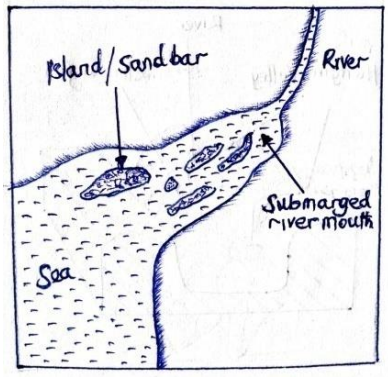
- **Arcuate delta**: this is the most common type of delta. They have a shape of an inverted cone and many tributaries. It is characterized by; coarse sediments such as gravel and sand, many distributaries, triangular shape with rounded seaward shape. Examples are the Rufigi delta in Tanzania, Tana delta in Kenya, the Semiliki delta on L. Albert, the Nile delta in Egypt, the Niger delta in Nigeria.

Illustration



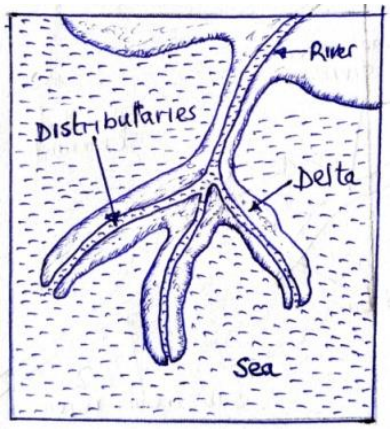
- **Estuarine/cuspate delta**: this is the type of delta built by rivers depositing materials/sediments in a submerged/drowned river mouth. The deposits form sand bars and islands with several distributaries winding around them. They take the shape of an estuary because they don't extend into the sea but simply builds within the submerged river mouth. Examples are the Congo delta and the Rufigi delta which are currently covered by mangrove swamps.

Illustration



- **Bird's foot delta**: this type of delta appears or resembles a bird's foot. It is formed by a river carrying very fine materials such as silt into the water where wave energy is too low to remove it. It is characterized by; a few very long distributaries bordered by levees extending some distance into the sea, fine materials mainly silt and resembles a bird's foot. Rivers with mouths in L. Victoria generally have bird's foot type of deltas

Illustrations



- **Note; Lacustrine delta**: this is an inland delta where a river joins a lake for example where the Victoria Nile joins the L. Albert and where R. Omo joins L. Turkana.

ECONOMIC IMPORTANCE OF DELTAS

- Deltas attract dense settlements due to the fertile soils which support crop cultivation e.g. Cairo city has one of the largest concentration of people in Africa.
- Deltas contain fertile alluvial soils which has given rise to large scale crop farming to provide food to feed the population e.g. in the Nile delta region, etc.
- Many deltas contain petroleum deposits such as the Niger delta in Nigeria, Rufigi delta in Tanzania, Semiliki delta at L. Albert. This promotes mining industry and provides a source of fuel energy.
- Deltas act as tourists' attractions hence earn foreign exchange to these respective countries such as the Nile delta in Egypt, etc.
- Deltas have got lagoons which contain silt good for fish hence attracting fishing activities which is a source of proteins and income to the people.
- Many deltas especially those in the tropics contain mangrove trees which is a source of poles used in the building and construction industry.
- The swamp vegetation covering deltas is used as a raw material in the craft industry where people earn income when the crafts products are sold,

However,

- Deltas tend to harbor disease-carrying vectors such as mosquitoes which cause malaria, snails which transmit bilharzias disease.
- Deltas are flat low-lying plains hence are subject to flooding causing damage to crops, settlement, etc, hence leading to losses.
- Deposition at the river mouth impedes navigation
- The saline water from the sea reduces the productivity of deltas for agriculture.

REVISION QUESTIONS

RIVER CAPTURE

River capture is also known as river piracy. It refers to the diversion of part or the whole of a river course into the system of another river which is adjacent and more powerful.

- River capture occurs when a stronger river flowing in a deep channel diverts the water of adjacent but a weak river.

CONDITIONS THAT INFLUENCE RIVER CAPTURE/PIRACY

- The existence of a powerful and weak river adjacent to each other,
- the existence of weak and soft rocks between the two rivers enabling the stronger/pirate river to erode backwards towards and capture the weak river,
- the pirate river may be flowing on a much steeper gradient than a weak river to accommodate more volume of water from a weaker river,
- The pirate river must have more active head ward erosion than the weaker river.

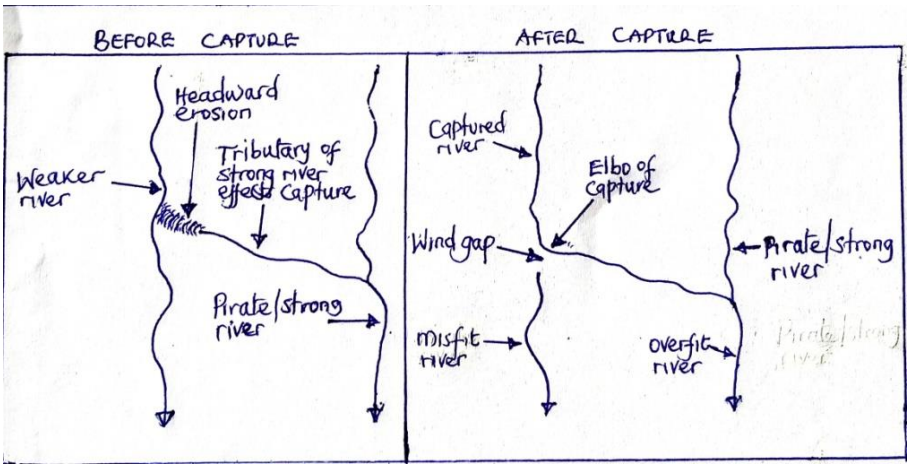
When these conditions prevail, the more powerful river cuts back its channel due to head ward erosion until it diverts the water in a weak river to flow into the valley of a stronger capturing river.

- In East Africa, river capture has taken place on the following rivers; R. Tiva captured one of the tributaries of R. Athi in Kenya, in Tanzania, R. Ruaha captured Pawaga drainage system, R. Aswa, Agago and Pager captured many streams which flowed to L. Kyoga.

FEATURES OF RIVER CAPTURE

The features formed by river capture are; Elbow of capture, wind gap, misfit stream, Overfit River, rejuvenated valley and knick point.

Illustration

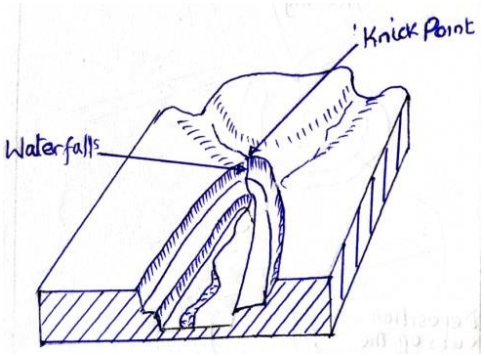


- **Wind gap**: this is an almost a dry valley of the beheaded river/stream below the point of river capture. The wind gap is a dry valley covered with old river gravel and other alluvial deposits which may later be colonized by swamps vegetation. A wind gap is formed when water from the beheaded/ misfit river is diverted to a strong Pirate River leaving behind an almost empty valley.
- **Elbow of capture**: this is a sharp bend found at a point where the water of a captured river/stream is diverted into the water of the pirate river/stream. It is formed from the fact that at this point, the weak river undercuts its bank at a sharp angle as it diverts to join a capturing river.
- **Misfit River**: this is also called a beheaded river/stream. This is a river/stream that has lost its headwaters due to diversion of its water to a pirate river/stream. It therefore contains less water as compared to the size of the channel it is flowing in. the river is called misfit because its water does

not match the valley in which it is flowing. In this case, because its water does not fill the entire valley, it is referred to as "misfit" or "under fit".

- **Overfit River**: this is the pirate stream/river which captures the weak stream/river. The pirate stream/river is extremely enlarged due to additional volume of water from the weaker river. Therefore the volume of water becomes too big for the original valley before the capture. This pirate stream/river is also a misfit but in this case it is referred to as an Overfit.
- **Rejuvenated valley**: this is a much deeper valley near the point of river capture. It is formed when the erosive power of the pirate stream/river increases downwards due to increased volume of water after capture. The increased volume of water increases the speed hence renewing the river's erosive activity thus forming a rejuvenated valley.
- **Knick point/water falls**: this is a break in slope downwards near the point of capture. It is formed when the river pirate is rejuvenated or incises/erodes its valley due to increase in water volume and speed from the weak river/stream.

Illustration



CONDITIONS WHICH INFLUENCE RIVER CAPTURE

- **The presence of two adjacent rivers** is the most important condition. This enables one of the rivers to capture the headwaters of the other river. For example; R. Aswa was near R. Agago, R. Moroto, and R. Pager which R. Aswa eventually captured.
- **One of the two rivers flowing adjacent to the other** should be more powerful than the other (difference in river/stream power) this enables the powerful river to erode its bed faster by headward erosion so as to capture the waters of the adjacent weak river. For example R. Nile which is more powerful river captured the weak waters of R. Okole, R. Tochi and R. Arocha basing on this principle.
- **Difference in rock hardness is another condition**. A river flowing over soft rocks deeply and easily cuts its valley by headward erosion into the valley of the other one flowing over hard rocks hence capturing it in the due course. An Example is R. Wassa flowing on soft rocks eroded and captured R. Nyaboroga in Fort Portal in western Uganda.
- **River rejuvenation** may also cause river capture. A rejuvenated river regains its strength and extends its valley through headward erosion into that of a weaker adjacent river.
- River capture may also occur due to **the influence of earth movements** such as uplift or down warping along the course of the river. A river flowing over a down warped channel may extend its valley by headward erosion and capture the waters of an adjacent weaker river flowing over an uplifted channel. Example R. Birira in southwestern Uganda captured the nearby R. Rwizi. Down warping and uplift led to river reversal. As a result, R. Katonga and R. Rwizi captured the weak rivers in western Uganda.
- **The influence of rock joints** can also cause river capture. Under circumstances where a pirate river/stream is flowing over a well jointed rock, its able to undertake headward erosion to deepen its valley and eventually captures the nearby river flowing on massive rocks.

- **Steep gradient** can also cause river capture. In circumstances where a pirate river/stream is flowing down a much steeper gradient than its victim's gradient, enables a pirate river to under take headward erosion towards the valley of a weak river leading to its capture. A good example is found in Ghana where R. Nsaki a tributary of R. Densu in the Akwapim hills (steeper gradient) captured R. Bisiasi (gentle/lower gradient.
- **Under circumstances where a pirate river flows in a short and direct course** while the adjacent weak river flows in along and circuitous course, the direct (pirate) river extends its course by headward erosion and eventually captures the one in a circuitous course.
- **Where there is direct flow of both the pirate and the weak river in the same direction**, river capture may occur. Under this condition, the pirate river easily carries out headward erosion and captures the weaker river/stream.

REVISION QUESTIONS

RIVER REJUVENATION

Rejuvenation refers to a process by which a river's erosive ability/power is renewed as a result of increased energy. Such a river is said to be made "young" again and starts behaving like a river in the youthful stage. The river starts to flow faster and with increased energy even if the river is in its mature or old stage. The river increases erosion ability on its bed creating potholes on its bed and incises its valley deeper through vertical erosion

CONDITIONS WHICH INFLUENCE REJUVENATION

- **Lowering of the base level**; a base level is the lowest level to which a river can erode and the universal base level is the sea. When there is a major climatic change involving drastic lowering of base level, may be due to glaciation which removes water form the seas causing a fall in sea level, the river's gradient is steepened inducing water to flow faster causing a lot of erosion hence river rejuvenation.
- **Earth movements** also cause rejuvenation. When up warping/uplift occurs along the course of the river/stream, it creates a steeper river gradient. This causes water to move faster hence increasing erosion of the river channel thus rejuvenation.
- **River capture** is factor to cause river rejuvenation. River capture is where part of a river course of a weak river is diverted and joins another powerful adjacent river. Therefore river capture adds more water into the capturing river. This increases the erosive energy and speed of the water hence renewing the erosion of the pirate river valley thus rejuvenation. For example the lower R. Tiva in Kenya was rejuvenated after it captured the upper R. Tiva.
- **Climate changes** especially the heavy rainfall received can lead to rejuvenation. Heavy rainfall leads to increase in the volume of water in the river channel which automatically results into a renewal of the river's erosive energy hence rejuvenation.
- **Changes in the nature of the basement rock** is also responsible for rejuvenation of rivers. When a river which was originally flowing over hard rocks, approaches soft rocks, its erosive power is renewed, so, the river erodes much of the soft rocks, thus deepening and widening its channel resulting into river rejuvenation.

- **Human activities** like dam construction, also leads to river rejuvenation. Construction of dams hold water in the reservoir behind the dam but at the same time water released out through turbines flows out with high speed thus renewing the rivers erosive energy.

PROCESSES THROUGH WHICH RIVER REJUVENATION OCCURS

- **Solution/corrosion**; this is the process through which the flowing water in a river channel dissolves some rocks such as rock salts limestone along its banks or bed.
- **Hydraulic action**; this is a process where a river erodes its channel and bed by use of the force of flowing water. Hydraulic action is very active on rocks which have been loosened by weathering.
- **Abrasion/corrosion**: this involves the wearing away of the river bed and its banks by use of the load it's carrying as a grinding tool. The load may include boulders, sand and stones. This happens when the load (boulders) are swirled around by water and therefore grind the river bed to remove rock fragments.
- **Attrition**: this involves double action of the load to erode the channel and also reduction in size as the load keep bumping onto each other thus becoming easy/light for the river to transport.

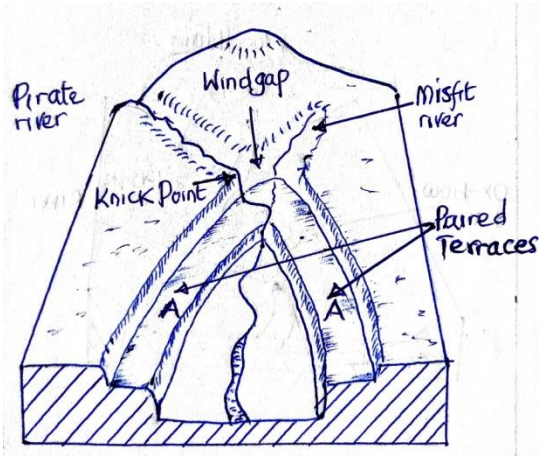
All the above processes help and facilitate river rejuvenation.

LANDFORMS FORMED BY RIVER REJUVENATION

There are several landforms that have been formed through river rejuvenation and they include; Paired terraces, knick points, incised meanders and valley within valley.

- **Paired terraces**: A river terrace is a bench or a step cut in the side of a river valley. They are formed in that when the base level falls, a river renews its erosive activity and incises the channel. Some parts of the former floodplain and underlying rock floor may not be removed by lateral erosion and thus remain on the valley sides as terraces. If renewed rejuvenation takes place, the process is repeated and a second pair of terraces at a lower level than the first one is created. Therefore if several falls in base level occur, a succession of terraces results and separated from each other by knick points. Such terraces may be of equal height and are therefore called paired terraces. Examples are found on R. Semiliki, R. Rwizi, R. Mpanga, R. Kafu, R. Nyando and Ngaila east of Kisumu in Kenya.

Illustration

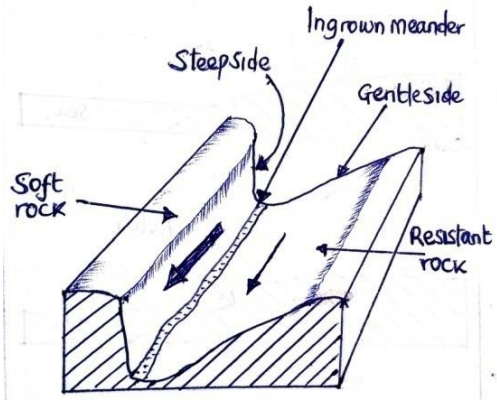


- **Knick points**: this is a break of a slope in the long profile of a river valley caused by a fall in the base level. It's formed when the river renews its erosive ability due to a fall in the base level. When there is a successive falls in the base level, successive knick points are formed on the bed of the river. Examples can be seen on the Mwachi River in Kenya.
- **Incised meanders**: this is a curved bend of a river whose floor has been greatly deepened so that the river now winds between steep walls. Many incised meander result from the rejuvenation of an

already meandering river due to a fall in the base level. There are two types of incised meanders formed depending on the rock resistance i.e.

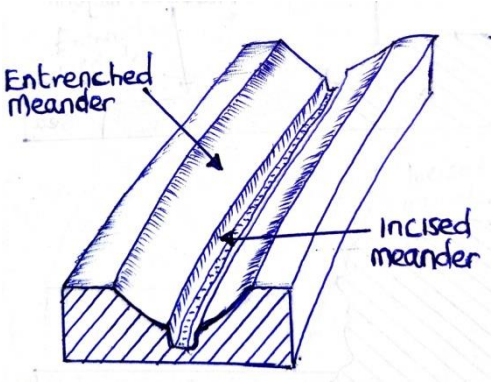
- **Ingrown incised meander**: this is a meander whose river valley is steeper on one side than the other forming an appearance known as a symmetrical cross-profile. It has a valley with a steep side/bank facing a side with a gentle slope on the opposite side of the valley. It develops where a meander valley have rocks of different resistance on the in-facing sides such that when the base level falls erosion highly under-cuts the side with soft rocks to form a cliff/steep side while the side with resistant rocks is eroded slowly to form a gentle slope in the opposite side. Examples can be found on R. Rwizi in Mbarara and on R. Mwachi northwest of Mombasa.

Illustration



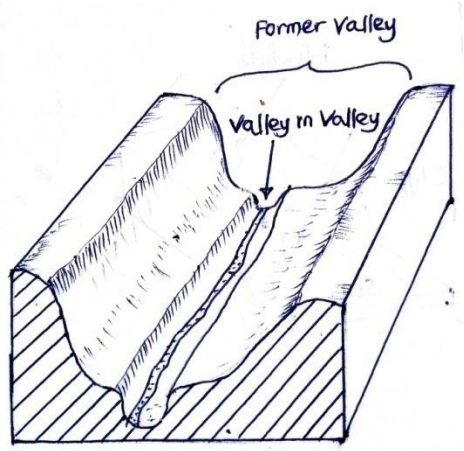
- **Entrenched incised meander**: this is a meander whose valley has a steep-sided symmetrical profile. It's formed in areas where the meander has rocks of the same resistance especially soft rocks such that when the base level falls quickly, rapid vertical erosion occurs on both sides of the river valley creating a V-shaped cross-profile. Under such circumstances there is no difference between the in-facing slopes of the river valley. Examples can be found on R. Mara and R. Kuja in western Kenya.

Illustration



- **A valley within a valley**: this is also known as a gorge within a valley. It is formed due to rapid rejuvenation along rivers as a result of extremely a large fall in the base level. This leads to rapid down cutting of a river with its valley creating another valley usually deeper and narrower than the former valley above. That is why is called a valley within a valley. Examples can be found along R. Mpanga in Fort Portal, R. Semiliki and R. Nyando.

Illustration



- **A gorge**: this is another feature formed due to river rejuvenation. (Refer to gorge formation).
- **Waterfall:** this is also another rejuvenation landform. (Refer to waterfall formation).

DRAINAGE PATTERNS

Drainage pattern refers to the physical outlay of a river and its tributaries on the earth's surface. There are many drainage patterns depending on the factors that influence it such as the nature of the slope and rock over which the river flows, the amount of rainfall received in the catchment area, earth movements, etc. Drainage patterns can be divided into two broad groups' namely accordant drainage pattern and discordant drainage pattern.

ACCORDANT DRAINAGE PATTERNS

These are the drainage patterns in which the flow of rivers and their tributaries follow the slope of land and the rock structure. These drainage patterns include radial, dendritic, trellis, annular, centripetal, parallel, hooked/barbed, fault guided, pinnate.

Radial drainage pattern: this is where several rivers flow outwards in different directions from the same dome/cone/upland shaped watershed normally a volcanic cone/mountain.

- Examples of radial drainage pattern are found on Mt. Elgon where rivers such as Sironko, Manafwa, Namatala, Sipi, Simu, Nzoia, Suan. It is found on Mt. Rwenzori where rivers such as Mubuku, Nyamugasani, Lume and Ruanoli originate. It's also found on Mt. Kenya where rivers such as Ewaso, Nyiro, Tana, Thiba and Sinyai radiate from.

Illustration



Conditions for the development of radial drainage pattern

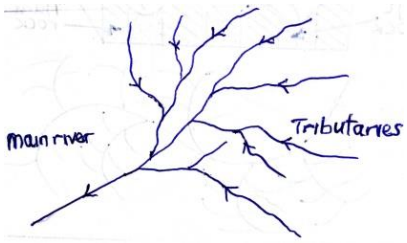
- Presence of a dome/cone shaped highland such as a volcanic cone/highland to provide the source of rivers,
- nature of slope which is steep accelerates down ward movement and erosion of rocks hence creating channels along which rivers flow,
- the cone shaped highland with uniform rock structure and hardness which enables the rivers to erode on any side of the highland,

- Heavy rainfall received/snow/glacier melting in the catchment area is important for the development of radial drainage pattern.

Dendritic drainage pattern: this is a drainage pattern where the main river and its tributaries resemble a structure of a tree and its branches. It's a tree-like pattern of drainage in which tributaries converge on to the main river from many directions and usually join the main river at an acute angle i.e. less than 900.

- Examples are found on rivers such as Rufigi, Victoria Nile, Malagarasi, Ruvuma, Nyando, Congo River, Nzoia, Kagera, Katonga, Aswa, Okok, Mara, etc.

Illustration



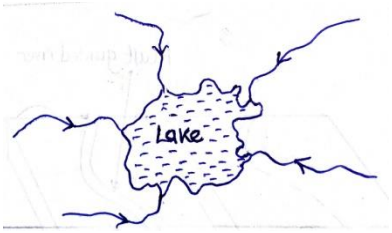
Conditions for the development of dendritic drainage pattern

- Develops in areas with uniform rock structure and hardness (homogenous rocks) and are common on the crystalline granitic rocks. This enables the river to erode with minimum effort,
- develops on a gently dipping/sloping landscape of sedimentary strata/layers where each tributary flows in a valley proportional to its size and maintains its flow,
- Develops in areas which receive heavy rainfall which is reliable to support the development of numerous rivers, develops in regions with a common catchment areas i.e. the numerous tributaries cover a large catchment area.
- In relation to slope, all consequent and insequent streams flow in the direction of initial slope of the area over which the pattern was established, consequent streams develop headward erosion and minor/insequent streams develop in the similar manner and join the main tributaries to complete the pattern.

Centripetal drainage pattern: this is a drainage pattern in which rivers from different directions flow and converge towards a common basin. It's common in basins/depressions due to down warping and faulting.

- Examples are well developed on L. Baringo in Kenya where rivers such as Loboi, Ol Mukutani and Ol Arabel flow towards the lake. It is also found in L. Victoria where rivers such as Kagera, Katonga, Nzoia and Sio flow towards the lake from different directions. Other examples include rivers which flow into the rift valley lakes such as L. Tanganyika, L. Turkana, L. Nakuru and Bogoria.

Illustration



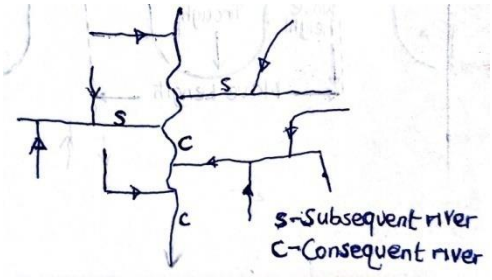
Conditions for the development of centripetal drainage pattern

- Develops in areas with a basin/depression/valley where the rivers release their water to form a lake.
- It develops in regions with rocks which offer equal resistance to erosion (homogenous rocks).
- It develops in regions which receive heavy rainfall to support the numerous rivers flowing towards a depression.

Trellis/rectangular/rectilinear drainage pattern: this is a drainage pattern in which rivers and streams display a rectangular shape with tributaries joining the main river at approximately at right angles (900).

- Examples are found on R. Mayanja, R. Kato-Wassa in Kichwamba, R. Aworanga, R. Pager, R. Aswa, R. Tiva, the lower Tana river joined by rivers such as Lac Galole and Lac Kanoni, R. Kerio and its tributaries such as Yeptos and Kinyach, rivers joining R. Galana between Lugards fall and Koito.

Illustration



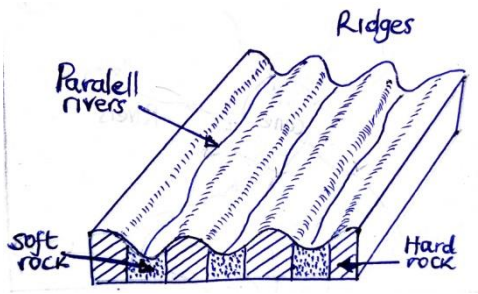
Conditions for the development of trellis drainage pattern

- Trellis drainage pattern develops in areas with jointed/rocks with fault lines so that it flows along fault lines joining at right angles.
- Can also develop due to gently sloping /steep slopes.
- It develops in areas of heterogeneous rocks i.e. areas with both soft and hard rocks.
- It requires an area with a large catchment area since the tributaries are wide-spread/spaced.
- Heavy and reliable rainfall is important to support the development of obsequent, consequent and insequent rivers which constitute trellis drainage pattern.
- They can also develop where river capture has occurred for example along R. Aswa, R. Tiva, R. Galana, R. Tochi, R. Pager, etc

Parallel drainage pattern: this is a drainage pattern where rivers flow parallel to each other for along distance with limited chances of joining hence the name parallel.

- Examples are R. Nkusi and R. Hoima in western Uganda flow parallel to each other before joining L. Albert, in Kenya they are found in Aberdare ranges and west of Mau ranges where tributaries of R. Athi such as R. Nairobi, the Thirika, Komu and Ruiru flow parallel to each other. They are common on ridges and slopes of escarpment.

Illustration



Conditions for the development of parallel drainage pattern

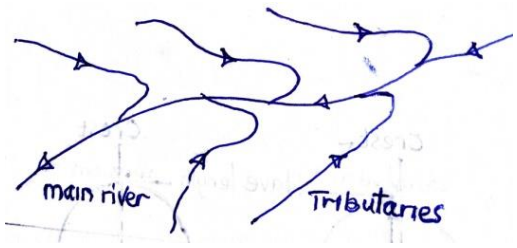
- They develop in areas with steep and elongated slopes such as escarpments.
- They develop in areas with alternating band of soft and hard rocks such that soft rocks are eroded to form river channels while resistant rocks form a divide which tend to prevent rivers from joining.
- The rivers should be flowing in the same direction each forming its channel through headward erosion of the soft rocks.

- There must be a pronounced divide made up of hard rocks between the rivers to prevent the chances that the rivers could join each other.
- It also develops in faulted regions in that rivers tend to flow by eroding the lines of weakness (fault lines) created by earth movements.

Hooked/Barbed drainage pattern: this is a drainage pattern where rivers appear to flow in the opposed direction to the main river before joining it at acute angle. It is mostly found in areas which experience river reversal or river capture like in western Uganda and northern Tanzania. It's better known as reversed drainage system.

- Examples are found along R. Kafu, Katonga, Rwizi and Kagera.

Illustration



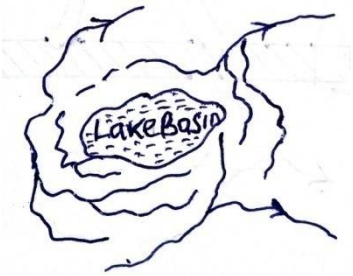
Conditions under which barbed/hooked drainage pattern develops

- Existence of alternating dips of slopes. The river joining the main river originates from a higher point opposite to the source of the main river. After a change in gradient as a result of uplift in the direction of the main river, the joining rivers reverse to join the main river because they fail to climb or erode through the steep gradient created ahead of them.
- River capture where a strong river arrests/diverts the waters of a weak river can also lead to the formation of a barbed drainage pattern.
- A big catchment area and a reliable rainfall support the existence of this drainage pattern.

Annular drainage pattern: this is a pattern where tributaries join the main stream/rivers at sharp angles but in a series of curves. It is developed in volcanic regions where rivers flow sharply in a series of curves around craters or calderas. It is some times confused with the centripetal pattern but the two are different.

- Examples are around Ngorogoro caldera, etc.

Illustration

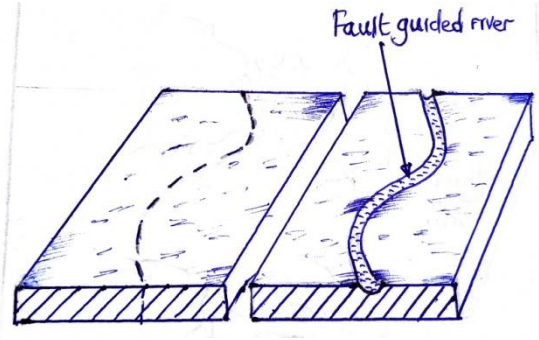


Conditions under which annular pattern develops

- A faulted zone with numerous fault lines or dissected plateau or done with alternating hard and soft rocks favours its development
- A large catchment area with reliable rainfall to support the growth of the river and sustain their erosive power to erode the soft rocks in concentric curves is necessary.
- Rivers must flow in concentric curves. The subsequent streams around dissected domes erode valleys in less resistant strata to flow out wards. Streams generally follow circular curves around the dome conforming to the weaker rock outcrop and flow outwards. Examples are found around Lake Bosumutwi in Ghana where R. Banko and Buonim flow in concentric curves. It's found in the black hills of South Dakota in USA.

Fault guided drainage: this type of drainage develops in faulted areas. Rivers flow down along the fault lines. They flow along this fault lines because they are relatively weak and easily be eroded. The actual pattern that develops is determined by the pattern of the faults lines.

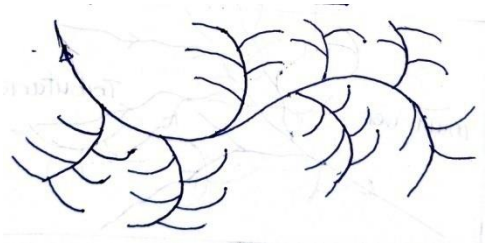
- Examples are R. Aswa in northern Uganda and R. Malewa, Gilgil and Kerio in Kenya, Wassa valley, follow the fault lines.



Pinnate drainage pattern: This is drainage pattern which resembles a shape or appearance of a feather. It constitutes so many small rivers or tributaries which joins the main river at extreme acute angles.

- Examples are found in Kerio valley of western Kenya.

Illustration



Conditions which favour pinnate drainage pattern

- It develops in regions which have uniform rock resistance to erosion. Usually it develops on soft rocks which con easily be eroded by even small rivers of minimum competence thus forming multiple small tributaries.
- It also develops in areas whose slope dips gently and with minimum rock permeability. This enables the existence of small streams despite low water volume or content.
- Large catchment area and heavy reliable rainfall supports and sustains the flow of rivers throughout the year.

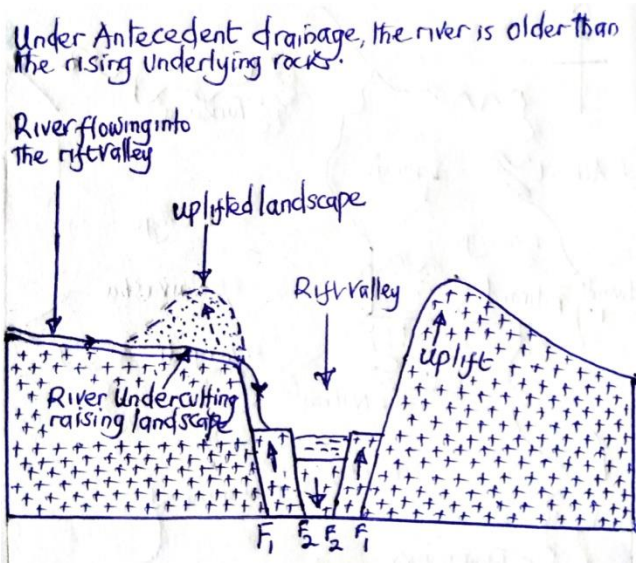
DISCORDANT DRAINAGE PATTERN

These are drainage patterns which show no relationship with the underlying rock-type and structure. The patterns are opposed to the dominant rock types and structures.

- These drainage patterns are;

Antecedent drainage pattern: this is a drainage pattern in which a river maintains its course on a landscape which was later uplifted by earth movements. The river continues to flow in its original direction by continuous down-cutting of the uplifted landscape into a gorge.

Illustration



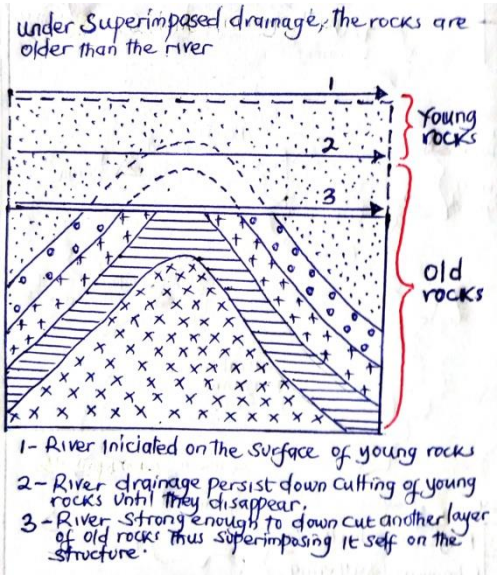
Conditions that favour antecedent drainage pattern to develop

- The uplift must be slow enough for the river to maintain its course in the path of the raising ground or land.
- The river also must be powerful enough to erode vertically down wards as fast as the rising ground/land to develop a gorge.
- Examples of antecedent drainage pattern occur along R. Malagarasi, R. Ruaha in the Iringa highlands in Tanzania, R. Lukuga which flows into Zaire basin, R. Nile at Murchison fall gorge, R. Malewa and Gilgil in Kenya.

Superimposed drainage pattern: this is a drainage pattern which develops on one set of rocks which are then eroded down to uncover an older and different landscape to which they are not related. If the drainage pattern from above is maintained and is not affected by the rock structure of the current rocks then it is superimposed.

- Examples are R. Zambezi superimposed on the Karro rock, R. Nile superimposed at Sabaloka in Sudan.

Illustration



Influence of relief on drainage patterns:

- The gentle slope enables a river to develop numerous tributaries hence developing a dendritic drainage pattern.
- The mountain/cone/highland/dome/ steep slopes influence the development of radial drainage pattern. Rivers radiate from the cone following all direction. Examples are on Mt. Elgon.
- Down warped landscapes/depressions/basins influence rivers flowing from different directions to converge into them hence forming centripetal drainage pattern e.g. rivers such as Nzoia, Kagera, Katonga and Malaba flow into a common point that is lake Victoria.
- Areas with Alternating dips slopes encourage the development of barbed/hooked drainage pattern. Examples are
- Areas with Gently sloping dips with minimum rock permeability encourage the development of pinnate drainage pattern e.g.
- Areas with steep and elongated slope between two or more river valleys encourage the development of parallel drainage pattern e.g. in the escarpment of a rift valley where rivers flow by the side of each other but with limited chances of joining each other.

Influence of climate/rainfall on drainage pattern:

- Areas with heavy rainfall support the development of barbed drainage pattern to supply water to the tributaries.
- Heavy rainfall received around volcanic mountain cones, encourage the development of radial drainage pattern e.g. Mt Elgon
- Heavy rainfall received in down warped basins influence the development of centripetal drainage pattern e.g. around Lake Victoria basin.
- Heavy rainfall also provides water for the numerous rivers hence the development of pinnate drainage pattern e.g.

Influence of rocks structure on drainage pattern:

- Jointed and faulted rocks have influenced the development trellis drainage pattern.
- Alternate bands of soft and hard rocks side by side influence the development of parallel drainage pattern.
- Homogenous or crystalline igneous rocks influence the development of dendritic drainage pattern.
- Difference in rock hardness (heterogeneous) of rocks, occurring in alternate belts of hard and soft rocks lying at right angles to the slope; influence the development of trellis drainage pattern.
- Permeable rocks such as limestone favours development of seasonal drainage patterns.
- These rivers exist for a short period of time in rainy season and disappear during dry season.

Influence of earth movement on drainage pattern:

- Volcanicity leads to the formation of dome shaped cones with uniform structure e.g. Mt Elgon hence influencing development of radial drainage pattern.
- Down warping of some areas forming lake basins have influenced the development of centripetal drainage pattern e.g. Lake Victoria.
- Faulting causes escarpments which have influenced the formation of Parallel River guided valleys hence development parallel drainage pattern.
- Faulting influences the formation of fault lines where some rivers flow through hence influencing the development of rectangular drainage pattern e.g. R. Aswa.
- Volcanicity influences the formation of craters with a series of curves where streams flow in circular around the crater hence development of annular drainage pattern.
- Occurrence of uplift on the river course influence the tributaries to appear to flow on the opposite direction to the main river thus developing hooked /barbed drainage pattern.

ECONOMIC IMPORTANCE OF RIVERS IN EAST AFRICA

- Rivers have water falls which are important sites for dams used in the production of HEP. These features are associated with the upper courses of the rivers examples the Owen Falls Dam on the R. Nile is an important power generating dam supplying the electricity needs for Uganda, Kenya and Tanzania, the seven fox dam in R. Tana, etc.
- Rivers contain flood plains and deltas areas which have very fertile alluvial soil deposits brought about by periodic flooding. These areas have attracted crop cultivation and settlement e.g. the flood plains along R. Manafwa in Eastern Uganda, the Nile delta, etc.
- Rivers provides navigation/water transport routes especially in the lower sections where the channels are wide and free from rapids and water falls. Examples are R. Nile in Egypt and Sudan, R. Tana is navigable for about 400km from its mouth, etc
- Large rivers are important sources of fish hence support fishing activities. Fish is consumed as food. Examples are R. Nile R. Tana, R. Rufigi, R. Ruvuma, etc.
- Rivers are source of water of supply to the settlement found along river channels, for livestock and for industrial use. Examples are the Nile breweries at Jinja gets its water from R. Nile, in Sudan and Egypt, the Nile water is used for irrigation farming in the Gezira scheme, industrial use at the Sennar dam, Aswan High Dam and domestic use.
- Rivers support mining activities especially at the lower courses of rivers where deposition takes place. Examples are the small scale mining along the river valleys of Elgeyo Marakwet district of Kenya, diamond mining in the orange river of South Africa, sand mining for glass and building industry; clay mining for brick making, boulders are also extracted from the river banks along R. Mobuku in Kasese and also along the R. Nile at Jinja all for construction and building purposes.
- Rivers promote tourism since they contain features such as water falls e.g. Murchison falls, gorges, Bujagali, etc are all important tourist attraction in East Africa hence a source of foreign exchange used for development of the economy.
- Rivers have influenced the development of urban centers especially at crossing points of rivers examples are Jinja town in Uganda began its growth and development as being near the point of a source of R. Nile and Nalubale dam.
- Rivers have supported many agricultural practices such as the growing of sugarcane; rice and cotton are done along rivers. Examples are rice is grown along the flood plains of R. Manafwa in Eastern Uganda; rice is also grown along R. Tana in Kenya, etc.

However,

- During heavy rains, rivers tend to flood thereby destroying crops and settlements of the surrounding areas e.g. flooding of R. Rufigi in Tanzania, R. Manafwa in Tororo, etc.
- River flooding increases the spread of water borne diseases like cholera, bilharzia, etc, which attack people and sometimes leading to death
- The presence of water falls and rapids along the rivers hinder navigation along rivers e.g. the upper course of R. Nile, etc
- Deep and fast flawing rivers are a social barriers/hindrance between the people living in the opposite sides of the river for example r. Nile divides Uganda into east and west hence limiting socialization, trade and other benefits.

COASTAL GEOMOPHOLOGY

A **coast** refers to the zone of land bordering the sea or any other extensive tract of water like the ocean. Coasts are constantly changing and the nature of the coast depends on the interaction of land and sea waves, sea level changes and the occurrence of coral rocks.

- **Coastline** refers to the margin of land sometimes taken to mean the limit at which wave action takes place.
- **A shore** refers to the strip of land that lies between high water and low water tides/levels.
- **A shoreline** refers to the line where the shore and water meet.

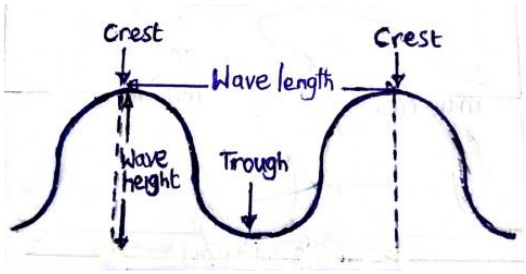
SEA WAVE/MARINE ACTION

Water waves are defined as ripples or oscillations on any water body. Water waves do not involve the movement of water horizontally but it is only the shape of the ripples that moves forward while the water only keeps oscillating up and down.

- Waves are caused by moving wind, earthquakes, volcanic activity like Tsunamis or eruptions and moving objects in water such as boats, whales, hippopotamus, etc.

The structure of a wave

Illustration



- **Wave crest** is the highest point of a wave.
- **Wave trough** is the lowest point of a wave.
- **Wave height** is the distance between a crest and trough (usually 5-6 m).
- **Wave length** this is the distance from one wave crest/trough to another.
- **Wave amplitude** this is a half height of a wave.
- **Wave velocity** is the speed of movement of a wave crest in a given time or distance covered by a particular wave crest in a given time.
- **Wave frequency** is the number of crests that pass a particular observation point per minute.

BREAKING WAVE

This is the most important beginning of wave action on the coast. Before waves break, there will never be any impact they create on the coastline. However, when they break, the energy that they carry pushes water forward towards the coastline to cause either erosion or deposition.

- Water in waves moves in circular orbits i.e. it rises up to the apex of the wave crest and down to the wave trough and the system continues comfortably. As the waves move towards the coast and the depth reduces, these waves come into contact with the bottom of the sea. This contact automatically generates friction which retards the velocity of waves at the bottom. The velocity of water particles at the crest remains normal and this creates frictional interference at the base which will steepen the crest as it moves ahead and, in the process, it will be thrown forward hence breaking the waves.

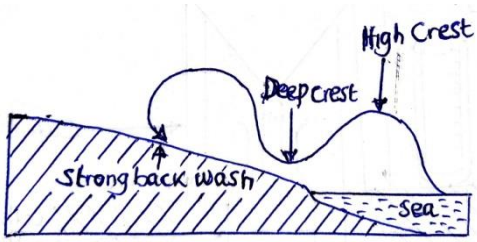
These breaking, pushes/throws the water to the coast/cliff and this water push is called a swash or a send. The water that flows back to the sea after a swash is called a backwash.

TYPES OF WAVES

There are two types of sea waves i.e. destructive waves and constructive waves.

- **Destructive waves**; these waves are sometimes referred to as strong waves/erosive waves/plunging waves. These are waves characterized by a strong backwash i.e. when water is thrown to the coast as waves break; it flows back to the sea with a strong force capable of eroding

materials from the coast to the sea. They have high frequency, high crest and deep trough. They are responsible for erosion at the coast.



- **Constructive waves**: these are also referred to as weak waves/depositional waves/spilling waves. They are characterized by a strong swash i.e. powerful throw of water up the coast with all materials it carries and deposits them there. However, it has a weak backwash such that deposited materials cannot be easily washed back to the sea. They also have a low frequency, low crest and shallow trough. They are responsible for deposition at the coast.

Therefore, the waves have two major effects on the coast namely; erosion of the coast and deposition of eroded and transported materials at the coast.

WAVE EROSION

Waves do not do its work of erosion the same way rivers do. Waves erode the coast when they break and throw water to the coast/shore. Strong waves are more erosive when they break than weak waves.

WAVE EROSIONAL PROCESSES

There are four processes through which waves erode the coast namely; solution/corrosion, hydraulic action, abrasion/Corrasion and attrition. The first three processes operate on the coastal rocks while attrition operates on the load itself.

- **Solution/corrasion**: this is common on the coasts composed of soluble rocks e.g. limestone and rock salts. The solvent action of water dissolves these rocks and are eroded and transported away in solution. This process is active at Mombasa where the coral limestone rocks exist.
- **Hydraulic action**: this is the direct use of a powerful swash/force by breaking waves to pound against the cliff face. When water hits the jointed cliff face, air within the cracks/joints is compressed and as water retreats during the backwash the air suddenly expands and generates shock waves that weaken rock particles and easily eroded by a backwash. This process is very active at the front of the headlands at Kasenyi landing site, etc.
- **Abrasion/Corrosion**: this occurs when waves use the materials they are carrying such as pebbles, boulders, sand and shingle as a grinding tool to wear down the coastal rocks. These materials are hurled strongly by the destructive waves on the cliff-face resulting into breakdown of some rock particles and eventually eroded.
- **Attrition**: this process occurs when boulders and pebbles dashed against the coast are themselves broken into small and finer particles as they nock each other. The materials carried by waves are pounded against the cliff and in the process they knock each other and break up hence reduce in size and subsequently erosion of the coastline.

FACTORS WHICH INFLUENCE WAVE EROSION

- **The energy of waves**: this is the most important factor that determines wave erosion. Waves commonly derive their energy from strong winds which is transformed into waves. The stronger the waves the more erosive they are and vice-versa.
- **Nature of rocks at the coast**: wave erosion is more pronounced on areas at the coast with weak and soluble rocks or in jointed and unconsolidated rocks. The hard and resistant rocks remain standing as headlands while easily eroded rocks become bays.

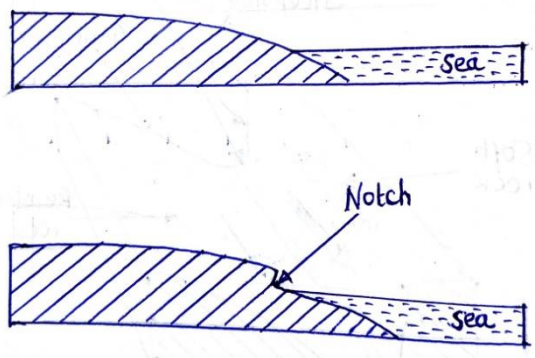
- **Exposure of the shore to wave attack**: coasts which are totally exposed to wave attack are easily manipulated by wave attack while those sheltered by coral reefs and islands eroded rocks become bays with lagoons.
- **Quantity and the size of load used as an abrasive tool**: when materials such as boulders, sand, etc, are in abundance, the coastline will easily be eroded through Corrasion process while the absence of such materials may result into meager erosion.
- **Stability of the coast**: this is the frequency of events that create waves. Coasts which are commonly visited by strong winds and earth movements like earthquakes are highly attacked by waves hence much erosion as compared to calm coasts that may generate weak waves which are less destructive.
- **Sea level changes**: when there is a drop in the sea level of the sea, the coast is exposed to more wave attack hence more erosion while a rise in the sea level the coast is submerged which reduces on wave attack.
- **The slope of the coast**: coasts which have gentle slopes are not heavily eroded by waves because break and water is just thrown on the beach and simply flows back to the sea. However coasts which are steep, act as barriers to water thrown by the breaking waves hence are terribly eroded.
- **Human activities**; such as dredging estuaries, building drainage canals and beaches interfere with the waves energy to erode. Groynes erected to prevent sand from being washed from the beach and break water, all hamper wave ability to erode.

LANDFORMS OF WAVE EROSION

There are several features formed by wave erosion and they are; Notch, cliff, wave-cut platform, caves, blow hole, Geo, arches, headland, bays, stack and stamps.

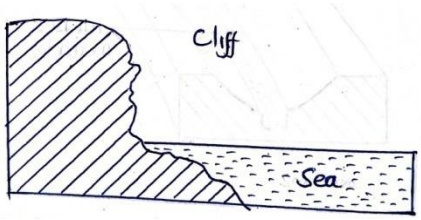
- **Notch:** this is a small opening on a gently sloping coastland. It is formed by undercutting at the coast by hydraulic and abrasion action of waves. It is the first stage in the development of a cliff. It shows where sea wave erosion has temporally stopped. Notches develop where there are less resistant rocks and joints in the rocks. Examples can be found at the Kilindini harbor entrance and Mtwara.

Illustration



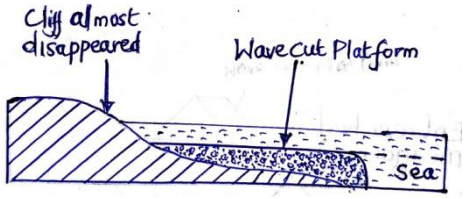
- **Cliff:** this is a steep slope rock-face facing the sea/lake and lies above high tide level. It is formed due to continued undercutting of the notch by hydraulic action of waves causing the notch to enlarge to form a steep face called a cliff. Examples are found at Fort Jesus near Mombasa, Watamu and Alhidina-Visram high school in Mombasa; botanical beach and Kasenyi in Entebbe; Fort Garaza near Kilwa in Tanzania.

Illustration



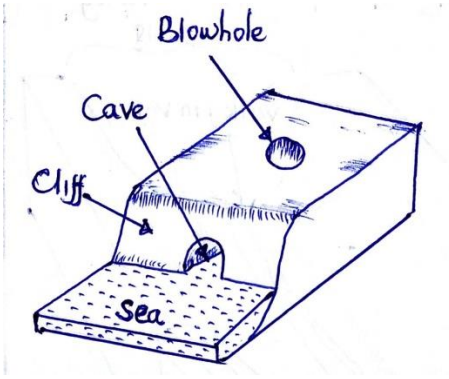
- **Wave-cut platform**; this is a beach-like feature sloping seaward and often found below the cliff. It is formed between high and low water tides. The abrasive and hydraulic action of waves wears down the cliff and the continental shelf creating a wave-cut platform. Examples can be seen south of Mombasa and the coast of Dar-se-salaam. The oceanic hotel at Mombasa is built on the wave-cut platform. However, most wave-cut platforms are under water and only exposed during low tide.

Illustration



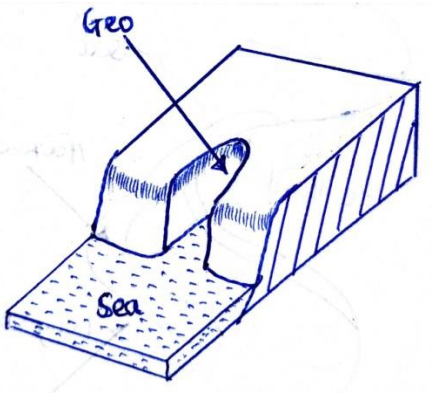
- **Cave;** this is a cylindrical tunnel or hollow found or drilled at the base of the cliff. It is wide at the entrance and narrow at the end. It is formed where rocks are weak, faulted or jointed. When breaking waves throw water on the cliff face with cracks, air trapped in the cracks is suddenly compressed. As the waves retreat during the backwash, the air trapped within the cracks suddenly expands explosively which weakens and breaks up the rocks. Repeated action on cracks leads to the formation of a cave. Examples are found at Entebbe palm resort beach, Kasenyi, Lutembe beach, Nsamisi and botanical beach; Tiwi beach and Kilifi in Kenya; oyster bay in Dar-es-salaam.

Illustration



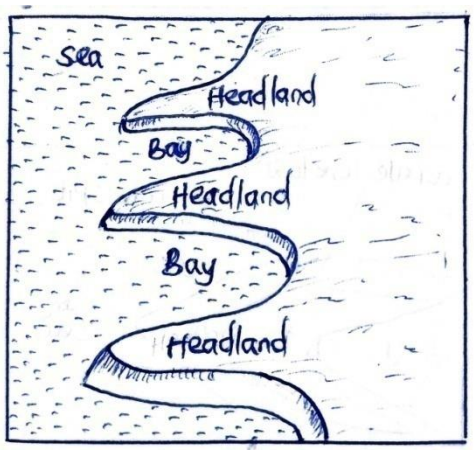
- **Blow hole/Gloup**: this is a cylindrical tunnel/opening which extends from the end of the cave to the top of the cliff surface. It is formed when waves breaking at the back of a cave splash water upwards to the cave roof. The splashing weakens the rocks at the roof of a cave which later are progressively removed until a hole is formed on top of the cliff referred to as a blowhole. Examples are found near Malindi, Mamangina drive at Mombasa; botanical beach at Entebbe peninsular, etc
- **Geo**: this is a narrow steep-sided inlet found along a cliff. It is formed when the entire roof of a cave collapses due to continuous hydraulic action and solution erosion. Before its formation a blow hole is usually formed at the end of a cave and this is followed by the collapse of the entire roof of a cave.

Illustration



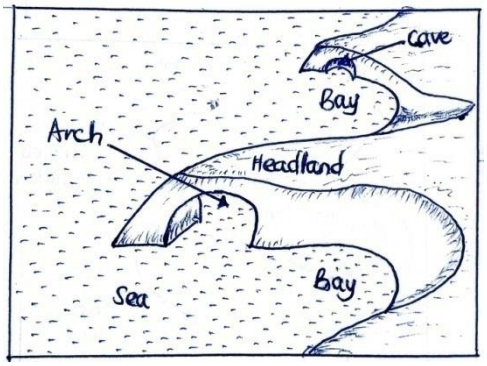
- **Headland**: this is a projection of land into the sea/lake. They are generally hard rocks or piece of land protruding into the sea. Headlands are formed in the areas of alternating hard and soft rocks such that soft rocks are eroded away leaving hard rocks which are resistant to erosion to remain protruding into the sea.
- Examples of headlands are Entebbe peninsular, Mweya peninsular, Kasenyi headland on Lake Victoria, Watamu on the Kenyan coast, Walukuba peninsular, Tiwi beach, Kilifi at Mombasa.
- **A bay:** this is a water projection into the adjacent land at the coast. It is a wide extension of a sea/lake waters into the land. They are formed in areas of alternating soft and hard rocks at the coast. Soft rocks are quickly eroded by strong swash and backwash movement of waves to form bays.
- Examples can be seen at Kasenyi, Sango bay, Murchison bay, etc in Uganda; Kavirondo bay, Homa bay, etc in Kenya; Speke gulf and Emin Pasha Gulf in Tanzania.

Illustration



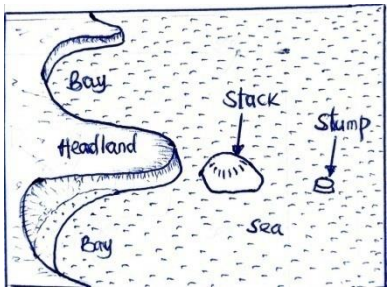
- **Arch**: this is a hole or inlet formed across the headland. It is formed in two ways; first it may form when a cave is driven into the side of a headland and ultimately opens out on the other side leaving a bridge like feature on top. Secondly it can also form when two caves develop from both sides of the headland and then ultimately meet creating a water passage through the headland and a bridge-like feature on top.

Illustration



- **Stack:** this is an isolated rock feature completely detached from the mainland. It is formed when the roof of an arch on the headland collapses thus leaving a certain piece of headland detached or isolated from the mainland. Examples can be found at Kasenyi, rock pillar stack near Entebbe airport, Malindi and Mombasa at the Kenyan coast. Etc.
- **Stamp:** this is a related landform to a stack. It is a stack which has been worn-down by wave action and therefore lies below the water level especially at high tide and can be slightly seen at low tide isolated far from the headland.

Illustration



LANDFORMS OF WAVE DEPOSITION

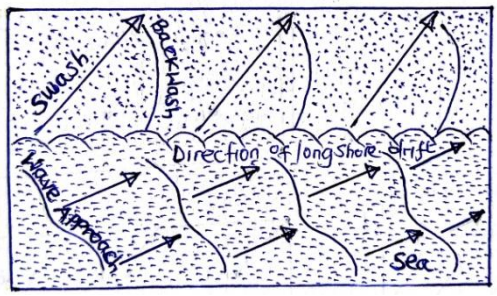
While some parts of the coast are being eroded by constant pounding of waves, the sea is also carrying other materials removed from other areas where it is deposited. Wave deposition is brought about by constructive waves.

Mechanism through which waves transport and deposit eroded materials:

There are two main mechanisms through which eroded materials are transported and deposited along the coast namely long shore drift and beach drifting.

- **Beach drifting**: this refers to the movement of sand and other marine sediments carried along by waves parallel to the shoreline.
- **Longshore drift**: this refers to the transport of sand and sediments along the beach by waves breaking at oblique angle to the shoreline. This often occurs where waves strike the shore at an oblique angle. The swash force carries materials diagonally up to the shore and the backwash drags the coast materials fragments back down at right angles to the shoreline.

Illustration



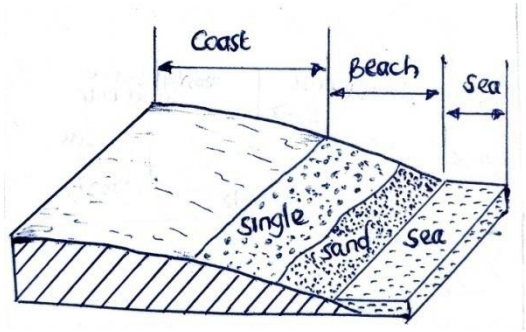
Other factors that facilitate wave transport and deposition of eroded material

- The wind direction should be on-shore so that the eroded materials are carried and deposited by the on-shore wind along the shoreline.
- The availability of a lot of weathered and eroded materials to be transported and deposited at the coast by the constructive waves facilitates the formation of various coastal landforms such as beaches, spits, sand bars, etc.
- The presence of a shallow continental shelf to provide a platform upon which the transported materials by the constructive waves are deposited to enables the formation of landforms.
- The nature of the coastline which is gently sloping, favours deposition and accumulation of material leading to a possible formation of depositional features.
- The existence of deposits of coral platforms from accumulation of shells from dead polyps favours the formation of coastal landforms e.g. a long the east African coast.
- The activity of river deposition at the river mouth has facilitated the formation of depositional features at the coast such as the deltas e.g. Rufigi delta at the coast of East Africa.
- Human activities such as dumping soil and rock boulders at the coast/shoreline to construct piers, etc has influenced formation of coastal depositional landforms.

As the above processes continue, the coastal materials such as sand and pebbles are gradually moved and finally deposited to produce a cross-section of wave depositional landforms such as beaches, sandbars, spits, tombolos, mudflats, salt marshes, coastal dunes, lagoon, a berm.

- **Beach:** this is a gentle slope at the coast where there is accumulation of sand and pebbles on the shores. A beach is usually formed between low and high water tides when constructive waves, produced by long shore drift deposits materials such as mud, pebbles, shingle, sand coral fragments and boulders along the coast to form a gently sloping landform called a beach. Examples are Ggaba beach, Lido beach, Kasenyi beach, Aero beach, Nabugabo beach, Lutembe beach, Munyonyo beach, Nyali beach at Mombasa, etc.

Illustration

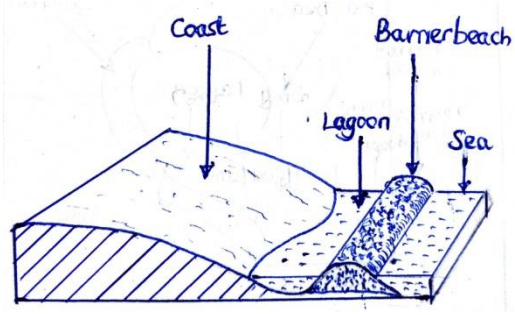


Types of beaches

- **Normal beach**: this type of beach develops on the shoreline. They are formed due to deposition of sand and shingle at the bay between headlands by long shore drift of the constructive waves. Examples are Lutembe, Ggaba, Lido, Bugonga beaches at L. Victoria.
- **A storm beach**: this is accumulated deposits of sand, pebbles and shingle beyond high tide mark level. They are formed by strong long shore drift or strong swash which throws materials beyond the high water tide/shore line.
- **Beach rock**: this is a hard crust-like deposit which projects above the sand on the beaches. Beach rocks are composed of sand, shells and pebbles cemented together by calcium carbonate.
- Examples of beach rocks are largely confined at the coast of Kenya.

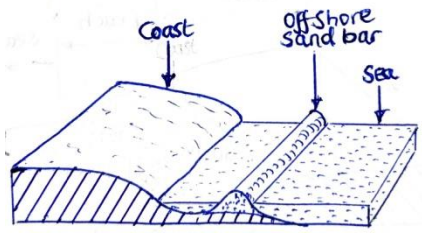
- **Barrier beach**: this is along ridge of sand deposits parallel to the coast and separated from it by a lagoon. It's formed when waves break some distance away from the shore. Barrier beaches are formed on gently sloping coasts by long shore drift. When the sea waves break before reaching the shoreline, they deposit materials in the sea which accumulate to form a barrier beach. Their formation is favoured by a fairly low tidal range and shallow off-shore zone. Examples are found in Madagascar, Benin and Nigeria coasts.

Illustration



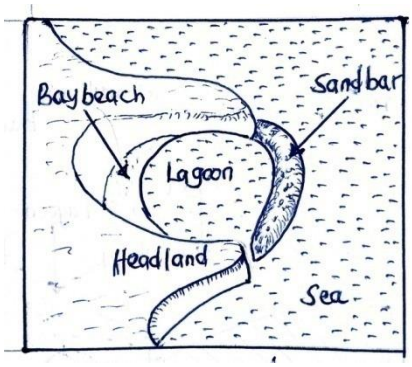
- **Sandbar:** this is a ridge of sand, mud; gravel and shingle deposited off-shore and lies parallel or almost parallel to the coast. They are formed on gently sloping coasts with irregular shorelines. Their formation is associated with both beach drifting and long shore drifting currents which deposits materials at a further point in the sea. The long shore waves break before the shoreline and at the same time the beach drifting drags materials from the shore towards the sea resulting to accumulation of materials on the sub-shoreline known as break point.
- Repeated processes of this kind form a sandbar behind resulting into the formation of lagoons, mudflats and marshes. Sandbars are classified into bay bars, Barrier Island, off-shore sandbars and foreshore bars.
- **Off-shore sand bars**: these are sandbars which develop on a very gently sloping coast where waves break off-shore depositing the materials carried by them. They are usually temporally submerged.

Illustration



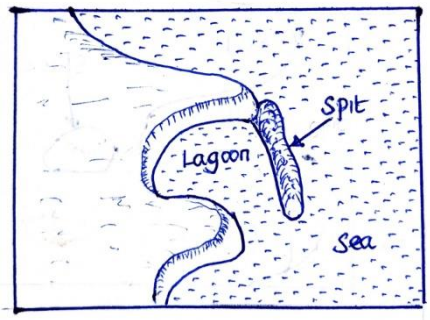
- **Foreshore sandbars**: these are sandbars formed by constant accumulation of sand causing off-shore sandbars to rise above the water surface. Examples of bars are found at Lamu along the coast of Kenya.
- **Bay bar**: this is formed when a spit continues to elongate through additional deposition of materials from one headland to another enclosing the bay, for example Nabugabo was cut-off from Lake Victoria by a bay bar.

Illustration



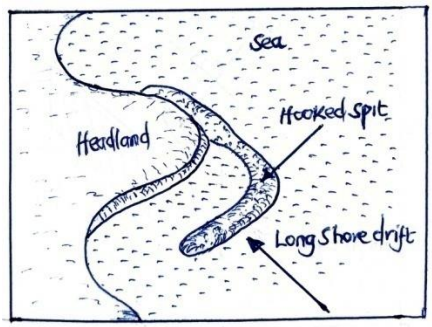
- **Spit**: this is a low, long and narrow accumulation of sand and shingle with one end attached to the mainland while the other projecting into the sea/lake or across an estuary. It is formed when alongshore drift deposits materials especially along indented coastline or coasts broken by river mouths. Examples are Kabo spit and Tonya spit on L. Albert. When a spit joins two headlands it forms a bay bar.

Illustration



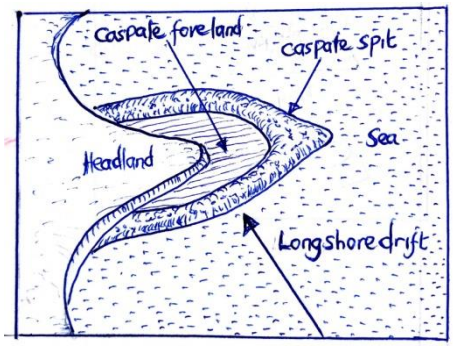
- Spits are grouped into two categories i.e. hooked spits and cuspate spits.
- **Hooked spit**; this is a spit whose seaward end is forced to curve towards the land/bay or an estuary. It si formed when either strong waves reach a point on the shore from different direction, or when the waves move to the shoreline at an oblique angle, or when deposition is laid in a curve format following the influence of the estuary which tend to push the deposits up to a certain point in the sea. Examples can be found at Kibangi on the shores of L. Albert.

Illustration



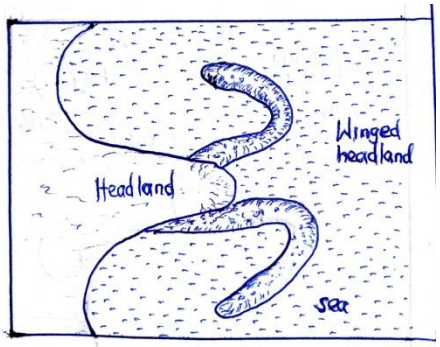
- **Cuspate spit**: this is formed either when two spits develop and grows towards the sea and later converge in the water or when a simple spit grows and curves until it joins the headland or shoreline at both ends. Example is Larree in Madagascar.

Illustration



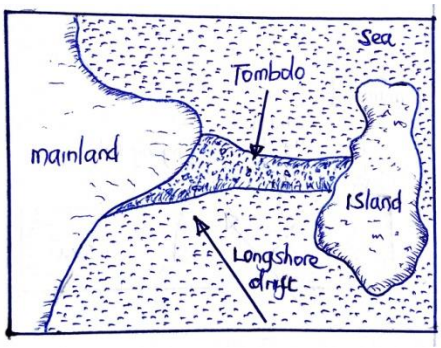
- **Winged headlands**: these are also formed by wave deposition. They appear as two depositional features attached on one headland. They are formed when spits develop at both sides of the headland. This occurs when waves reach the headland from the opposite sides/directions.

Illustration



- **Tombolo**: this is a ridge/depositional feature of sand and pebbles joining an island to the mainland. It may be a bar/spit which joins an island to the mainland or joins an island to an island. It is formed when along drift deposits materials like sand and shingle between the mainland and the island. Examples are found along L. Victoria where a Tombolo used to join Lambu Island, though this was destroyed in 1960. A Tombolo is also found along the coast of Kenya near Guogoni north of Malindi. Others are found in the north east coast of Somalia and Cape Verde region in the republic of South Africa.

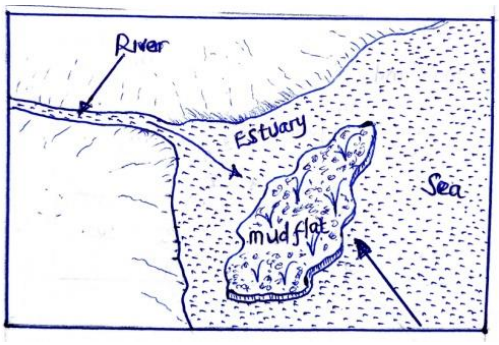
Illustration



- **Coastal sand dunes**: these are extension of sand deposits from the beach inland. Sand dunes form when coastal winds blow sand from the beaches further inland and this sand may be trapped and stabilized by grass and other vegetation to form sand dunes. Examples are common along arid coasts.
- **Mudflats**: this is a mud platform built of deposits of mud, silt and alluvium. It usually develops when rivers and waves deposit materials along gentle coasts especially in bays and estuaries. They are usually exposed during low tides and submerge during high tides. Mudflats are well developed

at the mouth of R. Rufigi, at Tanga, Lamu in Kenyan coast and at Murchison bay in Luzira on L. Victoria shores.

Illustration



- **Salt marshes**: these are actually coastal swamps usually called mangrove swamps. They develop after salt tolerant vegetation colonizes mudflats. Such vegetation tends to have trees with aerial roots fixed in the mud for support. Salt marshes are commonly found along the East Africa coast at Mombasa and Dar-es-salaam ports.
- **Lagoon:** this is a depositional lake enclosed by sandbars or spits at the shores. It is formed when, during the high tides, long shore drift deposits materials like sand and shingle some distance into the sea but towards the shoreline. So during the low tides/drop in sea level, the deposited materials/spit trap the water between the sea and the shoreline hence forming a lagoon. Examples lagoons are found at Mombasa, L. Nabugabo along the shores of L. Victoria. On L. Albert behind Toya and Kabira spits north of Butiaba port.
- **Berm**: this is a ridge-like depositional feature formed from accumulation of heavier stones and boulders at the upper limit of a beach. They are formed by strong swash current capable of transporting bigger materials towards the shoreline. Example can be found on the east coast of Lake Albert and along Wami delta on the coast of Tanzania.

ECONOMIC IMPORTANCE OF COASTAL LANDFORMS

- Beaches promote sand mining e.g. sand is mined at Kasenyi, Lutembe and Nabugabo. This sand is used as a raw material in the building and construction industry, glass making industry, etc.
- Beaches have provided recreational grounds for holiday makers, tourists which has attracted the development of hotels such as Entebbe botanical beach hotel, Speke resort at Munyonyo, Palm resort beach, etc.
- Bays and headlands provide breeding places for fish hence attracting fishing activities and the development of fish landing sites examples can be found at Kasenyi, Kajansi along Lake Victoria, etc.
- The magnificent depositional and erosional features like cliffs, headlands at fort Jesus and the white sand beaches like Nyali and Diani along the coast of east Africa attract tourists leading to foreign exchange earning used for development.
- Mudflats along the East African coast at Mombasa have attracted mangrove forests and swamps where lumbering and crafts making respectively have developed. Mangrove forests also provide poles for building.
- Bars and spits are important in the protection of the coastland against destruction by strong waves and sea floods. Coastal landforms both wave erosional and depositional support research and study especially on oceanology and fieldwork. Examples can be found at Kasenyi landing site, Lambu landing site lido beach, etc.
- **However,**

- Depositional materials such as snail's shells have led to the spread of diseases like bilharzias especially along the East African coast and Lake Mburo. Mosquitoes also inhabit mangrove swamps hence spreading malaria.
- Strong swash currents are characterized by flooding of the shoreline and the fishing villages. Flooding is a common problem along Lake Victoria and Kyoga which displaces and destroys property.
- Sandbars and stamps below the water level are a major cause of accidents on water bodies for instance boats tend to capsize on hitting such submerged obstacles.
- Waves sometimes bring dangerous poisonous snakes onto the shoreline. For instance at the coast of Srilanka and India snake bites are among the leading cause of death to the people living along the coast. In East Africa these snakes are common along L. Victoria and Kyoga during the flood seasons.

REVISION QUESTIONS

Qn*: To what extent has long shore drift influenced the development of depositional coastal features in East Africa?*

Approach: *Define the term long shore drift, explain how constructive waves lead to deposition, identify and describe the resultant landforms, give other factors, relevant examples and diagrams must be given, conclude.*

Qn: *Describe the processes responsible for the formation of erosional coastal landforms in East Africa.*

Approach: *define erosional coastal landforms, briefly describe the various processes by which waves erode coasts, identify erosional landforms and explain how each is formed, relevant examples and diagrams must be given.*

SEA LEVEL CHANGES/EUSTATISM

Sea level changes refer to the rise and fall of the sea level in relation to the land along the coastal areas. It is also defined as the vertical movement of land relative to the sea along the coastal regions. Overtime the sea level is not static but keeps on fluctuating i.e. rising and falling. When the sea level changes occur worldwide, it is referred to as eustatic changes (eustatism) and when it occurs on a local scale is called isostatic change (Isostacy)

TYPES OF SEA LEVEL CHANGES

There two types of sea level changes i.e. either positive eustatism or negative eustatism.

- **Positive eustatism/change** occurs when the sea level rises in relation to the land at the coast and this result into the submergence of the coasts/coastal landforms. The submergence of coastal/marine features is called transgression.
- **Negative eustatism/changes** occur when the sea level falls in relation to the land at the coast and results into emergence/appearance/exposure of the coastal landforms. The emergence/exposure of coastal marine features is known as marine regression.

CAUSES OF SEA LEVEL CHANGES

The causes of sea level changes include climate, glaciation and deglaciation, change in ocean waters temperature, earth movements (up-warping of the coastal areas, volcanicity on the ocean floor, plate tectonism, faulting in coastal areas), deposition of sediments, isostatic readjustments, global warming, and human activities, etc.

- **Glaciation during the last ice age**, temperatures dropped to freezing point and large quantities of water froze into ice on high mountains and at the Polar Regions resulting into a fall in the sea level. Deglaciation on the other hand during the inter-glacial period characterized by warm temperatures, melted the thick ice sheets hence releasing melt water back into the sea leading to a rise in sea level.

Earth movements are some of the major causes of sea level changes in the following ways;

- **Up-warping/uplift** along the coastal areas and down warping of the ocean basins leads to the fall in sea level while up-warping of the ocean basins and down warping of the coastal areas leads to a rise in the sea level. **The occurrence of volcanicity** on the ocean floor forming volcanic cones displaces water in the ocean basins up-wards causing a rise in the sea level. **During plate tectonism**, the divergence of plates at mid-ocean ridges leads to sea/ocean floor widening hence a fall in sea level while the convergence of tectonic plates leads to contraction/narrowing of the ocean floor hence a rise in sea level. **Faulting** in the coastal areas leads to down thrust of some parts causing a fall in sea level.
- **Deposition of sediments/sedimentation** is another cause for the rise in sea level. Many rivers and streams such as R. Tana, R. Rufigi, etc, pour large quantities of alluvial deposits into the ocean basins (Indian Ocean). This deposit displaces water and after along period of time the water gradually rises upwards in the sea hence a rise in sea level. At the same time the large quantities/volume of water poured into the sea by many rivers such as the Nile into the Mediterranean Sea, Rufigi and Tana in to the Indian Ocean, etc, leads to a rise in the sea level. **The development of coral reefs** in the sea/ocean displaces water hence leading to a rise in sea level like in the East African coast.
- **Heavy rainfall during the pluvial periods** is another cause of sea level changes. It's believed that hydrogen atoms from the sun (solar winds) often enter the earth's atmosphere and when they combine with oxygen in the ozone layer, they form water molecule/rainfall. This water may then enter the sea either directly from the sky or indirectly through the springs and rivers hence causing a rise in sea level overtime. On the other hand desiccation/ prolonged drought lead to excessive evaporation leading to a fall in sea level. East Africa for instance has been experiencing prolonged drought and reduced rains for the last 50 years. This has led to a fall in the water level of L. Victoria and the Indian Ocean hence exposing many features which were submerged.
- **Influence of global warming** is another cause of sea level changes. The steady rise in the average temperatures over the surface of the earth is affecting climate by melting ice and snow in the polar regions and on high mountains like Himalayas, Rwenzori, Kilimanjaro, Kenya, etc. this melt water moves into the oceans through rivers hence a rise in sea level. Intense heat generated from volcanic activity below the sea bed causes water expansion, its volume increases and the sea level inevitably adjusts upwards within the basin inform of sea level rise. The Krakatau volcano along the coast of Japan generated a lot of heat that caused a great expansion of water in the Sea of Japan and this was followed by flooding of the coastal locations along China and Japan. On the other hand during winter, water molecules tend to contract because of low temperature. This leads to relative drop in the sea level. Global warming also accelerates evaporation and aridity which reduces water on the earth surface hence a fall in sea level.
- **Isostacy** is another cause of sea level changes. Isostacy refers to the upward and downward movement of the land relative to that of the sea. The continents are made up of lighter Sial rock materials floating on the denser Sima rocks in the interior. The wearing down of the continental landmass by weathering and erosion reduces the pressure on the semi-liquid interior. This causes the level of land to rise relative to that of the sea hence a fall in the sea level. On the other hand deposition of weathered materials and other debris increases the downward pressure and causes the continents to subside. In this case, the level of the land falls relative to that of the sea hence a rise in the sea level. Isostacy therefore causes both a rise and fall in sea level.
- **Human activities like dumping** of sewage, garbage, construction of artificial piers and hotels on water bodies have led to a rise in the sea level. Throughout the world thousands of tones inform of

hardcore stones and soils are deposited in the sea especially near beaches to construct hotels. In Uganda Speke resort hotel at Munyonyo was partly constructed in L. Victoria through similar process.

ISOSTACY: This refers to the rise or fall of sea level in relation to the land but on a local/minor scale. It is a result of local earth movements. Isostacy changes affect particular coasts or are from isostatic adjustments due to loading and unloading of the crust. Isostatic changes also involve both positive movements inform of submergence and negative movements inform of emergence. When the level of the sea rises, the areas around the coast become submerged/covered by water/flooded. While, when the level of the sea falls, areas along the coast which were submerged/covered by water emerge/are exposed/appear. Therefore, there basically two kinds of coasts i.e. submerged coasts resulting from a rise/positive change in sea level and emerged coasts resulting from a fall/negative change in sea level.

LANDFORMS DUE TO SEA LEVEL CHANGES

These are in two groups i.e. landforms produced by submergence/rise in sea level and those produced by emergence/fall in sea level.

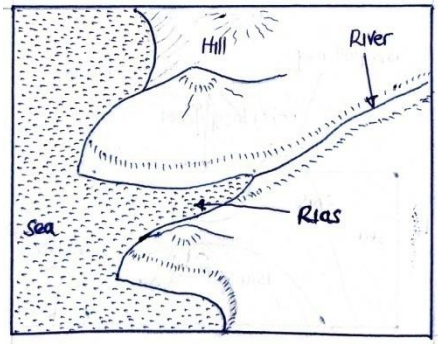
Coastal landforms due to submergence/rise in sea level:

When the level of water of the sea rises, coastal areas which were originally dry, becomes submerged or covered with sea rising water. This results into the formation of submerged features on either highland coasts or on lowland coasts.

Features on submerged highland coasts:

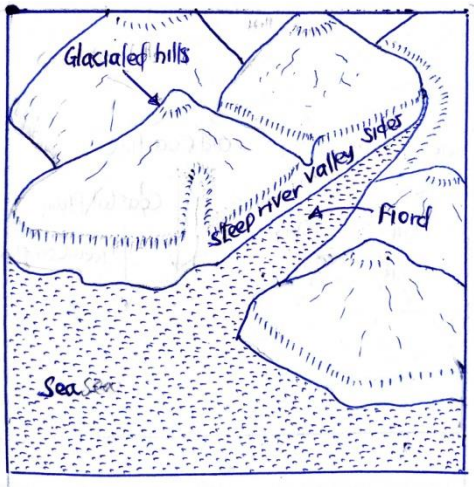
- **Rias**; this is a drowned river valley in the highland area. It is funnel shaped, wider and deeper at the entrance and narrow and shallower inland. Rias is formed when the coasts where hills and river valleys meet the sea at right angles. After a rise in the sea level, the lower parts of the river valley are completely flooded to form a rias. For instance Dar-es-salaam port is situated in a rias, Kilindini harbor on which Mombasa port is situated is a rias, it's also found at Tanga, river Mtwara in Tanzania, Lamu and along the coast of L. Victoria near Mwanza, etc.

Illustration



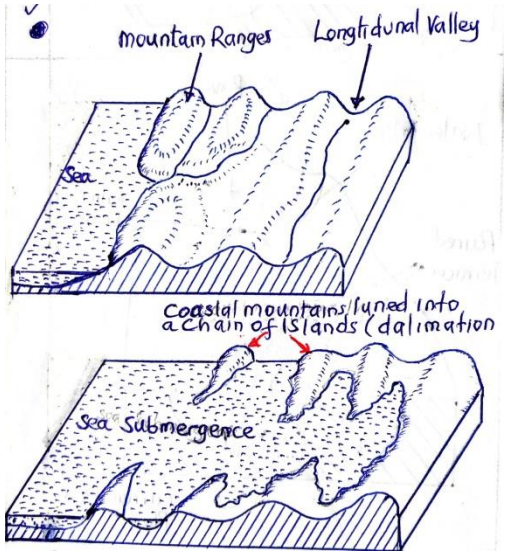
- **Fiord**: this is a drowned glaciated river valley in the highland areas/regions. It is formed when a river valley initially deepened by glacial erosion below the sea level is submerged by the rising sea level. Fiords are characterized by very steep sides and deeper at the head and shallower seawards. Fiords are not found in east Africa, they are only found in coasts said to have been glaciated for examples the coast of Norway, Sweden, British Columbia, southern Chile, etc.

Illustration



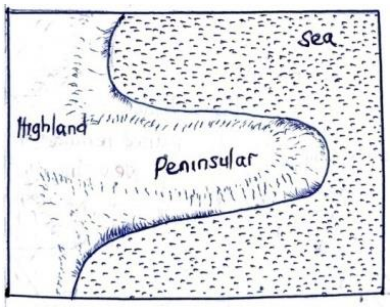
- **Dalmatian /longitudinal coast**: this is a coast with a chain of ridges/highlands running parallel to it. They are formed on coasts where elongated hills/ridges and valleys were originally lying parallel to the sea. When the rise in sea level occurred, the valleys were flooded or submerged leaving the ridges appear as chains of islands running parallel to the coastline. Examples are found at east coast of Pemba and Zanzibar islands, Bamburi hills south of Bukoba, at Mwanza on the shores of L. Victoria.

Illustration



- **Peninsular:** this is a piece of land projecting into the sea. It is formed where highlands occur at right angles to the coasts when there is a rise in sea level; the valleys are submerged leaving the highland projecting into the sea. Peninsulas appear like headlands at the coast/lake. Example is the Entebbe peninsular on L. Victoria.

Illustration



- **Islands:** if the highland region is submerged, the top parts which are not covered by water form islands e.g. the Ssese Is., Kome Is., Sigulu Is., and Ukerewe islands in L. Victoria.

Features on submerged lowland coasts:

- **Estuaries**; these are drowned river valleys in the lowland areas. Unlike rias which are formed in highland areas, estuaries develop in lowland areas where flooding of the river mouth takes place as it enters the sea. A rise in sea level causes the flooding of the river mouth forming estuary. Examples are found on R. Rufigi and Wani along the coast of east Africa.
- **Creeks**; these are former sea inlets which have been drowned to form larger inlets. When there is arise in sea level, the small river valleys in the lowlands are drowned to form creeks. The major difference between estuaries and creeks is that the creeks develop on smaller river valleys with shallow waters. Examples are found at the east African coast i.e. Mtwapa creeks.
- **Mudflats**: these are deposits of fine silt and alluvial materials from drowned rivers or deposited by swash during a rise in sea level. Mudflats are usually colonized by vegetation to form mangrove swamps or marshes along the East African coast. Such deposits may enclose water and separate it from the rest of the sea to form a lagoon.

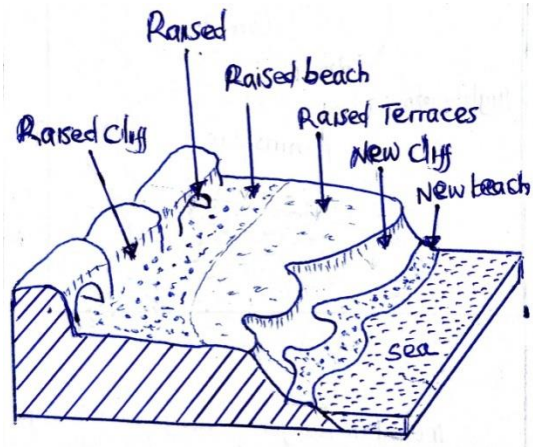
Coastal landforms due to a fall in sea level/negative eustatism

When the level of the sea falls, areas which were originally submerged/covered by water are exposed/emerges/appears out of the sea level. This exposes coastal features which occur both on highland coasts and lowland coasts.

Features on emerged highland coasts;

- **Raised beaches**; these are beaches which appear to be on a raised level as compared to the current level of the sea. When the level of sea falls, a new beach is formed at the lower level by constructive waves through long shore drift. The original/first beach which was formed during the time when there was a rise in the sea level is left at a higher level as a raised beach. Sometimes a raised beach may be colonized by vegetation such as at Kasenyi the raised beach has been colonized by vegetation, but mining of sand in the area is a proof that it existed. Other examples are Diani and Shimoni beaches along the East African coast, Mamangina Drive in Mombasa, Timbuani, etc.

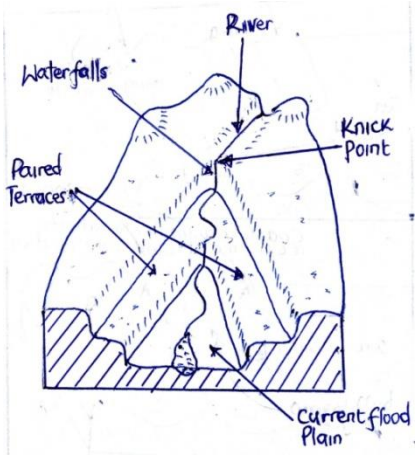
Illustration



- **Raised cliffs**; these are features which remain standing above the current sea level. They are exposed by a fall in the sea level. Originally the cliff's formation was a result of wave action. As waves hit the coastline through hydraulic action and abrasion, a notch develops between the high tide and low tide. A notch is gradually enlarged due to continued wave action to form a cliff. When a fall in sea level occurs, cliff are left on a raised elevation hence the name raised cliffs.

- **Rai sed caves, geos and blow holes**; originally these features were formed due to continuous wave action. **Caves** were formed due to continuous wave erosion by hydraulic action, abrasion and solution against the jointed coastal rock creating large holes on a cliff face called caves. When the waves erode the roof of the cave up to the surface a **blow hole** is formed while when the roof of the cave collapses it forms a narrow inlet called **a geo.** When sea levels falls, all these features will appear high above the current sea level hence raised caves, geos and blow holes. Examples can be seen at Shelly beach hotel and Bagamoyo in Tanzania, Lamu and Malindi in the Kenyan coast.
- **Raised terraces;** these are terraces found at a higher elevation as compared to the current sea level. Raised terraces are former wave-cut platforms which are no longer in contact with the sea and were created by materials eroded from the cliff. When sea level falls, another terrace near the current sea level is created and the old terraces which are no longer in contact with the sea level are left high above the current sea level as raised terraces. Example can be seen at Lutembe along L. Victoria shores.
- **Paired terraces and water falls**; these are also formed due to a fall in sea level. When a fall in sea level occurs, river rejuvenation may follow. A river is forced to renew its down cutting in order to reach the new base level. This rejuvenation leads to the formation of paired river terraces, knick points and water falls. In glaciated regions gorges may be formed.

Illustration



Emerged features in lowland areas

- **Coastal plain**; in lowland areas a fall in sea level leads to formation of coastal plains and an even shoreline without bays and headlands. Coastal plains are formed part of the continental shelf emerges from the sea after a fall in sea level. Such a coastland has no bays or headlands and examples are found at the north coast of the Gulf of Mexico.
- **Fiards;** these are drowned up-shaped glacial troughs/valleys formed along lowland coasts. They have a broader U-shaped profile than the fiord. For example the coast of south East Sweden, coast of Nova Scotia and Maine.

REVISION QUESTIONS

Qn: *(a) Explain the causes of sea level changes in the coastal areas.*

(b) Examine the landforms resulting from sea level changes.

Approach: *define sea level change, state the types of sea level changes, identify and describe the formation of both submerged and emerged coastal landforms on highland and lowland areas, diagrams and relevant examples must be given where applicable in East Africa and outside East Africa*.

CORAL LANDFORMS IN EAST AFRICA

Coral landforms/reefs are offshore limestone rock platforms/reefs which are attached to the sea coast but separated from the sea by a lagoon. Coral reefs are formed by continuous deposition and accumulation of shells or skeletons of tiny marine organisms known as coral polyps. Polyps are minute living organisms which live in millions of colonies and their bodies contain skeletons/shells made of calcium carbonate got from sea water. When the coral polyps die, the soft parts of their bodies decay and the shells/skeletons drop and accumulate on the continental shelf where they are compacted, cemented and compressed together as they increase in number to form a hard rock known as coral reef.

PROCESSES OF CORAL REEF FORMATION

- When coral polyps die their shells/skeletons rich in calcium carbonate drop and accumulate on the continental shelf. Over time the coral deposits increases in weight, become compressed, compacted and consolidated into hard rocks called coral reefs.
- The process of cementation and consolidation of coral reefs is facilitated by other organisms such as sea algae and echinoderms. After millions of years, coral rocks grow into very big size of rocks called coral reefs/coral islands and are seen at low tides/water levels.
- The nature of the coral reefs/landforms formed depends on the position and the shape of the landmasses on which they have been accumulated hence creating different types of coral reefs.

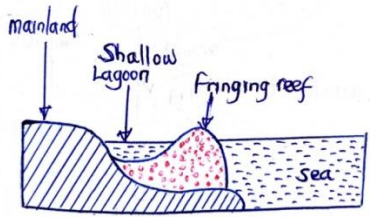
CONDITIONS THAT FAVOUR DEVELOPMENT OF CORAL REEFS

- **Warm temperatures** of tropical climate between $20℃ - 30℃$ are ideal for the growth of coral reefs. For this reason coral reefs grow only in areas around 30° North or South of equator and on the Eastern sides of continents where there are no cold ocean currents e.g. in the Indian ocean near the East African coast.
- **Salty and well oxygenated seawater** with salinity level of 27-40 parts per 1000 parts of saline ocean water provides adequate calcium carbonate taken up from the sea by coral polyps used to build and harden their skeletons/shells.
- Existence of **clear and silt free water** away from the river mouths where fresh water and alluvium may destabilize coral foundation.
- **Calm and stable water** free from strong waves such as typhoons, ocean currents, etc, which would drift away the dropping shells/skeletons hence deterring coral reef formation.
- **Shallow continental shelf** with a depth of not more than 60 meters allows sunlight penetration into the sea bottom thus facilitating photosynthesis by plankton on which the polyps feeds on.
- **Abundant planktons** within the sea which coral polyps feed on and survive ensure steady growth and development of coral reefs.
- **Presence of solid rock bed and a wide continental shelf** along the coast is ideal to provide a platform on which the fringing and barrier coral reefs grow.
- **Abundant coral polyps** in the sea water such that when they die in large quantities, deposits of skeletons of calcium carbonate quickly accumulate to form coral landforms.
- **The warm Mozambique Ocean current** that washes the East African coast helps to maintain the warm temperatures of the Indian Ocean hence enables polyps to survive.
- **Occurrence of sea level changes** encourages death of polyps and accumulation of coral reefs. Increase in the sea level encourages coral deposition and growth while a fall in sea level reduces the space for polyp's multiplication and exposes the coral reefs hence limiting the growth of coral reefs.
- **Presence of submerged oceanic ridges/volcanic islands** in the sea which provide bases for the growth of coral reefs especially atoll reefs that grows far away/deep in the sea.

TYPES OF CORAL REEFS

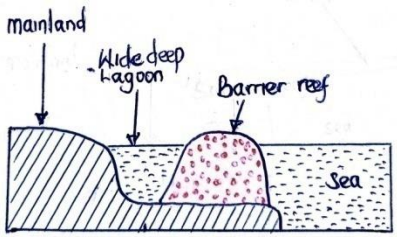
- **Fringing reefs**: this is a narrow coral platform of about one kilometer wide joined to the coast or separated from it by a shallow lagoon which may disappear at low water tide. It's formed when accumulations of shells form a narrow limestone rock a few meters away from the coast. It's characterized by a steeper seaward slope, approximately 500-2000m wide and may disappear at high tides.
- Examples of Fringing reefs can be seen at the east African coast near Kilifi, Tiwi, and Mombasa in Kenya; oyster bay at Dar-es-salaam in Tanzania.

Illustration



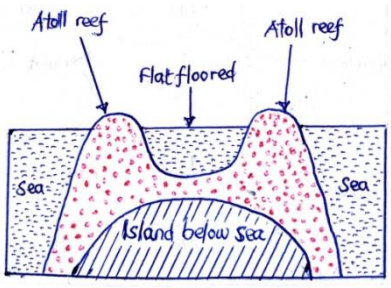
- **Barrier reef**: this is a wide coral platform formed much further or several kilometers from the coast than the fringing reef. It's separated from the coast by a wide and deeper lagoon. It's formed when accumulations of shells form a large limestone rock several kilometers away from the coast. It is characterized by a wider platform extending into the sea; a permanent lagoon; and its platform may not be exposed during low tides. Barrier reefs can be sited at Mayotte Island between Mozambique and Madagascar.

Illustration



- **Atoll reef**: this is a broken ring of small coral reefs islands surrounding a fairly deep and flat-floored lagoon. The ring of broken islands, form a circular shape around a central lagoon with narrow channels inlets. They are formed far away from the coast and some are found in the middle of the sea. They are usually above the sea level and feature as islands. For example Aldabara reefs lies 700km off the coast of East Africa. Others are Chumbe Island near Zanzibar and Shimon in southern Kenya.

Illustration



NOTE: Basing on the above conditions its therefore true that coral reefs grow but their formation is not yet understood, especially, the formation of a barrier reef and atolls has created a lot of controversy as they have been found to grow in very deep ocean waters, in some areas exceeding a depth of 1000m

where polyps can not survive as in this depth there is no plankton on which they would feed. As a result theories have come up to explain their formation.

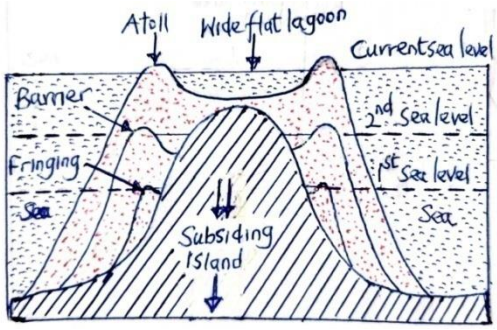
THEORIES OF CORAL REEFS FORMATION

There are three main theories which were advanced namely; Darwin's theory (subsidence theory), Darley's theory (Deglaciation theory) and Murray's theory (Antecedent theory).

Darwin's theory/subsidence theory

- This theory was put forward by Charles Darwin. The theory suggests that a volcanic eruption formed a volcanic Island on the ocean floor which gradually subsided due to isostatic adjustments. According to the theory, Coral polyps colonized the edge of the subsiding volcanic island hence formed a fringing reef separated from the coast by shallow lagoon. The continued subsidence of the volcanic island increased the depth of the water causing the fringing reefs to grow upwards and outwards keeping pace with the rate of subsidence of the volcanic island to form a barrier reef. As the volcanic island become completely submerged, circular reefs known as atoll reef forms.

Illustration



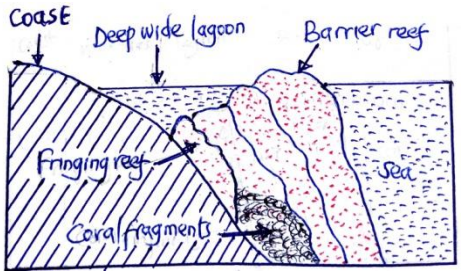
Relevance of the theory

- The theory is relevant because there was actual submergence of the east African coastline evidenced by the rias at Mombasa, mudflats in submerged coastal areas.
- Volcanic islands are also present off the coast of East Africa in the Indian Ocean.

John Murrays/antecedent theory

- This theory was advanced by Sir John Murray to explain the formation of barrier reefs and atoll reefs. He refuted the idea of subsidence as put forward by Charles Darwin. According to him the fringing reef originally existed and growing along the coast separated from the coast by a shallow lagoon. This fringing reef was attacked/pounded by sea waves and eroded. The fragments of the eroded fringing reef were gradually deposited and accumulated on the seaward side of the reef. To him the coral reef tends to grow more to the seaward side forming a barrier reef. He further suggests that while this is happening, the corals on the inner/lagoon side of the reef are deprived of food and so they die. He supposed that the dead coral in the lagoon was dissolved to form a deep wide lagoon separating a barrier reef form the coast.
- Murray further explained the formation of atoll. According to him there existed stable submarine platforms of volcanic origin on which pelagic deposits including corals accumulated at a depth not more than 60m. It's on this platform that corals begin to grow. The corals grow more on the seaward side where there is plenty of food (plankton) while the polyps on the inside of the reef died due to lack of food and their shells/corals dissolved into lagoon water forming a wide deep lagoon inside the reef. This process therefore changed the fringing reef into a barrier reef and finally into atoll.

Illustration



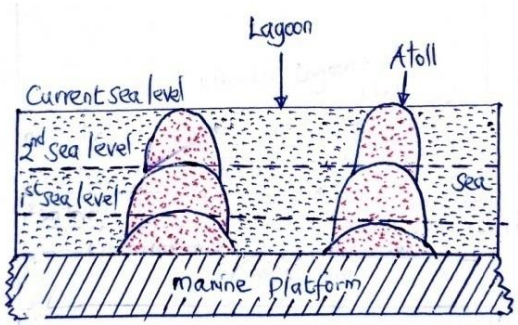
Relevance of Murray's theory

- Murray's theory helps to explain why barrier reefs and atolls are found in deep water of over 60m deep.

Daly's theory of deglaciation

- This theory was advanced by Daly. The theory is based on the changing sea level during and after the ice age. According to Daly before the ice age/glaciation, fringing reefs colonized the edge of the marine island due to warm water and other conditions that favour coral reef growth. During the ice age/glaciation, a lot of water freezed into ice sheets and caused a fall in the sea level. Marine erosion removed and leveled the top of the pre-glacial reefs and the islands to form a wave-cut platform. During deglaciation(return of warm conditions), there was increase in the sea level and new fringing reefs began to grow on wave cut platform upwards and outwards keeping pace with rate of rising sea level to form barrier reefs and later into atolls when the wave-cut platform was completely submerged.

Illustration



Relevance of Daly's theory

- Daly's theory explains the why the lagoons at the coastal areas of East Africa have fairly flat floors.
- It also proves that the coast of East Africa has experienced sea level changes as evidenced by submergence landforms such as rias, raised cliffs estuaries, creeks and emergence landforms such as raised beaches, cliffs, etc.
- The theory explains why some coral reefs occur at great depth.

Note: no theory satisfactorily explains the origin of the coral reefs. However, Daly's theory is nearer to the answer as it explains many factors, which are not explained by other theories.

ECONOMIC IMPORTANCE OF CORAL REEFS

- Coral reefs are tourist attractions because of their uniqueness, beautiful shapes and colour hence a source of foreign exchange for Kenya and Tanzania e.g. fringing reefs at Bamburi, Shungyu in Dar-es-salaam, etc.
- Coral reefs contain limestone which is extracted for the production of cement used in the construction works for example Bamburi cement in Mombasa.

- Coral reefs are a source of education materials and research for example in the field of oceanography, geomorphology and search for oil prospects.
- Coral reefs weather down into fertile soils which support crop cultivation for example coconuts, mangoes, citrus fruits and cashew nuts and cloves in Mombasa, Pemba and Zanzibar islands.
- Coral reefs contain limestone used in the manufacturing of fertilizers rich in calcium carbonates hence boosts crop cultivation.
- Fringing reefs have favoured the development of ports by protecting harbors from destructive waves e.g. ports such as Mombasa and Dar-es-salaam.
- Fringing reefs protect beach swimmers from dangerous marine animals such as sharks, crocodiles and hippopotamus because they find it difficult to cross the reefs to the beach. This promotes tourism
- Fishing is carried out in the coral lagoons such as the fishing of fish, crabs, lobsters, etc along Mombasa and other parts of the coast. This is a source of diet and income when fish is sold.
- Lagoons are also used for recreational purposes such as swimming and sun bathing especially by the tourists.
- Coral reefs are potential areas for mining oil because they contain a lot of fats which sip down and accumulate into rock strata/layers to form oil wells.

However,

- Coral reefs weather down into poor sandy soils which discourage growing of other crops apart from coconuts and cloves.
- Fringing and barrier reefs are obstacles to marine transport and fishing because they form hard projections of rocks which wreck ships, fishing boats and nets.
- Coral lagoons are colonized by mangrove vegetation during the low tides which may become breading places for vectors such as mosquitoes leading to spread of malaria.
- Fringing reefs limit the size/space of docking area such as at Mombasa and Dar-es-salaam ports which results into congestion of ships and delays.

REVISION QUESTIONS

Qn*: Account for the formation of coral reefs landforms in East Africa.(25marks)*

Approach*: define coral landforms, describe the process of their formation, identify the types of coral reefs, and explain the conditions which favour their formation in East Africa, explain the theories put forward to explain the formation of coral landforms.*

Qn: *Examine the relevance of Darwin's theory to the understanding of the formation of coral landforms in East Africa. (25marks)*

Approach*: define coral reefs, describe the process of their formation, identify the types, and explain the conditions which favour their formation, explain the formation of coral reefs with reference to Darwin's theory.*

LAKES IN EAST AFRICA

A lake is a body of water contained in/occupying a sizable hollow/basin/depression on the earth's surface. Lakes vary in size, shape, depth and salinity. Some lakes are very small while others are of great size. Some lakes are permanent while others fill-up or dry-up/drain-off quickly.

Characteristics of lakes in East Africa

- They are shallow e.g. Lake Victoria and Lake Kyoga
- They are hollow e.g. lake George and Edward
- They are small size e.g. Kabaka's lake
- They are deep e.g. lake Tanganyika

- They are fresh water e.g. Lake Victoria.
- Some lakes like Katwe are salty.
- They are irregular e.g. Lake Victoria.

Classification of lakes

Lakes are commonly classified according to the mode of formation of the basins/depressions they occupy. The main categories of lakes according to the mode of formation of their basins include;

- Lakes produced by tectonism (earth movements and volcanicity);
- Lakes produced by erosion;
- Lakes produced by deposition;
- Lakes produced by human activities.

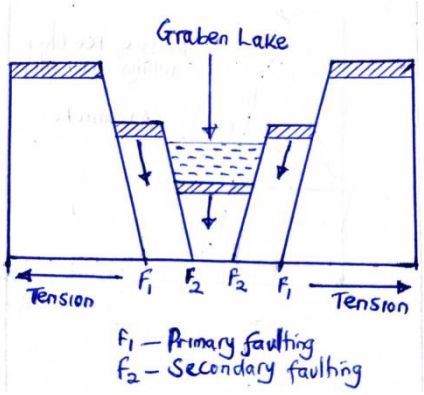
TECTONIC LAKES

These are lakes whose formation is associated to the forces originating from the interior of the earth which led to the earth movements such as faulting, volcanicity and warping. These earth movements are responsible for the formation of various depressions occupied by water to form lakes. These are subdivided into lakes due to faulting lakes, volcanic lakes and down warped lakes.

Faulting lakes: these are lakes formed from faulting. Faulting is the fracturing of the hard rocks of the earth's crust followed by displacement relative to one another along fault lines. Faulting has led to the formation of rift valley lakes and tilt block lakes.

- **Graben lakes/rift valley lakes**; these are lakes contained in the depressions at the floor of the rift valley. They are characterized by the following; long, deep and narrow, salty because they have inlets but no outlets, they are hollow shape, they are found in steep banks e.g. L. Tanganyika, L. Albert.
- They are formed when diverging convective currents produced tensional forces on the earth crust. The tensional forces pulled apart the crustal layers producing normal faults which divided the crust into three blocks. The central block sunk under its own weight while the side blocks remained stable in their original position forming a rift valley. As convective currents continued, secondary faulting resulted into the formation of secondary depression on the floor of the rift valley called a graben. These depressions were later filled with water from rivers, under ground streams, springs and rain water to form graben lakes. Examples include L. Albert, L. Tanganyika, L. Turkana, etc.

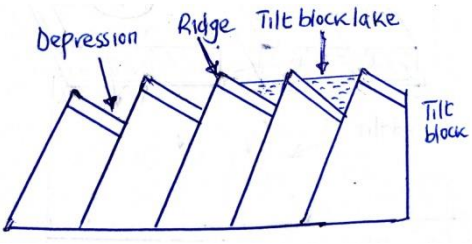
Illustration



- **Tilt block lakes**; these are lakes which occupy depressions between ridges/upland on a tilted block landscape. They were formed when tensional/compressional forces led to the formation of several parallel faults dividing the crust into several blocks. This was followed by general uplift of blocks;

however, each crustal block was subjected to different rates of uplift and then tilted to the same direction forming ridges and depressions on top of the tilted blocks. E.g. L. Ol-bolossat in the Aberdare ranges in Kenya.

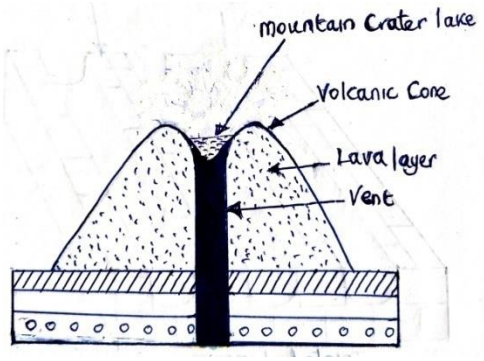
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Volcanic lakes; these are lakes formed by volcanic activity. Volcanicity is a total process through which the molten rock materials are ejected, cooled, crystallized and solidify on the earth's crust to form a depression that are filled with water hence forming a lake. These lakes include: crater lakes, caldera lakes, explosion crater lakes and lava dammed lakes.

- **Mountain Crater lakes**; this is a mass of water contained in a small funnel shaped depression on to top of a dormant volcano. These are lakes formed by violent eruption whose acidic lava blocks the vent in the course of eruption. Pressure then builds up beneath the volcano and violent eruption takes place blowing off the top of a volcano creating a small funnel shaped depression called a crater. The crater is then filled with water from rain or glacial melt water to form a crater lake. Examples can be found on Mt. Elgon, Mt Muhavura, Mt. Longonot, and Menengai Lake on Mt. Kenya.

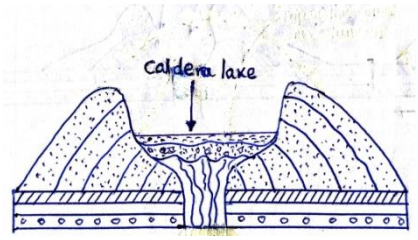
Illustration



- **Caldera lakes**; these lakes found/contained in a wide circular steep sided depression on top of the volcano. It is formed as a result of the destruction of the upper part of the volcano either by violent eruption or cauldron subsidence. When a volcano is formed magma may solidify in the vent creating a lot of pressure below the point of solidification. In the subsequent eruption, the pressure causes a violent eruption which destroys the whole summit of the volcano leaving a wider circular depression called a caldera.

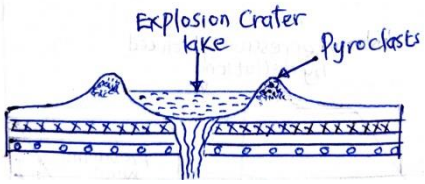
  Caldera formation by cauldron subsidence occurs when eruption ceases to take place. Magma is depleted and a wide space called chasm is left beneath a volcano. The weight of the upper cone becomes great and the materials collapse to fill up the chasm creating a wide depression on top of the volcano called a caldera, L. Longonot, L. Suswa, L. Ngozi. When rain water fills the caldera it becomes a caldera lake. Examples are Ngorogoro Lake in Tanzania. Alternatively cauldron subsidence may leave behind a huge depression which when filled with water from rain forms a caldera lake.

Illustration



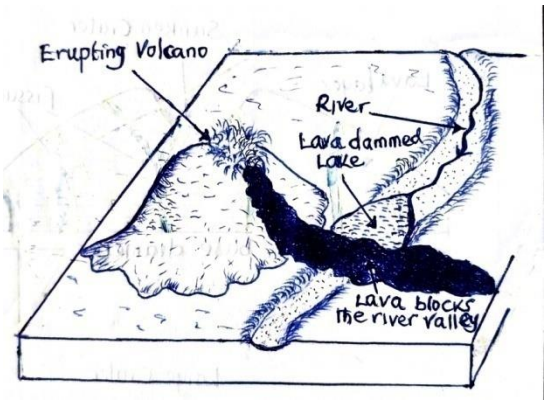
- **Explosion crater lakes**; these are lakes found/contained in explosion craters or circular depressions on the earth's/ ground surface. They are formed when a lot of pressure is built up by magma resulting into violent eruption that blows off the ground surface open creating a shallow circular depression called explosion crater. Later water from rain, rivers or underground streams may fill the depression to form an explosive crater lake. Examples are over 200 in the western arm of the rift valley alone such as Katwe, L. Nyamunuka, L. Nyamsingire, L. Nyungu, L. Rutoto, etc, in Port Fortal.

Illustration



- **Lava dammed lakes**; these are lakes formed where volcanic mountains lie adjacent to a river erupts and the lava flows and solidifies across a river valley. The solidified lava then blocks the flow of the river forcing the river to pond behind the lava. The water which collects behind the lava forms a lava dammed lake. Examples of such lakes include Bunyonyi, Mutanda, Muhehe, Muhoho, Butera, Kayumba, etc, in western Uganda.

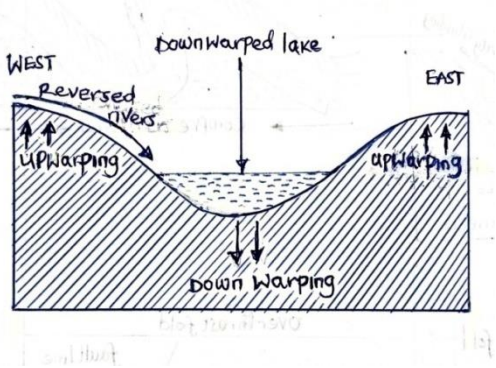
Illustration



- **Down warped lakes**; these are lakes formed through a combination of down warping and drainage reversal. Warping is the gentle uplift and sinking of the earth's crust. Down warping was probably caused by sinking convection currents that exert a drag force, pulling the upper layers of the crust down wards hence forming a depression/basin. These lakes are characterized by the following; they are shallow, they are saucer shaped, have irregular shoreline, have fresh water, surrounded by extensive swamps, have numerous islands.
- In East Africa, down warping occurred in the central plateau creating Victoria and Kyoga basins/depressions. At the same time, there was uplift of the western axis during the rift valley formation that forced rivers that initially flowed to the river Congo system to reverse their flow and back pond filling their former valleys and basins. This led to the formation of down warped lakes like Victoria, Kyoga, Wamala, Mburo, Opeta, etc.

- Lake Kyoga was formed by reversal and back ponding of river Kafu when it filled its former valleys and those in its tributaries and that is why lake has finger-like projections.
- Lake Victoria was formed when rivers Katonga and Kagera reversed their direction of flow and filled up the down warped/Victoria basin that is now occupied by L. Victoria.
- The back ponding of R. Rwizi in south western Uganda led to the formation of lakes such as Kachira, Nakivale, Mburo and Kijanebalola.

Illustration



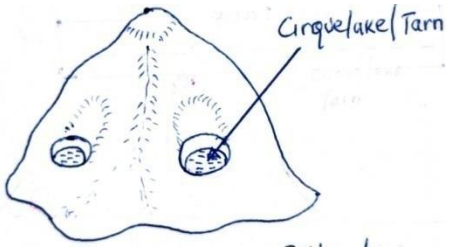
EROSIONAL LAKES

These are lakes produced by erosion by water, glaciation/ice and wind. These agents of erosion are responsible for the creation of depressions which are filled with water to form lakes. Such lakes include; glacial erosional lakes and wind erosional lakes.

Glacial erosional lakes; these lakes are formed on the glaciated mountain side when the glacier tends to erode the mountain sides by plucking and freezing creating circular depressions called cirques. These lakes are divided into three i.e. tarns, ribbon lakes and rock basin lakes.

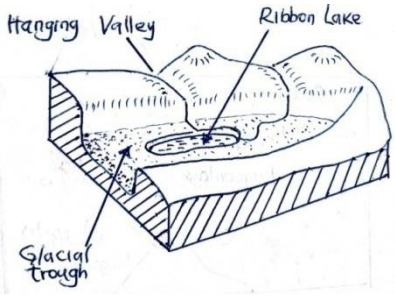
- **Tarns/cirque/corrie lakes**: these are lakes contained in a semi-circular steep sided rock depression. They are formed through the process of plucking, abrasion and basal sapping. A cirque lake is formed when a pre-glaciated hollow is widened and deepened by plucking and abrasion to form a semi-circular-steep-sided depression on the mountain side called a cirque/tarn. When cirques are filled with glacial melt water and rain, cirque lakes/tarns are formed. Examples are; Lac du Speke, Lac Catherine, Lac vent, Lac Noir on Mt. Rwenzori, L. Teleki, L. Michelson, Hannington tan, and Gallery tarn on Mt. Kenya and the most prominent on Mount Kilimanjaro is Mawenzi tarn.

Illustration



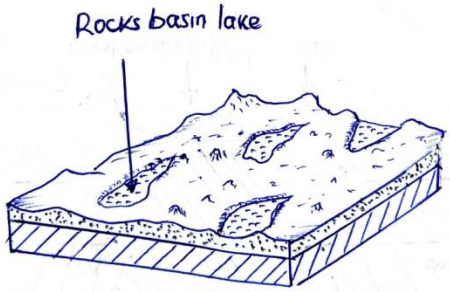
- **Glacial trough/ribbon lakes**; these are lakes that occupy elongated hollows excavated by glacier on the floor of the glacial trough (U-shaped valley). They are formed when soft rocks materials at the floor of the glacial trough are removed by abrasion and plucking creating elongated depressions within the valley. It's because of their elongated shape they are called ribbon lakes. The depressions are later filled with glacial melt water or rain water they form ribbon lakes e.g. L. Michelson and Sacred tarn on the slopes of Mt. Kenya.

Illustration



- **Rock basin lakes**; these are small lakes contained in the small depressions in the slopes of the glaciated highlands. They are formed on the surface of a U-shaped valley where soft rocks or jointed rocks are eroded by the glacier to create small depressions hence are referred to as rock basins. These depressions are filled with water from melting ice or rain water to form rock basin lakes e.g. L. Carr, L. Michaelson on Mt. Kenya Lac Alice on Mt. Rwenzori

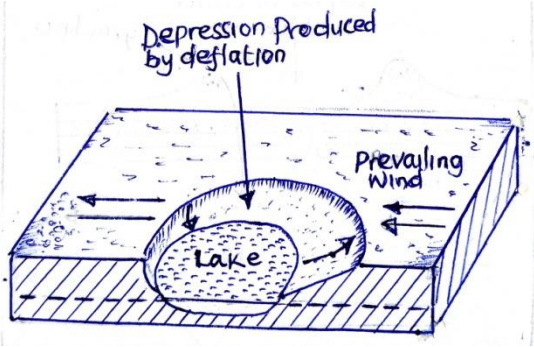
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Wind erosional lakes; these are lakes often found in arid and semi-arid regions.

- These lakes are formed when wind in desert areas erodes off some rocks/sand until a depression is excavated deep below the water table. The eddy wind currents scoop out loose sand and drain it away depositing it as sand dunes. When these depressions are filled with water, a lake is formed. Such lakes are not true lakes but are more of muddy swamps. The more prominent lakes of this kind develop when an aquifer (water bearing rock) is exposed and such lakes are called oasis. A good example is the Qattara Depression in Egypt.

Illustration



DEPOSITIONAL LAKES

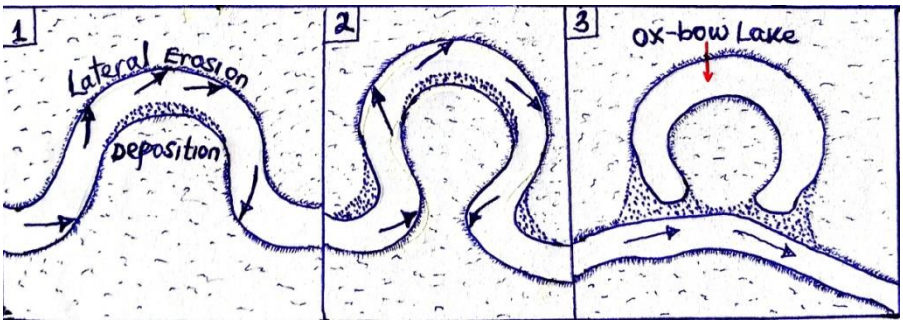
These are lakes formed when depositional materials block channels or an inlet of some kind. There are three forms of lakes produced by deposition namely; river depositional lakes, wave depositional lakes and glacial depositional lakes.

River depositional lakes; these are lake produced by river deposition and they are ox-bow lakes and delta lakes.

- **Ox-bow lakes**: This is a mass of water contained in the horse-shoe shaped depressions produced by meander cut-off. They are formed when erosion is concentrated on the concave bank cutting

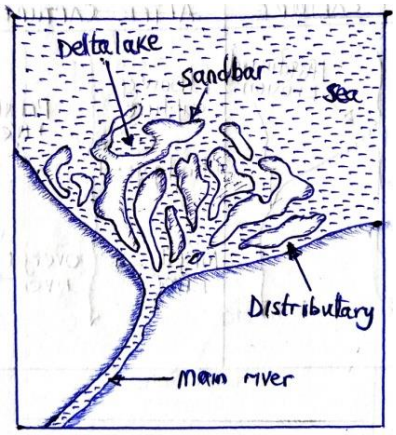
deeper and causing the adjacent bends to be closer while deposition takes place on the out side banks of the meander. Further erosion leads to the meanders cutting through each other or during the flood times the meander breaks though the meander loop/neck. The meander is then cut-off then sealed off by alluvium deposits to form ox-bow lake. Examples are found along rivers such as Tana, Athi, Ngaila, Nyando, Rwizi, Semiliki, Malaba, Ruvu, etc.

Illustration



- **Delta lakes**; these are small and shallow lakes formed in deltas in the lower course of the river. They are formed when large alluvium deposits form embankments that block the water within the distributaries. As a result the distributaries back pond to form delta lakes. Examples are L. Mangomeni, L. Mwananyama, L. Mchengu on R. Rufigi in Tanzania, etc.

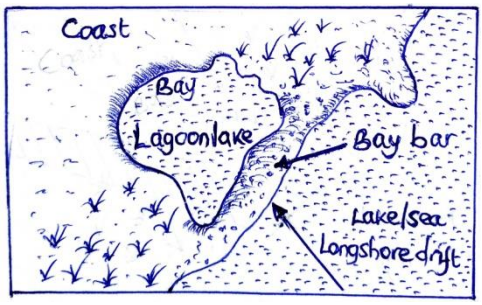
Illustration



Wave depositional lakes; these are also called

Lagoon lakes: this is a mass of water that has been cut off from the main water body. They are formed when spits and sand bars formed by long shore drift enclose a bay cutting it off from the sea. When the water level of the sea drops, the water that is cut-off by the sand bar may be trapped behind it hence forming a lake. For example L. Nabugabo along the shores of L. Victoria was cut-of from Nabugabo bay by a sand bar and Ikimba in North West Tanzania was also cut-off from L. Victoria. Tonya point Lagoon on L. Albert that is about 1km square.

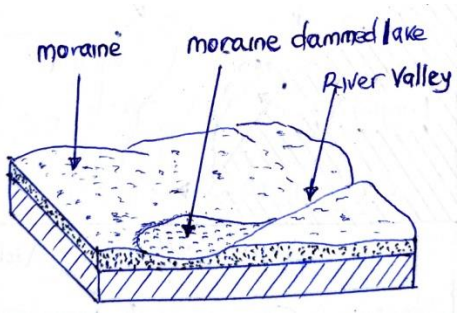
Illustration



Glacial depositional lakes; these are lakes formed by deposition of terminal moraine across a river valley. These lakes are divided into moraine dammed lakes and kettle lakes.

- **Moraine dammed lakes**; these are lakes formed when terminal moraine is deposited across a river valley and blocks water of the valley which back ponds and collects behind the moraine to form a lake. The water later overflows to continue its movement. Examples include Tyndall tarn on Mt. Kenya, lake Horhnel, L. Alice on Mt. Kenya, and Lac Griss on Mt. Rwenzori.

Illustration



- **Kettle lakes**; these lakes are formed when a mass of ice enclosed in the terminal moraine melts away due to hot temperatures leaving behind a depression on top of the terminal moraine (where the block of ice was). The depression is then filled with melt water to form a small lake known as Kettle Lake. Examples are Lake Mahoma on mount Rwenzori and Lake Ellis on Mt. Kenya.

Illustration



MAN-MADE LAKES

These are lakes in East Africa and indeed other parts of the world that have come into existence through man's activities.

- Man has **built dams**/artificial walls across rivers which has created the rising of water level up-stream (where the river comes from) to produce a lake. Examples include a lake due to damming of R. Pangani in Tanzania; Nalubale dam in Uganda, seven forks dam in Kenya and others are L. Nasser in Egypt, L. Kariba in Zimbabwe, L. Volta in Ghana, etc.
- Man has also **dug valley dams** to store water for irrigation, watering their animals, for fishing and cultural purposes. Such lakes include L. Kibimba, L. Kiyanja near view Hotel in Mbarara, Kabaka's lake in Kampala
- Man's activities of mining especially sand and clay mining has resulted into man made lakes e.g. the former lake kajjansi and around kajjansi.

OTHER TYPES OF LAKES

- **Solution lakes**; these are lake formed in limestone areas. They are formed when rain water in combination with Carbondioxide reacts with limestone rocks. The weak carbonic acid solution formed from the reaction dissolves limestone rocks leaving empty cavities/caverns underground. When the empty cavities are filled with water they form lakes. With time the roofs of the empty

cavities/caverns will collapse and these lakes will be exposed to the surface. These are also called Karst lakes. Examples can be found at Nyakasura in Fort Portal, Tanga in Tanzania, in Tororo, etc.

- **Landslide lakes**; these are lakes formed due to the damming of river valleys by debris from mass wasting /landslides. Such lakes are temporal. When landslides occur, they may temporally block a river valley forming a lake at the upstream side. For example river Mbaka in Tanzania in 1955 was blocked by landslides from Tukuyu hill forming a lake that lasted for 8hours.
- **Flood plain lakes**; these are lakes that result from deposition of alluvium in the lower stage of the river. They are formed when levees prevent the flood water from returning to the river hence making water accumulate behind the levees. Such lakes are shallow and vary in size. Examples can be seen west of Dar-es-salaam in the lower valley of river Sinza and river Luhanga forming Lake Magomenu and Lake Mwananyamara respectively.
- **Oasis Lakes**: these are more prominent and permanent desert lakes. They develop when an aquifer (porous water bearing rocks) is exposed. These lakes are known as oasis.

ECONOMIC SIGNIFICANCE OF LAKES

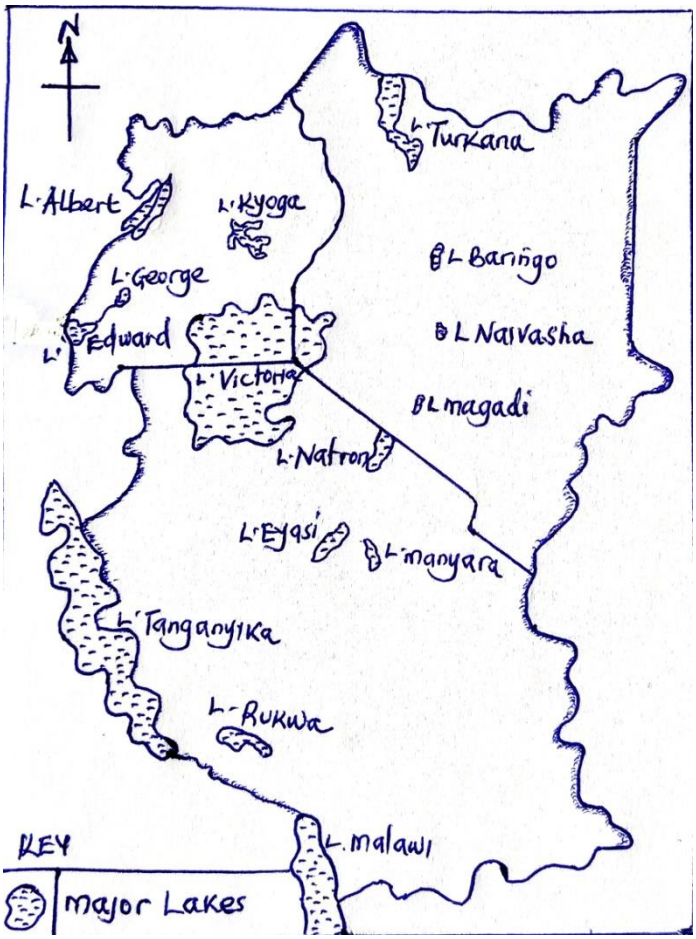
- Lakes provide water for domestic and industrial use. For instance water is pumped from L. Victoria, at Ggaba, etc, to several factories and thousands of homes in Kampala alone.
- Lakes supply water for irrigation purposes for example Lake Kyoga supports the Olweny scheme; L. Kibimba on R. Kitumbezi supports Kibimba rice scheme, etc.
- Lakes are a source of fish through fishing and therefore, contribute to the protein content in the people's diet. Small and large scale fishing is carried out on major East African lakes. Tilapia and Nile perch are the major fish caught in lakes such as Victoria, Kyoga, Tanganyika, etc. fish processing factories have also developed in the region.
- Lakes offer the cheapest mode of transport by use of canoes, boats and bigger vessels used to navigate through the lakes. For example marine vessels on Lake Victoria operate between Jinja and Port Bell in Uganda, with Mwanza in Tanzania and Kisumu in Kenya.
- Some lakes contain minerals and therefore support the mining industry. For example salt is mined from L. Katwe; soda ash is extracted from Lake Magadi. Soda ash is used for making soap, glass and salt. Sand mining is also done on lake beaches such as Kasenyi and Lutembe along the shores of Lake Victoria and is used in the construction industry as well as in the glass making industry.
- Lakes promote the development of art and crafts industry in the country. Lake Kyoga for example has dense cover of swamp vegetation such as papyrus and reeds that are used to make mats, baskets, huts and other products for sale.
- Lakes have modified the climate of Uganda and east Africa at large. Lakes through evapo-transpiration tend to recharge the atmosphere hence promoting the formation of convectional rainfall and temperature regulation. Areas around Lake Victoria and Lake Kyoga receive heavy rainfall and have moderate temperatures. This support dense settlement and crop cultivation.
- Lakes have promoted tourism development that brings in foreign exchange to the country. For example Lake Mburo and Lake Edward harbor crocodiles, bushbucks, hippopotamus, and snakes among other animals. This has made the government to gazette them into national parks for example queen Elizabeth national park on Kazinga channel and Lake Mburo national park in Mbarara; lake Nakuru in Kenya has a sanctuary of flamingoes.
- Lakes are water reservoirs which regulate the flow of many rivers. These rivers are dammed to provide hydro electric power used for domestic and industrial purposes. Lakes therefore ensure sufficient supply of water to the dammed rivers. For example Lake Victoria acts as a reservoir to the Nile River dammed at Jinja (Owen falls dam).

- Lakes are important damping zones for sewage and other industrial wastes. For example sewage in Kampala is directly channeled into Lake Victoria swamps and most industrial wastes are always damped there as well.
- Lakes provide materials which promote research and education. For example Entebbe Fisheries Research institute on Lake Victoria. Others are students in various institutions conduct field studies on coastal landforms, etc.
- Lakes are potential source of energy to the energy sector. Papyrus plants on the lake shores are believed to contain methane gas which can be harnessed to produce bio-gas highly needed for domestic and industrial use.
- Some lakes contain fertile alluvial soils derived from the deposition by rivers and streams flowing into them. The fertile soils support cultivation of crops. For example Lake Victoria basin favours coffee and bananas growing, sugar cane estates at Lugazi and Kakira.

However,

- Lakes harbor disease vectors such as water snails which transmit bilharzia and mosquitoes that cause malaria. This has contributed to high death rates in the regions of Bukedi and Busoga which are around Lake Victoria.
- Lakes are associated with floods during the wet season that displace homesteads and destroy people's property. For example the Elnino rains of 1997-1998 made Lake Victoria and Kyoga to over flood and displaced thousands of villages.
- Lakes also harbor dangerous marine animals such as crocodiles, hippopotamus, snakes among others, which a big threat to the people living around lakes. These animals can destroy crops, livestock and cause death to the people.
- Lakes which lie within the interstate boundaries have promoted interstate conflicts due to territorial claims. For example Uganda and Kenya have always conflicted over the ownership of Migingo Island in Lake Victoria.

A SKETCH MAP OF EAST AFRICA SHOWING THE MAJOR LAKES



REVISION QUESTIONS

Qn: To what extent has tectonism been responsible for the formation of lakes in East Africa?

Qn: Asses the relative importance of lakes to the economies of East Africa.

Qn: To what extent are the processes o erosion and deposition responsible for the formation of lakes in East Africa?

Qn: To what extent has vulcanicity been responsible for the formation of lakes in East Africa?

TIPS TO HELP YOU PASS GEOGRAPHY

- Immediately after receiving a question paper, say a short prayer.
- Read through the instructions at least three times to a void misinterpretation.
- Read the question thrice to interpret it correctly.
- Start with the question you understand better.
- Number your work properly to avoid losing marks.
- While explain features in geography paper one, follow the following steps given bellow:
- State the feature,
- Describe the feature, explain the process for the formation of the feature,
- Give examples illustrate.
- Avoid abbreviations and check your grammar in the text.
- UACE geography is marked out of impression; therefore you MUST be aware of writing a good essay with examples and neat diagrams.

GUIDELINES TO EXAMINERS

Section A marking is basically factual

Section B and C are marked by impression unless otherwise stated. The impression marking is based on the following;

- **O –** …… Completely irrelevant.
- **1 – 8**……Rudimentary facts / scattered facts.
- **9 – 11**…..O' level answer.
- **12 – 14**…Basic A' level answer.
- **15 – 17**…Good but not outstanding.
- **18 – 20**…A very good answer.
- **21^{+}**…....Excellent answer.

Note:

- Read through all the candidates work while making comments before the award of marks.
- Be consistent and accurate in your marking.
- Avoid unnecessary crossing of marks.
- Do not write in or cross the work of candidates.
- Alteration of marks in case of errors can only be done by the team leader.

What is required in the candidate's essay?

- Facts
- Orderly/logical
- Depth of analysis
- Maturity

Common questions start with the following:

Explain – how, and why?

Elaborate:

Discuss – one sided:

Examine:

Account:

Describe – giving how the features looks like:

To what extent – start with ….

- To a **larger extent**….
- Then to a **smaller extent ….**
- Evaluate the extent.
- Ensure maturity of content.

Distinguish/differentiate:

- Look at one item first and,
- Use the word **" where as"** or **" while"**
- Look at the last item.

LONG QUESTIONS AND SHORT QUESTIONS

QN. To what extent was faulting responsible for the formation of relief landforms in East Africa? (25marks)

Approach:

- Define faulting,
- Give the background of faulting,
- Give the areas affected by faulting in East Africa,
- To a **large extent** faulting is responsible for…..
- Identify landforms of faulting (i.e. rift valley, escarpments, Horst Mountains, tilt blocks, etc)

- Describe the shape of each landform,
- Explain the processes of each landform,
- Give example and,
- Illustrate its diagram.
- To a **smaller extent** other factors;
- Volcanicity; (define, give the origin, identify volcanic landforms, process for each, example and illustration)
- Glaciation; (define, give the origin, identify glaciation landforms, process for each, example and illustration)
- Weathering; (define, give processes, identify relief landforms due to weathering, give processes, examples and illustration).
- Evaluate the essay.

SHORT QUESTIONS

QN. Account for the formation of stamps in East Africa.

Approach:

- Define **a stamp**
- Describe where a stamp is formed…at a coast
- Define the coast
- Describe the process for the formation of a stamp
- Define waves
- Identify the causes of waves
- State a stamp is produced by wave erosion
- Give processes of wave erosion (hydraulic action, abrasion, solution, attrition).
- Formation of stamp starts with headland and bay, cliff, cave, arch and finally a stamp.
- Give the processes in each of landforms above.
- Give examples and
- Illustrations in each case.

QN. Examine the influence of tectonism in the formation of highlands in East Africa.

Approach:

- Define **tectonism** (these are earth processes including volcanicity)
- Give the origin of tectonism
- State the areas affected by tectonism
- Give the examples of processes (i.e. faulting, volcanicity, folding, warping)
- Define highlands i.e. land that rises above 1000m
- Highlands by faulting (i.e. Block Mountains, tilt block, escarpment, etc) in each highland identify, give the shape, process, example and illustration.
- Highlands by volcanicity (i.e. cumulo domes, volcanic plug, composite volcano, shield volcano, etc).
- Highlands by folding (i.e. anticlines)
- Highlands by up warping (i.e. plateaus)

QN. Examine the influence of tectonic movements in the formation of highlands in East Africa.

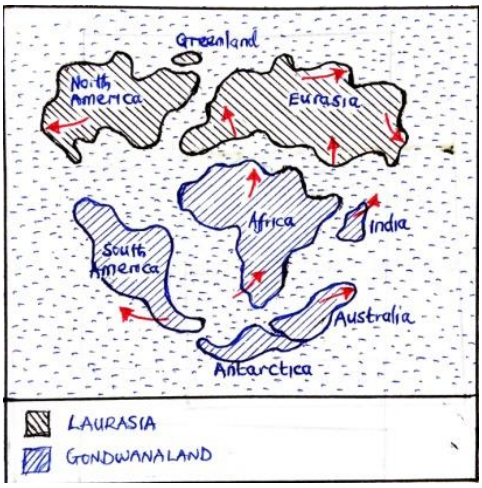
Approach:

- Define **Earth movements** (these are large scale processes)

- Give the origin of tectonic movements
- State the area affected by tectonic movements.
- Give the examples of earth movements (i.e. faulting, warping, folding, Earthquakes, without volcanicity)

QN. Examine the influence of tectonism in the formation of depressions in East Africa.

- Define tectonism
- Give the origin of tectonism
- Give examples of tectonism processes
- Define depression
- Depressions by faulting (rift valley,



SOILS

Soil refers to a thin loose surface layer of unconsolidated material on the earth's surface which supports plant growth. Soils include; clay, laterites among others.

It is made up of weathered rock material, humus, water, air and living organisms.

Components of soil

1. Organic matter

This is material derived from decomposed plant and animal remains to form humus which is obtained from decomposing litter. Organic matter is a source of plant nutrient for example sulphur, phosphates, nitrates.

2. Living organisms for example burrowing animals like moles, nitrogen-fixing- bacteria which are important for soil aeration.

3. Soil air; Soil consists of air for example oxygen, nitrogen and carbon dioxide responsible for increasing soil acidity. Soil air is responsible for respiration and germination.

4. Soil water. This enters the soil from rain or irrigation and moves down by gravity or upwards by capillary action.

5. Inorganic material (mineral). This is material derived from breakdown of parent rock. The major mineral elements include carbonates, aluminum, iron, magnesium, sulphates etc.

PROCESSES OF SOIL FORMATION

These are activities that take place to produce soil out of mineral matter, parent rock and organic matter. Numerous processes are involved in the formation of soil.

These processes include:

1.Weathering: This is the physical disintegration and chemical decomposition of rocks in situ mainly under the influence of rainfall and temperature to produce soil.

Usually physical weathering causes breakdown of parent rock leading to formation of regolith (loose particles). This occurs due to exfoliation, block disintegration, aridity-shrinkage etc common in semi-arid areas forming thin, shallow, immature soils.

Chemical weathering on the other hand causes decay and decomposition of rock materials into fine particles to form deep mature well-developed soil. This is through chemical reaction between water and rock minerals through chemical weathering processes for example hydrolysis, carbonation, hydration among others.

2. Leaching

This process involves the removal (transfer) by water of soluble nutrient (mineral compounds) from upper layer of the soil vertically to the underlying layers.

The compounds removed are transferred vertically from horizon A to the underlying horizon B of the soil profile.

These include salt, carbonate, silicate, magnesium and so on.

This process is common in areas of heavy rainfall and porous rocks.

Materials carried by this process leave behind impoverished (poor) soil on top.

Leaching results in the formation of complete soil profile but with horizon A having a colour ranging from brown, yellowish to reddish due to concentration of oxides after leaching the silica.

Examples occur on the hill tops of Buganda and Busoga.

3.Eluviation: This involves the movement of soil material in solution or suspension from one place to another within the soil.

Eluviation may be vertical or lateral depending on the direction of movement of soil water.

Unlike leaching which refers to movement of materials mainly in solution from, Eluviation is mainly the movement of colloids (small particles).

Eluviation also leads to the development of poor horizon A since most of the nutrients are washed down to deeper layers.

4.Illuviation

This is the precipitation and accumulation of leached and eluviated materials in the underlying layers of the soil. It is the natural result of eluviation and leaching.

This process takes place in the horizon B of the soil profile making it a dark layer hence this horizon is also known as the illuvial layer.

This process is influenced by the presence of impermeable rocks of the underlying layer.

This process leads to the development of mature, deep and well-developed soil common in humid areas.

5.Humification

This is the process through which organic matter is decomposed to form humus.

Organic matter involves remains of plants, litter and animals.

This process is therefore common in areas rich in vegetation with heavy rainfall that provides water that acts as a medium of chemical reaction and hot temperature that speeds up the reaction.

Chemical breakdown is further facilitated by organisms like bacteria.

The process leads to the development of deep and mature soils with humus in horizon A, giving it a dark colour.

6.Mineralization.

This is the process of soil formation that occurs under extreme conditions in which de composition of organic matter may extend further than humification.

The organic matter may be so broken down and changed that it is converted into mineral substances for example water, silica among others.

This takes place mainly in horizon A of the soil profile but can also occur anywhere provided organic matter is available.

7. Laterisation

This is process by which rocks are converted into laterite soils. It is dominant in areas where silica, magnesium, iron and aluminum are concentrated in the upper layers of the soil.

Silica and magnesium are leached out of the top layers leaving behind aluminum and iron which give the soil reddish –brown colour.

Iron and aluminum concentrate fast into small particles which later fuse (join) together and harden under dry condition giving rise to a hard pan known as duricrust which is laterite soil.

The process is common in tropical areas with deep and intense chemical weathering and the resultant soils have well developed profiles and deep soils.

8. Calcification

This is the process of soil formation common in dry areas.

It involves the upward movement of capillary water.

Because of excessive evaporation in dry areas, water containing dissolved calcium moves upwards through the soil (that's by capillary actions) and when it reaches the surface of the earth, it evaporates.

As the water evaporates dissolved calcium is deposited on the surface, leading to formation of soil rich in calcium known as pedocals (thin and shallow).

9. Salinization:

This process occurs in the semi-arid areas of East Africa.

In this process rain water dissolves soluble minerals from rocks and transports them.

Due to the hot and dry conditions, water evaporates, leaving behind salt layers that form saline soil. The resultant soils are thin and shallow with undeveloped soil profile.

10. Podzolization

This process involves the removal of iron and aluminum oxides together with humus. It is more common in cool climates where precipitation is in excess of evapo-transpiration and where soils are well drained or sandy.

The resultant soil is called podsol with 2 distinct horizons.

Horizon A is leached and drained of coloured minerals and horizon B is reddish brown because of deposition of iron and aluminum compounds from Horizon A.

11.Gleization

This is a soil formation process that occurs in wet and cool climate environments where there is impeded drainage like swamps. Water logging/impeded drainage excludes oxygen so that chemical reduction takes place changing the reddish coloured iron oxides to ferrous compounds which produce bluish or greenish grey clays which are very sticky for example at Kajjansi and Lweza in the Lake Victoria basin. This leads to development of shallow soils.

Questions

a) **What is meant by the tern "soil"?**

Approach

- Define soil
- Identify components of soil
- Identify soil types for example laterites, podsols etc.

b) **Describe the process of soil formation common in E.Africa**

- Define process
- Identify and explain (describe) the processes of soil formation

FACTORS (CONDITIONS) THAT INFLUENCE SOIL FORMATION

These are the controls that determine the nature and rate of soil formation. These factors/conditions either accelerate or retard the rate of soil formation. They include;

1. Climate:

- Climate influences soil formation through its elements of rainfall and temperature that lead to decomposition and breaking of rocks to produce soils. Climate influences soil formation in the following ways:
- In equatorial climate characterized by hot and wet/humid conditions, heavy rainfall and hot temperatures promote chemical weathering, leaching, and eluviation etc leading to formation of deep mature soils. For example around the northern shores of Lake Victoria (Mukono, Entebbe etc)
- In tropical climate areas characterized by alternating wet and dry conditions, physical and chemical weathering occurs leading to formation of fairly deep mature soils for example in Northern Uganda.
- In arid and semi-arid climatic areas, the hot temperatures lead to skeletal, immature soils.
- The hot and dry conditions also result into capillary attraction leading to formation to alkaline soil for example in Moroto and Northern Kenya.
- In montane climate areas characterized by cold conditions and limited surface water which limit weathering and other soil forming processes, shallow, immature soils are formed from decayed glaciers.

2. Relief (topography)

This is the general appearance of the landscape. The nature of the slope will influence soil formation because relief controls rate of weathering, erosion and deposition of materials.

The influence of relief includes the following:

- Steep slopes: On steep slopes e.g on mountain slopes, erosion of weathered material is dominant leading to shallow, skeletal and immature soils.
  On steep slopes erosion also exposes the rock to further weathering and soil formation.

- Gentle slopes: On gentle slopes some deposition takes place and the rate of soil removal is also limited. At the same time water percolates leading to chemical weathering, leaching, eluviation etc that lead to development of deep, mature and well drained soils e.g in Kericho in Kenya and Mukono areas in Uganda.
- Flat landscape: Generally flat landscape e.g Nyika plateau and Serengeti plains tend to promote leaching leading to formation of poor soils.
- Valleys and low lands: In valleys and low lands extensive deposition takes place leading to deep mature soils.

However, in lowlands with impeded drainage, there is partial decomposition of organic matter leading to peat soils.

3. **Nature of parent rock.**

Parent rock forms the basis upon which soil formation processes e.g weathering, eluviation, leaching, illuviation etc operate.

Different parent rocks like igneous, sedimentary and metamorphic rocks give rise to different types of soils depending on rock characteristics like permeability, porosity, mineral composition, jointing, colour, texture etc.

- **Hardness and softness of the rock**

Hard rocks e.g granite resist weathering and erosion leading to formation of shallow, skeletal and immature soils.

While soft rocks like limestone lead to deep, fine and dry soils because they are easily weathered.

- **Permeability/porosity of the rocks**

Permeable rocks allow water to percolate (pass through) leading to dry porous soils e.g those formed from limestone rocks.

While impermeable rocks like clay limit water percolation leading to poor drainage thus forming peat soils.

- **Colour of the rock**

Dark coloured rocks like basalt, pumice absorb a lot of heat leading to a lot of physical weathering thus forming deep and mature soils, while light coloured (shiny) rocks like sandstone absorb very limited heat leading to limited physical weathering thus forming shallow skeletal and immature soils.

- **Jointing of rock**

Well jointed rocks are easily weathered by chemical processes as water percolates through joints leading to deep soils.

Massive (unjointed) rocks on the other hand limit percolation leading to shallow soils.

- **Mineral composition**

Mineral composition determines the acidity, alkalinity, fertility etc of the soil for example quartz rocks give rise to sandy, shallow and infertile porous soils.

While volcanic rocks lead to deep mature fertile soils and lateritic rocks give rise to thin mature soils.

- **Age of the rock**

Old rocks that have been exposed for long to weathering processes produce well developed, deep soils

However, rocks of recent formation (young rocks) that have not been exposed for long to weathering processes produce shallow, immature soils

4. Living organisms

This involves the effect of plans and animals on soil formation.

- **Plants;** in areas of thick vegetation cover, plant roots lead to disintegration of rocks forming soil particles.

Also, the thick vegetation cover leads to thick humus layer when the leaves drop and decompose. This leads to formation of deep, mature and fertile soils.

- **Man:** Man, through activities like ploughing, mining and construction leads to the breaking of rocks into small particles leading to formation of soils.

- **Bacteria,** worms and other living organisms decompose organic matter to form humus which leads to deep, mature fertile soils.
- **Burrowing animals** for example moles break the rock into soil particles.

5. Drainage

Water-logged conditions in poorly drained areas retard weathering and other soil formation processes hence resultant soils are shallow with immature soil profile.

Well drained areas on the other hand lead to formation of developed, mature soils because they encourage chemical weathering and other processes of weathering hence resulting into deep well-developed mature soils.

6. Time

Time in this context refers to the duration of interaction of soil formation processes and factors. The longer the duration of interaction of soil formation processes, the more developed, deep and mature will be the soils.

The shorter the period of interaction of soil formation process the less developed and more immature the soils will be.

Question 1

To what extent is the nature of parent rock responsible for the formation of soils in E. Africa?

Approach

- Define soil
- Bring out types of soils
- Bring out the role of parent rock (with stand point)
- Bring out the roles of other factors

Question 2

To what extent has topography influenced the process of soil formation?

Approach

- Define soil
- Bring out soil types
- Bring out role topography/relief
- Bring out other factors.

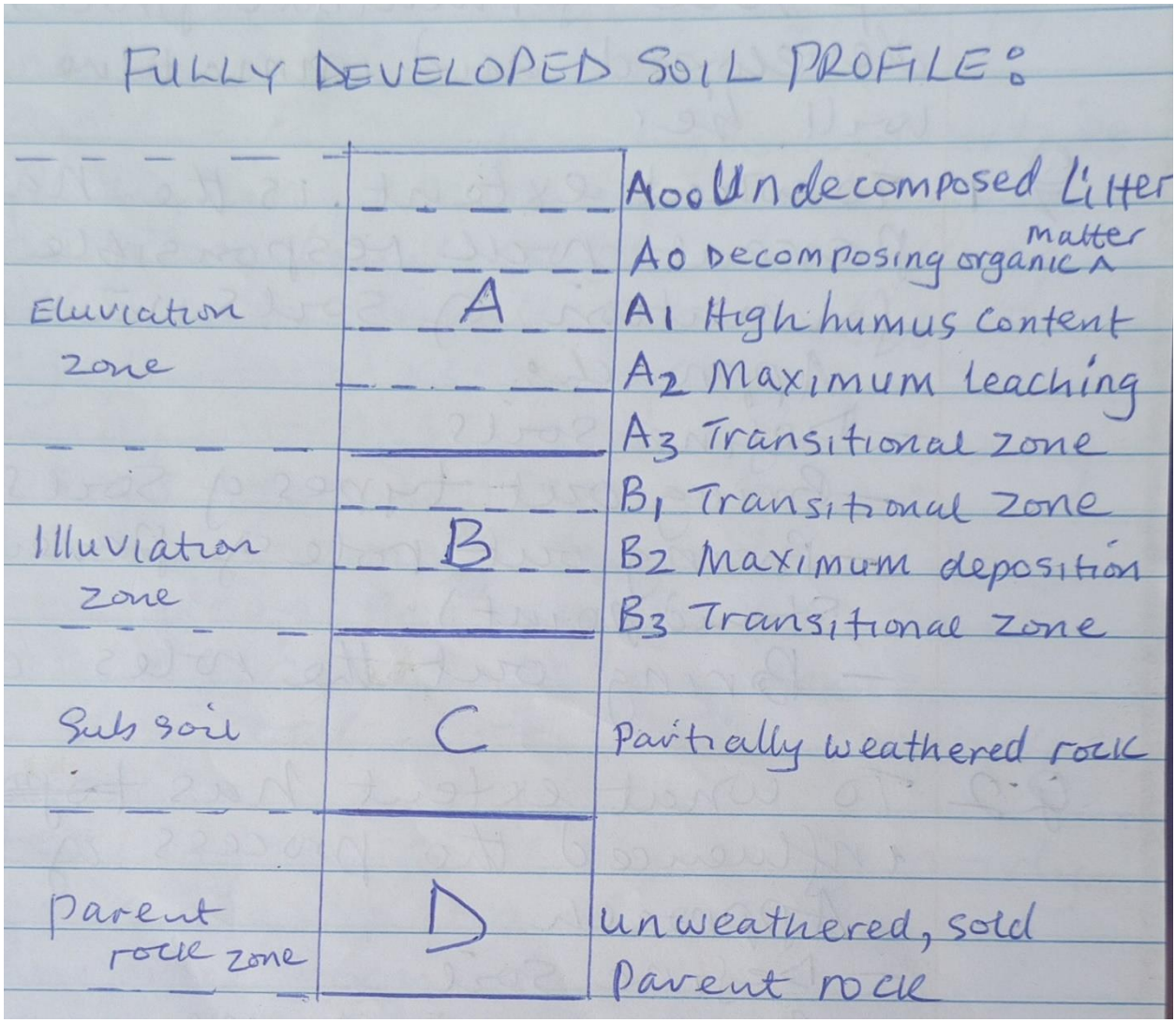
SOIL PROFILE

A soil profile refers to the vertical arrangement or section through the soil from the surface of the earth to the bedrock.

It is composed of soil layers or sections referred to as horizons which are differentiated in terms of colour and texture or mineralogy.

Soil profiles differ from place to place however an ideal or fully developed profile is composed of 4 horizons (layers) ranging from A to D as shown below:

FULLY DEVELOPED SOIL PROFILE



HORIZON A:

This is the topmost layer sometimes referred to as top soil.

The top designated as A00 consists of un decomposed litter of dead leaves and vegetation.

It is followed by A0 composed of decomposing organic matter.

This is followed by A1 which has high humus content and which gives the horizon its dark colour.

This is followed by A2 which is poor in nutrients because of the effects of leaching (removal in solution of nutrients) and eluviation (movement of material/clay particles in suspension) and which make it light coloured/bleached.

This is followed by A3 which is a transition Zone.

HORIZON B

This is the soil layer found below horizon A

It is the zone where nutrients and materials removed from horizon A accumulate through the process known as illuviation.

It is therefore often richer in nutrients than the zone above it and may be darker in colour.

It is sometimes characterized by a hard pan caused by accumulation of large quantities of clay and other nutrients.

HORIZON C

This layer is found below horizon B. it consists of recently weathered material or partially weathered parent material (regolith) resting on the bedrock.

HORIZON D

This is the last layer of the soil profile and is found below horizon C. It consists of solid parent rock or bedrock.

Question

What is meant by the term soil profile?

- Define soil profile
- Draw diagram showing a fully developed soil profile clearly indicating all the horizons in detail.
- Explain the components of each horizon.

FACTORS (CONDITIONS) THAT INFLUENCE THE DEVELOPMENT OF SOIL PROFILE

1. Climate

Climate through its elements of rainfall and temperature influences the formation of soil profile.

- Areas with hot and humid conditions (equatorial climate) experienced throughout the year have fast rates of chemical weathering leading to formation of deep, mature soil profiles for example the Lake Victoria basin.
- Arid and semi-arid climatic regions with low rainfall, hot temperature and scarcity of vegetation, experience physical weathering majorly leading to the formation of thin soils with incomplete soil profiles. Such soil profiles lack humus due to scarcity (absence) of vegetation cover for example in Chalbi desert in Northern Kenya, Turkana region.

2. Nature of parent rock

- Rock hardness and softness

Hard rocks which are not easily weathered lead to formation of thin soils and poorly developed soil profiles.

While soft rocks which are easily weathered lead to development of a deep soil profile.

- **Rock joints**

Rocks with joints and cracks are easily broken down to produce deep, mature soil profiles as compared to the rocks without joints.

- **Rock colour**.

Dark coloured rocks, absorb heat and are therefore easily weathered to produce deep and well-developed profiles, while light colored rocks like quartz reflect away heat hence are weathered slowly leading to formation of thin soils with poorly developed soil profiles e.g in Northern Kenya.

- **Rock mineralogy**

Basic igneous rocks and previously weathered sedimentary rocks are difficult to weather down thus forming a shallow, immature soil profile.

While soils weathered from volcanic rocks tend to form a deep mature soil profile because they allow leaching, water percolation, chemical weathering etc.

- **Rock permeability**

Permeable/porous rocks are easily weathered by chemical processes to produce deep/mature roil profiles

While impermeable rocks e.g quartzite produce shallow, undeveloped immature soil profile because they limit leaching and other processes.

3. Nature of vegetation

Thick vegetation cover leads to formation of humus which is added to horizon A of the soil profile forming dark coloured top soil and deep mature profiles.

Plant roots also lead to the disintegration of rocks and formation of soil profiles. Forested areas therefore tend to have deep soils with well-developed profiles while areas with no vegetation cover have thin skeletal soils and poorly developed profiles.

4. Drainage

Well drained areas encourage rapid chemical weathering and develop deep soils with mature (complete) soil profiles.

In water logged/poorly drained areas, soil forming processes are slow leading to the formation thin soils with incomplete soil profiles. Such soils occur in wetlands areas e.g along river Katonga and Mpologoma, and around shores of Lake Kyoga in Gogonyo etc.

5. Living organisms.

Living organism's e.g ants, earth worms etc through their burrowing action easily breakdown the rock as they make passages underground.

The created passages encourage weathering leaching hence leading to formation a deep mature soil profile.

Large herds of animals e.g cattle elephants also trample on the rocks breaking them to form soils.

Wastes of animals that's to say urine, have uric acid which causes rapid decay of rocks to form mature soil profile.

6. Human activities

Man through mining, quarrying, road construction, agriculture etc causes breakdown of rocks thus forming mature deep soil profile.

7. Topography/relief

- Steep slopes e.g the highlands of Rwenzori, Elgon, Kenya etc; escarpments, cliffs etc are characterized by high rate of soil erosion.

The weathered rocks/materials are quickly washed away resulting into thin/shallow and immature soil profile.

High rates of run-off (erosion) limit water percolation into the ground hence limiting chemical weathering, illuviation, eluviation etc resulting into poor profiles.

- On gentle slopes erosion is slower, there is a lot of deposition of soil eroded on steep slopes and rain water percolates to assist in soil profile development leading to deep mature soil profile.

- On valleys and lowlands there is extensive deposition and water logging occurs which hinders the development of a complete profile hence poor and immature profile is formed.

8. Time

Ample time is required for the formation of mature, fertile and deep soils and well-developed profile. If the time is short, immature soils will form with poorly developed soil profile.

Question1

Explain the conditions which have influenced soil profile development in various parts of East Africa.

Approach

- Define soil profile
- Draw a well-developed soil profile and explain components of each horizon.
- Explain the conditions/factors for soil profile development.

Question 2

Examine the processes responsible for the development of soil profile.

Approach

- Define soil profile
- Draw a full developed soil profile
- Explain all the horizons
- Explain processes of soil profile development

N.B processes are the same processes of soil formation but must be linked to soil profile e.g weathering, illuviation among others.

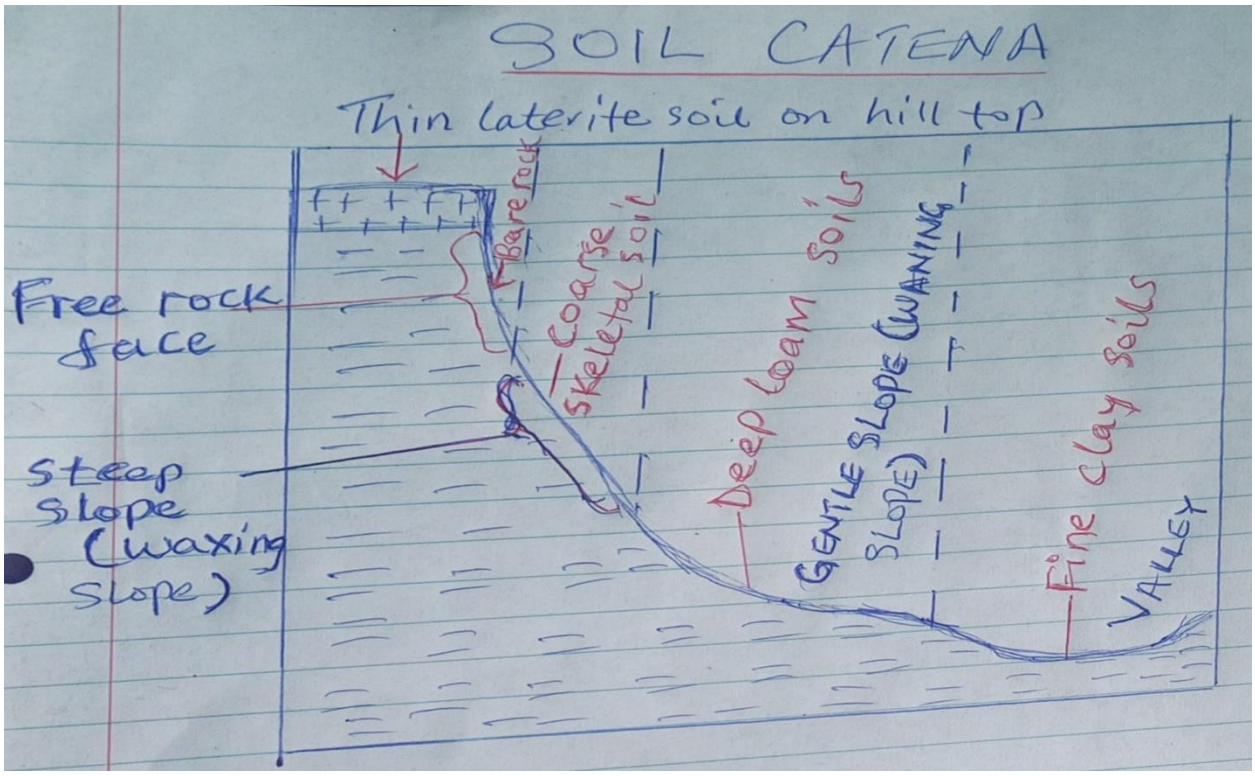
SOIL CATENA

A soil catena is a sequence of soil types along a slope.

It is a horizontal arrangement of differing soil types along a slope from the hill top (summit) to the valley bottom due to soil water movement.

Soil catena usually has 4-5 (4 to 5) different types of soil with different characteristics in terms of colour, depth, texture, water content etc as they progress from the hill top to the valley bottom.

Differences in soil catena are caused by variations in relief and drainage.



An example of soil catena will show the following soil types.

On the hill top: Here there is laterite/lateritic capping characterized by thin skeletal reddish-brown soils.

On the free rock face/vertical slope: Here there is bare rock.

On the steep slope (waxing/convex slope); Here there is thin, coarse, young skeletal soil due to intense erosion.

On the gentle slope (Waning/concave slope): there is deep loam soil due to maximum deposition of weathered material and high rates of chemical weathering.

In the valley, there is fine clay soil.

Question:

What is meant by the term soil "catena"?

Approach

- Define soil catena

- Write on the components of soil catena
- Draw well labelled diagram of a typical soil catena.

FACTORS (CONDITIONS) THAT INFLUENCE THE DEVELOPMENT OF SOIL CATENA IN EAST AFRICA.

The factors or conditions that influence the development of soil catena in East Africa include:

1. Relief

The nature of relief/slope influences the development of soil catena e.g the

flat topped hills of Buganda and Busoga landscape have well developed soil catena.

The hill tops have laterite capping due to leaching.

Steep slopes have stony thin (skeletal) soils due to erosion on the upper slopes.

On the gentle slopes there is deep loam soil because the rates of weathering and erosion are balanced.

While in the valley there is water logged clay/peat soil because of deposition.

2. Drainage

The moisture content of the soil influences the development of soil catena as soil changes along the slope.

Well drained mature soils are found on the gentle slope while thin, stony, dry, immature soils are found on the steep slopes. Clay/boggy soils are generally found in the valley bottoms where drainage is poor that's (Water-logged).

3. Nature of parent rock

The nature of parent rock varies in terms of hardness, jointing, porosity etc.

These aspects of parent rock will determine the rate of erosion and drainage along slope and the resultant nature of soils.

For instance, the **hard** parent rocks are likely to form lateritic soils and free face (very steep slopes) with bare rock due to their resistance weathering.

The **soft** rocks on the gentle slopes are likely to enhance balanced erosion and deposition and deep weathering forming deep mature soils. This will have a well-developed soil catena compared homogenous rocks.

4. Vegetation cover

Soil catena development is faster in areas without vegetation cover since the rate of erosion is faster.

However, thickly vegetated areas slow down the rate of erosion and therefore the rate of catena development.

In other words, areas without vegetation cover develop complete soil catena while areas of thick vegetation cover develop incomplete (poor) catena.

5. Climate: Heavy rainfall in East Africa for example in areas around Lake Victoria encourages leaching leading to development of laterite soil at the hill top.

Heavy rainfall also encourages erosion on the upper slope leading to the development of coarse thin soil, and deposition on the lower slopes and valleys leading to development of deep soil.

6. Human activities

Man's activities such as deforestation, settlement, agriculture etc result in the removal of surface vegetation which encourages erosion on the upper slopes and and deposition on the lower slopes.

However, activities like quarrying lead to the removal of surface soils in the Catena.

Time

Soil catena is developed well only over a long period of time.

Question

With reference to specific examples, account for the development of soil catena in East Africa.

Approach

- Define soil catena
- Draw diagram showing soil catena and explain its components
- Explain the factors which have led to development of soil catena, giving relevant examples.

MAJOR SOIL TYPES OF THE WORLD

:

The major soils of the world have been classified into **3** major categories

- Zonal soils
- Intrazonal soils
- Azonal soils

ZONAL SOILS

Zonal soils are mature soils with well-developed soil profile due to prolonged action of climate and vegetation.

MAIN FEATURES (CHARACTERISTICS) OF ZONAL SOILS

1. Zonal soils largely result from climatic factors that's climate greatly influences their formation and maturity.
2. They develop under conditions of good drainage or well drained soils.
3. They develop mostly on gentle slopes and flat landscape.
4. They are mature soils with well-developed profiles due to prolonged action of climate and vegetation.
5. They are mainly divided into 2 groups that's pedocals with calcium carbonate content (develop under conditions of low rainfall) and pedalfers (which are rich in aluminum and iron.

6. The resultant soil type is closely related to the nature of weathering that takes place under specific types of climate.
7. In low latitude areas, the hot humid conditions give rise to latosols, tropical black earths. In mid latitude climates, humid conditions are associated with the development of podsols and brown earths.

In areas receiving seasonal rainfall chernozem soils develop e.g on the Canadian parries.

In arid and semi-arid conditions, chest nut coloured soils develop.

In high latitude climates, Tundra and Arctic brown soils develop.

TYPES OF ZONAL SOILS

Zonal soils are classified into 2 categories i.e pedocals and pedalfers.

a) PEDOCALS

This is a type of zonal soil which has high calcium carbonate content and develops under conditions of low rainfall.

Pedocals include the following types of soils.

1. Red desert soils

Red desert soils are thin, stony or sandy, yellowish grey or reddish brown with no humus.

These soils lack humus or organic matter because of dry climate which limits vegetation growth and slows down the rate of chemical weathering and soil forming processes.

As a result, red desert soils lack moisture and nutrients for agriculture.

2. Chernozem soils

Chernozem soils develop in cool temperate continental climatic conditions. They have a loose and crumbly texture and are rich in humus and so they are black coloured.

The humus and dark colour are derived from the thick grass cover that decays to provide organic matter on top soil since there is insufficient leachinng to carry the humus deeper.

Chernozem soils are found in areas of tall and mid grass e.g the prairies of Canada.

3. Chestnut soils

These are similar to chernozem soils because they are also formed in areas with cool temperate continental climate except that they form in areas which are drier and therefore have short grass.

Chestnut soils are lighter coloured and in most cases brown e.g in the drier regions of the prairies of Canada. This is because the vegetation is relatively sparse leading to limited organic matter (humus).

Examples are also found in drier parts of the pampas of Argentina, high plains of USA, steppe lands of Russia etc.

b) PEDALFERS

These are zonal soils which have high content of aluminum and iron.

Pedalfers are made up of three units:

- **Ped** meaning soil

- **al** symbolizing aluminum
- **fer** symbolizing iron

Pedalfers include the following types of soil

1. **Tropical black soils**

These are soils found in tropical savannah regions.

In savannah regions with grassland vegetation soils with thin dark layer of humus form when grasslands die and decay especially during the dry season.

However, in the wet season, leaching arising from heavy rainfall removes the silica from the upper profile leaving behind red-coloured oxides of iron and aluminum.

2. **Laterite soils (latosols)**

These are reddish-brown soils that occur mainly in humid tropical climatic regions with heavy rainfall and hot temperatures.

They are sometimes called murram and are largely infertile.

Laterites form any type of rock which is leached.

Laterite is formed due to deep weathering of rocks arising from heavy rainfall and hot temperatures, silica is leached and eluviated from the top layers of the soil profile to the underlying layers.

This process leaves behind iron and aluminum to accumulate through lateralization. This is when an increase in temperature causes evaporation and drawing up of iron and aluminum compounds to the surface from the underlying layers through capillary action.

As these materials accumulate on the surface they from a hard pan of reddish brown soils called laterite (duricrust)

3. **Podzols**

These soils develop in humid cool temperate climates. They have a black, dark-brown or grey colour surface layer. They are associated with coniferous forests.

The strong leaching that takes place in the soils is called podzolization. Most of the minerals such as iron, aluminum, calcium are leached down to horizon B, giving it a brown colour while in horizon A their removal gives it a greyish colour.

These soils are therefore not rich in humus so they are acidic and infertile. They are common in British Columbia (Canada).

4. **Temperate brown soils**

These soils form in cool temperate western margin climatic regions which have natural vegetation of deciduous forests.

These soils are rich in humus derived from decaying vegetation.

Here precipitation exceeds evapotranspiration to allow leaching. This is common in Britain.

5. **Tundra soils**

These develop in areas of low annual temperatures, permanently frozen subsoil and a slow rate of evaporation.

Chemical weathering and biological activity are both very limited and water logging is common in summer. Such soils mainly occur in Greenland and Siberia.

FACTORS/CONDITIONS FAVOURING THE FORMATION OF ZONAL SOILS IN EAST AFRICA.

Laterite soils are the dominant zonal soils in East Africa and their formation is attributed to the following conditions/factors:

1. **Relief:**

Low-lying relief allows percolation of rain water to enable deep weathering and leaching to take place. The soluble silica is removed from horizon A by water leaving behind iron and aluminum to form laterite zonal soils.

2. **Nature of parent rock**

- Mineral content of rocks: The presence of rocks containing large quantities of silica, iron and aluminum encourages rapid chemical weathering under hot and wet conditions.

It is this iron and aluminum left behind in horizon A after leaching of silica that forms lateritic zonal soils.

- Presence of soft and permeable rocks encourages leaching and eluviation to take place. This leads to formation of reddish brown soil (laterites).

3. **Climate**

Humid climate characterized by heavy rainfall ad hot temperatures favours formation of zonal soils e.g laterites.

The heavy rainfall provides a medium through which chemical leaching takes place as well as leaching of silica from horizon A.

The hot temperatures intensify the rate of chemical reactions and allow evaporation to take place leading to accumulation of iron and aluminum oxides on the surfaces which harden to form laterites (zonal soils).

4. **Drainage**

Zonal soils e.g laterites form in well drained areas with sufficient moisture content that allows deep weathering, leaching and lateralization to take place.

5. **Vegetation**

Presence of thick vegetation cover e.g equatorial forests and woodlands favours formation of laterite zonal soils.

Vegetation cover provides humic acid that facilitates deep weathering. It also holds the soils together and increases on the water retention capacity within the soils.

6. **Human activities**

Limited human activities e.g quarrying, road construction and crop cultivation favour formation of zonal soils e.g laterites.

Such activities need to be minimized because they interfere with the operation of soil formation processes like weathering, leaching etc.

7. **Time:**

Zonal soils e.g laterites form after a considerable period of time. This is because the parent rock requires a long period of exposure to deep weathering, leaching, lateralization etc.

Questions

Account for the formation of zonal soils in East Africa.

Account for the formation of laterite soils in East Africa.

Clue:

In East Africa, the major type of zonal soil is laterite soil. (Both questions require same points/approach)

INTRAZONAL SOILS

These are soils whose formation is influenced by relief and parent rock other than other factors of soil formation e.g climate, vegetation etc.

Examples include bog/peat soils in marshy areas of East Africa such as the coastal regions; meadow soils in the river flood plains (silt and mud), saline soils in semi desert areas etc.

MAIN FEATURES/CHARACTERISTICS OF INTRAZONAL SOILS:

1. Relief and parent rock are the main controlling factors in the development (formation) of intrazonal soils.
2. They depend on a specific kind of parent rock such as limestone, on the presence of large amounts of salts and on the presence of large amount of water.
3. Intrazonal soils are divided into 3 groups i.e calcareous, hydromorphic, and halomorphic soils.
4. Calcareous soils develop from limestone parent rock. There are 2 types of calcareous soils i.e terra rossa and rendzina soils.

Terra rossa soils develop from limestone (calcareous rocks) under low rainfall e.g in semi-arid areas.

While rendzina soils develop from limestone rock but under humid conditions.

5. Hydromorphic soils are soils that develop in areas with excessive water or water logging e.g meadow, peat soils.
6. Halomorphic soils are soils in with high levels of soluble salts which render them saline. They develop from parent rocks that contain considerable quantities of soluble salts.

TYPES OF INTRAZONAL SOILS

1. Calcareous soils

These soils develop from limestone parent rocks (material).

There are 2 types of calcareous soils. These include:

a) **Terra rossa soils**: These are reddish residual soils which form in limestone depressions under semi-arid conditions or during prolonged summer in Mediterranean areas. Rainfall leads to carbonation weathering and the silicates are leached out of the soils to leave a residual deposit rich in iron hydroxide.

These soils occur extensively in Australian valleys.

b) **Rendzina soils**. These are dark coloured soils formed from limestone parent rocks under humid conditions.

Rainfall supports the growth of grasses and woodlands which decompose to form humus. This gives horizon A the dark or balck/dark-brown colour.

2. Hydromorphic soils

These are intrazonal soils with a constantly high-water content e.g gley and peat soils.

The types of hydromorphic intrazonal soils are:

a) Peat soils

These are soils which develop where drainage of water is blocked, precipitation is retained and decomposition of organic matter is slowed.

These soils occur in low-lying areas and valley which are poorly drained or water logged with almost no air and very few organisms.

Due to absence of oxygen and organisms, decomposition is limited hence peat simply accumulates in the area.

There are various forms of soil formed from peat. These include:

- **Fen peat soils: These** are soils that develop where the water contains much calcareous material derived from limestone or other alkaline salts. Such soils are black and spongy.
- **Meadow soils:** These are soils that develop a long plains where there is flooding and accompanied by deposition of silt and mud e.g in the swampy regions around the shores of Lake Kyoga and Victoria.
- **Bog peat soils**: These are soils that develop in moist areas such as high moorlands where humic acid developed by plant decays.
- **Dry peat soils**: These are soils that develop in moist sandy heathlands.

Examples of peat soils are found in swampy areas of East Africa.

Outside East Africa peat soils e.g bog peat are found in northern USA Canada, Russia etc.

b) Gley soil:

This type of hydromorphic soil occurs in saturated soils where pore spaces are filled with water hence excluding air.

The lack of oxygen leads to reduction (chemical weathering process) of iron compounds to form grey- blue coloured soils i.e gley soils.

3. Halomorphic soils

These are intrazonal soils with high levels of soluble salts which render them saline.

They include:

a) **Saline soils:** These are soils which have soluble salts in considerable quantities.

They mainly develop in areas with hot dry climates where in the absence of leaching, mineral salts are brought to the surface by capillary action e.g in areas of Marsabit, Kalahari, Namib and Semliki lowlands.

The parent rock or underground water contains a lot of carbonates, bicarbonates, sulphates e.g common salt. On reaching the surface the water evaporates, leaving a thick crust salts forming saline soils.

CONDITIONS /FACTORS FAVOURING THE FORMATION OF INTRAZONAL SOILS

1. **Nature of parent rock**

Limestone rocks rich in calcium carbonate are easily weathered down to form rendzina soils in humid tropical environments and terra rossa soils in hot and dry environments.

Permeable parent rocks e.g limestone are vulnerable to chemical weathering agents like carbonic acids which decompose the rocks to form calcareous soils.

2. **Climate**

In hot regions such as deserts and semi-deserts, saline soils are formed. This is because the hot temperatures induce high evaporation rates which draw mineral salts from the underlying layers to the surface where accumulation of salty soils takes place.

In humid tropical environments, the high rainfall accelerates chemical decomposition of limestone rocks to form soils rich in calcium carbonate e.g rendzina soils.

3. **Nature of relief/landscape:**

Low-lying areas e.g valleys and basins favour the formation of intrazonal soils. These regions accumulate a lot of water and deposition of organic matter takes place forming highly acidic soil e.g peat or bog soils.

4. **Drainage:**

Poorly drained areas with excess moisture favour the development of intrazonal soils.

Swampy and marshy areas have excess moisture that causes partial decomposition of organic matter to form acidic soils e.g peat or bog soils.

5. **River deposition**

Intrazonal soils are also formed on flood plains where meandering rivers deposit rock materials.

Water accumulates in these broad valleys called flood plains.

This allows further decomposition of organic matter to take place.

This is why soils in these valleys are thick and dark.

6. **Time**

For intrazonal soils to develop, the present rock should have interaction with soil formation processes and factors for considerably a short period of time. This is why they are regarded as immature soils with shallow soil profiles.

AZONAL SOILS

These are soils which are young and without a clear profile.

Their development is not complete because they have not been exposed to soil forming processes for long.

CHARACTERISTICS /MAIN FEATURES OF AZONAL SOILS

1. Azonal soils are young and therefore lack clear soil profile.
2. They are skeletal soils i.e with shallow profiles and show characteristics of their original parent rock materials which are weathered and resist change.
3. They are derived from unconsolidated materials e.g alluvium and volcanic ash which are weathered and resist change.
4. They are divided into 2 groups' i.e lithosols and regosols.
5. They include soils such as:

- Scree soils on mountain slopes
- Mudflat soils/marine clays
- Fluvial-glacial soils e.g till, out wash sands and gravel..
- Wind-blown soils especially sand sheets, dunes and loess.
- Lava or ash soils, cinder and pumice (volcanic soils)

Question

Distinguish between zonal and A zonal soils

Approach

- Define each separately
- Give characteristics of each separately
- Give types of each separately.

TYPES OF AZONAL SOILS

1. Volcanic soils

These consist of young deposits of lava, ash and cinder and pumice derived from recent volcanic activity.

These materials weather easily and are transported down to the lower slopes by rain-wash. These soils are found in Kisoro in South Western Uganda.

2. Alluvial soils:

These are soils derived from a mixture of sand, silt and clay transported and re-deposited along water courses by running water.

Flooding helps in the deposition of other materials making up alluvial soils. Such soils are found near the rivers responsible for their deposition e.g along rivers Tana, Athi, Nzoia all in Kenya and river Manafa, Mpanga and Rwizi in Uganda.

3. **Marine soils.**

These are soils derived from oceans and seas and deposited along low-lying coasts in the form of mudflats, sandbanks, etc by wave deposition e.g along Indian Ocean there is mudflat soil at Rufigi and Ruvuma river mouths.

4. **Glacial soils:**

These are soils derived from glacier and consist of deposits of till, out wash sands, gravel and clay. They are deposited by glacier and are found along slopes of mountains Kenya, Rwenzori and Kilimanjaro.

5. **Mountain soils:**

These consist of fragments of rock which disintegrate by physical weathering through exfoliation or frost action. They are also known as scree soils. Examples are on the upper slopes of mountain Kenya, Rwenzori, and Kilimanjaro etc.

6. Wind deposited soils.

These are soils formed as a result of wind erosion and deposition. These include sand sheets, dunes and loess and are common in arid and semi-arid regions for example Karamoja, Turkana etc.

FACTORS (CONDITIONS) THAT INFLUENCE THE FORMATION (OCCURRENCE) OF AZONAL SOILS

The factors favouring /influencing the formation of azonal soils include:

1. Weathering of parent rock leading to formation of screes on the mountain slopes. These soils usually show characteristics of their original parent rock material and resist change.
2. Tectonic deposit of lava through volcanic action (eruption) leads to formation of lava/ash soils, cinder and pumice soils.
3. Wave action (erosion and deposition) leads to formation of marine deposits.

The resultant soils include mudflat soils/marine clay soils.

4. Wind action (erosion and deposition) leads to formation of wind-blown soils like sand sheets, dunes and loess.
5. Glacial action (Erosion and deposition) results into formation of fluvio- glacial soils e.g tills, outwash sands, gravel and re-sorted clays deposited in glacial lakes.
6. River action (Erosion and deposition) leads to formation of alluvial soils along rivers e.g river Tana, Manafa etc.
7. Climate

- ❖ High amount of rainfall causes river floods that lead to deposition of alluvium in the lower course of the rivers.
- ❖ High rainfall also causes erosion on steep slopes and deposition of material in the lowlands leading to formation of alluvial soils.

- ❖ Temperature changes on mountain slopes influence physical weathering hence leading to formation of scree soils.

8. Relief: This mainly influences erosion.

On steep slopes for example erosion is rapid leading to formation of thin stony soils.

Gentle slopes and lowlands experience deposition hence formation of new soils which are deep.

9. Human activities e.g quarrying and mining lead to the breaking of parent rock into smaller particles and dumping of waste materials lead to formation of azonal soils.

- ➢ Deforestation, bush burning and overgrazing expose the parent rock to weathering processes and lead to formation of young soils e.g screes.
- ➢ These activities also expose the land to strong winds which pick soils, transport them and deposit them in a new area to form wind-blown soils for example loess, dunes etc.

10. Time: A zonal soils are immature soils and this mainly depends upon the short period of time entailed in their course of formation.

Question

Account for the formation of Azonal soils in East Africa.

MAJOR SOIL TYPES IN EAST AFRICA

Soils in East Africa can be divided into 4 major categories:

a) Volcanic soils

These are soils formed from volcanic materials

These soils may be coarse or fines textured and usually have high silt content.

They have dark- grey or black horizon A and are very porous with high organic content and high water storage capacity.

They are generally fertile soils and are found in the following areas:

A berdare and Mau escarpments, mountain Kenya region, Mountain Elgon region, Kigezi

highlands, mountain Kilimanjaro region and parts of Karamoja. These are largely zonal soils.

b) Laterite soils/Latosols:

These are soils that form from any type of rock which has been leached. They form under conditions of rapid weathering brought about by hot temperature and heavy rainfall.

Laterite soils/latosols are reddish-brown in colour and occur mostly in equatorial and savannah climate.

They consist of either iron or aluminum oxides which are formed as hard pans (duricrusts) or soft clays (bauxite).

In East Africa laterites are found on flat topped hills of Buganda and Busoga

CHARACTERISTICS OF LATERITES SOILS

1. They have limited humus content due to rapid action of organisms such as bacteria and maximum leaching.
2. They are highly porous /permeable
3. They vary in texture from clay to loamy sand.
4. They mainly occur in humid regions i.e hot and wet areas.
5. They are reddish-brown in colour
6. They form above the water table where constant alternate wetting and drying prevail.
7. Their A horizon contains much aluminum and iron oxides.
8. They form distinct /complete soil profiles.
9. They are very hard and will not soften when wetted.
10. They are highly deficient in plant nutrients.

Laterites soils/latosols are sometimes referred to as red/yellow earths.

FORMATION OF LATERITE SOILS/LATOSOLS

Laterite soils/latosols form due to excessive rainfall and hot temperatures which bring about decay and intense weathering of rocks. Such conditions exist in tropical countries e.g Kenya, Uganda, Tanzania, etc.

The silica present in the rocks is removed by water and transferred from Horizon A to Horizon B of the soil profile through the process of leaching and eluviation.

The iron and aluminum compounds which are insoluble remain concentrated behind in Horizon A giving it a reddish-brown color.

When temperatures rise the iron and aluminum compounds on the surface of the soil profile fuse together and harden to form a hard layer known as dursicrust or may remain in the presence of surface moisture (Water) to form soft clays known as bauxite e.g at Kajjansi clay works. The duricrust is thus laterite soil.

FACTORS /CONDITIONS INFLUENCING (FACILITATING) FORMATION OF LATERITE SOILS/LATOSOLS

1. **Relief:**

Latosols form under conditions of low and gentle relief. Such relief allows percolation of water and leaching of horizon A and deposition in horizon B, leaving behind insoluble iron and aluminum compounds to form latosols/lateritic soils.

2. **Nature of parent lock**

Laterites form in areas where rocks contain aluminum and iron as insoluble compounds and silica and magnesium as soluble compounds that are leached downwards.

3. **Drainage:**

Laterites form in well drained areas that allow leaching and chemical weathering.

4. **Climate**:

Hot temperatures are required for the increasing rate of chemical reactions

Hot temperatures also help in the drying and fusing particles into duricruts in horizon A.

Heavy rainfall on the other hand facilitates deep, chemical weathering and leaching of soluble materials leaving behind insoluble compounds of aluminum and iron oxides.

5. **Vegetation:**

Availability of vegetation (forests) facilitates formation of latosols. This is because vegetation helps hold the soil and allow weathering to occur leading to decaying of organic matter.

6. **Human activities:**

The formation of laterites is interrupted by human activities. Therefore, for proper formation of laterite soils, human activities e.g construction, quarrying etc must not prevail.

7. **Time:**

Laterites require a long geological period of time to allow the formation and accumulation of oxides e.g many laterites in East Africa were formed during the tertiary weathering period.

IMPORTANCE OF LATERITE SOILS/LATOSOLS

1. The sticky and plastic character makes latosols suitable for brick making and pottery e.g at Kajjansi clay works.
2. They support the growth of swampy vegetation that promotes art and craft industry
3. They favour growth of grass used for grazing
4. They aid in construction work e.g road surfacing and tarmacking.
5. They encourage the construction and engineering work due to a hard pan beneath giving them a good foundation for building.
6. The porous nature makes latosols infertile hence not suitable for agriculture.
7. They are used for tourism and study purposes.

Questions1

Account for the formation of lateritic soils/latosols

Approach

- Define latosols/lateritic soils
- Identify areas where they occur.
- Explain the formation
- Give characteristics
- Explain the factors/conditions **for the formation.**

Question 2

What is meant by the term lateritic soils?

Approach

- Define lateritic soils
- Explain the formation
- Give characteristics
- Question 3

Examine the factors influencing the formation of zonal soils in East Africa

Approach

- **Define zonal soils**
- **Give examples (major zonal are latosols).**
- **Give factors of soil formation but relating to formation of mature soils/zonal soils.**

c) CLAY SOILS

These soils are found in broad river valleys or plains that are water -logged.

They are usually grey in colour because of their periodic or permanent saturation by water.

These soils occur along Yala, Tana, and Turkana rivers in Kenya and along Katonga, Kagera and Rwizi rivers in Uganda.

d) SANDY SOILS

These soils have low organic content. They are often stony and rocky

They are best developed along the East African coast and Northern Eastern Kenya. They are moderately fertile.

Question:

With reference to specific examples account for the formation of the different soil types in East Africa

Approach

- Define soil
- Identify the different soil types and where they occur in East Africa.
- Explain the factors/conditions for soil formation linking them to different soil types e.g zonal, a zonal, intrazonal soils

N.B

- Soil types in East Africa fall under 3 major categories that's Zonal, Intrazonal and Azonal soils.
- Explain each category for example zonal soils where they occur and types of zonal soils like pedocals, pedalfers, loamy soils etc.
- Do the same for Azonal and intrazonal soils
- Then give the factors.

SOIL EROSION

Soil erosion refers to the removal or wearing a way of top soil by agents such as wind, running water, and glaciers and so on. The detached soil is then transported by the agents from one place to another.

CLASSIFICATION OF SOIL EROSION

Soil erosion can be classified into two broad categories that's

1. **Geological erosion**

This is the normal erosion which takes place under natural conditions without human interference/activity.

- It is a slow process in which the removal of surface soil may be matched by the formation of new soil.
- Geological erosion is mainly controlled by vegetation cover which slows down the speed of running water. It therefore occurs in areas with vegetation cover that's natural environments.
- Geological erosion is not dangerous to the soil. It is instead vital as it aids the process of soil formation. It occurs in forsted areas of Mabira, Bwindi, Budongo etc.

2. **Accelerated erosion**

This is the type of erosion that occurs when geological erosion is speeded up by man through his activities.

- In this case the rate of soil loss is greater than the rate of soil formation.
- This type of erosion is therefore very dangerous to the soil and it is the one that must be controlled.
- Accelerated erosion takes place in all environments that have been disturbed by man for example mountain slopes in districts of Bududa, Mbale, Manafa, Kigezi highlands in Kabale, Kisoro, Kotido, Moroto in North Eastern Uganda, Miombo woodlands of Tanzania etc.

Question

Distinguish between geological and accelerated erosion

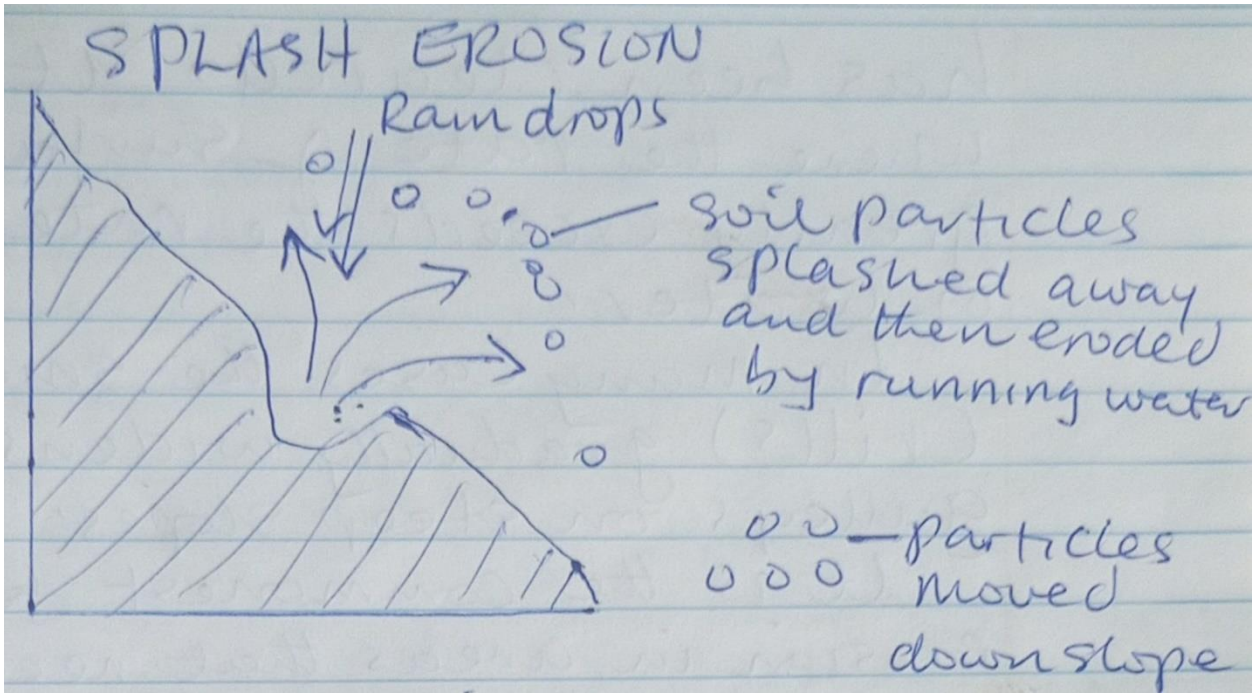
PROCESSES/FORMS /TYPES OF SOIL EROSION

The main agents of erosion are moving ice, wind and running water. These agents carry out soil erosion in a number of ways or processes which include.

1. **Splash erosion**

This is the type of erosion caused by the impact of rain drops. Heavy rain drops destabilize the soil as hey strike the ground.

The loose soil particles are scattered in various directions. The splashed materials/soils are later pulled away by gravity and fed into running water which washes them away.



This is common in arid and semi-arid areas.

2. **Sheet erosion**

This is a uniform removal of a thin layer of top soil by running water, wind, and glacier among others.

It involves slow movement of a thin layer of soil over a wide or extensive area.

Sheet erosion occurs on gently sloping land which is bare. It therefore occurs in areas where vegetation has been removed /destroyed.

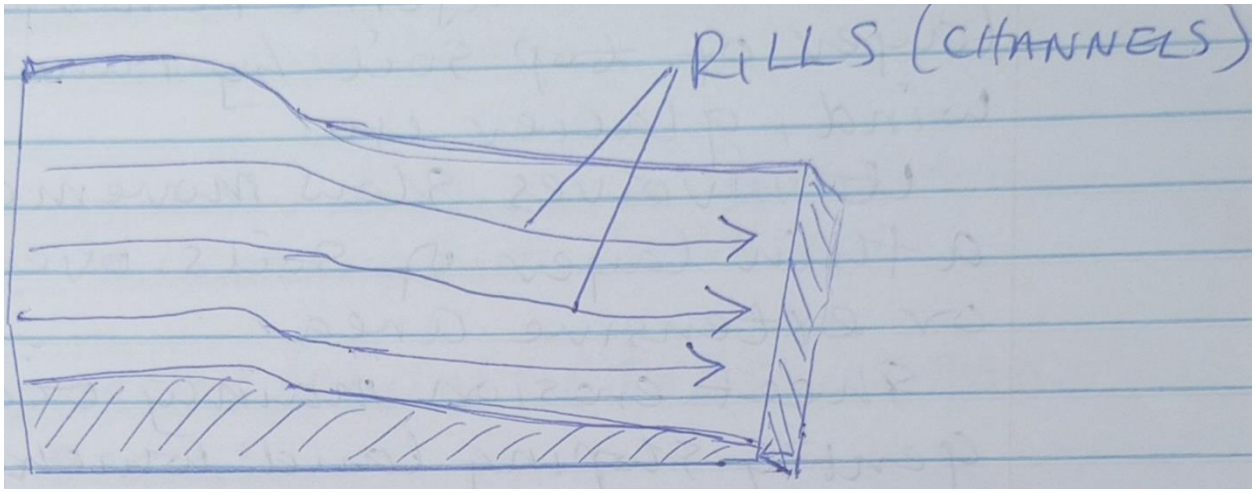
This process may not be immediately noticed but its effects can be noticed in form of soil particles deposited on the lower slopes e.g in Mbarara, Bushenyi, Kabale among others.

3. **Rill erosion**

This is the process of erosion in which numerous small channels are formed and these channels guide soil movement due to the influence of running water on the surface of the slope.

Rill erosion mainly occurs on gentle slopes where vegetation has been cleared. It occurs where the rate of surface run off greatly exceeds the rate of infiltration of water.

In many cases the small channels (rills) gradually widen to form gulleys on steep slopes.



It is the commonest form of soil erosion in areas that receive frequent rainfall e.g around the shores of Lake Victoria in Mukono, Wakiso among others.

4. **Gulley erosion**

This is the form of soil erosion where soil is washed away downslope through deep, wide channels or grooves.

It is common along steep or gentle slopes receiving very heavy rainfall.

Under gulley erosion, the deep channels are dug into the ground that's gullies result into creation of wasteland.

It occurs on steep slopes where vegetation has been removed for example Kondoa region of Tanzania, Baringo in Kenya, Kabarole, Kabale, Kisoro among others in Uganda.

5. **Wind erosion/deflation erosion**.

This is the removal of soil particles from one area to another by wind.

It is common on generally flat dry areas with little or no vegetation cover for example Chalbi desert of Kenya, Kotido and Moroto in Karamoja region and so on.

In this process small particles are blown away by wind and later deposited to form sand dunes.

FACTORS INFLUENCING SOIL EROSION/CAUSES OF SOIL EROSION IN EAST AFRICA

The causes of soil erosion are both physical and human in nature.

PHYSICAL CAUSES/FACTORS

1. **Climate**

- Rainfall: Heavy torrential rainfall in the hilly areas such as Kabale, Kapchorwa, Mbale, Bundibugyo leads to sheet, rill and gulley erosion.
- Strong winds experienced on bare ground especially in semi-arid areas erode the loose surface soil through deflation by wind for example in North Eastern Uganda, Northern Kenya, and central Tanzania and so on.

2. **Relief**

Steep slopes increase the rate of surface run-off increasing the rate of soil removal downslope by running water for example on the steep slopes of mountain Elgon in Bududa, Kapchorwa, Kigezi highlands in Kisoro, Kabale among others

Gentle slopes on the other hand encourage uniform removal of thin top layer in form of sheet erosion.

3. **Vegetation**

Areas covered by thick vegetation in form of forests for example Mabira, Budongo, have limited soil erosion while areas which lack vegetation for example Chalbi, desert in Kenya experience erosion caused by both wind and running water because their soils are directly exposed to agents of erosion.

4. **The nature of soils**

The young volcanic soils are mainly porous hence offer less resistance to wind and running water hence promoting soil erosion for example on the slopes of mountain Elgon in districts of Bududa, Manafa, Kapchorwa,and in Kigezi highlands.

Also areas dominated by sandy soils which are porous e.g in Kotido, Moroto, Turkana regions among others are prone to erosion.

5. **Biotic factors:**

For example harvest ants, and wild animals lead to loss of natural vegetation cover leaving soil bare and less resistant to erosion especially in the dry spell.

Also animals (wild animals) trample on the ground and carry large quantities of soil in their hooves hence accelerating erosion for example in game parks like Queen Elizabeth National Park.

Burrowing animals, locusts among others eat the vegetation and expose the soil to wind and running water to erode it.

6. **Crustal instabilities;**

For example earthquakes disorganize soil particles preparing them for massive flows through landslides, soil erosion for example in Bundibugyo and Kabarole.

HUMAN FACTORS/CAUSES

7. **Deforestation:**

Deforestation carried out by man especially on steep slopes through activities like agriculture, lumbering, road construction among others accelerates soil erosion by reducing the protective cover and encouraging runoff in form of sheet, gulley erosion for example in Kabale, Kisoro, Elgon slopes and so on.

8. **Overstocking/overgrazing**

Keeping animals beyond the carrying capacity of land leads to emergency of bare patches of land as a result of overgrazing. Such land is prone to erosion by wind and water leading to deflation,

sheet, rill erosion within the normal pastoral communities for example in Karamoja and Turkana regions.

9. Monoculture.

This practice of growing the same type of crop on the same piece of land for several years makes the soil exhausted and loose thus easily eroded away.

This is common in areas of plantation agriculture for example Kikuyu land in Kenya, North Eastern Tanzania, Central Uganda like in Buikwe, Mukono among others.

10. Bush burning

Bush burning especially by cultivators and pastoralists leads to destruction of natural vegetation cover exposing top soil to agents of erosion for example running water in Nakasongola, Moroto and Kotido among others.

11. Poor farming practices:

For instance up and down ploughing on hill slopes for example in Kikuyu, Kabale, Ntungamo, Kapchorwa accelerates erosion creating gullies and rills.

12. Mining and quarrying

Open cast mining exposes land to erosion because it involves removal of vegetation for example limestone mining in Tororo and quarrying in Buganda region.

13. Swamp reclamation:

Swamp reclamation for various activities like settlement, industrialization agriculture exposes the soil to agents of erosion for example running water causing erosion.

14. Construction of transport and communication routes.

This involves removal of vegetation and exposure of soil to agents of erosion for example wind, running water among others.

Road cutting are avenues of running water leading to gulley erosion for example Fort portal-Bundibugyo road, Kabale-Kisoro road and so on.

THE EFFECTS /IMPACTS OF SOIL EROSION IN EAST AFRICA

NEGATIVES

1. Soil erosion leads to loss of ferile top soil leaving behind poor and less productive soils. This results into low crop yields hence famine for example in Kotido, Moroto, central Tanzania, Kondoa region among others.
2. Soil erosion results into creation of bare ground with very little or no vegetation cover leading to desertification characterized by reducing amounts of rainfall for example in Kotido, Moroto, Turkana among others.
3. It leads to siltation of wetlands, valley dams, rivers and lakes. This affects quality of water for domestic and industrial use and also limits fishing activities for example river Manafa is silted by eroded soil from mountain Elgon slopes.

4. Silting of rivers and wetlands by soil erosion leads to flooding in lowland areas for example due to silting of river Manafa areas of Butaleja district have consequently suffered from severe flooding.
5. Water and wind erosion causes pollution of water sources and air hence increasing the costs of water purifications for example rivers originating from mountain Elgon for example river Manafa carry large quantities of sediments especially during rainy seasons due to soil erosion resulting into water and air pollution.
6. Soil erosion causes lowering of water table hence reduction of surface water leading to drying up of boreholes, wells, dams and swamps for example in Nakasongola, Pallisa, Kotido among others.
7. It creates gullies and wastelands which are a hinderance to communication as well as cultivation for example in Kigezi highlands (Kabale and Kisoro) Bududa Kindoa district of Tanzania among others.
8. It results into pasture shortage in pastoral areas for example Karamoja region in Kotido and Moroto, Turkana land in Kenya.

This is because soil erosion removes the fertile top soil and prevents the growth of vegetation cover.

9. Soil erosion causes landslides (mud flow) especially in hilly areas during periods of heavy rainfall leading to death of people and destruction of property for example in Bududa, Bulambuli on the slopes of mountain Elgon among others.
10. Soil erosion leads to displacement and consequently migration of people in the affected areas to other areas hence affecting economic and social life for example many people have migrated from Bududa, Bulambuli, Manafa among others to other areas.

POSITIVE EFFECTS

11. Soil erosion leads to removal of the top soil thus exposing the parent rock to agents of weathering leading to the formation of new soils therefore facilitates soil formation for example on slopes of Mountain Elgon and Rwenzori.
12. It leads to deposition of fertile soils in low-lying areas which favour the cultivation of certain crops along flood plains that's alluvium supporting rice, sugar cane, yam cultivation for example at Doho in Butalega.
13. Soil erosion creates unique and beautiful features like gullies, u- shaped valleys on highlands etc. which attract tourists hence earning foreign exchange for respective countries.
14. The materials deposited by soil erosion for example sand, mud, can be extracted for building and construction.
15. Soil erosion is important for research and study purposes leading to acquisition of skills and knowledge.
16. Soil erosion exposes minerals which are then easily mined by man for example some of the limestone rocks in Tororo are exposed by erosion near the surface.

QUESTIONS

Examine the causes and effects of soil erosion in East Africa

Approach

- ❖ Define soil erosion
- ❖ Bring out processes of soil erosion e.g gulley, rill, sheet etc
- ❖ Bring out causes, linking to process and giving examples of places of occerrence.
- ❖ Explain effects both positive and negative
- ❖ Conclude**.**

Question 2

"Man's activities more than anything else are responsible for the extensive soil erosion in East Africa" Discuss

Note

This is a 2 sided question

Approach

- ❖ Define soil erosion
- ❖ Bring out processes of soil erosion
- ❖ Bring out man's activities (with stand point). Man's activities include
- ❖ Deforestation
- ❖ Overgrazing
- ❖ Poor farming methods etc.
- ❖ Bring out other factors (however side)
- ❖ For example
- ❖ River flooding
- ❖ Nature of relief
- ❖ Mass wasting among others
- ❖ Conclude.

MEASURES TO CONTROL SOIL EROSION (SOIL CONSERAVTION MEASURES)

The measures being undertaken to control soil erosion in East Africa include:

1. **Strip cropping:**

This is the practice of planting alternate strips of grass and crops across a slope and at right angles to the natural direction of a slope. The grass strips intercept raindrops and runoff limiting the erosive activity along the slope for example in Kigezi highlands.

2. **Contour ploughing**

This refers to the ploughing of the slope along contour lines and planting the crops across the mountain in rows along the same elevation (height). This reduces the speed of water runoff downslope, limiting soil erosion for example in Kigezi highlands, Kenya highlands.

3. **Terracing:**

This refers to cutting of a series of wide flat steps along steep slopes on which crops are cultivated. Terracing slows down run off, giving water more time to infiltrate into the soil thus reducing its erosive power as done.

4. **Afforestation and re-afforestation**

Afforestation involves planting trees in areas where no forest cover existed before.

While re-afforestation involves re-planting of trees in areas where forests once existed but have been destroyed.

The trees act as windbreaks and protect the soil against the strong winds and running water.

Tree leaves intercept the impact of raindrops reducing on splash erosion

Tree roots bind the soil particles together making them more resistant to erosion for example on the slopes of mountain Elgon, Kilimanjaro etc.

5. **Crop rotation**

This involves alternating crops on the same piece of land from season to season. It overcomes the problem of monoculture for example alternating legumes with cereals results into conservation of soil nutrients that behind soil particles together making it difficult for agents of erosion to operate for example Northern Uganda.

6. **Mulching**

This involves covering of soil surfaces with materials such as dry grass, coffee husks ground nut husks among others. This minimizes moisture loss, improves soil water infiltration and when mulch is dry, decomposes and forms organic matter that binds soil particles together protecting it from erosion.

7. **Planting of cover crops**

Cover crops like potatoes etc have their leaves spread over the soil surface preventing the impact of raindrops, strong winds and surface runoff.

8. **Controlled grazing and paddocking**

This involves keeping a sizeable number of animals through establishment of ranches. This practice overcomes overstocking and overgrazing. Hence minimizes loss of vegetation hence controlling soil erosion. This is being done in Nyabushozi in Mbarara

9. **Mass education**

This involves the use of the media to sensitize people about the danger of soil erosion and the control measures to adopt for example farmers are educated on best farming practices.

10. **Agro-forestry**

This is the growing of trees and crops on the same farms. In most case trees are fruit –bearing trees.

These tress protect the soils from erosion

11. **Construction of ridges (tidal ridges)**

This involves the piling up of top soil to form a raised ground. This increases soil depth. Crops are then grown on the ridges (piled soils) for example on mountain slopes in Sukuma land.

12. **Construction of gabions**

These are structures with wires and filled with stones. It is commonly used where gulley erosion occurs and along river valleys. This controls soil erosion as it traps soil.

Questions: 1

Examine the steps being taken to control soil erosion in East Africa

Approach

- Define soil erosion
- Explain process of soil erosion
- Hint on the causes of soil erosion
- Explain the measures to control soil erosion with examples

Question: 2

Account for the high rate of soil deterioration in East Africa.

Approach

Define soil deterioration as decline in the productive value of the soil, leading to low crop yield

NB: Causes of soil deterioration (Causes of soil erosion)

MAJOR CONCEPTS IN THE STUDY OF SOILS (PROPERTIES OF SOIL)

1. **Soil texture**

Soil texture refers to the different sizes of soil particle.

The particles are measured in terms of diameter for example soils with texture of 0.2 - 0.2 mm are said to be fine sand while soils with texture above 2.0 millimeters are classified as very coarse soils for example gravel, and sand.

Soil texture determines the water transmission and retention capacity.

This means that if particles are big, the spaces between the particles will be larger hence water will be transmitted quickly to the lower layers. Such soils have low fertility.

Very fine –textured soils for example clay may be poorly drained, water logged and not well suited for agriculture.

2. **Soil structure**

Soil structure refers to how soil particles are held together into larger pieces or aggregates.

Soil particles are held together by tiny particles or organic and mineral substances in solution known as colloids.

Soil structure can be in various forms for example granular and crumb structure, prismatic structure, platy structure (flat-horizontal layers).

3. **Soil consistency**

This refers to the quality of stickness of wet soil, to the plasticity of moist soils and to the degree of coherence or hardness of dry soils.

Soil consistency is measured for wet soil/moist soil and dry soil samples.

To test the stickness of wet soils, hold wet soil between your thumb and fingers. Separate your thumb and finger and observe the extent to which soil sticks to your skin.

To test plasticity of soil, roll a small amount of soil into a rod shape. If the plasticity is high then the soil can be rolled into a long their rod (shape).

When soil is dry consistency is expressed by various levels of hardness ranging from loose to extremely hard.

4. **Soil porosity**

This refers to the total volume of space within a given volume of soil.The ratio is expressed in percentage. Water and air share the pore space in the soil. The smaller pores generally contain water and the large ones contain air.

5. **Soil PH**

This is the measure of acidity or alkalinity of the soil. Most soils have PH of between 4 and 8.

Soils with PH values about 8 have excess salts that is they are extremely alkaline while soils with PH value below 4 are very acidic.

Question

Distinguish between soil texture and soil structure

NATURAL VEGETATION

This refers to the naturally existing plant cover or the community of plant in an area. Or it is the natural plant life that florishes in a particular area on it's own without human interference.

There are a number of natural vegetation types in the world which include:

- Forests.
- Grasslands
- Desert/Semi desert vegetation

- ❖ Alpine Vegetation

FORESTS

The forest Vegetation is categorised into two ie tropical and temperate forests.

TROPICAL FORESTS

These forests located within the tropics and they are sub divided into four;

Equatorial rain forests that are found between latitudes 5 degrees North and South of the Equator and they are well developed in the;

-Amazon basin of South America

-Congo basin of Central Africa

-West African coast line (Nigeria, Ghana Ivory coast)

-Highlands of Malaysia, Burma, Indonesia etc

TROPICAL MONSOON RAIN FORESTS

These are located within the tropics between 5 degrees North and South of the Equator. They are well developed in areas like Burma, India, Thailand, Northern Vietnam, Philippines etc.

TROPICAL MONTANE FORESTS

These are confined on slopes of high mountains normally at a height of about 1800m above sea level. Common tree species include; camphor, cedar, podarcap etc.

In East Africa they are found on slopes of mountains like Rwenzori, Kenya, Kilimanjaro, Elgon etc.

MANGROVE FORESTS

These are also swamp vegetation because of growing in marshy areas. They grow along coastal margins between 5degrees North and South of the equator.

They are developed in the marshy river Deltas of the Amazon and the Niger in Nigeria.

TEMPERATE FORESTS

These are located in the temperate regions outside the tropics. They are grouped into two ie deciduous temperate forests and coniferous temperate forests. They grow between 30 degrees and 70 degrees North and South of the equator.

The deciduous temperate forests are further broken down into cool temperate and Mediterranean deciduous forests. The Mediterranean deciduous forests are mainly located on the Mediterranean shoreline of Algeria, Morrocco, Spain, Italy, France, Tunisia, South African coastline and Western coastline of Australia while the temperate deciduous forests are well developed in Central and Western Europe, Korea, China, Northern Japan, Chile, News land.

GRASSLANDS

These constitute the largest vegetation cover on the Earth and they are divided into two ie Tropical and Temperate grasslands.

Grasslands are known by different names in different parts of the world as shown in the table below;

AREA	NAME
North America	Prairies
South America	Pampas
South Africa	Highveld
Rest of Africa	Savannah
Australia	Downs

The Tropical grasslands are commonly referred to as Savannah grasslands and they located between 5 and 15 degrees North and South of the equator. They are well developed in India, Africa etc while the temperate grasslands are found in the interior of continents with temperate longitudes between 35 and 60 degrees North and South of the equator.

DESERT VEGETATION

This type of vegetation is divided into two namely;

- Tropical desert vegetation

- Cold desert vegetation

The Tropical desert vegetation is found in various parts of the world ie North America, Northern Mexico, Sahara region of Africa, Namibia, etc

The cold desert vegetation (Tundra) is mainly found in Asia, Northern parts of Europe in the Scandinavian countries, Canada and Green land.

ALPINE VEGETATION

This is divided into two ie

- Tropical Alpine vegetation
- Temperate Alpine vegetation

Tropical Alpine vegetation is sometimes referred to as montane vegetation it consists of clearly demarcated vegetation zones from the foot hills to the summit while the temperate Alpine vegetation is found in the temperate region like the Alps in Western Europe, Northern California, and British Columbia in the Appalachian Mountains etc.

NATURAL VEGETATION IN EASTAFRICA

E**a**st Africa has a variety of natural vegetation types that include the following;

- **Tropical/Equatorial rain forests**
- **Savannah vegetation**
- **Semi- desert vegetation**
- **Montane vegetation**
- **Swamp vegetation**

TROPICAL/EQUATORIAL RAIN FORESTS

These are generally found around areas of Lake Victoria region, parts of western Uganda like Bwindi, Maramagambo along the East African coast (mangrove) and within islands of Lake Victoria.

CHARACTERISTICS OF TROPICAL RAIN FORESTS

- ✓ The trees are tall ranging from 65 meters and above due to competition for light.
- ✓ The trees are closely packed together that is to say, forests tend to be dense
- ✓ There is limited under growth because of the dense cover limiting sun light penetration to the ground.
- ✓ The tree are associated with climbing plants like epiphytes
- ✓ The trees tend to be ever green because of the too much water in the environment.
- ✓ The trees have broad leaves to give out excessive water and trapping enough sunlight for photothyensesis.
- ✓ The main trees have buttress roots for extra support to the huge thanks and great height.
- ✓ They form levels of layers with the top most layer's leaves joined together to form canopies
- ✓ The main trees have thick backs
- ✓ The trees are normally found in mixed stands that is to say, there is a variety of species in one locality like Mahogany, Ebony
- ✓ The trees comprise of hard wood timber like ebony, mahogany etc.

CONDITIONS FAVOURING THE GROWTH OF TROPICAL RAIN FORESTS

- Climate: heavy rain fall amounts of over 1500mm for instance in the Lake Victoria shore region for provision of sufficient water which enables growth of tall trees with broad leaves.
- Hot temperatures of up to 27 degrees that help trees grow luxuriantly.
- Tropical forests grow well in areas of relatively low altitude of less than 2000m above sea level that is associated with hit temperatures required for tree growth.
- Well drained fertile soils that enable the growth of thick forests.
- High humidity that reduces the rate of evaporation.
- Low population density which enables the growth of tree with minimal human interference.

- Relief: gently sloping landscape where the soils seem to be thick to give support to the tall and huge trees.

Revision questions

Account for the growth of tropical rain forests

Account for the characteristics of tropical rain forests.

Examine the factors favouring the growth of tropical rain forests.

MONTANE VEGETATION

This type of Vegetation is basically found in mountainous parts of East Africa such as mt Kenya, mt Kilimanjaro, Mt Elgon, Mt Ruwenzori etc. In such areas there is altitudinal zonation of Vegetation type it an increase in altitude calls for a change in Vegetation.

The vegetation alignment is as follows;

- **Savannah vegetation**
- **Tropical/Equatorial rain forests**
- **Temperate forests**
- **Bamboo forests**
- **Heath and Moorland**

Illustration

SAVANNAH VEGETATION ZONE

This found at an altitude of 1000-1500m above sea level. It is characterised by;

Grasses such as elephant grass, umbrella shaped trees, shirt scattered trees, trees shade off their leaves during dry seasons etc on the Lee Ward side and Savannah vegetation turns into thicket with stunted trees, scrubs and bushes.

EQUATORIAL FOREST ZONE

These come after Savannah vegetation at an altitude of between 1500 and 2500metres above sea level. The trees in this zone tend to;

- ✓ Grow broad leaves.
- ✓ Trees have buttress roots.
- ✓ There is little or no under growth.
- ✓ A variety of tree species exit.
- ✓ There is existence of climbing plants such as lianas etc.

CONDITIONS FAVOURING THE GROWTH OF TROPICAL MONTANE FORESTS

- Heavy and well distributed rain fall of over 1500mm that favour the growth of luxuriant trees.
- Hot temperatures of over 20 degrees to support the growth of rain forests.
- Relatively low altitude between 1500-2500 metres above sea level attracts the growth of rain forests since at such an altitude temperatures are conducive for the growth of thick forests.
- Mountainous areas are drained by numerous rivers eg Mt Ruwenzori has rivers like Mpanga, Mobuku etc which continuously provide water and silt materials necessary for tree growth.
- Presence of fertile soils that support the growth of rain forests.

TEMPERATE FOREST ZONE

This is found at an altitude of between 2500 and 3500metres above sea level. The vegetation in this zone is characterised by;

- ✓ Tall trees with small straight branches.
- ✓ Ever green trees like spruce, pine.
- ✓ Trees are cone shaped as an adoption to snow conditions.
- ✓ The trees have small pointed leaves.
- ✓ The trees have a short gestation period.
- ✓ There are soft wood tree species e.g. pine, podarcap etc

Trees appear in pure stands.

CONDITIONS FAVOURING THE GROWTH OF TEMPERATE FORESTS

- Moderate rain fall above 1000 mm encourages the growth of trees like podacarp.
- Cold temperatures ranging between 10-19 degrees encourage the growth of temperate tree species like pine.
- Presence of thin and relatively fertile soils which encourage the growth of small straight tree species like camphor, pod carp etc.
- Steep slopes along mountains which encourage the growth of Temperate tree species.

BAMBOO FOREST ZONE

This vegetation zones lies at an altitude of between 3500 and 4000 metres above sea level.

CHARACTERISTICS OF BAMBOO FORESTS

- ✓ They tend to grow in a single layer.
- ✓ They grow in pure stands.
- ✓ They tend to be yellow (have yellowish stems)
- ✓ Their stems are segmented
- ✓ They have prop roots for aeration
- ✓ They have small rough / tough leaves
- ✓ There is limited or hardly any under growth

CONDITIONS FAVOURING THE GROWTH OF BAMBOO FORESTS

- Plentiful supply of precipitation in form of fog, Dee, mist, little rain fall which supports the growth of bamboo trees.
- Cold temperatures of about 10 degrees encourage the growth of cold resistant tree species like bamboo.
- Thin skeletal soils to support the growth of light trees such as bamboo.
- Low humidity resulting into the growth of the bamboo trees.
- Limited human activities such as crop cultivation, settlement hence the existence of bamboo trees.

HEATH AND MOORLAND ZONE

This vegetation zone is found at an altitude of between 4000 -4500 metres above sea level. It is characterised by;

- ✓ A variety of beautiful plants in form of grasses and flowers e.g. mosses, ferns, berries etc.
- ✓ Grasses such as sedges have small leaves.
- ✓ Plants grow short.

CONDITIONS FAVOURING THE GROWTH OF HEATH AND MOORLAND VEGETATION

- Presence of thin soils that support the growth of short plants such as sedges.
- Occurrence of little rain fall coupled with other forms of precipitation like snow, fog, mist which encourage the growth of heath and moorland vegetation.
- Moorland vegetation is adapted to cold conditions of 10 degrees c hence it's growth.
- High altitude of over 4000 meters above sea level that is associated with low temperatures that are essential for the growth of the Heath and moorland vegetation.
- Limited human activities such as cultivation, settlement etc thus creating room for Heath and moorland vegetation.
- Government policy of forest conservation in mountainous areas resulting into survival of the Heath and moorland vegetation.

SWAMPY VEGETAION

These are also regarded as wetland vegetation because it grows in seasonally or permanently water logged areas. It is categorised into three ie

- ❖ **Mangrove swamps**
- ❖ **Papyrus swamps**
- ❖ **Grassland swamps**

MANGROOVE FORESTS (salt water forests)

These are salt water swampy forests and are found along the East African coast.

CHARACTERISTICS OF MANGROOVE FORESTS

- ✓ They adapted to grow in salt water
- ✓ They have short stems
- ✓ They have broad leaves which tend to be ever green
- ✓ The forests tend to be dense. (trees are compacted together)
- ✓ They normally put down new Roots from the branches.
- ✓ They are found in the muddy flats along the East African coast eg Mombasa, Kilwa.

The papyrus swamp vegetation is found along rivers such as Katonga, Kafu, Semiliki,etc and on lake shores of lake Victoria, lake Kyoga, lake Wamala.

SAVANNAH VEGETATION

This is found in most of the parts of East Africa. However it may be of various types like savannah woodland, savannah grass land and Savannah scrub/dry Savannah.

11.Savannah woodland

This type covers several parts of East Africa including Miombo woodland in western Tanzania and in parts of northern Uganda and parts of western Uganda.

CHARACTERISTICS OF SAVANNAH WOODLAND

- ✓ More or less continuous cover of trees
- ✓ They may be clusters or dense tracks
- ✓ They raise up to an average height of 15m (8 – 20 m)
- ✓ There are a variety of tree species which may include Acacia, and other drought resistant species.
- ✓ There is dense under growth of grass as well as the creepers or the bushes and this is because there is enough light near them.
- ✓ They are mainly deciduous in nature that is to say they tend to shade off their leaves during the dry season to minimize water loss.
- ✓ Several trees tend to be short with small leaves and thick backs.
- ✓ They tend to be with flat a top that is to say they form umbrella shapes eg Acacia.

- ✓ Most trees tend to be fire resistant that is to say they can easily revive after being burnt.

12. savannah grassland /wooded grassland

This is found in the fringes of Savannah wood land and covers much of central Uganda and the East African coastal region as well as parts of northern Tanzania. Savannah grass land is dominant in areas that experience rainfall between 500-1000mm. it is also found much in central and northern Uganda, Serengeti Plain of northern Tanzania as well as coastal regions.

CHARACTERISTICS OF SAVANNAH GRASS LAND / WOODED GRASS LAND

- ✓ There are short scattered trees / dotted trees.
- ✓ The grasses range from natural pasture to elephant grass
- ✓ Grass vary in height from very short to tall grass that may grow as high as 5m that is to say they range between 3-5m
- ✓ The grass tends to dry up during the dry season, change to brown color and during the wet season they become green again. This means that there is alternating period of green and brown color grasses.
- ✓ The trees are drought resistant.

Dry savannah /scrub vegetation

This type of vegetation is a characteristic of semi arid areas of East Africa for example northern and Eastern Kenya, central and parts of northern Tanzania, northern Eastern Uganda, parts of western Uganda and the Masai steppe and Nyeri desert in Kenya.

CHARACTERISTICS

- ✓ Very short grass with thickets in between.
- ✓ Tress are deciduous in nature (shade off leaves)
- ✓ Existence of thorny bushes and scrubs.
- ✓ Trees are short and drought resistant
- ✓ Trees tend to be fire resistant
- ✓ Trees grow small leaves to reduce transpiration.
- ✓ During dry seasons, the grasses dry up and turn brown only to regenerate during rainy season.

CONDITIONS FAVOURING THE GROWTH OF SAVANNAH VEGETATION

- Climate; Moderate rain fall amounts ranging between 750-1500mm per annum and hot temperatures over 25 degrees lead to increased evaporation and scarcity of water enabling trees to grow thick barks to reduce water loss.
- Low and un reliable rain fall of about 350-500mm favour the growth of drought resistant trees like Baobab and Acacia and hot temperatures of over 26 degrees for the growth of trees with small twisted leaves to reduce loss of moisture through rapid transpiration and evaporation.
- Soils, fairly fertile soils with low water retention capacity to enable the growth of drought resistant plants such as Cactus, Acacia, Baobab.
- Sandy soils with limited water retention capacity result into poor plant growth.
- Relief; low lying lands are associated with trees intermixed with scrubs for instance Otze and Kei in Northern Uganda.
- Human activities such as def forestation, over grazing, bush burning etc lead to reduction of forests into wood lands, wood lands into grasslands and finally into scrub.

SEMI-DESERT VEGETATION

This is a type of that occurs in semi-arid areas of East Africa such as central Tanzania, Ankole-Masaka dry corridor, lake Albert flats, North-Eastern Kenya, Turkana region of Kenya and North-Eastern Uganda in the districts of Kotido, kaboong etc.

Semi-desert vegetation is characterised by;

- ✓ Drought resistant plants ie xerophytes such as Cactus due to limited water.
- ✓ The trees are umbrella shaped to preserve soil moisture around the root zone.
- ✓ The plants store water in their tissues during the short rainy season for use during the long drought season.
- ✓ Most plants are thorny such as Cactus having small waxy leaves to reduce transpiration as well as to deter animals which may feed on them.

✓ Plants have very long tap roots to enable them tap underground water and also keep them firm in the ground against strong winds.
✓ The grasses are short and turf.
✓ The trees are short and scattered to avoid competition for water and sun light.
✓ The seeds of semi-arid plants lie dormant during the dry season until the rains come.
✓ The grasses turn yellow/brown during the drought season and turn green during the short rainy season.
✓ Semi-desert plants like cactus have swollen stems that contract later as moisture is slowly lost through transpiration.

CONDITIONS FAVOURING THE GROWTH OF SEMI-DESERT VEGETATION

- **Li**ttle and un reliable rain fall usually below 350 mm results into the growth of short trees with tinny leaves.
- Low humidity usually below 30% which leads to growth of semi-desert vegetation characterised by short grasses and tap rooted trees.
- Low lands/plain lands associated with hot temperatures resulting into growth of drought resistant plants.
- Low altitude of below 500m above sea level is associated with hot temperatures hence semi-desert vegetation.
- The hot temperatures of 30 degrees plus enable the growth of semi-arid plants.
- Existence of thin and shallow infertile soils in areas like Turkana land, Masai land and Kalamoja favour the growth of stunted trees and short grasses.
- Over grazing by wild animals especially in national Parks like Kidepo valley, Queen Elizabeth etc where the existing vegetation is eaten up leading to it's modification into semi-desert plants.
- Fairly drained soils favour the growth short stunted trees.
- Government policy of gazzeting some areas as national parks e.g. Serengeti, Kidepo valley etc leads to existence of semi-desert vegetation.

LAND USE ACTIVITIES ASSOCIATED WITH NATURAL VEGETATION IN EASTAFRICA.

A number of economic activities and uses can be associated with natural vegetation and these include

- Pastoralist / grazing, it involves the keeping of live stock such as goats, sheep etc in the Savannah grass lands and semi-desert pastures due to presence of pastures.
- Equatorial rain forests exist in areas where there is abundant and we'll distributed rain fall coupled with hot temperatures such conditions are conducive for the growth of crops such as coffee, cocoa, banana, tea etc which require similar conditions.
- Varied vegetation types such as equatorial rain forests, semi-desert vegetation provide potential sites for film shooting.
- Extensive Savannah grass lands serve as habitants for wild animals like Antelopes, water bucks, giraffes etc that attract hunting activities.
- Lumbering especially in regions of tropical rain forests and savannah woodland to provide timber and fuel wood that is to say charcoal and fire wood.
- Tourism / wild life conservation, this has been practiced in national parks, game reserves and game centuries for example the Tsavo national park, Kidepo national park, the Selous reserves in southern Tanzania.
- Bee keeping, this is very common in savannah wood lands and the steppe savannah region in northern parts of masindi, very common in Nakasongola, Masai steppe region and the Miombo wood land I Tanzania.
- Charcoal burning to produce fuel for use or sell to the urban dwellers. It has led to the depletion of trees, pollution of the environment and desertification.

NATURAL VEGETATION AS A BASIS FOR LAND USE PLANNING

- The distribution of natural vegetation provides a basis for land use planning in that the existence of rain forests can be a basis for gazzeting of forest reserves, game reserves and natural parks e.g. the Ruwenzori, Elgon national parks were established in the rain forest region. Crops which require high rainfall and temperatures can be grown e.g. sugarcane at Jinja were grown just near the Mabira forests. Some parts of the forest have been cut down to plant sugarcane, tea, coffee, cocoa, etc. Furthermore lumbering can be undertaken to provide wood.

- Savannah vegetation has a potential for a variety of economic activities. Grass can enhance rearing of animals , therefore modern ranches like Kisozi , Ankole-Masaka ranching scheme ,Njeru stock firm ,Maruzi, Batiima (Kabale firm) have been put in place due to presence of inexhaustible pastures. The seasonal rainfall within the Savannah vegetation zone has influenced the growing of seasonal crops like millet, Groundnuts, beans, maize, in Busoga, Buganda and commercial crops like tobacco cotton are grown especially in the North.

- The extensive savannah grasslands and the wild game within are generally attractive to tourists. Therefore the land use planers have used this as a basis to gazette some area as game parks and reserves e.g. the Queen Elizabeth, L.Mburo, Murchison falls National parks were located in Savannah grass lands to promote tourism.

- Desert and semi desert vegetation is used as another potential to develop the film and tourism industry. The rare plant species like Acacia and a variety of wild animals and landforms that exists there have led to demarcation of some areas as game parks e.g. the Kidepo valley national park within Kotido district. Drought resistant and quick maturing plants like millet and sorghum have been introduced in these areas of human consumption.

- The montane vegetation indicates the existence of cool climate and this has been a basis for land use planners to introduce the growing of crops like wheat, barley and pyrethrum on highland areas of mt. Elgon and Ruwenzori where temperatures are low.

- The existence of natural soft wood trees in the temperate forests has been used as a basis to plant softwood trees /forests along the Kabale, Bwindi and near Maramagambo forest in South Western Uganda. These provide soft wood to manufacturers of matchsticks, soft boards, papers, etc.

- The existence of health and moorland has been used as a basis to encourage Alpine grazing, exotic breeds which require cool temperatures as kept on the highlands of Kabale and besides the vegetation is rare and is reserved as a tourist asset.

- Swamp vegetation has provided a basis for the establishment and location of the arts and craft industry. The brick and tiles manufacturing industry was

located at Kajjansi due to presence of clay from the swamps around. This is because from swamps there is clay which can be used for pottery arts and brick making. The papyrus grass is also a potential to process biogas energy to supplement on the Hydro-electricity.

- Fishing is also recommended in areas with swamp vegetation e.g. along R. Mayanja, Katonga another shores, Katonga, another shores of L.Victoria and Kyoga. The most common type of fish caught is the mudfish and Tilapia.
- However, the distribution of natural vegetation does not always provide a basis for land use planning. There is no clear and accurate demarcation between the vegetation type and in some cases different vegetation types co-exist in one area. This makes it difficult for planners to identify a specific vegetation type on which to base their policy planning and besides, today natural vegetation has greatly been modified by man and is no longer truly natural therefore depending on it for planning maybe fruitless.
- Secondly , the distribution of natural vegetation occurs naturally without man's interference ,wherever land use planers comes in ,it implies the involvement of man and therefore other factors have to be considered when planning land use projects . These factors include economic viability, population, minerals, climate, soils, physical landscape, accessibility and others e.g. the presence of rainforests with a good timber not entirely responsible for lumbering which takes place there. There should be market for timber technology to exploit the timber, accessibility to the forest and capital to pay the workers before lumbering can take place.
- In conclusion, the distribution of natural vegetation may be the first consideration for land use planning, but it's not independent of other factors. It may also have its limitation and therefore other factors are considered as well.

FACTORS AFFECTING THE NATURAL VEGETATION (DISTRIBUTION)

- Climate, it is an important factor affecting the distribution of natural vegetation and more so elements of rainfall and temperature as well as sunshine all of which affect the plant growth .

- Heavy rainfall of 1800mm is responsible for the growth of tropical rain forests as well as other luxuriant types. Such forests may include Mabira forest in Buikwe district, Budongo forest in Masindi district, and Bugoma forest in Hoima district.
- Moderate rainfall ranging between 700 – 1000mm is responsible for the growth of savannah more especially savannah grass land and wood land. This is very common in Miombo wood lands in Tanzania, savannah grass land in Serengeti, wooded savannah in the plains of Kasese.
- In addition, climatic conditions characterized by cold temperatures is responsible for montane forests that is to say Alpine / temperate forests, heath and moorland as its common in mountains like Elgon, Ruwenzori, and highland areas like Kigezi highland.
- It should be noted that the cool temperatures are responsible for the Bamboo forests as it is the case with Ecuya forests towards kisoro.
- Limited rainfall less than 500mm is responsible for the steppe, scrubs and thickets especially in arid areas like Ankole Masaska corridle, Karamoja.
- The strength and duration of sunlight affects the growth of plants and that's why places with sufficient sunlight tend to have equatorial / luxuriant forests that include equatorial forests.
- While places of limited sunlight, plant growth is retarded resulting into stunted scattered tress as it is the case on the lee ward sides of some mountains.
- Soils, soil types are responsible for the different vegetation types. The soil mineral compositions, neutrality, depth, and PH determine the type that is likely to exist in a given area. Soils that are very fertile, deep and mature encourage the growth of luxuriant forests as it is the case with Mabira forest in Buikwe district, Budongo forest in Masindi district, Bugoma forest in Hoima district, and Maramagambo in Kanungu district.
- Areas that have fairly fertile soils favor the existence of the woodland and grass land and this is evidenced in some parts of Kasese, Serengeti plains, Ankole Masaka corridle among others.
- Soils that tend to be skeletal and immature tend to support stunted trees such as thickets and scrubs as it is seen in Miombo woodland in Tanzania.
- Drainage, the nature of drainage influences the vegetation type. Well drained areas will lead to the growth of tropical rain forests, wooded land and grass land.
- Poorly drained soils or water logged soils are responsible for the growth of swamp forests or mangrove forests.
- Moist or water logged conditions are responsible for the growth of Hydrophytes (water loving plants).

- On the hand, places of dry conditions or where soils have limited moisture give rise to the growth of Xerophytes (dry resistant plant species).
- Relief / Topography, it has influenced the distribution of vegetation in that it affects rainfall, temperatures as well as the nature of the soils which also in turn affect the mature of vegetation in a given area, that is to say, the steep slopes there is soil erosion and rain water drains away soil particles very fast as run off leading to development of scattered or short trees (stunted trees) because of the thin skeletal soils.
- On the other hand, on the gentle slopes the soils are well drained and the infiltration of water is sufficiently enough to accelerate the formation of fertile soils boosting the growth of tropical forests.
- On the lee ward side there is less rainfall because of the descending winds resulting into semi desert types like wood land, grassland, thickets, and scrubs.
- The PH condition, the alkalinity or acidity of an area also determines the growth of plant types. Areas that have a lot of salt (saline conditions) are responsible for the growth of mangrove forests at the coast.
- Biotic factor, this involves the role of animals and man. Animals such as buffalos, elephants have to some extent affected the nature of the vegetation cover. These animals are so destructive to the extent that they uproot trees leading to change of the existing natural vegetation that is to say wood land changed to grass land.
- On the other hand, mans activities like deforestation for settlement, agriculture among others, there is over grazing, over stocking all have led to the depletion of some types of vegetation meaning that mans activities also account for the distribution of vegetation in East Africa.
- Altitude, it has determined the natural vegetation zone especially in Highland areas of East Africa. Changes in altitude or height above the sea level have been responsible for the changes in vegetation types. In highland areas like Elgon, Kenya, Kilimanjaro, Kigezi among others, differences in height have created differences in the environmental conditions.

Conditions like rainfall, temperature, soil types all of which have affected the different vegetation types that is to say

A height above 1000m there is savannah wood land and grass land. This is because of the hot temperatures and limited rainfall as evidenced in Miombo wood land, Serengeti grass land.

At a height of about 2000m above the sea level, there are the tropical rain forests. This is because of the heavy rainfall about 1500mm as well as the moderate

temperatures of $25^0 – 27^0$c, in addition, soils are fairly deep and fertile to support the trees.

In the region about 3000m above the sea level, there are the montane or temperate forests and this is because of the reduced temperatures. The cold temperatures favor the growth of coniferous species and in such areas the trees are somehow luxuriant but as one goes higher, the trees become shorter or fairly stunted.

At a height above 4000m above the sea level, bamboo forests exist and this is because of the cold temperatures and reduced rainfall.

Above the bamboo that is to say above 4000m above the sea level, there exists mountain Heath and moorland. This type is characterized by very cold temperature types. The plants are mainly adapted to cold conditions and they tend to disappear as one goes higher.

MEDITERRANEAN VEGETATION

(a)Describe the characteristics of the Mediterranean type of vegetation

(b) Account for the growth of Mediterranean type of vegetation in Africa.

Mediterranean type of vegetation occurs on the west coast of continents in the mid-latitudes for example central California and central Chile; and low lands for example around the Mediterranean Sea basin and the Western Cape Province of South Africa.

Characteristics of Mediterranean vegetation;

- ✓ Mediterranean forests have broad leaves and evergreen trees such as oak, eucalyptus, cork oak, red wood due to wetter winters and withstand the hot and dry summer conditions.
- ✓ Much of the woody vegetation in Mediterranean has small shiny waxy leaves to reduce water loss by transpiration during the hot dry summer months.
- ✓ Trees and other Plants such as grape vine have long tap root system which enables the plants draw water deep underground during the hot dry summer season.
- ✓ Some plants have large fleshy bulbous roots to store water for use in hot and dry summer.
- ✓ Some trees such as cork oak have thick rough barks to store water during summer

✓ There are sweet-smelling herbs and shrubs such as lavender, rose Mary, thymes, thyme and oleander.
✓ Some trees are cone shaped especially pine.
✓ Some trees are short and flat topped for example cork oak.
✓ In some places there is mixed type of vegetation with deciduous trees, coniferous, tough grass and dwarf trees.
✓ Some trees have compact weedy stems, bushy scrubs with dense thickets
✓ Short grass is common especially in the Mediterranean basin, grass land of California, central valley e.t.c

Factors or conditions favoring the growth of Mediterranean vegetation

Climate-that is;
▪They have hot and dry summers of about 250 C.
▪Cool and moist rainy winters of about 120 C
▪Moderate annual rain fall of about 500mm -750mm and the onshore westerly winds blow in the winter bring cyclonic rain.
Soil-that is;
▪The ashy Mediterranean residual soils support Mediterranean vegetation
▪Limestone soils are porous and drain water easily hence deep root system of plants.
▪In the drier parts, soils are poor hence support Mediterranean scrub-like vegetation compos

DETERIORATION OF NATURAL VEGETATION IN EASTAFRICA

- **T**here is a rapid decline in natural vegetation coverage in East Africa due the following reasons:
- Rapid population increase which has caused population pressure on land due to the ever increasing demand for land for crop cultivation, settlement, industrial establishments, etc hence clearing away of the vegetation.
- Increasing demand for Bio mass energy resources e.g. fuel wood, charcoal has also lead to a decline of land under natural vegetation cover.
- Poor farming methods such as bush burning, over grazing have led to the rapid disappearance and clearance natural vegetation eg Savannah, semi-desert vegetation.

- Natural calamities like landslide which occur in mountainous areas of East Africa e.g. Bududa on the Mt Elgon slopes have led to destruction of natural vegetation.
- In addition soil erosion has led to destruction of vegetation especially in areas where the run-off causes rapid removal of vegetation e.g. Kigezi in Uganda, Baringo in Kenya and Kondoa in Tanzania.
- Burning of vegetation to provide room for growth of new pastures for nomadic pastoralism, dairying and ranching e.g. in Karamoja, kaabong, moroto, Masai land ,Turkana land etc
- Clearing of bamboo forests and grasslands and swamp vegetation e.g. papyrus for purposes of getting raw materials used in making art and craft products like papyrus mats has also resulted into destruction of natural vegetation.
- Lumbering in tropical rain forests to obtain timber for both domestic and commercial purpose.
- Swamp reclamation for agriculture, dairying, settlement, industrial development
- Mining and quarrying in Mwadui (Diamond), Tororo (limestone and vermiculite), Kilembe (Cobalt), Bamburi have resulted into clearing away of the existing natural vegetation hence it's destruction.